TEORICA 1

n.1

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1: INTRODUCTION

These notes were taken during prof. Röntsch' lessons in 2026 Spring semester. The main language is english, with some italian comments, except for the 2.nd chapter which is mainly written in italian.

Reference legenda:
M. S.C → means reference to MAGGIORE Book, chapter S.C
P&S S.C → means reference to Peskin & Schroeder book, chapter S.C

The course presents first an introduction on Lorentz and Poincaré' group, which are the symmetry group of a free particle in Minkoski space.

Then classical field theory is introduced and Noether theorem. Then we will go through each type of field (ie "particle") as FREE FIELD:
- SPIN ZERO FIELDS → scalar fields ϕ
- SPIN ONE FIELDS → Boson fields A_μ of photon } BOSONS
- SPIN 1/2 FIELDS → fermions ψ } FERMIONS

For each field, it will be presented first the classical theory and Lagrangian then the canonical quantization (→ promoting field into operator)

I feel the need to make a little clarification now that may help understandings = The WORLD IS DIVIDED into 2 types of particles = bosons (integer spin) and FERMIONS (half integer spin). These two types behave differently = fermions with fermi-Dirac statistics and bosons with Bose-Einstein statistics. this difference REFLECTS in the way we QUANTIZE our field = for fermions we will use ANTICOMMUTATOR, for BOSONS the COMMUTATOR. the reason of this difference is that we WANT fermions quantum fields to follow fermi statistics (antisimmetric) and BOSONS to follow Bose statistics (symmetric). So nothing strange = we simply want our QUANTIZED bosons and fermions fields to still be BOSONS and fermions.

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After having discussed Free fields we will study what happens when fields interact. (interaction between fields reflects into an additional term (the "interaction" term) in the Lagrangian describing our system.

When studying an interaction our aim is to compute the probability of the interaction to occur = this probability is quantified into the S-matrix and the cross-section. S-matrix is simply the overlap amplitude between the initial state and the final state after the interaction. and gives a measure of the probability to go from such initial state into such specific final state. The cross section is a sort of mean value (that will be discussed more rigorously after). To compute such scattering/interaction amplitude a powerful tool are FEYMAN DIAGRAMS, which will be presented.

However, when computing such amplitudes, we will discover that some of them DIVERGE: our rules to compute the amplitude apparently give infinite amplitude in some cases! So we are making a mistake ... our theory IS WRONG. What is wrong? The Lagrangian. RENORMALIZATION is the process with which we can CORRECT our theory (= Lagrangian) to give (with some rules of before) NON Divergent amplitude. We will discuss it at the end of the course.

Additionally, few lectures were spent to discuss ANOTHER POSSIBLE PROCEDURE of quantization = for all the course we, either promoty a classical field into a quantum field, used the CANONICAL quantization process. However, there is another way to quantize = the PATH INTEGRAL. A chapter will be dedicated to discuss briefly the path integral quantization of fields.

Apologizing in advance for the mistakes that will surely be present, I hope these notes will prove helpful in the study of this complex yet fascinating subject.
HAPPY STUDYING!

Domitilla A.

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2 Maggiore 2.2 - 2.7

LORENTZ & POINCARÉ SYMMETRIES

2.1 LORENTZ GROUP

Def = Lorentz group $O(3,1)$

It is the group of linear coordinate transformations applied on a 4-vector in Minkowski space

$$x'^{\mu} = \Lambda^{\mu}_{\ \nu} x^{\nu}$$

where $\mu = 0,1,2,3$ Minkowski

such that the Minkowski square of x^{μ} is unchanged:

$$x_{\mu}x^{\mu} \overset{!}{=} x'_{\mu}x'^{\mu}$$

Dove $x_{\mu}x^{\mu} = g_{\mu\nu}x^{\nu}x^{\mu} = (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2 =: X \cdot X$

Dove $g_{\mu\nu} = \text{diag}(1,-1,-1,-1)$ tensore metrico di Minkowski

Intervalli come $X \cdot X$ rimangono costanti quando passiamo da un sist di rif ad un altro, la distanza è Lorentz invariante.

$O(3,1)$ GRUPPO ORTOGONALE

In generale una trasf di coord $(x_1 \dots x_n, y_1 \dots y_m)$ che lascia invariata il prodotto $\sum x_i^2 - \sum y_i^2$ è detta GRUPPO ORTOGONALE $O(m,n)$

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EQ CARATTERIZZANTE

Come si traduce la richiesta $x_\mu x^\mu \overset{!}{=} x'_\mu x'^\mu$ sulla Λ?

$$x'^\mu x'_\mu = g_{\mu\nu} x'^\mu x'^\nu = g_{\mu\nu} (\Lambda^\mu{}_\rho x^\rho)(\Lambda^\nu{}_\sigma x^\sigma)$$

$$\text{!!}$$

$$x^\mu x_\mu = g_{\rho\sigma} x^\rho x^\sigma$$

allora

$$g_{\mu\nu} \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma \overset{!}{=} g_{\rho\sigma} \qquad (2,1)$$

↓ notazione matriciale ↓

$$\Lambda^T g \Lambda \overset{!}{=} g \qquad (2,1)$$

O(3,1) È UN GRUPPO

In questo gruppo ammette:

-) Il prodotto di due trasf di Lorentz è ancora una trasf di Lorentz
  $\Lambda \in O(3,1) \quad \Delta \in O(3,1) \quad \Lambda \cdot \Delta = L \in O(3,1)$

-) ∃ Identità $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu \in O(3,1)$

-) ∃ Inverso ∀ trasf e l'inverso è una trasf di Lorentz $(\Lambda^{-1})^\mu{}_\nu \in O(3,1)$

INVERSA

$$g_{\rho\sigma} = g_{\mu\nu} \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma = \Lambda_{\nu\rho} \Lambda^\nu{}_\sigma$$

applico $g^{\rho\gamma}$ a entrambi i lati:

$$g^{\rho\gamma} g_{\rho\sigma} = g^{\rho\gamma} \Lambda_{\nu\rho} \Lambda^\nu{}_\sigma$$

$$\delta^\gamma{}_\sigma = \Lambda_\nu{}^\gamma \Lambda^\nu{}_\sigma$$

$$\mathbb{1} = \Lambda^{-1} \Lambda$$

$$\Lambda^\rho{}_\nu \xrightarrow{T} \Lambda_\nu{}^\rho = (\Lambda^T)^\rho{}_\nu$$

$$\Lambda_\nu{}^\rho = (\Lambda^{-2})^\rho{}_\nu \qquad (2,2)$$

↓

L'inversa è la trasposta! Come in un normale gruppo ortogonale!

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TRASPOSIZIONE → cambio dx con sx $\Lambda^{\rho}{}_{\nu} \rightarrow \Lambda_{\nu}{}^{\rho}$

ricordo che Λ^{-1} è ancora una trasf di Lorentz posso riscrivere l'eq 2.1 anche per $g^{\mu\nu}$ inverso di $g_{\mu\nu}$

2.1 con Λ^{-1} al posto di Λ ↓

$$g_{\mu\nu} (\Lambda^{-1})^{\mu}_{\ \rho} (\Lambda^{-1})^{\nu}_{\ \sigma} \overset{!}{=} g_{\rho\sigma}$$

$$g_{\mu\nu} \Lambda_{\rho}{}^{\mu} \Lambda_{\sigma}{}^{\nu} = g_{\rho\sigma}$$ ↘ invito alto e basso con sommatoria su tutto

$$g^{\mu\nu} \Lambda^{\rho}_{\ \mu} \Lambda^{\sigma}_{\ \nu} \overset{!}{=} g^{\rho\sigma} \qquad (2.3)$$

PROPRIE VS IMPROPRIE

Dallo $g = \Lambda^T g \Lambda \rightarrow (\det \Lambda)^2 = 1 \rightarrow \det \Lambda = \pm 1$

$$\det \Lambda = \pm 1$$

$\det \Lambda = +1$

ROTAZIONI PROPRIE

Subgroup called $SO(3,1)$

$\det \Lambda = -1$

ROTAZIONI IMPROPRIE

An improper tx can always be written as a product of a proper Lorentz tx and a DISCRETE number (numero finito) di tx that is an INVERSION di un un numero DISPARI di coordinate $x^0 x^1 x^2 x^3$

ES Perché $P = (t, x y z) \mapsto (t, -x-y-z)$ inverte 3 coordinate

Time reverse $T = (t, \vec{x}) \mapsto (-t, \vec{x})$ inverte 1 coordinata

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ORTHOCHRONOUS VS NON-ORTHOCHRONOUS

In eq 2.1 we consider $\rho = 6 = 0$ first coordinate, we have

$$g_{00} = g_{\mu\nu} \Lambda^{\mu}_{\ 0} \Lambda^{\nu}_{\ 0}$$

$\downarrow g_{00} = 1$ $\quad$ $\downarrow$ sum over μ and ν from 0 to 3

$$1 = (\Lambda^{0}_{\ 0})^2 - (\Lambda^{i}_{\ 0})^2 \qquad \text{con } i = 1,2,3$$

that means that $(\Lambda^{0}_{\ 0})^2 \geq 1$

$\Lambda^{0}_{\ 0} \geq 1$

ORTHOCHRONOUS (+)

$\Lambda^{0}_{\ 0} \leq -1$

NON ORTHOCHRONOUS (−)

La trasf infinitesimale connessa con l'identità è ortocrona

$\Lambda = \mathbb{1} + \omega \quad \omega \ll 1$

is orthochronous

Una trasf non ortocrona può sempre essere scritta come prodotto di una trasf ortocrona con una inversione di un numero PARI di coordinate (es = inverto xy, xz, tz ecc ecc)

Noi tratteremo solo il gruppo di Lorentz connesso con l'identità cioè quello delle trasf PROPRIE E ORTOCRONE

$$SO(3,1)^{+}$$

In generale $O(3,1)$ è diviso in 4 parti disconnesse fra loro:

a) Proprie e ortocrone P, O → parte connessa con l'identità
b) Proprie, non ortocrone P, NO
c) Improprie, ortocrone NP, O
d) Improprie, non ortocrone NP, NO

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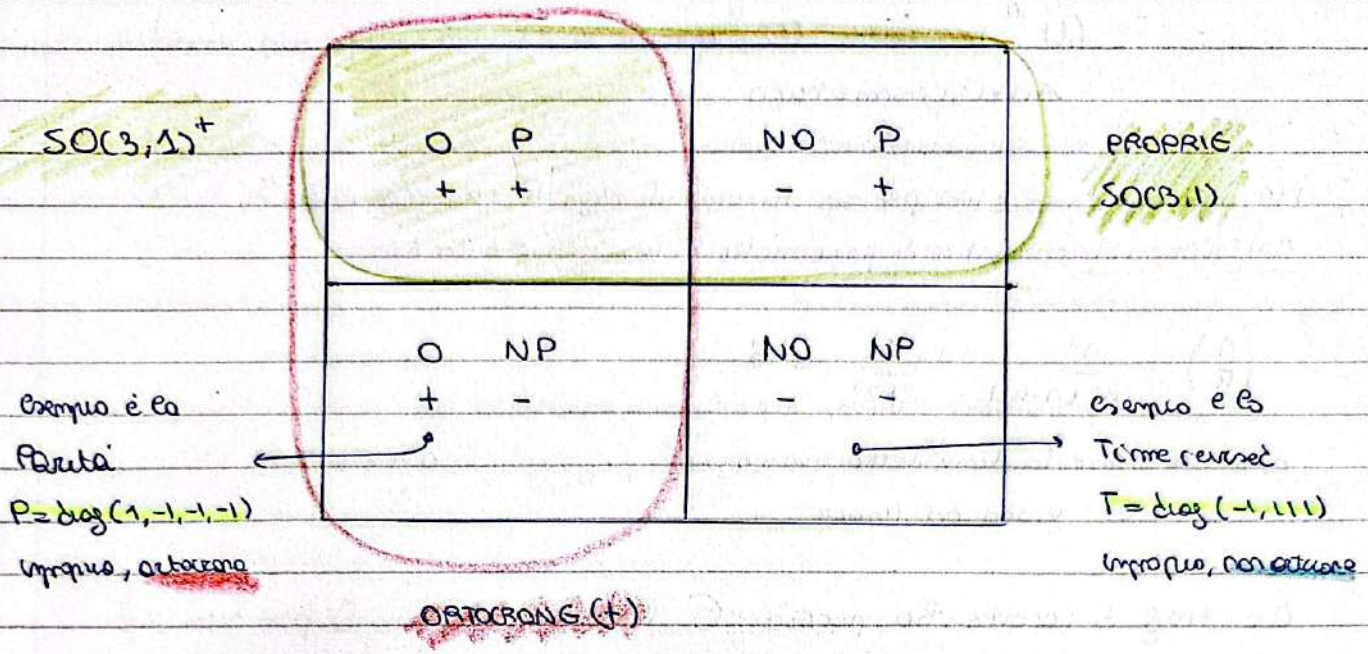

$SO(3,1)^+$ è la parte connessa con l'identità: e può venir approssimata intorno all'identità dalla sua algebra di Lie. Vedremo infatti che è un gruppo di Lie (cioè parametri continui e differenziabili)

BOOSTS AND ROTATIONS → LORENTZ $SO(3,1)^+$

Una Λ di Lorentz dipende da 6 parametri, 3 angoli (rotazioni) e 3 boosts. La condizione 2.1 mi fissa dei vincoli che fa sì che che la Λ non sia del tutto generale ma dipenda da 6 parametri. Vediamolo facendo i conti sulla Λ infinitesima ma sulla generale

infinitesimo $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu$

$(2.1) \rightarrow g_{\mu\nu}\Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma \overset{!}{=} g_{\rho\sigma}$ ↷ infinitesimo

$$g_{\mu\nu}(\delta^\mu{}_\rho + \omega^\mu{}_\rho)(\delta^\nu{}_\sigma + \omega^\nu{}_\sigma) \overset{!}{=} g_{\rho\sigma}$$

se faccio un po' di conti che skippo si arriva a dimostrare che

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$$\omega^{\mu}{}_{\nu} \overset{!}{=} -\omega^{\nu}{}_{\mu}$$

Antisimmetrico

Noi sappiamo che per un tensore di rango $k=2$ (come $\omega^{\mu}{}_{\nu}$) antisimmetrico, il n° di parametri liberi è dato da:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{4!}{2!\,2!} = 6$$

dove $n=4$ è la dim dello spazio su cui agisce (0123)
$k=2$ grado del tensore

Una trasf di Lorentz ha perciò 6 PARAMETRI. Si può dim che tali parametri corrispondono a

- 3 ROTATIONS on (x,y) (yz) (xz) planes forming a SUBGROUP OF $SO(3,1)$ called $SO(3)$, the subgroup of rotations in the spatial coordinates
  → parametrized by 3 angles θ_x θ_y θ_z $\theta_i \in [0, 2\pi]$

- 3 BOOSTS leaving (t^2-x^2) (t^2-y^2) (t^2-z^2) invariant, they are NOT A SUBGROUP and the parameter of the boost is the Velocity (which is not limited)
  → parametrized by 3 velocity $\vec{v} \in (-\infty, +\infty)$
  (o equivalently by 3 rapidities $\vec{\eta} \in (-\infty, +\infty)$)

FOCUS = 1 BOOSTS

Nel gruppo di Lorentz troviamo le 3 rotazioni $SO(3)$ sulle coordinate spaziali e poi 3 boosts. I boosts non sono altro che le trasformazioni che in fisica abbiamo sempre chiamato come trasf di Lorentz, tipicamente considerando un sist S' in movimento con vel v lungo x rispetto a S

$$\Lambda(v) = \begin{bmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$ boost lungo (t^2-x^2)

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Il boost più classico a tutti è proprio:

$$\begin{cases} t' = \gamma(t + vx) \\ x' = \gamma(x + vt) \\ y' = y \\ z' = z \end{cases}$$

[NB = $c = 1$ perciò $\beta = v$ per noi perciò $ct = t$]

[NB = rispetto a come definiti gli scorsi, ho considerato $v \to -v$]

I BOOST possono anche essere considerati come ROTAZIONI IPERBOLICHE dove le trasf di prima si possono riscrivere così:

$$\begin{cases} t' = \cosh(\eta)\, t + \sinh(\eta)\, x \\ x' = \sinh(\eta)\, t + \cosh(\eta)\, x \\ y' = y \\ z' = z \end{cases}$$

dove $v = \tanh(\eta)$ e $-\infty < \eta < +\infty$ con η nuovo parametro che chiamo η = RAPIDITY

RAPPRESENTAZIONI IN MINKOWSKI

ROTAZIONI $\theta \in [0, 2\pi)$

$$\begin{array}{c} \\ t \\ x \\ y \\ z \end{array} \begin{array}{c} \begin{array}{cccc} t & x & y & z \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos\theta & -\sin\theta \\ 0 & 0 & \sin\theta & \cos\theta \end{bmatrix} \end{array} \text{su } x$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & 0 & \sin\theta \\ 0 & 0 & 1 & 0 \\ 0 & -\sin\theta & 0 & \cos\theta \end{bmatrix} \text{su } y$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{su } z$$

BOOSTS $\eta \in (-\infty, +\infty)$

$$\begin{array}{c} \\ t \\ x \\ y \\ z \end{array} \begin{array}{c} \begin{array}{cccc} t & x & y & z \end{array} \\ \begin{bmatrix} -\cosh\eta & -\sinh\eta & 0 & 0 \\ -\sinh\eta & \cosh\eta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array} \text{su } x$$

$$\text{su } y \begin{bmatrix} \cosh\eta & 0 & -\sinh\eta & 0 \\ 0 & 1 & 0 & 0 \\ -\sinh\eta & 0 & \cosh\eta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- se voglio $-v$ invece che $+v$

$$\text{su } z \begin{bmatrix} \cosh\eta & 0 & 0 & -\sinh\eta \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\sinh\eta & 0 & 0 & \cosh\eta \end{bmatrix}$$

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2.2 LORENTZ ALGEBRA

Il gruppo di Lorentz è un gruppo di Lie perché le sue trasformazioni variano in modo continuo e differenziabile rispetto ai PARAMETRI $\vec{\theta}$ e $\vec{v}$

Un gruppo di Lie possiede un'ALGEBRA che approssima il gruppo nell'intorno dell'identità, per trasf infinitesime. Dall'algebra (generatori) è poi possibile risalire a tutto il gruppo cioè trasf FINITE semplicemente esponenziando i generatori.

Ricordo inoltre che un gruppo è un concetto astratto e che può avere diverse RAPPRESENTAZIONI = la matrice Λ è solo una delle tante rappresentazioni di un elemento di SO(3,1)$^\uparrow$, è la rappr nello spazio 4D di Minkowski che agisce tra 4 vettori.

RIPASSO - GROUP THEORY & LIE ALGEBRA

Representation of a group

A representation R of a group G is an operation that associates a: LINEAR OPERATOR (Matrix) $D_R(g)$ (acting on a linear space) to EVERY ELEMENT $g \in G$ of the group

$$R : g \longmapsto D_R(g)$$

group element — matrix

And R is such that the group structure is preserved, hence:

$$D_R(g_1) \cdot D_R(g_2) \doteq D_R(g_1 \circ g_2)$$

$\cdot$: prodotto tra matrici; $\circ$: prodotto del gruppo

A seconda dello spazio su cui agisco, ho rappresentazioni matriciali diverse

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Lie group

A group G is a Lie group if its components $g \in G$ are continuous and differentiable functions of parameters.

Lie Algebra

is a vector space $\mathbb{V}$ with a bilinear operation $[\,,\,]$

$$[\,,\,] : \mathbb{V} \times \mathbb{V} \to \mathbb{V}$$

such that

- $[V, W] = -[W, V]$ $\quad \forall\, V, U, W \in \mathbb{V}$
- $[V, [W, U]] + [W, [U, V]] + [U, [V, W]] = 0$

È lo spazio tangente centrato all'identità di G, ed è dotato da una operazione $[\,,\,]$ indotta dalla struttura di gruppo. Gli elementi di $\mathbb{V}$ dell'algebra spazio vettoriale sono i generatori

Per parametri infinitesimi $\theta_a \to 0$ una trasf di G centrata all'identità si scrive

$$D_R(g) = \mathbb{1} + i\,\theta_a\, T_R^a + O(\theta^2)$$

INFINITESIMAL TRANSFORMATION

dove T_R^a sono i GENERATORI nella rappresentazione R

I generatori formano l'algebra secondo le regole di commutazione

$$[T_R^a, T_R^b] = i\, f^{ab}{}_c\, T_R^c$$

dove abc indici sul n. di generatori - R indica per ricordare della rapp R.

COSTANTI DI STRUTTURA

Le costanti di struttura IDENTIFICANO l'algebra di Lie e NON dipendono dalla specifica rapp R scelta, sono una caratteristica del gruppo G in senso generale.

Il n° di generatori è una proprietà del gruppo = in qualsiasi Rappresentazione R ho sempre lo stesso n° di generatori.

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NB = il numero di generatori T_R^a è lo stesso in ogni coppr R (nel caso di Lorentz $a = 1,2,3,4,5,6$ sono 6 generatori) MA la FORMA del generatore DIPENDE dalla rappresentazione scelta

$[T_R^a]^i_j$ → i: indice della matrice che rappresenta T^a nella coppr R, scorre sulla dim della rappresentazione

j: indice del generatore, scorre sul n° di generatori = uguale x R

ES = in Minkoski le matrici sono 4Dimensioni

ES = SO(3,1) a scorre su 6 generatori

TRASFORMAZIONE FINITA

For a FINITE Transformation of a Lie group in the representation R we can write:

$$D_R(g) = e^{i\theta_a T_R^a}$$

NB = θ_a doesn't have to be small! They can be BIG! they just have to be SMOOTH!

COMPACT REPRESENTATION IN MINKOSKI

Per elelgebra di Lorentz SO(3,1)$^+$ IMPORTO di scrivere Λ come:

$$\Lambda \doteq e^{-i\frac{1}{2}\omega_{\mu\nu}J^{\mu\nu}}$$

FINITE LORENTZ TRANSF

→ general expression for every representation ←

dove $\mu,\nu = 0,1,2,3$ scorrono nei 4 indici di Minkoski

dove $\omega_{\mu\nu}J^{\mu\nu}$ contiene sia la somma sui generatori che l'informazione di cosa succede nello specifico mondo 4D di minkoski.

Since $\omega_{\mu\nu} = -\omega_{\nu\mu}$, also $J^{\mu\nu} = -J^{\nu\mu}$ are antysimetric (cause the contraction sum + antysim would lead to zero)

. Factor $\frac{1}{2}$ because of each generator counted twice in the sum

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To find the explicit expression of $\omega_{\mu\nu} J^{\mu\nu}$ in Lorentz, let's analyze how Λ acts upon 4-vector and scalars in Minkoski space.

SCALAR

ϕ è uno scalare di Lorentz sse

$$\phi' = \Lambda \phi \overset{!}{=} \phi$$ cioè se $\delta\phi = 0$ la variazione è nulla

4vector controvariant

$V'^{\mu} = \Lambda^{\mu}{}_{\nu} V^{\nu}$ finito $\qquad V'^{\mu} = V^{\mu} + \delta V^{\mu}$ infinitesimo

TRASF infinitesima:

$$\Lambda^{\mu}{}_{\nu} = \delta^{\mu}{}_{\nu} + \omega^{\mu}{}_{\nu} \longrightarrow V'^{\mu} = V^{\mu} + \omega^{\mu}{}_{\nu} V^{\nu}$$

$$\Lambda^{\mu}{}_{\nu} \overset{!}{=} \delta^{\mu}{}_{\nu} - \frac{i}{2} \omega_{\alpha\beta} (J^{\alpha\beta})^{\mu}{}_{\nu} \longrightarrow V'^{\mu} \overset{!}{=} V^{\mu} - \frac{i}{2} \omega_{\alpha\beta} (J^{\alpha\beta})^{\mu}{}_{\nu} V^{\nu}$$

dalla teo con l'algebra di Lie — elemento di matrice $\mu\nu$

$\Lambda = \mathbb{1} - \frac{i}{2} \omega_{\alpha\beta} J^{\alpha\beta}$

poiché

$$\delta V^{\mu} = \omega^{\mu}{}_{\nu} V^{\nu} \quad \rightarrow \text{dalla teo di prima} \qquad (2.6)$$

$$\delta V^{\mu} \overset{!}{=} -\frac{i}{2} \omega_{\alpha\beta} (J^{\alpha\beta})^{\mu}{}_{\nu} V^{\nu} \quad \rightarrow \text{dalla teo di Lie} \qquad (2.7)$$

se svolgo i conti uguagliando 2.6 = 2.8 trovo che in Lorentz

$$(J^{\alpha\beta})^{\mu}{}_{\nu} = i(g^{\alpha\mu} \delta^{\beta}{}_{\nu} - g^{\beta\mu} \delta^{\alpha}{}_{\nu}) \qquad 2.8$$

elemento di matrice di $J^{\alpha\beta}$ nel mondo di Minkoski dove $\mu, \nu = 0, 1, 2, 3$

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4vector COVARIANT

$$V'_\mu = \Lambda_\mu{}^\nu V_\nu \quad \text{finito} \qquad V'_\mu = V_\mu + i\,\delta V_\mu \quad \text{infint}$$

dove $\Lambda_\mu{}^\nu = g_{\mu\rho}\Lambda^\rho{}_\sigma g^{\sigma\nu} = (\Lambda^{-1})^\nu{}_\mu$

EXPRESSION IN TERMS OF ROTATIONS AND BOOST GENERATORS

La scrittura $\omega_{\mu\nu} J^{\mu\nu}$ è scomoda. Per l'interpretazione fisica è utile spacchettarla nei contributi di rotazione e boosts

Def $J^i = \frac{1}{2}\varepsilon^{ijk} J^{jk}$ generatore rotazione SO(3)
$i,j,k = 1,2,3$ spaziali

Def $K^i = J^{i0}$ generatori boosts
i spaziale

Def $\theta^i = \frac{1}{2}\varepsilon^{ijk}\omega^{jk}$ parametro angolo di SO(3) $\theta \in [0, 2\pi)$

Def $\eta^i = \omega^{i0}$ parametro rapidità di Boost $\eta \in (-\infty, \infty)$

allora se faccio i conti

$$\frac{1}{2}\omega_{\alpha\beta} J^{\alpha\beta} = \vec{\theta}\cdot\vec{J} - \vec{\eta}\cdot\vec{K}$$

$\vec{J}$: vettore $\vec{J} = (J^1 J^2 J^3)$, vettore $\vec{\theta} = (\theta^1, \theta^2, \theta^3)$

$\vec{K}$: vettore $\vec{K} = (K^1 K^2 K^3)$, $\vec{\eta} = (\eta^1 \eta^2 \eta^3)$

per cui

$$\Lambda = e^{-i\vec{\theta}\cdot\vec{J} + i\vec{\eta}\cdot\vec{K}} \qquad (2.12)$$

$-i\vec{\theta}\cdot\vec{J}$: termine di rotazione SO(3)

$i\vec{\eta}\cdot\vec{K}$: termine di Boosts

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LORENTZ ALGEBRA

It is possible to demonstrate that the generators J^i and K^i obey the following commutators rules:

$$[J^i, J^j] = i\,\varepsilon^{ijk} J^k \qquad (2.9)$$

$$[J^i, K^j] = i\,\varepsilon^{ijk} K^k \qquad (2.10)$$

$$[K^i, K^j] = -i\,\varepsilon^{ijk} J^k \qquad (2.11)$$

2.9 → Rotazioni sono chiuse e formano sottogruppo = due rotazioni sono una rotazione

2.10 → $\vec{K}$ trasforma come vettore sotto rotazioni.
J e K, rotaz e boost non commutano

2.11 → Boost non sono chiusi, non sono un sottogruppo = due boost sono una rotazione

Tuttavia è possibile separare l'algebra di SO(3,1) in due sottoalgebre indipendenti e chiuse tra esse. Def nuovi generatori

$$\vec{J}_\pm = \frac{1}{2}\left(\vec{J} \pm i\vec{K}\right)$$

allora

$$[J_+^i, J_+^j] = i\,\varepsilon^{ijk} J_+^k$$
$$[J_-^i, J_-^j] = i\,\varepsilon^{ijk} J_-^k$$
$$[J_+^i, J_-^j] = 0 \qquad \forall ij$$

L'algebra di SO(3,1) si è separata nella somma di due algebre di SU(2)

$$\mathcal{SO}(3,1) = \mathcal{SU}(2) \oplus \mathcal{SU}(2)$$

DIM 4 DIM 2 DIM 2

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FOCUS $SO(3,1) \in SL(2,\mathbb{C})$

Come $SU(2)$ è il doppio ricoprimento di $SO(3)$, cioè hanno algebra uguale ma lo spazio finito non lo è

$$\begin{array}{ccc} SU(2) & \xrightarrow{\text{omeomorf.}} & SO(3) \\ \uparrow \exp & & \uparrow \exp \\ \mathfrak{su}(2) & \xrightarrow[\text{isomorfismo}]{} & \mathfrak{so}(3) \end{array} \qquad \begin{array}{l} \frac{SU(2)}{\mathbb{Z}_2} = SO(3) \quad \text{spazi} \\ \mathfrak{su}(2) = \mathfrak{so}(3) \quad \text{algebre} \end{array}$$

Così $SO(3,1)$ è doppiamente ricoperto da $SL(2,\mathbb{C})$. Poi ovviamente l'algebra di $SL(2,\mathbb{C})$ è isomorfa a quella di $SU(2)$, per cui a livello di algebre $so(3,1)$ è isomorfo a due $su(2) \oplus su(2)$

$$\begin{array}{ccc} SL(2,\mathbb{C}) & \xrightarrow{\text{omeom}} & SO^+(3,1) \\ \uparrow \exp & & \uparrow \exp \\ \mathfrak{sl}(2\mathbb{C}) & \xrightarrow[\cong]{} & \mathfrak{so}(3,1) \end{array} \qquad \begin{array}{l} \frac{SL(2,\mathbb{C})}{\mathbb{Z}_2} = SO(3,1)^+ \quad \text{gruppi} \\ \mathfrak{su}(2) \oplus \mathfrak{su}(2) = \mathfrak{so}(3,1) \quad \text{algebre} \end{array}$$

Dove ricordo che $SU(2) \subset SL(2\mathbb{C})$ è un sottogruppo, e può essere visto come il sottogruppo delle sole rotazioni spaziali in $SL(2\mathbb{C})$.

$$SL(2\mathbb{C}) = SU(2) \times \mathbb{R}^3$$

(rotazioni; Boosts)

Inoltre $SL(2\mathbb{C})$ NON È UNITARIO, quindi le trasf di Lorentz non sono mutue centrate

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2.3 $SO(3,1)^+ \times SO(3,1)^+$ ON TENSORS

Possiamo anche trasformare tensori $T^{\mu\nu}$ con trasf di Lorentz. In questo caso ogni trasf che agisce sui tensori appartiene al gruppo PRODOTTO DIRETTO $SO(3,1) \times SO(3,1)$

$$T'^{\mu\nu} = \underbrace{\Lambda^{\mu}_{\rho}\Lambda^{\nu}_{\sigma}}\, T^{\rho\sigma}$$

matrice di $SO(3,1) \times SO(3,1)$ in Minkowski

In realtà, se Λ^{μ}_{ρ} e Λ^{ν}_{σ} sono associate alla stessa TRASFORMAZIONE g si può vedere che $\Lambda^{\mu}_{\rho}\Lambda^{\nu}_{\sigma}$ è ancora dentro solo $SO(3,1)$. Potrà scriversi con solo trasf di $SO(3,1)$:

LA RAPPRESENTAZIONE SUI TENSORI $T^{\mu\nu}$ È RIDUCIBILE

- Un tensore qualsiasi $T^{\mu\nu}$ può sempre scomposto nella sua parte sim e antisim

$$T^{\mu\nu} = A^{\mu\nu} + S^{\mu\nu}$$

$$A^{\mu\nu} = \frac{T^{\mu\nu} - T^{\nu\mu}}{2} \qquad S^{\mu\nu} = \frac{T^{\mu\nu} + T^{\nu\mu}}{2}$$

- La parte simmetrica può venir a sua volta scomposta in quella traccia più la parte traceless, o cui è stato rimosso la traccia

$$S^{\mu\nu} = \underset{\text{Traceless}}{S^{\mu\nu}} + \frac{1}{4} g^{\mu\nu} S$$

Infatti $S = S^{\mu}{}_{\mu} = g_{\mu\nu} S^{\mu\nu}$ dove S è uno scalare quindi Lorentz invariante l'operatore traceless è def come $S^{\mu\nu}$ a cui viene sottratto S

$$\underset{\text{Traceless}}{S^{\mu\nu}} = S^{\mu\nu} - \frac{1}{4} g^{\mu\nu} S \qquad \text{infatti così } \underset{\text{Traceless}}{S^{\mu}{}_{\mu}} = S - \frac{1}{4} 4 S = \emptyset$$

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- Allora un qualsiasi tensore può venir scomposto in:

$$T^{\mu\nu} = A^{\mu\nu} + S^{\mu\nu}_{\text{traceless}} + \frac{1}{4} g^{\mu\nu} S$$

generale — Antisimmetrico — Simmetrico traceless — Traccia del simmetrico

In Minkowski μ e ν vanno da 0, 1, 2, 3, ovvero 4 valori. Il n° di parametri del generico tensore di rango 2 in modo di dim = 4 è $4^2 = 16$.

il n° di parametri di un tensore antisim di rango 4 = 2 in modo n = 4 è 6 mentre i parametri liberi del sim-metrico sono 10. La traccia è uno scalare quindi 1 parametro, allora il traceless ha 9 parametri liberi.

$$4 \otimes 4 = 6 \oplus 9 \oplus 1$$

Tensore — Antisim — Sim traceless — Sim traccia

- Come è possibile scomporre il tensore 4×4 nella somma di tre parti $6+9+1$, così è possibile RIDURRE la trasf di Lorentz $\Lambda^\mu_\rho \Lambda^\nu_\sigma$ 4×4 che agisce su $T^{\mu\nu}$ nella somma di tre trasf di dimensioni minori $6+9+1$ che agiscono sulla parte sim, antisim ecc ecc
→ la rappr $SO(3,1) \times SO(3,1)$ fondamentale 4×4 è RIDUCIBILE nella somma di tre rappresentazioni $6+9+1$

→ qui - Decomposition of $SO(3)$ and $SO(3) \times SO(3)$
Visto + avanti sotto 2, 4 representations

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PRIMA = SOTTO IL CAP 2.3

SO(3) in 4D space - Decomposition of SO(3)

Prima di studiare le rappresentazioni di $SO(3,1)^+$ studiamo le rappresentazioni del SOTTOGRUPPO $SO(3)$. $SO(3)$ ha dim fondamentale 3, sono le rotazioni spaziali. Quale è la sua rappr in ambiente 4D ? Cerco la sua rappr come sottogruppo di Minkoski di $SO(3,1)^+$?

$SO(3)$ agisce come l'identità sul tempo e come rotazione sullo spazio. È quindi intuibile capire che si scompone in:

$$\underset{\sim}{4} = \underset{\sim}{1} \oplus \underset{\sim}{3}$$

Rappr di $SO(3)$ a 4D — scalare tempo — rotazioni spaziali

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N.3.

Le rappresentazioni di SO(3) sono etichettate da j del momento angolare

$j=0 \longrightarrow$ dim $\underset{\sim}{1}$

$j=\frac{1}{2} \longrightarrow$ dim $\underset{\sim}{2}$

$j=1 \longrightarrow$ dim $\underset{\sim}{3}$

$j=\frac{3}{2} \longrightarrow$ dim $\underset{\sim}{4}$

$j=2 \longrightarrow$ dim $\underset{\sim}{5}$

...

NB = solo per SO(3) la rappr $\underset{\sim}{4}$ si RIDUCE a $\underset{\sim}{1}+\underset{\sim}{3}$!! La rappr di SO(3,1) di dim $\underset{\sim}{4}$ è quella fond e non si può ridurre !!

$x^\mu = (x^0, \vec{x}^j)$

- x^0: time scalar which doesn't change under SO(3)
- $\vec{x}^j$: space, 3 coordinates which change as a vector under SO(3)

$\underset{\sim}{1} \oplus \underset{\sim}{3}$

$x^\mu = (x^0, x^1, x^2, x^3)$

they change ALL under a generic SO(3,1) tx !!

$\underset{\sim}{4}$

Decomposition of SO(3) x SO(3) in 4 dim space

- Su un tensore $T^{\mu\nu}$ agiscono due trasf di Lorentz. Considero le trasf di Lorentz ma solo il sottogruppo SO(3) x SO(3) in 4D. La sua decomposizione è:

$$\underset{\sim}{4} \times \underset{\sim}{4} = (\underset{\sim}{1} \oplus \underset{\sim}{3}) \times (\underset{\sim}{1} \oplus \underset{\sim}{3}) = \underset{\sim}{1}\times\underset{\sim}{1} \oplus \underset{\sim}{1}\times\underset{\sim}{3} \oplus \underset{\sim}{3}\times\underset{\sim}{1} \oplus \underset{\sim}{3}\times\underset{\sim}{3}$$

- $\underset{\sim}{4} \times \underset{\sim}{4}$: algebra di SO(3) in 4D
- $(\underset{\sim}{1} \oplus \underset{\sim}{3}) \times (\underset{\sim}{1} \oplus \underset{\sim}{3})$: scomposizione di SO(3) in 4D

Come scompongo 1×1, 1×3, 3×1 e 3×3? Sfrutto il parallelismo con il momento angolare

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2.3

- In generale, emendo $\underset{\sim}{n}$ una rappr di SO(3), so che corrisponde ad un momento angolare j_n. Una rappr $\underset{\sim}{n} \times \underset{\sim}{m}$ corrisponde alla SOMMA di due momenti agolari j_n e j_m

La somma di momenti agolari è data dalla regola:

$J_n + J_m = J_{TOT}$ dove $J_{TOT} \leftarrow |J_n + J_m|, |J_n + J_m - 1|, \dots, |J_n - J_m|$

- allora

1×1

$j_m = 0$ dim 1
$j_m = 0$ dim 1
$\rightarrow j_{TOT} = 0$ dim 1
$\Rightarrow \underset{\sim}{1} \times \underset{\sim}{1} = \underset{\sim}{1}$

È ancora una rappr di dim $\underset{\sim}{1}$

1×3 e **3×1**

$j_n = 0$ dim 1
$j_m = 1$ dim 3
$\rightarrow j_{TOT} = 1$ dim 3
$\Rightarrow \underset{\sim}{1} \times \underset{\sim}{3} = \underset{\sim}{3}$, $\underset{\sim}{3} \times \underset{\sim}{1} = \underset{\sim}{3}$

3×3

$j_n = 1$ dim 3
$j_m = 1$ dim 3
$\rightarrow j_{TOT} = 0, 1, 2$ (dim 1, dim 3, dim 5)
$\Rightarrow 3 \times 3 = 1 \oplus 3 \oplus 5$

- allora

$$\underset{\sim}{4} \times \underset{\sim}{4} = \underset{\sim}{1} \times \underset{\sim}{1} \oplus (\underset{\sim}{1} \times \underset{\sim}{3} \oplus \underset{\sim}{3} \times \underset{\sim}{1}) \oplus (\underset{\sim}{3} \times \underset{\sim}{3}) = \dots$$

$$= \underset{\sim}{1} \oplus (3 \oplus 3) \oplus (1 \oplus 3 \oplus 5)$$

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pg 3

Lo comparo con quanto avevo trovato prima per una generica trasf di Lorentz su $T^{\mu\nu}$ adimensionale (infatti SO(3) è sottogruppo di Lorentz)

$\underset{\sim}{4} \times \underset{\sim}{4} = \underset{\sim}{1} \oplus \underset{\sim}{6} \oplus \underset{\sim}{9}$

$\underset{\sim}{4} \times \underset{\sim}{4} = \underset{\sim}{1} \oplus (\underset{\sim}{3} \oplus \underset{\sim}{3}) \oplus (\underset{\sim}{1} \oplus \underset{\sim}{3} \oplus \underset{\sim}{5})$

Traccia S

scalare

corrisponde alla componente 00 cioè

$1 \times 1 \rightarrow T^{00}$

$j=0 \quad j=0$

Antisimmetrico $A^{\mu\nu}$

2 spatial vectors

can be written as 2 spatial vectors

$1 \times 3 \rightarrow T^{0i}$

$j=0 \quad j=1$

$3 \times 1 \rightarrow T^{i0}$

$j=1 \quad j=0$

Traceless symmetric $S^{\mu\nu}_{traceless}$

3x3 tensor

can be written as a 3x3 tensor spatial

$3 \times 3 \rightarrow T^{ij}$

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2.4 REPRESENTATIONS

Finora ho lavorato nella RAPPRESENTAZIONE FONDAMENTALE DI $SO(3,1)^+$, quella di $DIM = 4$ dove gli elementi del gruppo si manifestano come matrici 4×4 che agiscono su vettori dello spazio di Minkowski

in generale però ciascuna trasf $g \in G$ gruppo può avere rappr matriciali diverse su spazi diversi

$$g \begin{cases} \to [D_R(g)]^i_j & \text{dim } n \times n \\ \to [D_{R'}(g)]^\mu_\nu & \text{dim } n' \times n' \\ \to \dots & \dots \end{cases}$$

Il NUMERO DEI GENERATORI e la forma delle regole di commutazione (che è COST DI STRUTTURA) sono le stesse in tutte le rappresentazioni.

mentre

la MATRICE che rappresenta i generatori è DIVERSA a seconda della rappresent.

Nel caso completamente generale sia ϕ^i un oggetto/vettore $i = 1 \dots n$ dimensionale, la rappresentazione di Lorentz su tale generico mondo è una matrice $n \times n$:

$$\phi'^i = [\Lambda]^i_j \, \phi^j$$

↓

Matrice di forma generica dove ij scorrono su n indici del mondo generico in cui rappresento Lorentz

$$\phi'^i = \left[e^{-i \frac{1}{2} \omega_{\alpha\beta} J^{\alpha\beta}} \right]^i_j \phi^j$$

La trasf infinitesima sarà

$$\delta\phi^i = -\frac{i}{2} \omega_{\alpha\beta} [J^{\alpha\beta}]^i_j \, \phi^j \qquad (2.5)$$

↓

RAPPR di dim $n \times n$ del generatore $\alpha\beta$

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Representations of $SO(3,1)^+$

Since $SO^+(3,1)$ algebra can be decomposed in the product of two $SU(2)$ algebra by definition:

$$J_\pm^i = \frac{J^i \pm iK^i}{2} \qquad i = 1,2,3$$

$$\rightarrow \quad [J_+^i, J_+^j] = i\varepsilon^{ijk} J_+^k$$
$$[J_-^i, J_-^j] = i\varepsilon^{ijk} J_-^k$$
$$[J_+^i, J_-^j] = 0 \qquad \forall ij$$

riconosco due sottoalgebre indipendenti uguali all'algebra di $SU(2)$ perciò

$$\mathfrak{so}(3,1) = \mathfrak{su}(2)^+ \oplus \mathfrak{su}(2)^-$$

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L'algebra di SU(2) e quella di SO(3) sono la stessa, sono legate. Entrambe hanno rappresentazioni etichettate da j del momento angolare

$j = 0, \frac{1}{2}, 1, \frac{3}{2}, 2 \ldots$

- 0 → rappr banale → dim = 1
- $\frac{1}{2}$ → rappresentazione spinoriale → dim = 2
- 1 → rappr di dim 3
- $\frac{3}{2}$ → dim 4
- 2 → dim 5

Posso quindi RAPPRESENTARE la mia trasf di Lorentz assegnando j_+ alla sottoalgebra $SU^+(2)$ e j_- alla $SU^-(2)$ perché (in realtà per convenzione è il contrario, − a sx e + a dx)

algebra :	$SO(3,1)$	$=$	$SU^+(2)$	$\oplus$ $SU^-(2)$
rappresent :	(j_+, j_-)	:	j_+	j_-
dimensione :	$(2j_+ + 1)\cdot(2j_- + 1)$		$(2j_+ + 1)$	$(2j_- + 1)$

La rappresentazione $j = 1/2$ è detta SPINORIALE ed è la rappr fondamentale di $SU(2)$.

E' un teorema per cui "*All representations of SU(2) can be constructed from fundamental representation*"

Ovvero qualsiasi altra rappr $j > \frac{1}{2}$ può essere scomposta in rappr $\frac{1}{2}$ spinoriali. Fisicamente ciò significa che dato un sistema con uno spin qualsiasi (es: ferromagnete) lo posso scomporre in una collezione di spin 1/2, magari con interazioni tra le parti a spin 1/2

Es $\quad \underset{\text{due spin 1/2}}{\overset{j}{\frac{1}{2}} \times \overset{j}{\frac{1}{2}}} = \underset{\substack{\text{singoletto} \\ \text{antisim}}}{\overset{j}{0}} + \underset{\substack{\text{tripletto} \\ \text{simmetrico}}}{\overset{j}{1}}$

$$\left\{\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)\right\} \quad \left\{|\uparrow\uparrow\rangle, \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle), |\downarrow\downarrow\rangle\right\}$$

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Le rappresentazioni di $SO(3,1)$ che convogano $j_+ \circ j_- = 1/2$ sono dette rappresentazioni spinoriali di $SO(3,1)$.

Vediamo alcune rappresentazioni.

(1) (0,0) Scalar — dim = 1

This is the trivial representation where $j_+ = j_- = 0$ hence it has dimension = 1. This representation acts on SCALARS

$$\Lambda = e^{0} = \mathbb{1} \quad \text{so} \quad \Lambda\phi = \mathbb{1}\phi = \phi \quad \hookrightarrow \text{scalar}$$

(2) $(\frac{1}{2},0)$ or $(0,\frac{1}{2})$ Weyl Spinors — dim = 2

They are matrices Λ_L or Λ_R that act respectively on vector denoted ψ_L or ψ_R called LEFT (for $(1/2\,0)$) and RIGHT (for $(0,1/2)$) Spinor representation

They act on a world of dim $= (2\frac{1}{2}+1)(1) = 2$ so ψ_L and ψ_R are 2 dim vectors

$\Lambda_L \in (1/2,0)$ acts on ψ_L as $\psi_L' = \Lambda_L \psi_L$

$\Lambda_R \in (0,1/2)$ acts on ψ_R as $\psi_R' = \Lambda_R \psi_R$

Hence the matrices Λ_L and Λ_R are 2x2 matrices and are the exp of pauli matrices as generators

$SU(-)$ $SU(+)$

for $(1/2,0)$ L $\longrightarrow$ $\vec{J}_- = \frac{\vec{\sigma}}{2}$ (dim 2) so that $\vec{J} = \vec{J}_+ + \vec{J}_- = \vec{\sigma}/2$

$\vec{J}_+ = \emptyset$ $\rightarrow$ dim 1 $\quad \vec{n} = i\vec{\sigma}/2$

for $(0,1/2)$ R $\longrightarrow$ $\vec{J}_- = \emptyset$ $\quad \vec{J} = \vec{\sigma}/2$

$\vec{J}_+ = \frac{\vec{\sigma}}{2}$ so that $\vec{n} = -i\vec{\sigma}/2$

posto io so di $SU(2)$ ho come generatori σ^i, dico sono le algebre sono tre, ci 2 dim ordi $SO(3,1)$ ho 6 i core generatori.

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Λ_L

$\left(\frac{1}{2}, 0\right)$

$\downarrow \quad \downarrow$

$J_- \quad J_+$

$\left.\begin{array}{l} J_- \text{ ha dim 2 ed è un generatore di } SU(2) \longrightarrow \vec{J}_- = \frac{\vec{\sigma}}{2} \\ J_+ \text{ ha dim 1 ed è un gen di } SU(2) \longrightarrow \vec{J}_+ = \emptyset \end{array}\right\}$ allora $\mapsto \begin{array}{l} \vec{N} = \vec{J}_+ + \vec{J}_- = \frac{\vec{\sigma}}{2} \\ \vec{K} = i(\vec{J}_- - \vec{J}_+) = i\frac{\vec{\sigma}}{2} \end{array}$

$e^{-i\vec{\theta}\vec{J} + i\vec{K}\vec{\lambda}} = e^{-\frac{i}{2}\vec{\sigma}\cdot\vec{\theta} - \vec{n}\cdot\frac{\vec{\sigma}}{2}}$

Λ_R

$\left(0, \frac{1}{2}\right)$

$\downarrow \quad \downarrow$

$J_- \quad J_+$

$\left.\begin{array}{l} J_- = \emptyset \\ J_+ = \frac{\vec{\sigma}}{2} \end{array}\right\} \mapsto \begin{array}{l} \vec{N} = \frac{\vec{\sigma}}{2} \\ \vec{K} = -i\frac{\vec{\sigma}}{2} \end{array} \longrightarrow e^{-i\vec{\theta}\vec{J} + i\vec{n}\cdot\vec{u}} = e^{-\frac{i}{2}(\vec{\sigma}\cdot\vec{\theta}) + \vec{n}\cdot\frac{\vec{\sigma}}{2}}$

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Recalling that $\Lambda = e^{(-i\vec{\theta}\cdot\vec{J} + i\vec{m}\vec{n})}$ we have that

$\Lambda_L = e^{(-i\vec{\theta} - \vec{m})\cdot\frac{\vec{\sigma}}{2}}$ $\quad \Lambda_L \psi_L = \psi_L'$

$\Lambda_R = e^{(i\vec{\theta} + \vec{m})\cdot\frac{\vec{\sigma}}{2}}$ $\quad \Lambda_R \psi_R = \psi_R'$

(Λ_R: 2x2 Matrix; ψ_R: 2 vector)

where $\vec{\sigma} = (\sigma_1, \sigma_2, \sigma_3)$ Pauli Matrices

$$\sigma_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \qquad \sigma_2 = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \qquad \sigma_3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

(3) (1/2, 1/2) DIRAC REPRESENTATION — dim = 4

Dimension $dim = (2\frac{1}{2}+1)(2\frac{1}{2}+1) = 4$, it acts on 4 vectors Ψ that can be written in form of two components of Weyl Spinors components

$$\Psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} \qquad \Psi' = \Lambda_D \Psi \qquad \Lambda_D = \begin{pmatrix} \Lambda_L & 0 \\ 0 & \Lambda_R \end{pmatrix}$$

(Ψ: 4 vector; ψ_L, ψ_R: 2 vectors; Λ_D: 4x4 matrix)

(4) Vectors — dim 4

fundamental representation that acts on vector V^μ

(5) Tensors — dim 4x4 = 16

Representation of SO(3,1) x SO(3,1) acts on tensors

These are not Spinorial representations i.e they don't use spin $j = 1/2$

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2.4.1 Field Representations

Abbiamo classificato le rappresentazioni di $SO(3,1)$ in base al parametro $j = 0, \frac{1}{2}, 1, \dots$, ora siamo interessati a scoprire come è fatta concretamente la matrice Λ in queste varie rappresentazioni, in questo mondo.

I vari mondi dove ho definito Λ sono mondi vettoriali dove ho chiamato tali vettori CAMPI (ϕ campo scalare, ψ_L, ψ_R campi spinoriali di Weyl, Ψ campo di Dirac ...) sono campi perché dipendono dal punto x di applicazione ($\phi(x)$, $\psi_L(x)$, $\Psi(x)$, $V^\mu(x)$ → NB anche V^μ è un campo vettoriale che dip. dal punto x!)

Ci poniamo il seguente OBBIETTIVO = Dato la trasf di Lorentz $x \xrightarrow{\Lambda} x'$ cambio di coordinate, come varia il campo $\phi(x)$, $\psi_L(x)$, $\Psi(x)$ definito sul punto?

LORENTZ $x \xrightarrow{\Lambda} x'$ sulle coordinate ⟶ INDUCE sui campi:

$$\phi(x) \xrightarrow{\Lambda} \phi'(x')$$

$$\psi_L(x) \xrightarrow{\Lambda_L} \psi'_L(x')$$

$$\psi_R(x) \xrightarrow{\Lambda_R} \psi'_R(x')$$

$$\Psi(x) \xrightarrow{\Lambda_D} \Psi'(x')$$

A seconda della NATURA del campo esso trasformerà come uno scalare, come uno spinore di Weyl e come un vettore di Dirac

Cambio di prospettiva:

un oggetto è un VETTORE se sotto Rotazioni trasforma come un vettore
un oggetto è uno scalare se sotto Rotazioni trasforma come scalare cioè invariante
un oggetto è uno spinore di Weyl se ... come spinore di Weyl ...

ovvero classifico gli oggetti definiti sul mio mondo x in base a come trasformano

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OBBIETTIVO

$$\phi'_{(x)} = [\Lambda]\,\phi_{(x)} = \left[e^{-\frac{i}{2}\omega_{AB} J^{AB}}\right]\phi_{(x)}$$

$$\delta\phi_{(x)} = \phi'_{(x)} - \phi_{(x)} = -\frac{i}{2}\omega_{AB}\left[J^{AB}\right]\phi_{(x)} \qquad (2.5)$$

$\omega_{AB} \to 0$ infinitesimo

Sfruttare 2.5 per trovare l'espressione di $[J^{AB}]^{i}_{\ j}$ della matrice dei generatori sul modo del campo ϕ

NB = Nell'eq di sopra il campo è valutato sempre in x, sia ϕ' che ϕ, ovvero è valutato nello stesso VALORE di x

NB = se il campo è un vettore con più componenti ϕ^j allora eq 2.5 per componenti diventa

$$\delta\phi^i = \phi'^i - \phi^i = -\frac{i}{2}\omega_{AB}\left[J^{AB}\right]^i_{\ j}\phi^j \qquad (2.5)$$

METODO

Parto dalla trasf di Lorentz sulle coordinate $x'^{\mu} = \Lambda^{\mu}_{\ \nu} x^{\nu}$ dove conosco l'espressione esplicita della $\Lambda^{\mu}_{\ \nu}$ e dei suoi generatori

$$x'^{\mu} = \Lambda^{\mu}_{\ \nu} x^{\nu} = \left[e^{-\frac{1}{2}\omega_{AB} J^{AB}}\right]^{\mu}_{\ \nu} x^{\nu} \xrightarrow{\text{infinitesimo}} x^{\mu} \underbrace{-\frac{i}{2}\omega_{AB}(J^{AB})^{\mu}_{\ \nu} x^{\nu}}_{\delta x^{\mu}}$$

$$\delta x^{\mu} = -\frac{i}{2}\omega_{AB}(J^{AB})^{\mu}_{\ \nu} x^{\nu} \qquad (2.17)$$

dove $(J^{AB})^{\mu}_{\ \nu} = i(g^{A\mu}\delta^{B}_{\ \nu} - g^{B\mu}\delta^{A}_{\ \nu})$ in Minkowski

Essendo (a) 2.17 nel conto del campo $\delta\phi$ vedo come (a) induce la trasf di Lorentz del campo

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(1) SCALAR FIELDS

a) Considero un punto P labeled by x^μ in one frame S and by x'^μ in another frame S'

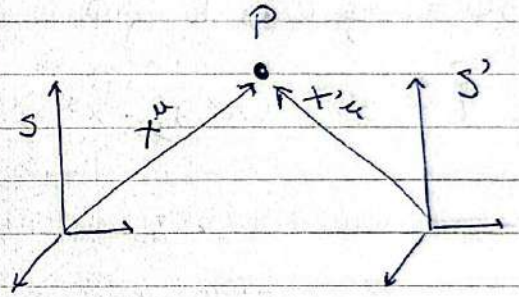
If $\phi(x)$ is a scalar, $\phi(x)$ and $\phi'(x')$ are the same value in the two frames (SR1) S and S'. IMPONGO la condizione di campo scalare

$$\phi'(x') \overset{!}{=} \mathbb{1}\,\phi(x) = \phi(x)$$

↓

trasf di Lorentz $(0,0)$

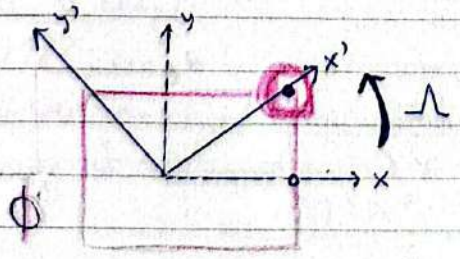
$\Lambda = \mathbb{1}$



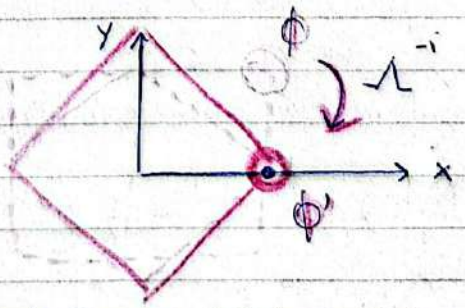
Ovviamente, affinché $\phi'(x') = \phi(x)$ il campo $\phi' \neq \phi$ DEVE cambiare in modo da COMPENSARE il cambiamento sulle $x \to x'$ Lorentz.

Se considero $\phi'(x)$ e $\phi(x)$ allora sono DIVERSI $\phi'(x) \neq \phi(x)$ perché sto valutando due campi in forme diverse nello stesso VALORE di x, ma se $x\phi'$ corrisponde alla trasformata di Lorentz di ϕ, allora sto valutando $\phi'(x)$ e $\phi(x)$ in due PUNTI DIVERSI, solo che accidentalmente ad due SR1 assumono lo stesso valore numerico delle coordinate.

b) Calcolare la trasf di Lorentz SUL CAMPO $\phi \xrightarrow{\Lambda} \phi'$ significa calcolare la variazione $\phi'(x) = \Lambda\,\phi(x)$ sullo stesso x (valore)



Trasformazione del punto $x \to x'$ a campo "fisso"



Trasformazione del campo scalare a punto "fisso"

$\Lambda x \leftrightarrow \Lambda^{-1}\phi$) equival.

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a) $\delta\phi = \phi'(x) - \phi(x) =$ → $x' = x + \delta x$ infinitesima

$= \phi'(x' - \delta x) - \phi(x) =$ → 1° ordine con Taylor (NB)

$= \phi'(x') - \delta x^\rho \partial_\rho \phi - \phi(x) =$ → NO $\phi'(x') \overset{!}{=} \phi(x)$ perché scalare

$= -\delta x^\rho \partial_\rho \phi =$

$=$ ↙ (2.17) $\delta x^\rho = -\frac{i}{2}\omega_{\alpha\beta}(J^{\alpha\beta})^\rho{}_\sigma x^\sigma$

$= \frac{i}{2}\omega_{\alpha\beta}(J^{\alpha\beta})^\rho{}_\sigma x^\sigma \partial_\rho \phi =$ ↙ $(J^{\alpha\beta})^\rho{}_\sigma = i(g^{\alpha\rho}\delta^\beta_\sigma - g^{\beta\rho}\delta^\alpha_\sigma)$

$= -\frac{i}{2}\omega_{\alpha\beta} i(x^\alpha \partial^\beta - x^\beta \partial^\alpha)\phi$

so

$$\delta\phi = -\frac{i}{2}\omega_{\alpha\beta}[L^{\alpha\beta}]\phi \qquad J^{\alpha\beta}$$

where $L^{\alpha\beta} = i(x^\alpha \partial^\beta - x^\beta \partial^\alpha)$ (2.18)

sono i generatori della trasf di Lorentz nel caso scalare

$\Lambda_0 = e^{-\frac{i}{2}\omega_{\alpha\beta}L^{\alpha\beta}}$ sugli scalari $\phi(x)$

$\phi'(x) = \Lambda_0\, \phi(x)$

Definisco $p^\mu = i\partial^\mu$ → $p^0 = i\partial^0 = i\frac{\partial}{\partial t}$ Momento

↳ $p^i = i\partial^i = -i\partial_i = -i\frac{\partial}{\partial x^i}$

allora

$L^{\alpha\beta} = x^\alpha p^\beta - x^\beta p^\alpha$ xché $\partial^i = \frac{\partial}{\partial x_i}$ $\partial_i = \frac{\partial}{\partial x^i}$ $x_i = (t, -x, -y, -z)$

$x^i = (t, x, y, z)$

Where the spatial part is the orbital angular momentum $\vec{L}$

$\vec{L} = (L^1, L^2, L^3)$ $\quad L^i = \frac{1}{2}\varepsilon^{ijk}L^{jk} = \varepsilon^{ijk}x^j p^k$

NB $\phi'(x) = \phi(x - \delta x) \overset{\text{Taylor 1° ordine}}{=} \phi(x) - \delta x^\rho \partial_\rho \phi$

$\phi'(x) = \phi'(x' - \delta x) = \phi'(x') - \delta x^\rho \partial_\rho \phi'$ ⌐ allora $\partial_\rho \phi = \partial_\rho \phi'$ ⌙

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NB = Passare attraverso la trasf sulle coord $x \to x'$ per trovare la trasf su campi è solo UN ARTIFICIO matematico. In realtà, a livello formale, il campo ϕ è la BASE della trasf di Lorentz nelle rappr scalare, la trasf Λ su ϕ esiste allo stesso piano gerarchico di quella su x, Λ su ϕ NON è una conseguenza di Λ su x! Sono gerarchicamente equivalenti! solo rappr diverse

(2) WEYL FIELDS

a) Considero lo stesso punto P labelled as x in S and x' in S' dove $x' = \Lambda x$ trasf di cambio di SRI di Lorentz.

Gli spinori di Weyl per definizione sono oggetti che, passando da un SRI ad un altro, trasformano come vettori 2D cioè compongo la condizione di spinore:

$$\psi'_L(x') \stackrel{!}{=} \Lambda_L \psi_L(x)$$

$$\psi'_R(x') \stackrel{!}{=} \Lambda_R \psi_R(x)$$

ψ' trasformato nel x' trasf — ψ e x con trasformati

dove $\Lambda_L = e^{(i\vec{\theta} - \vec{n})\vec{\sigma}/2}$ $\quad$ $\Lambda_R = e^{(i\vec{\theta} + \vec{n})\cdot\vec{\sigma}/2}$

b) Ma a noi interessa vedere come Lorentz agisca SOLO sullo spinore $\psi_{L/R}$, non quando trasformo entrambi x e ψ. Se trasf entrambi sto facendo un SRI cambio volutato nello stesso punto? Se cambio solo ψ guardo a come si trasforma la FORMA stessa del campo.

↓

Trasformato sì anche lui

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•) Considero quindi una variazione infinitesima di $\psi'_L(x) \to \psi_L(x)$ valutata nella stessa coordinata x come valore numerico (quindi punti diversi)

$$\delta\psi_L = \psi'_L(x) - \psi_L(x) = \quad \searrow\ x' = x + \delta x \text{ per coerenza infinit}$$

$$= \psi'_L(x' - \delta x) - \psi_L(x) = \quad \searrow\ \text{Taylor al 1° ordine dove } \partial\psi = \partial\psi'$$

$$= \psi'_L(x') - \delta x^\rho \partial_\rho \psi_L(x) - \psi_L(x) = \quad \swarrow\ \psi'_L(x') = \Lambda_L \psi_L(x) \text{ condizione di spinore}$$

$$= \left(\Lambda_L - \mathbb{1}\right)\psi_L(x) - \partial x^\rho \partial_\rho \psi_L(x) =$$

ora faccio una generalota, vado a scrivere definire Λ_L che agisce su ψ_L nella forma:

$$\Lambda_L := e^{-\frac{i}{2}\omega_{\mu\nu}S^{\mu\nu}}$$

dove ad es ho chiamato $(S^{\mu\nu})^r_s$ nella base spinoriale come $(S^{\mu\nu})^r_s = [S^{\mu\nu}]^r_s$ di spin ed è una matrice 2x2 perché ψ vive nello 2 dim.

$$\Lambda_L \simeq \mathbb{1} - \frac{i}{2}\omega_{\mu\nu}S^{\mu\nu}$$

•) allora per la trasf infinitesima ho:

$$\delta\psi_L = \left(-\frac{i}{2}\omega_{\mu\nu}S^{\mu\nu} - \partial x^\rho \partial_\rho\right)\psi_L(x)$$

ω conosco sono le matrici di Pauli $S^{\mu\nu}$

È il momento orbitalico visto prima $L^{\mu\nu}$

per un confronto con 2.5 $\delta\psi_L = -\frac{i}{2}\omega_{\alpha\beta}J^{\alpha\beta}\psi_L$ Vedo che

$$\Lambda_{L_R} = e^{-\frac{i}{2}\omega_{\alpha\beta}J^{\alpha\beta}}$$

tic $\Lambda^L_A \psi^L_R(x) = \psi'^L_A(x)$ (2.19)

$$J^{\alpha\beta} = S^{\alpha\beta} + L^{\alpha\beta}$$

spin, orbitalico

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dove il momento di spin è

$$S^i = \frac{1}{2}\varepsilon^{ijk} S^{jk} = \frac{\sigma^i}{2} \qquad S^{i0} = -\frac{i\sigma^i}{2} \quad \Big| \text{ Left}$$

$$S^i = \frac{1}{2}\varepsilon^{ijk} S^{jk} = \frac{\sigma^i}{2} \qquad S^{i0} = \frac{i\sigma^i}{2} \quad \Big| \text{ Right}$$

NB = $\Lambda_{L,R} = \exp\left(-\frac{i}{2}\omega_{\alpha\beta} J^{\alpha\beta}\right)$ dove:

a) Per la trasf congiunta $\psi \to \psi'$ $x \to x'$ cioè $\Big| \to J^{\alpha\beta} = S^{\alpha\beta}$

$$\psi'_{L,R}(x') = \Lambda_{L,R}\, \psi_{L,R}(x)$$

b) Per la trasf solo $\psi \to \psi'$ cioè $\Big| \to J^{\alpha\beta} = S^{\alpha\beta} + L^{\alpha\beta}$

$$\psi'_{L,R}(x) = \Lambda_{L,R}\, \psi_{L,R}(x)$$

(3) DIRAC FIELDS (1/2, 1/2)

A Dirac Field is defined as $\Psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}$ and has 4 components, 2 from ψ_L and 2 from ψ_R. It undergoes de Lorentz tx

$$\Psi' = \Lambda_D \Psi \qquad \text{with } \Lambda_D = \begin{pmatrix} \Lambda_L & 0 \\ 0 & \Lambda_R \end{pmatrix}$$

so nothing new. We have the definitions of Λ_L and Λ_R for both transformation $\psi \to \psi'$ and $\psi \to \psi'$ & $x \to x'$. Λ_D definition is a direct consequence of them.

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PARITY

I campi di Dirac si intrecciano per il loro comportamento sotto parità.
La trasf di PARITÀ

$$P : (t, \vec{x}) \longmapsto (t, -\vec{x})$$

È una trasf di $O(3,1)$ MA NON DI $SO(3,1)^+$!! Quindi non può venir descritta dall'algebra di Lie! Con i generatori!

Boosts generators $\vec{K}$ are a spatial 3vector so it transforms under parity as

$$\vec{K} \xrightarrow[P]{} -\vec{K}$$

While rotations generators $\vec{J}$ are a time like 3vector so they are preserved:

$$\vec{J} \xrightarrow[P]{} \vec{J}$$

Thus Parity switches the two algebra of $SO^+(3,1)$ (proper orthocronous) Lorentz

$$\vec{J}_+ \overset{P}{\longleftrightarrow} \vec{J}_-$$

Thus under Parity ψ_L and ψ_R are switched, Parity switches Right and left Spinors → Spinors are not preserved under parity!

$$(\tfrac{1}{2}, 0) \overset{P}{\longleftrightarrow} (0, \tfrac{1}{2})$$
$$\psi_L \overset{P}{\longleftrightarrow} \psi_R$$

Indeed, a Dirac field under parity changes in another DIRAC FIELD! It changes yes, but it is STILL A DIRAC FIELD

$$\underline{\psi}(x) \xrightarrow{P} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \underline{\psi}(x)$$

So $\underline{\psi}(x)$ provides a representation of both Lorentz $SO^+(3,1)$ and Parity transformation

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(4) VECTOR FIELDS & PSEUDOVECTOR

vector fields V^μ as x^μ are the fundamental basics of Lorentz transformation and we already know how they transforms = is the Λ repr in Minkoski space:

$$V'^{\mu}_{(x')} \overset{!}{=} \Lambda^{\mu}{}_{\nu}\, V^{\nu}_{(x)}$$

NOTE that I change both V' and x' !!

the repr of $\Lambda^\mu{}_\nu$ on ONLY $V \to V'$ while x is constant is the same of a SCALAR FIELD, because the coordinate function $V^\mu(x) = \phi(x)$ is a scalar field / function of x

PARITY

$$V^\mu = (V^0, \vec{V}) \xrightarrow{P} (V^0, -\vec{V})$$

PSEUDOVECTOR

A pseudovector is a vector under Lorentz, so it undergoes the same

$$\omega'^{\mu}_{(x')} \overset{!}{=} \Lambda^{\mu}{}_{\nu}\, \omega^{\nu}_{(x)}$$

BUT for parity the behaviour is different, parity inverts time in pseudovectors

$$\omega^\mu = (\omega^0, \vec{\omega}) \xrightarrow{P} (-\omega^0, \vec{\omega})$$

NOTA = ECCO perché si chiamano pseudovettori e pseudotensori! Sono tensori e sono vettori sotto Lorentz, cioè quelle trasf che DEFINISCONO cos'è un vettore o tensore, MA sotto Parità, che non fa parte di Lorentz proprio, si comportano diversamente ... sono vettori, si ma solo sotto Lorentz proprio, sotto Lorentz improprio no! (i.e. $\vec{x} \to -\vec{x}$ si comportano come vettori sotto Lorentz)

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SCHEMATIC OF LORENTZ REPRESENTATIONS

general Lorentz transformation $\Lambda = e^{-\frac{i}{2}\omega_{\mu\nu} J^{\mu\nu}}$

acting on 4D vectors as $x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu}$

where $\Lambda^{\mu}{}_{\nu} = \left[e^{-\frac{i}{2}\omega_{\alpha\beta} J^{\alpha\beta}} \right]^{\mu}{}_{\nu}$ $\quad \left[J^{\alpha\beta} \right]^{\mu}{}_{\nu} = i(g^{\alpha\mu}\delta^{\beta}_{\nu} - g^{\beta\mu}\delta^{\alpha}_{\nu})$

SCALAR FIELDS

$\phi \to \phi'$ & $x \to x'$	$\phi'(x') = \Lambda_0 \phi(x) \overset{!}{=} \phi(x)$	$J^{\alpha\beta} = 0$
$\phi \to \phi'$	$\phi'(x) = \Lambda_0 \phi(x)$	$J^{\alpha\beta} = L^{\alpha\beta}$

WEYL FIELDS

$\psi \to \psi'$ $\quad x \to x'$	$\psi'_{L,R}(x') \overset{!}{=} \Lambda_{L,R} \psi_{L,R}(x)$	$J^{\alpha\beta} = S^{\alpha\beta}$
$\psi \to \psi'$	$\psi'_{L,R}(x) = \Lambda_{L,R} \psi_{L,R}(x)$	$J^{\alpha\beta} = S^{\alpha\beta} + L^{\alpha\beta}$

VECTOR FIELDS

$V \to V'$ $\quad x \to x'$	$V'^{\mu}(x') \overset{!}{=} \Lambda^{\mu}{}_{\nu} V^{\nu}(x)$	$(J^{\alpha\beta})^{\mu}{}_{\nu} = i(g^{\alpha\mu}\delta^{\beta}_{\nu} - g^{\beta\mu}\delta^{\alpha}_{\nu})$
$V \to V'$	$V'^{\mu}(x) = \Lambda_0 V^{\mu}(x)$	$J^{\alpha\beta} = L^{\alpha\beta}$

single coordinate: $V \to V'$ IS A SCALAR!

TENSOR FIELDS

A vector fields but with double $\Lambda^{\mu}{}_{\nu}\,\Lambda^{\rho}{}_{\sigma}$ because is a transformation of $SO(3,1) \otimes SO(3,1)$ and not just $SO(3,1)$

↳ if I want to compute the transformation on tensor $V'^{\mu}(x)$ mixing coordinate, I will also have a spin component? Because Boson Vector Field Does have a spin part too! See quad 2.

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2.5 POINCARE' GROUP

DEFINIZIONE

Il gruppo di Poincare è un ampliamento del gruppo di Lorentz: Agisce sempre su vettori x_μ a 4 dim di Minkowski con in più le traslazioni

Lorentz Matrici Λ applicate al vettore x_μ tc

$$x'_\mu = \Lambda^\mu{}_\nu x^\nu$$

con la condizione di invarianza

$$ds'^2 \doteq ds^2$$

POINCARE Matrici Λ e TRASLAZIONI a tc

$$x^{\mu'} = \Lambda^\mu{}_\nu x^\nu + a^\mu$$

con la stessa condizione di invarianza

$$ds'^2 \doteq ds^2$$

Il gruppo di Poincare è quello delle proprietà di simmetria delle particelle nello SPAZIO VUOTO dove infatti valgono tutte le simmetrie rotazioni, boost e traslazioni.

Lorentz = Λ matrices, Linear operator
Gruppo di Lie NON abeliano con l'op prodotto tra matrici

Traslazioni = a cost, le traslazioni T sono un GRUPPO A SESTANTE MA AFFINE (non lineare)
Gruppo di Lie → può venire scritto come generatori
operazione è la SOMMA $a^\mu + b^\mu$ → Abeliano

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Lorentz group	Translation group
$x'^\mu = \Lambda^\mu{}_\nu x^\nu$	$x'^\mu = x^\mu + a^\mu$
↓ infinitesimo	↓ infinitesimo
$x'^\mu = x^\nu + \delta x^\mu$	$x'^\mu = x^\mu + \epsilon^\mu$
dove δx^μ dip da punto	con $\epsilon \to 0$ cost

ALGEBRA

Il gruppo di Poincaré è un gruppo di Lie perché unione di due gruppi di Lie = Lorentz e traslazioni. L'algebra di Lorentz la conosciamo già, studiamo quella delle traslazioni e poi come interagiscono le due algebre Lorentz + traslazioni

ALGEBRA TRASLAZIONI

Given the translation $x'^\mu = x^\mu + a^\mu$ posso scriverla con un generatore come

$$\phi(x') = e^{-i a_\mu P^\mu} \phi(x)$$

(spatial derivative)

dove $P^\mu = i\partial^\mu$ (vedi dim dopo).

⇒ Ora il generatore NON È uno scalare! Il gruppo delle traslazioni non è Lorentz ma affine e tensore trasf e generatori sono differenziali! Il generatore agisce SULLE FUNZIONI, agisce sulle COORDINATE DENTRO LE FUNZIONI DI X → generatore delle traslaz è DIFFERENZIALE NON MATRICIALE

Bisogna vedere poi anche la traslazione delle coord $x'^\mu = x^\mu + a^\mu$ con l'azione del generatore ma solo se PENSO alla $X^\mu(x)$ FUNZIONE COORDINATA stessa

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DIM sulle x^μ

$$x'^\mu = e^{-i a_\nu P^\nu} x^\mu = (\mathbb{1} - i \varepsilon_\nu P^\nu) x^\mu = x^\mu + \varepsilon_\nu \partial^\nu x^\mu =$$

$a_\nu = \varepsilon_\nu \to 0$ infinitesimo

$\partial^\nu x^\mu = g^{\nu\mu}$

$$= x^\mu + \varepsilon_\nu g^{\nu\mu} = x^\mu + \varepsilon^\mu \quad \text{ciao!}$$

ALGEBRA DI POINCARE'

$$[J^i, J^j] = i \varepsilon^{ijk} J^k \quad (*)$$

$$[K^i, K^j] = -i \varepsilon^{ijk} J^k$$

$$[J^i, K^j] = i \varepsilon^{ijk} K^k \quad (*)$$

Lorentz

$$[P^i, P^j] = 0 \quad (*)$$

$$[P^i, H] = 0 \quad (\emptyset)$$

Traslazioni dove $P^0 = i H$ Hamiltoniana

$$[J^i, P^j] = i \varepsilon^{ijk} P^k \quad (*)$$

$$[K^i, P^j] = i H \delta^{ij}$$

$$[J^i, H] = 0 \quad (\emptyset)$$

$$[K^i, H] = i P^i \quad (\flat)$$

Miste

$(*)$ = Tell us that $\vec{K}$, $\vec{P}$ and $\vec{J}$ rotate sotto $SO(3)$ come VETTORI

$(\emptyset)$ = a dicono che J^i e P^i commutano con Hamiltoniano si conservano, mentre K^i No! $P_0 = H$ è la traslazione temporale, e l'evoluzione presa è l'Hamiltoniano

$(*)$ = traslazioni spaziali su direzioni che commutano

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FIELD REPRESENTATION OF TRANSLATIONS

((Tutti i campi $\phi(x)$ (funzioni di x) sono SCALARI sotto traslazioni))

Perché? Anche il nostro modo è fatto così, matematicamente potrei dei campi non scalari sotto traslazione ma non esistono in natura → è la base della relatività per cui in ogni SR1 ogni fisico deve trovare la stessa, non ci sono punti preferiti dello spazio.

$$\phi'(x') \overset{!}{=} \phi(x) \qquad \text{ipotesi}$$

where $x' = x + a$ traslazione

Imponendo la condizione di scalare si dimostra che la traslazione agisce nel seguente modo

$$\phi(x') \overset{!}{=} e^{-ia_\mu P^\mu} \phi(x) \qquad \text{ipotesi}$$

$$\phi'(x) = e^{+ia_\mu P^\mu} \phi(x) \qquad \text{conseguenza}$$

where $P^\mu = i\partial^\mu$ conseguenza

DIMOSTRAZIONE

Voglio dim che $P^\mu = i\partial^\mu$ dall'ipotesi di ϕ scalare e $\phi(x') = e^{-ia_\mu P^\mu} \phi(x)$ come ordine.

a) Considero una variazione infinitesima traslazione $\varepsilon_\mu \to 0$ tic $x' = x + \varepsilon$

$$\delta\phi = \phi'(x) - \phi(x) =$$
$$= \phi'(x' - \varepsilon) - \phi(x) = \quad \text{Taylor 1° ordine}$$
$$= \phi'(x) - \varepsilon_\mu \partial^\mu \phi(x) - \phi(x) = \quad \text{ipotesi scalare } \phi'(x') \overset{!}{=} \phi(x)$$
$$= -\varepsilon_\mu \partial^\mu \phi(x) \quad (1)$$

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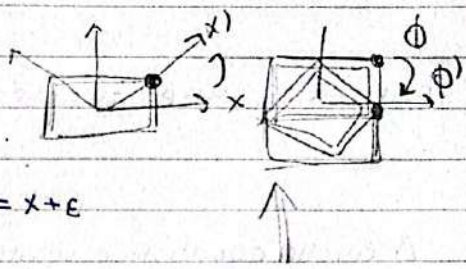
•) Per come abbiamo def il generatore vale !



$$\phi(x') = e^{-i\varepsilon_\mu p^\mu}\,\phi(x) \quad \rightarrow \quad x' = x+\varepsilon$$

$$\phi(x+\varepsilon) = e^{-i\varepsilon_\mu p^\mu}\,\phi(x)$$

Per cui se $\phi = \phi'$ ed $\varepsilon \to -\varepsilon$ è: → perché è uno scalare per cui $\phi \to \phi'$ e $x \to x'$ ognuno dall'opposto

$$\phi'(x'-\varepsilon) = e^{-i(-\varepsilon_\mu)p^\mu}\,\phi'(x') = e^{i\varepsilon_\mu p^\mu}\,\phi(x) \qquad (*)$$

Ma allora

$$\delta\phi = \phi'(x) - \phi(x) = \phi'(x'-\varepsilon) - \phi(x) \Rightarrow (*)$$

$$= e^{i\varepsilon_\mu p^\mu}\,\phi(x) - \phi(x) = \Rightarrow \text{infinitesimo}$$

$$= \phi(x) + i\varepsilon_\mu p^\mu \phi(x) - \phi(x)$$

$$= i\varepsilon_\mu p^\mu \phi(x) \qquad (2)$$

Ma allora confrontando ① e ② otteniamo:

$p^\mu = i\partial^\mu$

$$p^0 = i\frac{\partial}{\partial t} =: H \qquad \text{hamiltoniano}$$

$$p^j = i\partial^j = -i\partial_j = -i\frac{\partial}{\partial x^j} \qquad \text{spaziale momentum}$$

NB ! Qui il punto cruciale diverso da Lorentz = in Lorentz avevo definito il generatore con l'azione sulle ϕ cioè

$$\phi'(x) = e^{-\frac{i}{2}\omega_{\mu\nu}S^{\mu\nu}}\,\phi(x)$$

ora invece il gen e' ho def o portare dall'azione sulle x dentro ϕ ! Ed infatti è così

$$\phi(x) = e^{-i\varepsilon_\mu p^\mu}\,\phi(x)$$

NB = Per campi scalari, qualsiasi trasformazione vista sulle ϕ è l'opposto di quella agente sulle coord x !!

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$p^\mu = i\partial^\mu$ → $p^0 = i\frac{\partial}{\partial t} =: H$ hamiltonian

→ $p^j = i\partial^j = -i\partial_j = -i\frac{\partial}{\partial x^j}$ spatial momentum

NB ! Qui il punto cruciale dunque da Lorentz = in Lorentz ovvero definito il generatore con l'azione sulle ϕ cioè

$$\phi'(x) = e^{-\frac{i}{2}\omega_{\mu\nu}S^{\mu\nu}}\phi(x)$$

ora invece il gen e'ho def a partire dall'azione sulle x dentro ϕ ! Ed infatti e'dunque

$$\phi(x') = e^{-i\varepsilon_\mu P^\mu}\phi(x)$$

NB = per campi scalari, qualsiasi trasformazione vista sulle ϕ è l'opposto di quella agente sulle coord x !!

Lorentz

$$\phi(x') = e^{-\frac{i}{2}(-\omega_{\mu\nu})L^{\mu\nu}}\phi(x)$$ sulle x con $-\omega_{\mu\nu}$

$$\phi'(x) = e^{-\frac{i}{2}(\omega_{\mu\nu})L^{\mu\nu}}\phi(x)$$ sulle ϕ con $+\omega_{\mu\nu}$

TRANSLATION

$$\phi(x') = e^{-i\varepsilon_\mu P^\mu}\phi(x)$$ sulle x con $+\varepsilon_\mu$

$$\phi'(x) = e^{-i(-\varepsilon_\mu)P^\mu}\phi(x)$$ sulle ϕ con $-\varepsilon_\mu$

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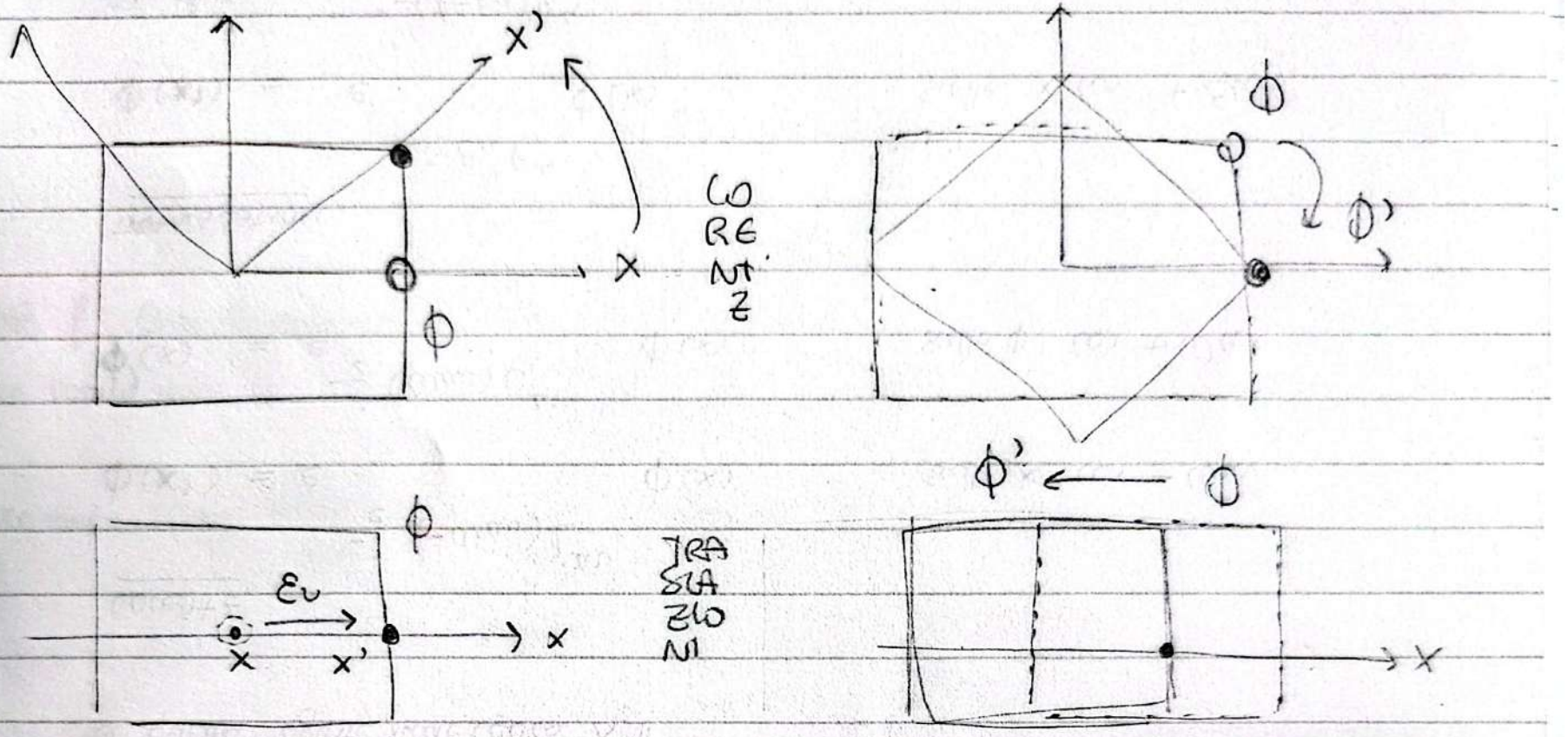

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POINCARÉ REPRESENTATION ON ONE-PARTICLE STATES

A livello quantistico a intreemocá studiare le particelle. Vedremo come una particella sarà l'eccitazione di un campo $\phi(x)$. Per ora iniziamo a studiare come il gruppo di Poincare viene rappresentato su un modo vettoriale sotto dello spazio di Hilbert e come base scelgo quelle dei ket di particelle LIBERE

$$|\vec{p}, s\rangle \in \mathcal{H} \qquad \{|\vec{p}, s\rangle\} \text{ basis of } \mathcal{H}$$

momento della particella — all other quantum numbers of the particle

L'algebra di Poincaré è l'algebra dello spazio VUOTO (quale spazio? Minkowski 4D) quindi descrive le proprietà di simmetria delle particelle in uno spazio vuoto (cioè senza interazioni, particella isolata). Una particella isolata ha due quantità conservate = massa e spin

TEOREMA DI WIGNER

Any symetry trasformation on an Hilbert space can be written in terms of unitary operators (NB = è possibile in realtà anche anti-unitary operator, the only symmetry which is anti-unitary is time reversal)

Di conseguenza, essendo i rappresentanti delle simmetrie i GENERATORI, the Generators of Poincare Group (H, P^i, J^i, K^i) are HERMITIAN OPERATORS so that the trasform Matrices exp(generators) are UNITARY MARRICES

Generators $H\ J^i\ P^i\ K^i$ ⟶ Hermitian

Finite trasformation: $\exp(H, J, K, P)$ ⟶ unitary

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Invariante di Casimir

Given a Lie Algebra, a Casimir invariant is an operators that commutes with all the generators of the algebra

Siccome l'algebra di Poincaré è quello delle particelle libere in uno spazio vuoto, che conservano massa, momo, spin, gli invarianti di Casimir dell'algebra di Poincaré' saranno legati alla massa e allo spin.

MASS INVARIANT

- $P_\mu P^\mu = m^2$ mass-square of particle

$$[m^2, P^\nu] = 0 \qquad [P_\mu P^\mu, J^i) = [P_\mu P^\mu, K^i] = 0$$

SPIN INVARIANT

We define the Pauli-Lubanski Vector =

$$W^\mu = -\frac{1}{2} \varepsilon^{\mu\nu\rho\sigma} J_{\nu\rho} P_\sigma$$

which generates the casimir invariant:

- $W_\mu W^\mu = W^2$

Additionally W^μ itself commutes with P^ν

$$[W^\mu, P^\nu] = 0 \qquad \forall_{\mu,\nu}$$

We can distinguish two types of particles = massive and non-massive

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MASSIVE representation $m^2 \neq 0$

Consider the case of a massive particle with $m^2 \neq 0$. We can always decide to calculate W^2 and m^2 in the REST FRAME of the particle since W^2 and m^2 are Lorentz invariant and each frame is equivalent (= yields the same value)

REST FRAME $p^\mu = (m, \vec{0})$ since $\vec{v} = 0$ velocity of particle

so in the rest frame

$$W^\mu = -\frac{1}{2} \varepsilon^{\mu\nu\rho 0} J_{\nu\rho} m = \frac{m}{2} \varepsilon^{0\mu\nu\rho} J_{\nu\rho}$$

- $W^0 = 0$
- $W^i = \frac{m}{2} \varepsilon^{0ijk} J_{jk} = m J^i$ zero angular momentum

true only in the rest frame case W^μ is not a Lorentz invariant !

So the Casimir invariant

$$W_\mu W^\mu = -m^2 \vec{J}^2 = -m^2 j(j+1)$$

on the eigenstates of $\vec{J}^2$

Always true in any frame - for a particle of mass m and spin j

We can show that the label j can be $j = 0, \frac{1}{2}, 1, \frac{3}{2} \dots$ with $j_z = -j, -j+1 \dots 0 \dots j$

Hence in QFT a massive particle is labeled by m and j and each representation j contains $2j+1$ degrees of freedom (j_z) (or $2j+2$ considering also the mass as a free degree)

Massless representation $m^2 = 0$

For a massless particle the definition of p^μ as $p^\mu = (m, m\gamma\vec{v})$ is no more valid and the rest frame doesn't exist

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we can write p^μ in the form:

$$p^\mu = (\omega, 0, 0, \omega) \qquad (t\ x\ y\ z)$$

this vector is invariant under rotation in xy plane hence under rotations around the z axis so under the rotations generated by J^3.

Massless particle can then be labeled by HELICITY which is the EIGENVALUE OF J^3 and can assume the values (Helicity is $\vec{J}$ projected in the direction of spatial $\vec{p}$ momentum)

$h = 0, \pm 1, \pm 2 \ldots$

Massless particles can be characterized by 1 degree of freedom: h.

Helicity → $h = \hat{p} \cdot \vec{J}$ is a pseudovector

under parity $h \xrightarrow{P} -h$ (parity inverts $\vec{p} \to -\vec{p}$ other $h \to -h$)
$\vec{J} \to +\vec{J}$

So only for parity conserving systems (example = electromagnetic forces) we can treat $h = -1$ & $h = +1$ the same! → the photon is part of electromagnetism so $h = +1$ and $h = -1$ is the same (left and right handed chirality states)

Se $m = 0$ non esiste il rest frame perché la particella sta viaggiando alla velocità della luce. We choose $p^\mu = (\omega\ 0\ 0\ \omega)$ so that

$W^0 = W^3 = 0$
$W^1 = \omega (J^1 - K^2)$
$W^2 = \omega (J^2 + K^1)$

so $W_\mu W^\mu = \omega^2 \left((K^2 - J^1)^2 + (K^1 + J^2)^2 \right)$

| Always true in any Frame cause W^2 is a Lorentz invariant

Noto che se faccio il limite di $W_\mu W^\mu$ per $m \to 0$ del caso massivo NON FINISCO MAI in $W_\mu W^\mu$ del caso non massivo! Perché $W_\mu W^\mu = m^2(-s(s+1)) \to 0$ mentre $W_\mu W^\mu = \omega^2(\ldots) \neq 0$!! Questo dice che oggetti massivi non potremo mai raggiungere la velocità della luce = oggetti massivi e non massivi sono diversi e sono da trattare diversamente

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3 Maggiore

3.1 3.2 3.3.1

CLASSICAL FIELD THEORY

3.1 MINIMUM ACTION PRINCIPLE

PRINCIPIO DI MINIMA AZIONE – CASO DISCRETO

Consider a system with N degrees of freedom $i = 1, 2, \dots N$ described by coordinates $q_i(t)$ (each coord can be de position of a particle or the position and momentum of the same particle, for example) We define the velocity coordinates as $\dot{q}_i(t) = \frac{\partial}{\partial t} q_i(t)$

Lagrangian:

$$L := L(\{q_i\}, \{\dot{q}_i\}) = \sum_{i=1}^{N} L_i(q_i(t), \dot{q}_i(t))$$

example = For N different particles is $L = \sum_{i=1}^{N} \frac{m_i}{2} \dot{q}_i^2 - V(q_i)$

Action =

$$S := \int_{t_A}^{t_B} dt\, L(\{q_i\}, \{\dot{q}_i\})$$

Principio di minima azione

Particles evolve in trajectories in coordinate space for which S is an extremum, usually minimum, with fixed coordinate of boundaries

$$\delta S \doteq 0 \qquad \text{with} \qquad \delta q(t_A) = \delta q(t_B) \doteq 0$$

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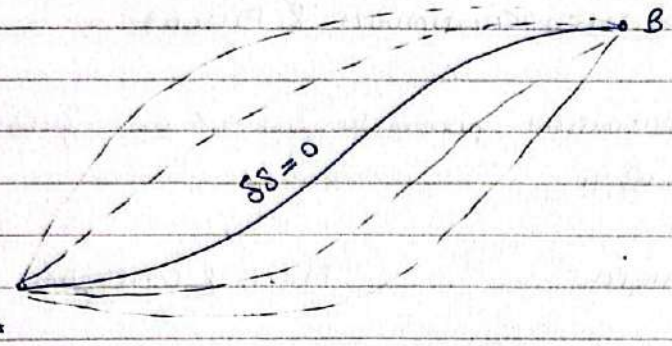

Euler Equations

$$\delta S = \delta\left(\int dt\, L(q_i, \dot{q}_i)\right) = \int dt\, \delta L =$$

$$= \int_{t_a}^{t_b} dt \sum_i \left(\frac{\partial L}{\partial q_i}\delta q_i + \frac{\partial L}{\partial \dot{q}_i}\delta \dot{q}_i\right) = \begin{array}{l}\text{in virtue of the derivative}\\ \text{as } \frac{d}{dt} \text{ commute with } \delta\end{array}$$

$$\therefore = \delta\frac{\partial q}{\partial t} = \frac{\partial}{\partial t}\delta q$$

$$= \int_{t_a}^{t_b} dt \sum_i \left(\frac{\partial L}{\partial q_i}\delta q_i + \frac{\partial L}{\partial \dot{q}_i}\frac{\partial}{\partial t}\delta q_i\right) \qquad (3.1)$$

Integrate by parts (= IBP) and thanks to the fixed boundaries $\delta q_a = \delta q_b \overset{!}{=} 0$ we can rewrite that such as

$$= \int dt \sum_i \left(\frac{\partial L}{\partial q_i} - \frac{\partial}{\partial t}\frac{\partial L}{\partial \dot{q}_i}\right)\delta q_i \overset{!}{=} 0$$

This must be $\overset{!}{=} 0$ for every δq_i variation so we get

$$\frac{\partial L}{\partial q_i} - \frac{\partial}{\partial t}\frac{\partial L}{\partial \dot{q}_i} = 0 \qquad \text{Euler Lagrange equations} \qquad (3.2)$$

Hamiltonian Formalism

$$p_i := \frac{\partial L}{\partial \dot{q}_i} \qquad \text{conjugate momentum}$$

$$H := \sum_i p_i \dot{q}_i - L \qquad \text{Hamiltonian}$$

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GENERALIZATION INTO CONTINUUM & FIELDS

We now use continuum parameters $i \to x$ and fields $\phi(x)$ as coordinates instead of q_i

DISCRETO		FIELD & CONTINUUM
i	$\longrightarrow$	x^μ
q_i	$\longrightarrow$	$\phi(x)$
$\dot{q}_i$	$\longrightarrow$	$\partial_\mu \phi(x)$
(just ∂_t)		all $\partial_{x^\mu} = (\partial_{x^0} \partial_x \partial_y \partial_z)$
$\sum_i$	$\longrightarrow$	$\int dx^3$ spatial

Lagrangian

$$L = \int d^3x \; \mathcal{L}(\phi, \partial_\mu \phi)$$

$\int d^3x$: just space; $\mathcal{L}$: Lagrangian density

Action

$$S = \int d^4x \; \mathcal{L}(\phi, \partial_\mu \phi)$$

$\int$: integration over all 4 coordinates
We now treat space time EQUALLY ♥♥ ‖

Minimum Action

$\delta S \stackrel{!}{=} 0$ varying over $\delta\phi$ with condition that fields are constant at boundaries ∂V of the volume V of integration $\int_V dx$

$\delta\phi \big|_{\partial V} \stackrel{!}{=} 0$

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Euler Lagrange equations

Repeating the procedure we get

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = 0$$

- no one stops us from defining more fields ϕ^α instead of just one ϕ

$$i \longrightarrow x, \alpha$$

(x: continuum, α: discrete)

$$q_i \longrightarrow \phi_\alpha(x)$$

so

$$L = \int dx^3 \, \mathcal{L}(\{\phi_\alpha\}, \{\dot{\phi}_\alpha\}) \qquad S = \int dx^4 \, \mathcal{L}(\dots)$$

$$\frac{\partial \mathcal{L}}{\partial \phi_\alpha} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_\alpha)} = 0$$

Euler Lagrange for a enim fields ϕ_α

TOTAL DIVERGENCE INVARIANCE

Considering a new Lagrangian $\mathcal{L}'$ that differs from the first $\mathcal{L}$ by a total divergence of a vector K^μ, the resulting dynamic i.e Euler Lagrange equations are invariant. Let's demonstrate that:

$$\mathcal{L}' = \mathcal{L} + \partial_\mu K^\mu$$

Totoe divergence of K^μ $\quad \partial_\mu K^\mu = \frac{\partial K^0}{\partial t} - \vec{\nabla}\vec{K}$

- Integrating the action we have

$$S' = \int_V dx^4 \left(\mathcal{L} + \partial_\mu K^\mu\right) = \int_V dx^4 \mathcal{L} + \int_V dx^4 \, \partial_\mu K^\mu$$

- $\int_V dx^4 \mathcal{L} = S$ action previous

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- thanks to Stokes theorem

$$\int_V d^4x\, \partial_\mu K^\mu = \int_{\partial V} d\Sigma\, n_\mu K^\mu$$

∂V: surface of V; $d\Sigma$: element of sup; n_μ: versore della sup in 4D

- but we imposed that the Fields are CONSTANT on the boundaries ∂V so also K^μ is constant so the term is CONSTANT, does not vary under δ variation so:

$$S' = S + \int_{\partial V} d\Sigma\, n_\mu K^\mu$$

S: ↓, S': ↓, $\int_{\partial V} d\Sigma\, n_\mu K^\mu$: ↓ constant

$$\delta S' = \delta S + \delta\left(\int_{\partial V} d\Sigma\, n_\mu K^\mu\right) = \delta S$$

$\delta\left(\int_{\partial V} d\Sigma\, n_\mu K^\mu\right) = 0$

$\Rightarrow \delta S' = \delta S$ so the Euler Lagrange equation of motion are the same, $\mathcal{L}'$ and $\mathcal{L}$ are physically equivalent states

HAMILTONIAN FORMALISM

Conjugate momentum $\Pi_\alpha(x) = \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi_\alpha)}$ ↳ time coordinate

Hamiltonian density $\mathcal{H}(x) = \sum_\alpha \Pi_\alpha(x)\, \partial_0 \phi_\alpha(x) - \mathcal{L}$

Hamiltonian $H = \int d^3x\, \mathcal{H}(x)$

In the Hamiltonian formalism the Lorentz invariance is less obvious : it seems that space and time are treated differently. IT IS NOT TRUE ! But we prefer to work in the Lagrangian formalism because the symmetry is more evident

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3.2 NOETHER THEOREM (CLASSICAL)

Let's consider some fields ϕ_i ; completely general (scalar, vector...) and a completely generic change in the coordinates $x^\mu \to x'^\mu$

INFINITESIMAL CHANGE IN THE COORDINATES

$$x'^\mu = x^\mu + \delta x^\mu = x^\mu + \varepsilon^a A^\mu_a(x) \qquad \text{with } \varepsilon^a \to 0$$

$$\boxed{\delta x^\mu = \varepsilon^a A^\mu_a(x)} \quad (3.4)$$ where $\mu = 0,1,2,3$ recall 4-dim indeces

a = arbitrary index that runs over the number of generators of the transformation

↓

(Scelgo di rappresentare così la variazione in modo del tutto arbitrario. Per ora a è un indice ARBITRARIO che scorre su parametri arbitrari. Poi poi saranno il n° di Generatori della trasf di coord.)

INFINITESIMAL CHANGE IN THE FIELDS

$\Phi'(x') - \Phi(x)$

$$\delta\phi_i(x') := \phi'_i(x') - \phi_i(x')$$ change in ϕ, fixed x' or x, same

$$\phi'_i(x') - \phi_i(x) = \varepsilon^a F_{i,a}(\phi_i, \partial\phi_i)$$ change in ϕ and in x both

↓

Anche qui scelgo di rappresentare arbitrariamente la variazione congiunta del campo $\phi_i \to \phi'_i$ sotto la variazione $x \to x'$ con una funzione $F_{i,a}$ dove a scorre sui generatori.

$$\begin{aligned}\phi'_i(x') &= \phi_i(x) + \varepsilon^a F_{i,a}(\phi_i, \partial\phi_i) \quad \swarrow\ x = x' - \delta x \\ &= \phi_i(x' - \delta x) + \varepsilon^a F_{i,a}(\phi_i, \partial\phi_i) = \\ &= \phi_i(x') - \partial_\mu\phi_i\, \partial x^\mu + \varepsilon^a F_{i,a}(\cdots) \quad \swarrow \text{usando } \delta x^\mu = \varepsilon^a A^\mu_a\end{aligned}$$

cioè

$$\boxed{\delta\phi_i = \phi'_i(x') - \phi_i(x') = \varepsilon^a \left(F_{i,a} - A^\mu_a \partial_\mu \phi_i\right)} \quad (3.5)$$

dove i = indice del campo , a = scorre sui generatori , μ = indice coordinato spaziotempo ovvio

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SYMMETRY TRANSFORMATION

Given a system described by $\mathcal{L}$ Lagrangian, a SYMMETRY is a TRANSFORMATION

$$x \to x + \delta x$$
$$\phi \to \phi + \delta\phi$$

such that $\mathcal{L}$ changes by (at most) a total divergence, i.e. the physics (Euler Lagrange equation) of the system is the same:

$$\mathcal{L}'(x', \phi', \partial\phi') - \mathcal{L}(x, \phi, \partial\phi) =^{!} \partial_\mu f^\mu \quad (3.6)$$

$$\delta S =^{!} \delta S' \qquad \delta S = dS' - dS =^{!} d^4x\, \partial_\mu f^\mu$$

NOTE = We will consider Lagrangian $\mathcal{L}(\phi, \partial\phi)$ depending directly from just ϕ and $\partial\phi$ and NOT from x, for the next calculations of the conserved current. This is just a little trick to semplify calculations, at the end the current j^μ we obtain is the same of the one we could have obtained considering the whole dependence $\mathcal{L}(x, \phi, \partial\phi)$

NOTE = $\mathcal{L}(\phi, \partial\phi)$ è funzione esplicita di solo ϕ e $\partial\phi$ ma dip. comp. anche da x perché $\phi(x)$ dip da x

CONSERVED CURRENT

- Given a Lagrangian $\mathcal{L}$ we have an action $S = \int d^4x\, \mathcal{L}$. The element of action is

$$dS = d^4x\, \mathcal{L}$$

- Let's consider the variation of the action under a symmetry transformation $x \to x + \delta x = x' \quad \phi + \delta\phi = \phi'$. Both d^4x and $\mathcal{L}$ vary:

$$\delta(dS) = \delta(d^4x)\mathcal{L} + d^4x\,(\delta\mathcal{L})$$

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In 1 dim: $\delta(dx) = dx' - dx = \frac{\partial x'}{\partial x} dx - dx = \frac{\partial x'}{\partial x} dx - dx = dx \frac{\partial}{\partial x} \delta x$

$\Rightarrow dx' = \frac{\partial x'}{\partial x} dx$

$\Rightarrow \frac{\partial x'}{\partial x} = 1 + \frac{\partial}{\partial x} \delta x$

$\downarrow$

$x' = x + \delta x$

In 4 dim: $\delta(d^4x) = d^4x \, \partial_\mu \delta x^\mu$

$\delta\mathcal{L} = \delta(\mathcal{L}(\phi, \partial\phi)) = \frac{\partial \mathcal{L}}{\partial \phi} \delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial\phi)} \delta(\partial\phi)$ → Hypothesis of $\mathcal{L}(\phi, \partial\phi)$ instead of more general $\mathcal{L}(x, \phi, \partial\phi)$ but the result will be the same as if $\mathcal{L}(x, \phi, \partial\phi)$

If more fields ϕ_i $i = 1 \dots N$ fields and $\partial_\mu \phi_i$ 4Dim derivation

$\delta\mathcal{L} = \frac{\partial \mathcal{L}}{\partial \phi_i} \delta\phi_i + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \delta(\partial_\mu \phi_i)$ → with implicit sum over i and μ

but now $\delta(\partial_\mu \phi_i) = \partial_\mu(\delta\phi_i)$ because δ and ∂_μ in that case commute because we were considering $\partial_\mu \phi$ already computed, nel senso che $\partial_\mu \phi$ è un oggetto già fatto e finito e sto variando $\delta(\partial_\mu \phi)$, quindi δ influenza solo ϕ, non sto variando anche come varia ∂_μ su ϕ!

$\delta\mathcal{L} = \frac{\partial \mathcal{L}}{\partial \phi_i} \delta\phi_i + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \partial_\mu(\delta\phi_i) =$ sum over $i = 1 \dots N$ n. of fields, $\mu = 0$ to 4D

$= \frac{\partial \mathcal{L}}{\partial \phi_i} \delta\phi_i + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \delta\phi_i \right) - \delta\phi_i \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \right) =$

$= \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \delta\phi_i \right) + \delta\phi_i \left(\frac{\partial \mathcal{L}}{\partial \phi_i} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \right) =$

$= \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \delta\phi_i \right)$

$= 0$ if ϕ follows Euler Cosgrge equations

$\delta(dS) = d^4x \, \partial_\mu(\delta x^\mu) \mathcal{L} + d^4x \, \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \delta\phi_i \right)$

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Since for our hypotheses $\mathcal{L}(\phi\, \partial\phi)$ does not dep directly on x, so that $\partial_\mu \mathcal{L} = 0$ we can write $\partial_\mu(\delta x^\mu)\mathcal{L} = \partial_\mu(\partial x^\mu \mathcal{L})$, conseq $\mathcal{L}$ constant under ∂_μ, so:

$$\delta(dS) = dx^4 \partial_\mu \left(\delta x^\mu \mathcal{L} + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \delta\phi_i \right)$$

- Now we impose to have a symmetry trasf so that $\mathcal{L}' - \mathcal{L} \overset{!}{=} \partial_\mu f^\mu$, that is the same of requiring $\delta dS \overset{!}{=} d^4x\, \partial_\mu f^\mu$ (with $\tilde{f}$ slightly different from f) because

$$dx^4 \partial_\mu \left(\delta x^\mu \mathcal{L} + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \delta\phi_i \right) \overset{!}{=} dx^4 \partial_\mu f^\mu$$

↳ symmetry transformation

so we got to define:

$$\partial_\mu j^\mu = 0 \qquad 3.9$$

with $$j^\mu = -\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \delta\phi_i - \mathcal{L}\, \delta x^\mu + f^\mu \qquad 3.10$$

NOETHER THEOREM

For each generator $a = 1 \dots N$ of a continuous Symmetry $x \to x'$ $\phi \to \phi'$ there is a CONSERVED CURRENT j_a

$$j^\mu_a = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \left(A^\mu_a \partial_\nu \phi_i - F_{i,a} \right) - \mathcal{L} A^\mu_a + f^\mu \qquad (3.11)$$

↓ implicit sum over all i Fields ϕ_i

$$\partial_\mu j^\mu_a = 0 \qquad \forall a = 1 \dots N \text{ n° of generators of the symmetry}$$

$$\cdot J^\mu_a = \frac{j^\mu}{\varepsilon^a}$$

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$$S = \int d^4x \mathcal{L} = \int dS \rightarrow dS = d^4x \mathcal{L}$$

$$S' = \int dx'^4 \mathcal{L}' = \int dx^4 \mathcal{L} + \int dx^4 \partial_\mu f^\mu \rightarrow dS' = dS + d^4x \partial_\mu f^\mu$$

hence $\delta dS = dS' - dS = d^4x \partial_\mu f^\mu$

- Now we impose t
same of requir

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DIMOSTRAZIONE

Per ricavare la 3.10 ovvero chiarire solamente che la trasf forma una trasf di simmetria per cui posso partire dalla 3.10

$$j^\mu = -\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \delta\phi_i - \mathcal{L}\,\delta x^\mu + F^\mu$$

ora sostituisco la $\delta x^\mu = \varepsilon^a A^\mu_a(x)$ (3.4)
e la equazione (3.5) $\delta\phi_i = \varepsilon^a (F_{i,a} - A^\mu_a \partial_\nu \phi_i)$

per cui

$$j^\mu = \varepsilon^a \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} (A^\nu_a \partial_\nu \phi_i - F_{i,a}) - \varepsilon^a \mathcal{L} A^\mu_a + f^\mu$$

raccolgo su ε^a (dividendo su ε^a, tanto j^μ è costante e la posso raccogliere per un fattore moltiplicativo globale che non cambia la proprietà di conservazione)

$$j^\mu_a = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} (A^\nu_a \partial_\nu \phi_i - F_{i,a}) - \mathcal{L} A^\mu_a + \mathcal{f}^\mu \qquad \text{CVD!!}$$

$\mathcal{f}^\mu = \frac{f^\mu}{\varepsilon^a}$, riscalato, tanto è solo una quantità arbitraria

dalla 3.5 segue:

$$\partial_\mu j^\mu_a = 0$$

CVD!!

CONSERVED CHARGE

Associated to a conserved current there is a conserved charge

$$Q_a := \int d^3x\, j^0_a(x)$$ (integrale spaziale $x_1 x_2 x_3$)

$\forall a$ generator

that satisfies $\dot{Q}_a = \partial_0 Q_a = 0$ doesn't change in time

DIM: $\partial_0 Q = \int d^3x\, \partial_0 j^0(x) = \int_{Vol} d^3x\, (-\partial_i j^i)(x) = \int_{\partial V} \vec{j}\cdot d\vec{s} = 0$

$\partial_\mu j^\mu = \partial_0 j^0 + \partial_i j^i = 0$

$\partial_\mu j^\mu = \partial_0 j^0 - \vec{\nabla}\vec{j} = 0$

$\vec{\nabla}\cdot\vec{j}$ → Stokes theorem

condizioni al contorno, campo nullo sul bordo ∂V quindi j nullo sul bordo

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3.2.1 Examples of symmetries

Internal symmetry

The first basic symmetry that we analyze is the one associated to $\delta x^\mu = 0$ ie $A^\mu{}_a = 0$ i.e the coordinate don't change

$x \longrightarrow x' = x$ $\quad A^\mu{}_a = \emptyset$ IDENTITY

In that easy case, setting $f^\mu \neq 0$ we obtain

$j^\mu = -\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \delta\phi_i = -\frac{\partial \mathcal{L}}{\partial(\partial\phi)} F_a$ conserved charge current

↳ sum over i

This symmetry is internal in the sense that depends only on ϕ_i the fields, it is related only to the properties of the fields ϕ_i considered. Since we are not changing coordinates x, is a symmetry internal, inside ϕ_i.

(example = when ϕ is complex, is related with the phase $e^{i\theta}$ of the field)

Space-time translation symmetry

a) Coordinate translation

$x^\mu \to x'^\mu = x^\mu + \varepsilon^\mu = x^\mu + \delta^\mu{}_\nu \varepsilon^\nu$ TRANSLATION

(delta di Kronecker)

so $A^\mu{}_a = A^\mu{}_\nu = \delta^\mu{}_\nu$ with $a = \nu$ the index of generators is the same of $\nu = 0123$ spacetime 4D

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o) FIELDS transformation

ALL FIELDS ARE SCALAR UNDER TRANSLATION so we have

$\phi_i'(x) \longrightarrow \phi_i'(x') \overset{!}{=} \phi_i(x)$ SCALAR

so $F_{i,a} = 0$

o) CONSERVED CURRENT

Scrivo la 3.11 per la carica conservata è solo che ora la chiamo T perché storicamente è il tensore degli sforzi

$$3.11 \rightarrow J^\mu_a =: T^\mu_\nu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \partial_\nu \phi_i - \delta^\mu_\nu \mathcal{L}$$

$a \rightarrow \nu$

$J =: T$

so

$$T^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \partial^\nu \phi_i - g^{\mu\nu} \mathcal{L} \qquad (3.14)$$

ENERGY MOMENTUM TENSOR

↓ implicit sum over i number of Fields ϕ_i

$$\partial_\mu T^{\mu\nu} = 0 \qquad (3.15)$$

CONSERVATION OF $T^{\mu\nu}$ IN THE FIRST μ INDEX

Un sistema isolato è un sistema che conserva il tensore degli sforzi = questa relazione mi dice che un sistema che ha simmetria traslazionale nello spaziotempo è sempre un sistema isolato = in qualunque punto dello spazio di un sistema isolato la fisica è uguale ($\partial_\mu T^{\mu\nu} = 0$ isolato ; $\partial_\mu T^{\mu\nu} = F^\nu$ detto forza, non isolato)

Questa è un'approssimazione, solo l'intero universo è veramente isolato, i campi ϕ in teoria si estendono all'∞, quindi considerarli isolati è un'approssimazione

$T^{\mu\nu}$ NON È SIMMETRICO! $\partial_\mu T^{\mu\nu} \neq \partial_\nu T^{\mu\nu}$ perché $T^{\mu\nu} \neq T^{\nu\mu}$

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•) CONSERVED CHARGE

The conserved charge related to spacetime translation symmetry is the 4-MOMENTUM

$$P^\nu = \int d^3x\, T^{0\nu} \qquad (3.16) \qquad \partial_0 P^\nu = 0$$

ENERGY

$$P^0 =: H = \int d^3x\, T^{00} = \int d^3x\, \mathcal{H} \qquad (3.17)$$

H → Hamiltonian; $\mathcal{H}$ → Hamiltonian density

3-MOMENTUM

$$P^j = \int d^3x\, T^{0j} = -\int d^3x \sum_i \Pi_i\, \partial^j \phi_i \qquad (3.18)$$

(j: index $j = 1,2,3$ of spatial coordinates)

$\sum_i$: sum over all fields $i = 1 \dots n$

Π_j: conjugate momentum $\Pi_j = \dfrac{\partial \mathcal{L}}{\partial(\partial_0 \phi_j)}$

if only one field ϕ

$$P^j = -\int d^3x\, \Pi\, \partial^j \phi \qquad \text{with } \Pi = \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi)}$$

SYMMETRIZATON $T^{\mu\nu} \to \tilde{T}^{\mu\nu}$ $P^\nu \to \tilde{P}^\nu$

$T^{\mu\nu}$ is not symmetric in $\mu \leftrightarrow \nu$, we want a symmetric Tensor $\tilde{T}^{\mu\nu}$, we can write it in the form:

$$\tilde{T}^{\mu\nu} = T^{\mu\nu} + \partial_\rho A^{\rho\mu\nu} \qquad \text{SYMMETRIC}$$

with $A^{\rho\mu\nu}$ tensor antisymmetric in $\rho \leftrightarrow \mu$, defined arbitrarily in order to satisfy $\tilde{T}^{\mu\nu} \overset{!}{=} \tilde{T}^{\nu\mu}$ (RICORDO $A^{\rho\mu\nu}$ impongo la simmetria su $\tilde{T}$)

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this new symmetric tensor satisfies the same conservation rule of $T^{\mu\nu}$, in fact

$$\partial_\mu \tilde{T}^{\mu\nu} = \partial_\mu T^{\mu\nu} + \partial_\mu \partial_\rho A^{\rho\mu\nu} = 0$$

$\partial_\mu T^{\mu\nu} = 0$ because of 3.15

$\partial_\mu \partial_\rho A^{\rho\mu\nu}$: sym, antisym

We can also define the related charge

$$\tilde{P}^\nu = P^\nu + \int d^3x\, \partial_\rho A^{\rho 0\nu} = \int d^3x\, \tilde{T}^{0\nu}$$

that satisfies the same conservation rule of P^ν in fact:

$$\partial_0 \tilde{P}^\nu = \partial_0 P^\nu + \int d^3x\, \partial_0 \partial_\rho A^{\rho 0\nu} = 0$$

$\partial_0 P^\nu = 0$

$\partial_0 \partial_\rho A^{\rho 0\nu}$: sym nop, anti nop

$\Rightarrow$ Thus $\tilde{T}^{\mu\nu}$ and $\tilde{P}^\nu$ are physically equivalent to $T^{\mu\nu}$ and P^ν

3.3 KLEIN-GORDON EQUATIONS

3.3.1 SCALAR REAL FIELD

Le eq di Klein gordon sono le eq di Eulero-Lagrange che descrivono una particella (o meglio un CAMPO) scalare, con massa m, e spin nullo. (Es = Bosone di Higgs) e LIBERE cioè senza interazione esterne o interne.

IPOTESI = $\phi(x)$ Reale e Scalare $\phi \overset{!}{=} \phi^*$

LAGRANGIANA: $$\mathcal{L} \overset{!}{=} \frac{1}{2}(\partial_\mu \phi \partial^\mu \phi - m^2 \phi^2) \qquad (3.17)$$

termine cinetico, termine di massa

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Abbiamo quantizzato un campo massivo scalare reale a spin nullo e libero, con solo cinetico e massa nella Lagrangiana e nessuna interazione. Risolvendo le eq di Eulero Lagrange per tale L ottengo le cosiddette Equazioni di Klein Gordon:

$$(\partial_\mu \partial^\mu + m^2)\,\phi = 0$$

3.118 EQ DI KLEIN-GORDON
(Free massive spinless field)

Tali equazioni sono eq delle onde infatti $\partial_\mu \partial^\mu = \frac{\partial^2}{\partial t^2} - \vec{\nabla}^2$ op di d'Alambert. Le soluzioni di tale eq sono ONDE PIANE della forma elementare:

$$\phi(x) = e^{\mp i\, p\cdot x}$$

PLANE-WAVE

$$\downarrow$$

where $p\cdot x = p^\mu x_\mu$ $\qquad p^2 = p^\mu p_\mu = m^2$

La soluzione più generale è quindi una sovrapposizione combinazione di onde piane $+$ e $-$, dove i coeff di $e^{+i\dots}$ e $e^{-i\dots}$ devono essere uno il complesso coniugato dell'altra affinché $\phi = \phi^*$ sia reale come quantizzato:

$$\phi(x) = \int \frac{d^3\vec{p}}{(2\pi)^3}\,\frac{1}{\sqrt{E_{\vec{p}}\,2}}\left(a_{(\vec{p})}\, e^{-i p\cdot x} + a^*_{(\vec{p})}\, e^{i p\cdot x}\right)\Big|_{p^0 = E_{\vec{p}}}$$

- x^μ quadrivettore
- integrale su $\vec{p}$ spaziale
- dep da $\vec{p}$ spaziale
- prodotto scalare tra $p^\mu x_\mu$

dove $E^2_{(\vec{p})} = \vec{p}^{\,2} + m^2$

OSSERVAZIONI

Se butto nell'eq di KG una singola onda piana $\phi = e^{\mp i p\cdot x}$ ottengo la relazione di dispersione:

cinetico ↑ massa ↑

$$E^2 = \vec{p}^{\,2} + m^2 \qquad \left(\text{in unità normali } E^2 = p^2c^2 + m^2c^4\right)$$

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L'eq di KG venne storicamente introdotta a partire dalla relazione energia impulso di una particella libera relativistica:

$$E^2 = p^2c^2 + m^2c^4$$

Sostituendo gli operatori $E \to i\hbar \frac{\partial}{\partial t}$ $\vec{p} \to i\hbar \vec{\nabla}$, nell'intento di unificare la relatività ristretta alla meccanica quantistica.

Il problema è che l'eq così ottenuta cioè proprio l'eq di KG $(\partial_\mu \partial^\mu + m^2)\phi = 0$ ha come soluzioni funzioni $\phi = e^{\pm i p \cdot x}$ che non possono essere considerate densità di prob come in MQ, quindi non descrivono particelle $|\phi|^2$ NON può essere una densità di prob. Per questo venne inizialmente scartata a favore dell'Eq di Schrödinger.

perché non sono con area finita, sono periodiche spaziali

Intanto l'eq di KG è BUONA ma bisogna interpretare ϕ come un CAMPO e non una funzione d'onda dove le particelle sono eccitazioni del campo.

Il secondo problema sono soluzioni a energia negativa $(e^{+ip\cdot x})$ oltre a quelle a energia positiva $(e^{-ip\cdot x}) \Rightarrow$ che, nell'interpretazione di campo, non sono un problema! Sono le ANTIPARTICELLE

In the hamiltonian formalism we can also define

$$\mathcal{L} = \partial_\mu \phi \partial^\mu \phi - m^2 \phi$$

(basta sostituire)

$$\Pi = \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi)} = \partial_0 \phi$$ conjugate momentum

$$\mathcal{H} = \Pi \partial_0 \phi - \mathcal{L} = \frac{1}{2}\left(\Pi^2 + (\vec{\nabla}\phi)^2 + m^2\phi^2\right) \quad (3.20)$$

$$H = \int d^3x \, \mathcal{H}$$ hamiltonian

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Symmetries of K-G Lagrangian

SIMMETRIE INTERNE

NO SIMMETRIA DI FASE

Since the field is a scalar, there is no symmetry of phase $\phi \to e^{i\alpha}\phi$ simply because we impose ϕ to be a scalar, otherwise the $\mathcal{L}$ would be invariant under such transformation

SIMMETRIA DISCRETA

$\mathcal{L}$ è invariante per inversione $\phi \to -\phi$ tuttavia è una simmetria discreta e quindi non si applica il teo di Noether e non ci sono correnti conservate annesse

SIMMETRIA TRASLAZIONALE

La Lagrangiana è invariante per traslazioni $x^\mu \to x'^\mu = x^\mu + a^\mu$ e il campo ϕ è uno scalare. Abbiamo visto come in tale caso la corrente conservata è il TENSORE DEGLI SFORZI

$$T^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial^\nu \phi - g^{\mu\nu}\mathcal{L} \quad \to \quad \mathcal{L} = \frac{1}{2}(\partial_\mu\phi\partial^\mu\phi - m^2\phi^2)$$

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = \partial^\mu \phi$$

$$T^{\mu\nu} = \partial^\mu\phi\partial^\nu\phi - g^{\mu\nu}\mathcal{L}$$

that satisfies $\partial_\mu T^{\mu\nu} = 0$

we find that $T^{00} = \frac{1}{2}\left((\partial_0\phi)^2 + (\nabla\phi)^2 + m^2\phi^2\right) = \mathcal{H}$ as expected

the conserved quantities are energy and momentum, the related charges conserved are energy and momentum

$$H = \int d^3x\, T^{00} = \int d^3x\, \mathcal{H} \qquad \text{carica conservata} = \text{energia}$$

$$P^\nu = \int d^3x\, T^{0\nu} \qquad \text{momento conservato}$$

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LORENTZ SYMMETRY (BOOST AND ROTATIONS)

The K-G Lagrangian $\mathcal{L} = (\partial_\mu \phi \partial^\mu \phi + m^2 \phi^2)\frac{1}{2}$ is also LORENTZ-INVARIANT because:

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi \partial^\mu \phi + m^2 \phi^2)$$

prodotto scalare, per def è Lorentz-invariante

campo scalare perciò $\phi = \phi'$ sotto trasformazioni di Lorentz

So now we want to find the conserved current and charges related to the conservation of Lorentz symmetry (as predicted by Noether theorem) We will find that the charge conserved is the RELATIVISTIC ANGULAR MOMENTUM

a) coordinate transformation of Lorentz

$$x^\mu \longrightarrow x'^\mu = x^\mu + \delta x^\mu \qquad \text{LORENTZ}$$

for Lorentz trx we have $\delta x^\mu = -\frac{i}{2}\omega_{\alpha\beta}(J^{\alpha\beta})^\mu{}_\nu x^\nu$ with $(J^{\alpha\beta})^\mu{}_\nu = i(g^{\alpha\mu}\delta^\beta{}_\nu - g^{\beta\mu}\delta^\alpha{}_\nu)$ so we have

$$\delta x^\mu = \frac{1}{2}\omega_{\alpha\beta}(g^{\alpha\mu}\delta^\beta{}_\nu - g^{\beta\mu}\delta^\alpha{}_\nu)x^\nu$$

$$= \frac{1}{2}\omega^{\rho\sigma} g_{\alpha\rho} g_{\beta\sigma}(g^{\alpha\mu}\delta^\beta{}_\nu - g^{\beta\mu}\delta^\alpha{}_\nu)x^\nu$$

$$= \frac{1}{2}\omega^{\rho\sigma}(g_{\alpha\rho} g_{\beta\sigma} g^{\alpha\mu}\delta^\beta{}_\nu x^\nu - g_{\alpha\rho} g_{\beta\sigma} g^{\beta\mu}\delta^\alpha{}_\nu x^\nu) =$$

$$= \frac{1}{2}\omega^{\rho\sigma}(\delta^\mu{}_\rho\, g_{\beta\sigma} x^\beta - \delta^\mu{}_\sigma\, g_{\alpha\rho} x^\alpha) =$$

$$= \frac{1}{2}\omega^{\rho\sigma}(\delta^\mu{}_\rho x_\sigma - \delta^\mu{}_\sigma x_\rho)$$

$$= \frac{1}{2}\omega^{\rho\sigma} A^\mu{}_{\rho\sigma}$$

($\omega^{\rho\sigma}$: previously was ε^a; $A^\mu{}_{\rho\sigma}$: previously was $A^\mu{}_a$)

$$\Rightarrow \delta x^\mu = \frac{1}{2}\omega^{\rho\sigma} A^\mu{}_{\rho\sigma} = \sum_{\rho<\sigma} \omega^{\rho\sigma} A^\mu{}_{\rho\sigma} \qquad \text{LORENTZ}$$

(3.21) with $A^\mu{}_{\rho\sigma} = \delta^\mu{}_\rho x_\sigma - \delta^\mu{}_\sigma x_\rho$ (Kronecker delta) — so now what we called $a \to (\rho,\sigma)$ is a compact index $\rho\sigma$ such that

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a) FIELD TRANSFORMATION

ϕ is a scalar so $\phi \to \phi' = \phi$ so $\boxed{F_{\rho\sigma} = 0}$

b) CONSERVED CURRENT

$$3.11 \longrightarrow j^{\mu}_{a} = j_{\rho\sigma}{}^{\mu} = +(\partial^{\mu}\phi)(\cancel{\partial_{\rho}\phi X_{\sigma}} - \cancel{\partial_{\sigma}\phi X^{\rho}}) - (\delta^{\mu}_{\rho} X_{\sigma} - \delta^{\mu}_{\sigma} X_{\rho})\mathcal{L}$$

$a = \rho\sigma$

$(X_{\sigma}\partial_{\rho} - X_{\rho}\partial_{\sigma})\phi$

so

$$j^{\alpha\beta\mu} = j_{\rho\sigma}{}^{\mu}\, g^{\rho\alpha} g^{\sigma\beta} = (\partial^{\mu}\phi)(\cancel{\partial_{\rho}\phi X_{\sigma}} - \cancel{\partial_{\sigma}\phi X_{\rho}}) g^{\rho\alpha} g^{\sigma\beta} -$$

$$- (\delta^{\mu}_{\rho} X_{\sigma} - \delta^{\mu}_{\sigma} X_{\rho})\mathcal{L}\, g^{\rho\alpha} g^{\sigma\beta} =$$

$$= \text{jumping some algebra} = \dots =$$

$$= \underbrace{(\partial^{\mu}\phi\, \partial^{\alpha}\phi - g^{\mu\alpha})\mathcal{L}}_{+T^{\mu\alpha}} X^{\beta} - \underbrace{(\partial^{\mu}\phi\, \partial^{\beta}\phi - g^{\mu\beta})\mathcal{L}}_{+T^{\mu\beta}} X^{\alpha} =$$

$$= -(X^{\alpha} T^{\mu\beta} - X^{\beta} T^{\mu\alpha})$$

Rescaling the current with a minus sign (always possible) we define:

$$j^{\alpha\beta\mu} := -(X^{\alpha} T^{\mu\beta} - X^{\beta} T^{\mu\alpha}) \qquad (3.22)$$

Conserved current unde Lorentz symmetry

c) CONSERVED CHARGE

The conserved charge related to the $j^{\alpha\beta\mu}$ current is the:

$$M^{\alpha\beta} = \int d^3x\; j^{\alpha\beta 0}$$

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TEORICA 1

n. 2

DOMITILLA ALOI

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3.3.1 REAL SCALAR FIELD - part 2

3.3.2 COMPLEX SCALAR FIELD

C.4 CANONICAL QUANTIZATION OF SCALAR FIELDS

4.1. REAL SCALAR FIELD

→ normal ordering motivation

4.2 COMPLEX SCALAR FIELD

C.5 SPIN-ONE FIELDS

5.1 CLASSICAL ELECTRODYNAMICS FIELD THEORY

5.2 QUANTIZATION IN COULOMB GAUGE

5.3 COVARIANT QUANTIZATION

C.6 SPINOR FIELDS

6.1 WEYL FIELDS

6.2 DIRAC FIELDS

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Considering only the spatial $\alpha \to i$ $\beta \to j$ with $ij = 1,2,3$ we have

$$M^{ij} = \int dx^3\, j^{ij0} = \int dx^3\, (x^i T^{0j} - x^j T^{0i}) = \int d^3x\, \partial_0\phi\,(x^i\partial^j - x^j\partial^i)\phi =$$

$$= -i \int d^3x\, \partial_0\phi\, L^{ij}\phi \quad = \quad \text{with } L^{ij} = i(x^i\partial^j - x^j\partial^i)$$

generator of angular momentum on scalar fields.

but $L^{ij} = -L^{ji}$ so

$$= -i \int d^3x\, \partial_0\phi\, \left(\frac{L^{ij} - L^{ji}}{2}\right)\phi =$$

$$= -\frac{i}{2} \int d^3x\, \big(\partial_0\phi(L^{ij}\phi) - \partial_0\phi\,(L^{ji}\phi)\big) =$$

Integrating by parts this only $\partial_0\phi = A$ $\phi = B$ and $L^{ij} = \partial$ derivative we have

$\int A\partial B = AB - \int \partial A \cdot B$ so

$$= -\frac{i}{2} \int d^3x \big(-\phi\,(L^{ij}\partial_0\phi) - \partial_0\phi(L^{ji}\phi)\big) = \quad \text{since } L^{ij} = -L^{ji}$$

$$= \frac{i}{2} \int d^3x \big(\phi(L^{ij}\partial_0\phi) - \partial_0\phi(L^{ij}\phi)\big) =$$

$\phi\,\partial_0(L^{ij}\phi)$

because ∂_0 L^{ij} commutes

Defining the scalar product:

$$\langle \phi_1 | \phi_2 \rangle := \frac{i}{2} \int d^3x\, \phi_1 \overleftrightarrow{\partial_0} \phi_2 = \frac{i}{2} \int d^3x \big[\phi_1\partial_0\phi_2 - (\partial_0\phi_1)\phi_2\big]$$

use with $f \overleftrightarrow{\partial_\mu} g := f\partial_\mu g - (\partial_\mu f) g$

Hence with $\phi_1 \to \phi$ $\phi_2 \to L^{ij}\phi$ we have

$$M^{ij} = \langle \phi | L^{ij} | \phi \rangle \qquad (3.25)$$

La carica conservata è il valore di aspettazione del momento angolare, perché essi conservano il momento angolare! Infatti i generatori di Lorentz su campi scalari sono il momento L^{ij} angolare.

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GENERAL CONSERVED QUANTITY OF K-G EQUATION

Given the scalar product

$$\langle \phi_1 | \phi_2 \rangle := \frac{i}{2} \int d^3x \; \phi_1 \overleftrightarrow{\partial_0} \phi_2 = \frac{i}{2} \int d^3x \left[\phi_1 \partial_0 \phi_2 - (\partial_0 \phi_1) \phi_2 \right]$$

if ϕ_1 and ϕ_2 satisfy the K-G equation, then $\langle \phi_1 | \phi_2 \rangle$ is a conserved quantity such that

$$\partial_0 \langle \phi_1 | \phi_2 \rangle = 0$$

DIMOSTRAZIONE

$$\partial_0 \langle \phi_1 | \phi_2 \rangle = \frac{i}{2} \int d^3x \; \partial_0 \left[\phi_1 \partial_0 \phi_2 - (\partial_0 \phi_1) \phi_2 \right] = \frac{i}{2} \int d^3x \left[\phi_1 \partial_0^2 \phi_2 - (\partial_0^2 \phi_1) \phi_2 \right] =$$

$$= \frac{i}{2} \int d^3x \left[\phi_1 (\vec{\nabla}^2 \phi_2) - (\vec{\nabla}^2 \phi_1) \phi_2 \right] = 0$$

(= : if ϕ_1 ϕ_2 satisfy KG $(\partial_0^2 - \vec{\nabla}^2)\phi + m^2 \phi = 0$)

(= 0 : integrate by parts)

Conseguenza

Dato un generatore J^α di una trasf di simmetria su un campo scalare ϕ, la quantità conservata è il valore di aspettazione di tale J^α sugli ket $|\phi\rangle$ del campo scalare

	generatore su ϕ scalari	quantità conservata
SIMMETRIA TRASLAZIONALE:	$P^\nu = i\partial^\nu$	$P^\nu = \langle \phi \| i\partial^\nu \| \phi \rangle = \int d^3x \, T^{0\nu}$
SIMMETRIA CORENTE:	$L^{\alpha\beta}$	$M^{\alpha\beta} = \langle \phi \| L^{\alpha\beta} \| \phi \rangle = \int d^3x \, j^{\alpha\beta 0}$

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3.3.2 COMPLEX (CHARGED) SCALAR FIELD

IPOTESI: $\phi(x)$ Complesso e scalare

$$\phi(x) \stackrel{!}{:=} \frac{1}{\sqrt{2}}(\phi_1(x) + i\phi_2(x)) \qquad \phi_1, \phi_2 \text{ reali}$$

LAGRANGIANA $\mathcal{L} \stackrel{!}{:=} \partial_\mu \phi^* \partial^\mu \phi - m^2 |\phi|^2$ dove $|\phi|^2 = \phi\phi^*$

this Lagrangian gives the Klein Gordon equation for ϕ and ϕ^*

$$(\partial_\mu \partial^\mu + m^2)\phi = 0$$
$$(\partial_\mu \partial^\mu + m^2)\phi^* = 0$$

and also ϕ_1 and ϕ_2 satisfy K-G equation for mass m

$$(\partial_\mu \partial^\mu + m^2)\phi_1 = 0$$
$$(\partial_\mu \partial^\mu + m^2)\phi_2 = 0$$

The solution is a superposition of waves:

$$\phi(t,\vec{x}) = \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left(a(\vec{p})\, e^{-ip\cdot x} + b^*(p)\, e^{ip\cdot x} \right)$$

$p \cdot x = p^\mu x_\mu$

dove ora nel 2° termine ho un coeff b_p diverso da quello che rispetto al campo reale non dove più esserci reale appunto

Ancora vale =
$E_p = \sqrt{m^2 + \vec{p}^2}$
$p^2 = p_\mu p^\mu = m^2$

In the hamiltonian formalism we can define a conjugate momentum

$$\Pi = \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi)} = \partial_0 \phi^* \qquad \Pi^* = \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi^*)} = \partial_0 \phi$$

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An hamiltonian density as

$$\mathcal{H} = \Pi \partial_0 \phi + \Pi^* \partial_0 \phi^* - \mathcal{L} = 2\partial\phi^* \partial_0 \phi - \mathcal{L} =$$

$$= \Pi^* \Pi + \vec{\nabla}\phi^* \cdot \vec{\nabla}\phi + m^2 \phi^* \phi$$

Thus an hamiltonian:

$$H = \int^3 dx \, \mathcal{H}$$

Symmetries of K-G complex lagrangian

We have all the traslational and Lorentz symmetries and preserved current and charges we already had for the scalar real ϕ field of chapter 3.3.1. Additionally now we also have an internal phase symmetry

PHASE U(1) SYMMETRY

The Lagrangian is invariant under rotations of phase U(1)

$$\phi(x) \longrightarrow e^{i\theta} \phi(x) \qquad \theta \text{ constant}$$

[NB θ is constant, hence we are considering a global symmetry not a local as Gauge symmetry when $\theta(x)$ depends on the point.]

a) INFINITESIMAL TRANSFORMATION OF FIELDS ϕ and ϕ^*

This is an INTERNAL SYMMETRY where only ϕ changes and not the coordinate. For an infinitesimal transformation of phase we have:

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$$\phi(x) \to \phi(x) + i\theta\,\phi(x) \qquad \text{with } \theta \to 0$$
$$\phi^*(x) \to \phi(x)^* - i\theta\,\phi^*(x)$$

so $\delta\phi = i\theta\,\phi(x)$ $\qquad$ $\delta\phi^* = -i\theta\,\phi^*(x)$

according to equation 3.5 of chapter 3.2, now we are calling
$\varepsilon^a \to \theta$ and $F_\phi \to i\phi$ $\qquad$ $F_{\phi^*} \to -i\phi^*$

-) CONSERVED CURRENT

This is an internal symmetry so the current is

$$j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)}\,\delta\phi_i = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\,\delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^*)}\,\delta\phi^* =$$

(sum over i index of fields, now we have two fields: ϕ and ϕ^*)

$$= -\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} F_\phi - \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^*)} F_{\phi^*}$$

(rescaling on θ, dividing per θ)

we get

$$j^\mu = -i\,(\phi\,\partial^\mu \phi^* - \phi^* \partial^\mu \phi) = i\phi^* \overleftrightarrow{\partial^\mu} \phi$$

-) CONSERVED CHARGE

$$Q = \int d^3x\, j^0 = i \int d^3x\, \phi^* \overleftrightarrow{\partial^0} \phi$$

which coincides with the scalar product of complex scalar field defined as:

$$\langle \phi | \phi \rangle := i \int d^3x\, \phi^* \overleftrightarrow{\partial_0} \phi$$

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4

CANONICAL QUANTIZATION OF SCALAR FIELDS

The quantization process of a system of coordinates q^i and momentum p^i in the Schrodyer picture consist of promoting $q^i \to \hat{q}^i$ $p^i \to \hat{p}^i$ into operators and imposing the commutation

$$[\hat{q}^i, \hat{p}^j] = i\,\delta^{ij}$$

In the Heisenberg picture is the same but $\hat{q}^i_H(t)$ $\hat{p}^j_H(t)$ depend on time and the relations are written for a specific time t, same time:

$$[\hat{q}^i_H(t), \hat{p}^j_H(t)] = i\,\delta^{ij}$$

the same principle can be applied to a scalar field theory where we take

$i \to \vec{x}$ — index become continuos

$q^i_{(t)} \to \hat{\phi}(t,\vec{x})$ — coordinate becomes field

$p^i_{(t)} \to \hat{\pi}(t,\vec{x})$ — conjyate momentum

So we want to PROMOTE ϕ and π into HERMITIAN OPERATORS which satisfy the commutation relation. In the commutation now $ij \to \vec{x}-\vec{x}'$ are replaced by a continuos index so delta di kronecher δ^{ij} becomes $\to \delta^{(3)}(\vec{x}-\vec{x}')$ e Dirac delta

$$[\hat{\phi}(t,\vec{x}), \hat{\pi}(t,\vec{x}')] \overset{!}{=} i\,\delta^{(3)}(\vec{x}-\vec{x}') \qquad (4.1)$$

same time — different position

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4.1 REAL SCALAR FIELDS (M 4.1.1)

A real scalar fields has Lagrangian

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi \partial^\mu \phi - m^2 \phi^2)$$

which gives the Klein-Gordon equations

$$(\partial^\mu \partial_\mu + m^2)\phi = 0$$

Which has the solution:

$$\phi(t,\vec{x}) = \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_{\vec{p}}}} \left(\hat{a}_{\vec{p}}\, e^{-ip\cdot x} + a^*_{\vec{p}}\, e^{ip\cdot x}\right)$$

quantization ↓ process of the field

$$\hat{\phi}(t,\vec{x}) = \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_{\vec{p}}}} \left(\hat{a}_{\vec{p}}\, e^{-ip\cdot x} + \hat{a}^+_{\vec{p}}\, e^{ip\cdot x}\right)$$

↓ hermitian operators

In the quantization process $\hat{\phi}$ becomes an operator given by the SUPERPOSITION of many operators $\hat{a}_p$ and $\hat{a}^+_p$

Recall that for scalar real field $\hat{\pi} = \partial_0 \hat{\phi}$, impose the commutation relation we see that $\hat{a}_p$ satisfies the commutation relation

$$[\hat{a}_{\vec{p}}, \hat{a}^+_{\vec{p}'}] = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{p}') \qquad (4.2)$$

$$[a_{\vec{p}}, a_{\vec{p}'}] = [a^+_{\vec{p}}, a^+_{\vec{p}'}] = 0 \qquad \forall \vec{p}, \vec{p}'$$

where $\vec{p}$ is a 3vector momentum

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Vacuum state and Fock space

what are $\hat{a}_{\vec{p}}$ and $\hat{a}^{+}_{\vec{p}}$ operators? I WANT to identify them as the annihilation and creation operators of an armonic oscillator. We IMPOSE or better DEFINE a VACUUM state $|0\rangle$ AD HOC in order to satysfy the annihilation property

$\exists\, |0\rangle$ a vacuum state such that $\langle 0|0\rangle = 1$ normalized

such that $\hat{a}_{\vec{p}}\,|0\rangle \stackrel{!}{=} \emptyset \qquad \forall \text{all } \vec{p}$

The generic state of the Fock space is obtained by applying the creator operator

$$\sqrt{2E_{\vec{p}}}\;\hat{a}_{p}^{+}\,|0\rangle = |\vec{p}\rangle$$

in general for N particles

$$|\vec{p}_1\,\vec{p}_2 \cdots \vec{p}_N\rangle = \sqrt{2\epsilon_{\vec{p}_1}}\,\sqrt{2\epsilon_{\vec{p}_2}} \cdots \sqrt{2\epsilon_{\vec{p}_N}}\;\hat{a}^{+}_{p_1}\hat{a}^{+}_{p_2}\cdots\hat{a}^{+}_{p_N}\,|0\rangle \qquad (4.3)$$

↓ N particle state — ↓ vacuum state

now I can create or destroy a particle by acting with $\hat{a}^{+}$ or $\hat{a}$:

IL PUNTO

QM non è possibile creare o distruggere particelle = dato un sistema il n° di particelle è definito, se voglio cambiare particelle devo cambiare Hamiltoniano del sistema.
In RELATIVITÀ NON È COSÌ: massa ed energia sono in tutti i casi possibile cambiare il n° di particelle convertendole in energia serve cambiare Hamiltoniano
→ ciò ed quantizzazione di ϕ permette di fare ciò! Ho detto alla MQ la proprietà di poter cambiare il n° di particelle come in relatività speciale!

factor $\sqrt{2E_{\vec{p}}}$ is for normalization of $|\vec{p}\rangle$ state

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The Fock states are orthogonal:

$$\langle \vec{p}_1 | \vec{p}_2 \rangle = \sqrt{2E_{p_1}}\sqrt{2E_{p_2}}\, \langle 0 | \hat{a}_{p_1} \hat{a}^\dagger_{p_2} | 0 \rangle =$$

$$= \sqrt{\quad}\sqrt{\quad}\, \langle 0 | a_{p_1} a^\dagger_{p_2} - a^\dagger_{p_2} a_{p_1} | 0 \rangle =$$

because this term is zero because $a_{p_1}|0\rangle = \emptyset$

$$= \sqrt{\quad}\sqrt{\quad}\, \langle 0 | [a_{p_1}, a^\dagger_{p_2}] | 0 \rangle =$$

$$= \sqrt{\quad}\sqrt{\quad}\, \delta^{(3)}(\vec{p}_1 - \vec{p}_2)\,(2\pi)^3 =$$

$$\langle \vec{p}_1 | \vec{p}_2 \rangle = 2E_{p_1}\,(2\pi)^3\,\delta^{(3)}(\vec{p}_1 - \vec{p}_2)$$

Action of $\hat{\phi}$ on zero vacuum state

$\hat{\phi}$ is an operator given by the composition of a_p and $a_p^\dagger$ operators, so we can compute the action of ϕ on $|0\rangle$ vacuum state

$$\hat{\phi}(x)|0\rangle = \int \frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2\epsilon_p}}\left(\hat{a}_{\vec{p}}\, e^{-ip\cdot x} + \hat{a}^\dagger_{\vec{p}}\, e^{ip\cdot x}\right)|0\rangle =$$

$\hat{a}_{\vec{p}}\, e^{-ip\cdot x} = 0$ because $a_p|0\rangle = \emptyset$

$\hat{a}^\dagger_{\vec{p}}\, e^{ip\cdot x} \Rightarrow a^\dagger_p|0\rangle = \frac{1}{\sqrt{2\epsilon_p}}|p\rangle$

$$\hat{\phi}(x)|0\rangle = \int \frac{d^3p}{(2\pi)^3}\frac{1}{2E_{\vec{p}}}\, e^{ip\cdot x}\,|\vec{p}\rangle$$

Lorentz invariance of $\phi(x)|0\rangle$

Looking at the expression of $\phi(x) = \int \frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2\epsilon_p}}\left(a_p e^{-ip\cdot x} + a_p^\dagger e^{ip\cdot x}\right)$ we may wonder if it is Lorentz invariant since apparently time and space are treated differently.

Eventually $\phi(x)$ IS NOT LORENTZ INVARIANT but this is no problem because $\phi(x)|0\rangle$ instead IS LORENTZ INVARIANT, and the operator gains sense only when acting on a state, ϕ itself has no physical meaning.

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Now we will demonstrate that $\phi(x)|0\rangle = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\epsilon_{\vec{p}}} e^{ip\cdot x} |\vec{p}\rangle$ is an invariant of Lorentz.

1 General Lorentz invariant quantity

•) Let's consider the quantity:

$$\frac{d^4p}{(2\pi)^4} \delta(p^2 - m^2)\, 2\pi\, \theta(p^0)$$

$\downarrow$ $p^2 = p^\mu p_\mu$ is Lorentz invariant; on shell condition $p^2 = m^2$

$\downarrow$ Heaviside function which tells the energy p^0 must be positive

$$\theta(p^0) = \begin{cases} 1 & p^0 > 0 \\ 0 & p^0 < 0 \end{cases}$$

Such quantity is Lorentz invariant because d^4p and $d\,p^2$ are Lorentz invariant by construction and $\theta(p^0)$ is Lorentz invariant and more, tells us we are dealing with particles with positive energy (Matematically we are considering orthochronous Lorentz transformation)

•) $\frac{d^4p}{(2\pi)^3} \delta(p^2 - m^2)\, \theta(p^0) = \frac{d^4p}{(\pi 2)^3} \delta(p^{0^2} - \vec{p}^2 - m^2)\, \theta(p^0) = (*)$

•) recalling $\delta(f(x)) = \sum_n \frac{\delta(x - x_n)}{|f'(x_n)|}$ where x_n are roots (zeroes) of $f(x)$

so that $\delta(p_0^2 - (\vec{p}^2 + m^2)) = \frac{\delta(p_0 - \sqrt{\vec{p}^2 + \vec{m}^2})}{2p_0} - \frac{\delta(p_0 + \sqrt{\vec{p}^2 - m^2})}{2p_0} =$

:

so

$(*) = \frac{d^4p}{(2\pi)^3} \frac{\delta(p_0 - \sqrt{\vec{p}^2 + \vec{m}^2})}{2p_0} =$

concee because this implies that p_0 has to be negative $p_0 = -\sqrt{\ }$ which is not possible for the $\theta(p^0)$

$$= \frac{dp_0\, d^3\vec{p}}{(2\pi)(2\pi)^3} (2\pi) \frac{\delta(p_0 - \sqrt{\vec{p}^2 + m^2})}{2p_0}$$

•) Integrating over dp_0 I compute the δ so $p_0 = \sqrt{\vec{p}^2 + m^2}$ so we obtain

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$$\frac{d^3\vec{p}}{(2\pi)^3}\,\frac{1}{2\vec{E}_p} \qquad \text{with } E_p = \sqrt{m^2+\vec{p}^2} \qquad (*)$$

This is a general quantity which is Lorentz invariant

[2] Comparing $\phi(x)|0\rangle = \int \frac{d^3\vec{p}}{(2\pi)^3}\,\frac{1}{2E_p}\, e^{ip\cdot x}|\vec{p}\rangle$ with $(*)$ we deduce that $\phi(x)|0\rangle$ is Lorentz invariant too

CVD

$\Rightarrow \phi(x) \sim d^3p\,\frac{1}{\sqrt{E_p}}$ NOT Lorentz invariant

$\phi(x)|0\rangle \sim d^3p\,\frac{1}{E_p}$ IS Lorentz invariant

Since creation operators $a_{p_1}^\dagger, a_{p_2}^\dagger$ COMMUTE among themselves

$[a_{p_1}^\dagger, a_{p_2}^\dagger] = 0 \quad \forall p_1, p_2$

allora the state $|p_1 p_2 \dots p_n\rangle$ IS SYMMETRIC under PARTICLE exchange $p_i \leftrightarrow p_j$

$|p_1 \dots p_i \dots p_j \dots p_n\rangle = |p_1 \dots p_j \dots p_i \dots p_n\rangle$

$\Rightarrow$ ovvero the Fock states follow the Bose-Einstein statistics

Chiaro come dimostreremo fra poco, particelle con campi scalari sono particelle a spin = 0, nullo perciò sono bosoni (part con spin intero)

Su cui impongo anche $\delta(p^2-m^2)$ and $\theta(p_0)$ (On shell condition and positive energy!)

HWT

$\phi(x)|0\rangle \cong |x\rangle$ è un autostato della posizione per

" ↑ "

$\cong \int d^4p\, |p\rangle\langle p|x\rangle = \int d^4p\, |p\rangle\, \frac{e^{ipx}}{(2\pi)^4}$

aggiungo $\delta(p^2-m^2)$ e $\theta(p_0)$ e faccio i conti, ottengo proprio $\phi(0)$!

$\sim \int \frac{d^3p}{(2\pi)^3}\,\frac{1}{2E_p}\, e^{ipx}\,|p\rangle =$

Hamiltonian and normal ordering

The hamiltonian density is $\mathcal{H} = \frac{1}{2}\left[\partial_0\phi^2 + (\vec{\nabla}\phi)^2 + m^2\phi^2\right]$
which leads to an hamiltonian

$$\hat{H} = \int \frac{d^3\vec{p}}{(2\pi)^3} \frac{E_{\vec{p}}}{2} \left(\hat{a}^+_{\vec{p}}\,\hat{a}_{\vec{p}} + \hat{a}_{\vec{p}}\,\hat{a}^+_{\vec{p}}\right)$$

VACUUM ENERGY

I can rewrite $\hat{H}$ as follows:

$$\hat{H} = \int \frac{d^3p}{(2\pi)^3} E_{\vec{p}} \left(\hat{a}^+_{\vec{p}}\,\hat{a}_{\vec{p}} + \frac{1}{2}\left[\hat{a}_{\vec{p}}, \hat{a}^+_{\vec{p}}\right]\right)$$

The $|0\rangle$ state is an eigenstate of $\hat{H}$ with the eigenvalue $E_0 = E_{vacuum}$ called Energy of the vacuum

$$\hat{H}|0\rangle = \int \frac{d^3p}{(2\pi)^3} E_p\, a^+_p a_p |0\rangle + \int \frac{d^3p}{(2\pi)^3} \frac{E_p}{2} \left[a_p, a^+_p\right]|0\rangle$$

$= 0$ because $a_p|0\rangle = \emptyset$ $\quad =$

$$\hat{H}|0\rangle = \int \frac{d^3p}{(2\pi)^3} \frac{E_p}{2} (2\pi)^3 \delta(0)\,|0\rangle = \therefore E_{vacuum}|0\rangle$$

← ma $(2\pi)^3\delta(0) = V$ è il Volume di integrazione che può essere ∞ o finito a seconda del fatto che stia o meno integrando su tutto lo spazio delle $\vec{p}$

allora

sico!

$$E_{vacuum} = \frac{1}{2} V \int \frac{d^3\vec{p}}{(2\pi)^3} E_{\vec{p}} \qquad \text{VACUUM ENERGY}$$

it is the superposition of all zero-energies of all particles momentum $\vec{p}$.

$$\rho_{vacuum} = \frac{E_{vac}}{V} \qquad \text{VACUUM ENERGY DENSITY}$$

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perché $\delta^{(3)}(\vec{p}) = \int \frac{d^3x}{(2\pi)^3} e^{i\vec{p}\cdot\vec{x}}$

allora $\delta^{(3)}(0) = \int \frac{d^3x}{(2\pi)^3} = \frac{\text{Volume}}{(2\pi)^3}$ ←

cioè $V = \delta^{(3)}(0)\,(2\pi)^3$

dove V è il volume nello spazio delle x! il vero volume fisico

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Such energy is INFINITE! It diverges! Because $E_p = \sqrt{\vec{p}^2 + m^2}$ so for large $\vec{p}$ it becomes linear $E_p \sim |\vec{p}|$ and the integral diverge!

$$E_{VOC} = \int \frac{d^3p}{(2\pi)^3} \frac{V}{2} \sqrt{\vec{p}^2 + m^2} \longrightarrow \infty$$

Per ovviare a questo problema introduco di integrare solo fino ad una scala CUT OFF Λ (UV CUT OFF = ultra violet cut off, taglio via l'UV) ma che ovviamente tende a ∞ se mando $\Lambda \to \infty$.

$$E_{VOC}^{\text{cut off}} = \int_0^{\Lambda} \frac{d^3p}{(2\pi)^3} \frac{V}{2} \sqrt{\vec{p}^2 + m^2} \sim \Lambda^2 \longrightarrow \infty \quad \text{IF } \Lambda \to \infty$$

Quindi E_{VACUUM} diverge è legittimo che faccia? No problema! RISCALO sottraendo E_{VACUUM}.

Infatti posso sempre aggiungere o togliere una costante alla $\mathcal{L}$ e alla $\mathcal{H}$ tanto le eq. del moto non cambiano

$$\mathcal{L} \longrightarrow \mathcal{L} + E_{VAC}$$
$$\mathcal{H} \longrightarrow \mathcal{H} - E_{VOC}$$
$$H \longrightarrow \int \frac{d^3p}{(2\pi)^3} E_p \left(\hat{a}_p^+ \hat{a}_p\right)$$

$\Rightarrow$ VACUUM ENERGY RESCALING $\hat{H}_{\text{Rescaled}} = \int \frac{d^3p}{(2\pi)^3} E_p \left(\hat{a}_p^+ \hat{a}_p\right)$

NORMAL ORDERING

The process of Rescaling eliminating the vacuum energy is EQUIVALENT to act a NORMAL ORDERING process to the classical hamiltonian

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normal ordering

Given an operator $\hat{O}$, the normal ordered $:\hat{O}:$ is an operator obtained by writing all creation operators in O to the left and all annihilation operators in O to the right

$$\overbrace{\hat{O}}$$

EX: $:\hat{a}_p \hat{a}_p^+: = a_p^+ a_p$

EX: $: a_p a_p^+ a_{p'}: = a_p^+ a_p a_{p'}$

Thus given the classical starting hamiltonian $H = \int \frac{d^3p}{(2\pi)^3} \frac{E_p}{2} (a_p^+ a_p + a_p a_p^+)$ if we APPLY To such H the normal ordering we obtain

$$:\hat{H}: = \int \frac{d^3p}{(2\pi)^3} \frac{E_p}{2} :(a_p^+ a_p + a_p a_p^+): = \int \frac{d^3p}{(2\pi)^3} E_p a_p^+ a_p$$

which is EXACTLY THE HAMILTONIAN RESCALED by SUBSTRACTING the Evacuum

Since we want to work in this H rescaled we can take this as general principle

"Given a classical Hamiltonian (or generic quantity) with FIELDS, we promote the fields into OPERATORS AND APPLY a NORMAL ORDERING"

= principle to quantize operators

COMMUTATIONS RELATIONS OF NORMAL ORDERED $\hat{H}$

$$[\hat{H}, \hat{a}_p^+] = \int \frac{d^3p'}{(2\pi)^3} E_{p'} (a_{p'}^+ a_{p'} a_p^+ - a_p^+ a_{p'}^+ a_{p'}) =$$

$= a_{p'}^+ a_p^+ a_{p'}$ because $a_{p'}^+$ and a_p^+ commute

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$$= \int \frac{d^3p'}{(2\pi)^3} E_{p'} a_{p'}^+ \underbrace{[a_{p'}, a_p^+]}_{= (2\pi)^3 \delta^{(3)}(p-p')} = E_p a_p^+$$

$[\hat{H}, \hat{a}_p^+] = E_{\vec{p}}\, a_p^+$ with $\hat{H}$ in normal ordering

$[\hat{H}, \hat{a}_p] = \dots = -E_p a_p$

EIGENSTATES AND EIGENVALUES

The eigenstate of H are $|\vec{p}\rangle$ states

$\hat{H} |\vec{p}_1, \vec{p}_2 .. \vec{p}_N\rangle = (E_{p1} + E_{p2} + \dots E_{pn}) |\vec{p}_1, \dots \vec{p}_n\rangle$

Translational symmetry = MOMENTUM OPERATOR

The Lagrangian of a scalar field is preserved under translation transform. The Noether theorem tells us there is a Noether conserved charge that is the momentum P^i and a Noether current $T^{\mu\nu}$, the energy momentum tensor.

$P^i = \int d^3x\, T^{0i}$ with $T^{0i} = +\partial^0\phi\, \partial^i\phi$ Noether charge

We want TO QUANTIZE this P^i into an operator $\rightarrow \hat{P}^i$. We follow the principle we acted for the hamiltonian = (1) converting the fields into operators (2) normal ordering such operators

a) $\hat{P}^i = \int d^3x\, :\hat{T}^{0i}: = \int d^3x\, :(+\partial^0\hat{\phi}\,\partial^i\hat{\phi}):$ =

$$= \int d^3x \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2E_k}} \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} :\left(\partial^0\left[\hat{a}_k e^{-ik\cdot x} + \hat{a}_k^+ e^{ik\cdot x}\right](+\partial^i)\left[\hat{a}_p e^{-ip\cdot x} + \right.\right.$$

$$\left.\left. + \hat{a}_p^+ e^{ip\cdot x}\right]\right): =$$

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•) calling $d^3x = [dx]$ $\quad \frac{d^3p}{(2\pi)^3\sqrt{2E_p}} = [dp]$ $\quad \frac{d^3k}{(2\pi)^3\sqrt{2E_k}} = [dk]$

and recalling that $k\cdot x = k^\mu x_\mu = k^0 t - \vec{k}\cdot\vec{x}$ with $k^0 = E_k$

$p\cdot x = p^\mu x_\mu = p^0 t - \vec{p}\cdot\vec{x}$ $\quad p^0 = E_p$

we get

$$\hat{P}^i = \int [dx][dp][dk]\ (-iE_k) :\left[a_k e^{-ik\cdot x} - a_k^+ e^{ik\cdot x}\right]: (-ip^i) :[$$

$$:\left[a_p e^{-ip\cdot x} - a_p^+ e^{ip\cdot x}\right]: =$$

$$= \int [dx][dp][dk]\ (-E_k p^i)\, e^{-i\vec{k}\cdot\vec{x}}\, e^{-i\vec{p}\cdot\vec{x}} :\left(a_{-k} e^{-iE_k t} - a_k^+ e^{iE_k t}\right)\cdot($$

$$\cdot\left(a_{-p} e^{-iE_p t} - a_p^+ e^{iE_p t}\right): =$$

→ Cobely with $-k$ and $-p$ instead of k and p because tanto integro da $-\infty$ a $+\infty$ e' uguale

•) all is independent from $\vec{x}$ of integration $\int d^3\vec{x}$, $\vec{x}$ compares only in which gives a delta

$$\int d^3x\ e^{-i(\vec{k}+\vec{p})\cdot\vec{x}} = \delta^{(3)}(\vec{k}+\vec{p})\,(2\pi)^3$$

so integrating also over $\vec{k}$ always over $\int \frac{d^3k}{(2\pi)^3}$ we extinguish the Delta and $\vec{k} \to$ becomes $-\vec{p}$ and $E_k \to E_p$ so

$$= \int \frac{d^3p}{(2\pi)^3}\ \frac{1}{2E_p}\ (-E_p\, p^i) :\left(a_p e^{-iE_p t} - a_{-p}^+ e^{iE_p t}\right)\left(a_{-p} e^{-iE_p t} - a_p^+ e^{iE_p t}\right):$$

•) the terms in the product:

$$p^i (a_p - a_{-p}^+)(a_{-p} - a_p^+) = \overbrace{p^i a_p a_{-p}}^{(1)} + \overbrace{p^i a_{-p}^+ a_p^+}^{(2)} + \dots + \dots$$

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(1) and (2) are odd when change $p \to -p$ so their integration over all $\int_{-\infty}^{\infty} d^3p$ gives zero (in general $\int_{-\infty}^{\infty} f(x) = 0$ if $f(x)$ is odd i.e $f(x) = -f(-x)$)

so we remain with

$$\hat{P}^i = \int \frac{d^3\vec{p}}{(2\pi)^3} (-p^i) : \left(-\hat{a}_p \hat{a}_p^\dagger - \hat{a}_p^\dagger \hat{a}_p \right) :$$

Hence

$$\hat{P}^i = \int \frac{d^3\vec{p}}{(2\pi)^3} \; p^i \; \hat{a}_p^\dagger \hat{a}_p$$

$\left(\to \hat{P}^i |\vec{p}\rangle = p^i |\vec{p}\rangle \right)$ ($\hat{P}^i$: operator, p^i: eigenvalue)

$\hat{P}^i$: Operator of momentum (Noether charge due to translation symmetry)

p^i: number, eigenvalue of $|p\rangle$ particle state

the noether momentum operator of the field is the sum of all momentum p^i of each particle $|p\rangle$

Lorentz Symmetry — Spin

The Lorentz symmetry gives the angular momentum as conserved current and the expectation value of angular momentum as Noether charge. For scalar field it is only the ORBITALIC L'is MOMENTUM, there is no SPIN

$\Rightarrow$ scalar fields creates particle with spin=0 $\to$ bosons

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RICORDA

$x^\mu = (x^0, x^1, x^2, x^3) = (t, x, y, z)$

$x_\mu = (x_0, x_1, x_2, x_3) = (t, -x, -y, -z)$

$\longrightarrow$ cioè $x^0 = x_0$, $x^j = -x_j$

$p^\mu = (p^0\ p^1\ p^2\ p^3) = (E_p, p_x, p_y, p_z)$

$p_\mu = (p_0\ p_1\ p_2\ p_3) = (E_p, -p_x, -p_y, -p_z)$

$\longrightarrow$ cioè $p^0 = p_0$, $p^j = -p_j$

Allora

$$p^\mu x_\mu = p^0 x_0 + p^1 x_1 + p^2 x_2 + p^3 x_3 = E_p t - p_x x - p_y y - p_z z$$

ovvero

$$p^\mu x_\mu = p^0 x_0 + p^j x_j = p^0 x^0 - p^j x^j = p_0 x_0 - p_j x_j$$

(sommo sui j)

Ne segue che quando faccio la derivata

$$\partial_j (p^\mu x_\mu) = \frac{\partial}{\partial x^j}(p_\mu x^\mu) = p_j = -p^j$$

$$\partial^j (p^\mu x_\mu) = \frac{\partial}{\partial x_j}(p^\mu x_\mu) = p^j$$

$$\partial_\mu = \frac{\partial}{\partial x^\mu} = (\partial_0, +\vec{\nabla})$$

$$\partial^\mu = \frac{\partial}{\partial x_\mu} = (\partial^0, -\vec{\nabla})$$

$$T^{0j} = \partial^0\phi\,\partial^j\phi = -\partial^0\phi\,\partial_j\phi \qquad \text{perché } \partial^j = -\partial_j$$

$$\partial^j\left(e^{-ip^\mu x_\mu}\right) = \partial^j\left(e^{-i(p^0x_0 + p^1x_1 + p^2x_2 + p^3x_3)}\right) = -ip^j$$

Così se svolgo i conti tutto torna:

$$P^j = \int d^3x\, :T^{0j}: = \int dx^3\, :(\partial^0\phi\,\partial^j\phi):$$

$$= \int dx^3 \int \frac{d^3k}{(2\pi)^3}\frac{1}{\sqrt{2\varepsilon_u}} \int \frac{d^3p}{\sqrt{2\varepsilon_p}(2\pi)^3} :\partial^0\left(a_u^\dagger e^{-iux} + a_u^\dagger e^{iux}\right)\partial^j\left(a_p e^{-ipx} + a_p^\dagger e^{ipx}\right):$$

$$= \int dx^3 \int [dk] \int [dp] :(-ik^0)\left(a_u^\dagger e^{-iux} - a_u e^{iux}\right)(-ip^j)\left(a_p e^{-ipx} - a_p^\dagger e^{ipx}\right):$$

$$= \int dx^3 \int [dk] \int [dp]\, (-k^0 p^j) :\Big[a_u^\dagger a_p e^{-i(k+p)x} - a_u a_p^\dagger e^{+i(p-k)x} - a_u^\dagger a_p e^{i(k-p)x} + a_u^\dagger a_p^\dagger e^{i(p+k)x}\Big]:$$

$$= \int \frac{d^3k}{(2\pi)^3\sqrt{2\varepsilon_u}} \int \frac{d^3p}{(2\pi)^3\sqrt{2\varepsilon_p}} (2\pi)^3 (-k^0 p^j) :\Big[a_u^\dagger a_p e^{-i(k_0+p_0)x}\,\delta(\vec{p}+\vec{u}) - a_u a_p^\dagger e^{-i(k_0-p_0)x}\,\delta(\vec{p}-\vec{u}) -$$

$$- a_u^\dagger a_p e^{i(k_0-p_0)x}\,\delta(\vec{p}-\vec{u}) + a_u^\dagger a_p^\dagger e^{+i(k_0+p_0)x}\,\delta(\vec{u}+\vec{p})\Big]: =$$

(vedi dietro)

$$= \int \frac{d^3p}{(2\pi)^3}\frac{1}{2\varepsilon_p}(-\varepsilon_p p^j) :\Big[\cancel{a_{-p} a_p} - a_p a_p^\dagger - a_p^\dagger a_p + \cancel{a_{-p}^\dagger a_p^\dagger}\Big]: =$$

(dispari) (dispari)

$$= \int \frac{d^3p}{(2\pi)^3}\, p^j\, a_p^\dagger a_p \qquad \text{CVD!}$$

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4.2 COMPLEX SCALAR FIELDS (M 4.1.2)

A complex scalar field has Lagrangian density:

$$\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - m^2 \phi \phi^*$$

which gives K-G for both ϕ and ϕ^* with solutions:

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} \left(a_{\vec{p}}\, e^{-ip\cdot x} + b^*_{\vec{p}}\, e^{ip\cdot x} \right)$$

$\phi^*(x) = \dots$

quantization ↓

promoting $\phi \to \hat{\phi}$ $\phi^* \to \hat{\phi}^+$ and $a \to \hat{a}$ $b^* \to \hat{b}^+$

$$\hat{\phi}(x) = \int \frac{d^3\vec{p}}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} \left(\hat{a}_{\vec{p}}\, e^{-ip\cdot x} + \hat{b}^+_{\vec{p}}\, e^{ip\cdot x} \right)$$

$$\hat{\phi}^+(x) = \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_{\vec{p}}}} \left(\hat{a}^+_{\vec{p}}\, e^{ip\cdot x} + \hat{b}_{\vec{p}}\, e^{-ip\cdot x} \right)$$

Then imposing commutation 4.1: $[\phi, \pi] = i\delta^{(3)}(x-x')$ with conjugate momentum $\pi = \partial_0 \phi^+$ and $\pi^+ = \partial_0 \phi$ we get:

$$[a_{\vec{p}}, a^+_{\vec{p}'}] = [b_{\vec{p}}, b^+_{\vec{p}'}] = (2\pi)^3 \delta^{(3)}(\vec{p}-\vec{p}')$$

$$[a_{\vec{p}}, a_{\vec{p}'}] = [a^+_{\vec{p}}, a^+_{\vec{p}'}] = [b_{\vec{p}}, b_{\vec{p}'}] = [b^+_{\vec{p}}, b^+_{\vec{p}'}] = 0$$

So exact relations as before for the real scalar field but doubled for also $b_{\vec{p}}$ in addition that for $a_{\vec{p}}$. More, $\hat{a}$ with $\hat{b}$ commute:

$$[a_{\vec{p}}, b_{\vec{p}'}] = [a_{\vec{p}}, b^+_{\vec{p}'}] = [a^+_{\vec{p}}, b_{\vec{p}'}] = [a^+_{\vec{p}}, b^+_{\vec{p}'}] = 0$$

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Vacuum state and Fock space

We impose the existence $\exists\, |0\rangle$ of a vacuum state $\langle 0|0\rangle \doteq 1$ normalized

such that : $\hat{a}_{\vec{p}}\,|0\rangle \stackrel{!}{=} 0$ $\Rightarrow$ particle and antiparticles have the same ground state

$\hat{b}_{\vec{p}}\,|0\rangle \stackrel{!}{=} 0$

And a Fock space generated by creation operators $a_{\vec{p}}^{+}$ and $b_{\vec{p}}^{+}$, exactly as before but now two copies :

$\sqrt{2E_{\vec{p}}}\;\hat{a}_{\vec{p}}^{+}\,|0\rangle = |\vec{p}\rangle_a$ $\Rightarrow$ particles and antiparticles

$\sqrt{2E_{\vec{p}}}\;\hat{b}_{\vec{p}}^{+}\,|0\rangle = |\vec{p}\rangle_b$ show different fock space

with orthogonality relations :

$${}_a\langle p_1|p_2\rangle_a = 2E_{p_1}(2\pi)^3\,\delta^{(3)}(\vec{p}_1-\vec{p}_2)$$

$${}_a\langle p_1|p_2\rangle_b = 0$$ $\Rightarrow$ stati a e b sono ortogonali $\forall p_1, p_2$

$${}_b\langle p_1|p_2\rangle_b = 2E_{p_1}(2\pi)^3\,\delta^{(3)}(\vec{p}_1-\vec{p}_2)$$

Hamiltonian formalism - normal ordering

$$\hat{H} = \int \frac{d^3p}{(2\pi)^3}\, E_{\vec{p}}\,(\hat{a}_{\vec{p}}^{+}\hat{a}_{\vec{p}} + \hat{b}_{\vec{p}}^{+}\hat{b}_{\vec{p}})$$

$\downarrow$ normal ordered (hence the $:\hat{H}:$)

Traslational symmetry - momentum

$$\hat{P}^i = \int \frac{d^3\vec{p}}{(2\pi)^3}\, p^i\,(\hat{a}_{\vec{p}}^{+}\hat{a}_{\vec{p}} + \hat{b}_{\vec{p}}^{+}\hat{b}_{\vec{p}})$$

$\downarrow$ normal ordered

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Lorentz symmetry → spin and angular momentum

Angular momentum L^{ij} is preserved. No spin $S=0$ cause scalar field

U(1) symmetry — particle and antiparticle

A complex field has also the phase U(1) symmetry which a real field doesn't have. The noether charge associated to such phase symmetry is:

$$Q = \int d^3x \; j^0 = i \int dx^3 \; \phi^+ \overleftrightarrow{\partial_0} \phi \qquad \text{classical}$$

↓ quantization of fields $\phi \to \hat{\phi}$ $\phi^+ \to \phi^+$ & normal ordering

$$\hat{Q} = \int d^3x \; \hat{j}^0 = i \int dx^3 \; :\hat{\phi}^+ \overleftrightarrow{\partial_0} \hat{\phi}: \; =$$

$$= \dots = i \int \frac{d^3\vec{p}}{(2\pi)^3} \; :(a_p^+ a_p - b_p b_p^+):$$

thus

$$\hat{Q} = \int \frac{d^3\vec{p}}{(2\pi)^3} \left(\hat{a}_p^+ \hat{a}_p - \hat{b}_p^+ \hat{b}_p \right)$$

$\hat{a}_p^+ \hat{a}_p$: n° of states created by $\hat{a}$

$\hat{b}_p^+ \hat{b}_p$: minus n° of states created by $\hat{b}$

For energy and momentum the 2 terms of a and b are summed with a + !! We call the particles created by $\hat{b}$ the ANTIPARTICLES.

Antiparticles have OPPOSITE CHARGE in respect to the particle's one, but all other properties (mass, momentum, energy, spin...) ARE THE SAME

Energy & Momentum = Energy and momentum of part. and antip. are summed

Charge = if the n° of particles is the same as the n° of antip, the total charge Q of the field is zero, part and antip charges are substracted

In the field $\phi(x)$:

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$$= \dots = i \int \frac{d^3\vec{p}}{(2\pi)^3} \; : (a_p^+ a_p - b_p b_p^+) :$$

thus

$$\hat{Q} = \int \frac{d^3\vec{p}}{(2\pi)^3} \; (\hat{a}_p^+ \hat{a}_{\vec{p}} - \hat{b}_p^+ \hat{b}_{\vec{p}})$$

$\hat{a}_p^+ \hat{a}_{\vec{p}}$ ⇓ n° of states created by $\hat{a}$. $\hat{b}_p^+ \hat{b}_{\vec{p}}$ ⇓ minus n° of states created by $\hat{b}$

For energy and momentum the 2 terms of a and b are summed with a + !! a call the particles created by $\hat{b}$ the ANTIPARTICLES.

Antiparticles have OPPOSITE CHARGE in respect to the particle's one, but all other properties (mass, momentum, energy, spin...) ARE THE SAME

Energy & Momentum = Energy and momentum of part. and antip. are summed

Charge = if the n° of particles is the same as the n° of antip, the total charge Q of the field is zero, part and antip charges are substracted

in the field $\phi(x)$:

a) POSITIVE frequency mode $e^{-iE_p t}$ = has a particle annihilation operator $\hat{a}_p$ as coefficient

b) NEGATIVE frequency mode $e^{iE_p t}$ = has an antiparticle creation operator $\hat{b}_p^+$ as coefficient

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5

SPIN-ONE FIELDS

A spin one field is a vector field A^μ (es = photon) the corresponding quantum field theory is the quantum electrodynamics. Thus we start by discussing the classical electrodynamics expressed as a field theory.

5.1 CLASSICAL ELECTRODYNAMICS FIELD THEORY (M. 3.5.1, 3.5.2, 3.5.3)

The E.M. field is a 4-vector called GAUGE POTENTIAL:

$$A^\mu = (\varphi, \vec{A})$$

φ: potenziale scalare; $\vec{A}$: potenziale vettore

Which gives the Field Strenght tensor (Faraday tensor):

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \qquad (5.1)$$

Which is an antysymmetric tensor which contains the electric $\vec{E}$ and magnetic $\vec{B}$ field components:

$$F^{0i} = \dots = -E^i \qquad (5.2)$$
$$F^{ij} = \dots = -\varepsilon^{ijk} B^k$$

$$F_{\mu\nu} = \begin{bmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & B_z & -B_y \\ -E_y & -B_z & 0 & B_x \\ -E_z & B_y & -B_x & 0 \end{bmatrix}$$

$$F^{\mu\nu} = \begin{bmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{bmatrix}$$

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Lagrangian and Maxwell eq of motion

In the vacuum with no free charges the Lagrangian of the EM fields is:

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \qquad (5.3)$$

which is of course Lorentz covariant, translational invariant. From such $\mathcal{L}$ we obtain the equations of motion:

$$\partial_\mu F^{\mu\nu} = 0 \;\rightarrow\; \partial_0 F^{0\nu} + \partial_j F^{j\nu} = 0 \qquad (5.4)$$

Such equations contain the MAXWELL EQUATIONS for the $\vec{E}, \vec{B}$ fields.

RICORDA

$$\partial_\mu = \left(\frac{\partial}{\partial x^0}, +\frac{\partial}{\partial x^1}, +\frac{\partial}{\partial x^2}, +\frac{\partial}{\partial x^3}\right) = \left(\frac{\partial}{\partial t}, +\vec{\nabla}\cdot\right)$$

$$\partial^\mu = \left(\frac{\partial}{\partial x_0}, +\frac{\partial}{\partial x_1}, +\frac{\partial}{\partial x_2}, +\frac{\partial}{\partial x_3}\right) = \left(\frac{\partial}{\partial t}, -\vec{\nabla}\right)$$

perché $x^\mu = (t, x, y, z) \rightarrow \frac{\partial}{\partial x_j} = -\frac{\partial}{\partial x^j}$ perché $x^j = -x_j$

$x_\mu = (t, -x, -y, -z)$

and $\vec{\nabla} = \partial_j = \frac{\partial}{\partial x^j}$

Se $\nu = 0$

$$\partial_0 F^{00} + \partial_j F^{j0} = 0 \rightarrow \vec{\nabla}\cdot\vec{E} = 0 \qquad 1^\circ \text{ Maxwell}$$

Se $\nu = i$

$$\partial_0 F^{0i} + \partial_j F^{ji} = 0$$

$$-\partial_0 E^i - \varepsilon^{ijk}\partial_j B^k = 0 \rightarrow \vec{\nabla}\times\vec{B} - \frac{\partial \vec{E}}{\partial t} \qquad 3^\circ \text{ Maxwell}$$

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Defining: $\tilde{F}^{\mu\nu} := \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}$ (5.5)

The eq of motion gives also zero divergence for $\tilde{F}^{\mu\nu}$ because

$$\partial_\mu \tilde{F}^{\mu\nu} = \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}\partial_\mu F_{\rho\sigma} = \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}\left(\partial_\mu\partial_\rho A_\sigma - \partial_\mu\partial_\sigma A_\rho\right) = 0$$

(ε: antisym in $\mu\rho$ and $\mu\sigma$; $\partial_\mu\partial_\rho$: sym in $\mu\rho$; $\partial_\mu\partial_\sigma$: sym in $\mu\sigma$)

So

$$\partial_\mu \tilde{F}^{\mu\nu} = 0$$

which gives the remaining 2 eq of Maxwell

Se $\nu = 0$ $\longrightarrow$ $\vec{\nabla}\cdot\vec{B} = 0$ 2° Maxwell

Se $\nu = i$ $\longrightarrow$ $\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}$ 4° Maxwell

Hamiltonian and conjugate momentum

From $\mathcal{L}$ we can write the conjugate momentum for each A^μ

$$\Pi^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_0 A_\mu)} \longrightarrow \begin{cases} \Pi^0 = \frac{\partial\mathcal{L}}{\partial(\partial_0 A_0)} \overset{\otimes}{=} 0 & \text{gauge invariant } \checkmark \\ \Pi^i = \frac{\partial\mathcal{L}}{\partial(\partial_0 A_i)} \overset{\otimes}{=} +E^i & \text{gauge invariant } \checkmark \end{cases}$$

$$\mathcal{H} = \Pi^\mu\partial_0 A_\mu - \mathcal{L} = \Pi^i\partial_0 A_i - \mathcal{L} \overset{\otimes}{=} \dots = \frac{1}{2}(\vec{E}^2+\vec{B}^2) + A_0(\vec{\nabla}\cdot\vec{E})$$

(+ termine non gauge invariante)

$$H = \int d^3x\,\mathcal{H} \overset{\otimes}{=} \dots = \int d^3x\,\frac{1}{2}(\vec{E}^2+\vec{B}^2) \quad \text{gauge invariant } \checkmark$$

⊗ questi risultati che han validità generale in ogni gauge li calcoleremo più avanti mettendoci in un gauge specifico

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Gauge invariance

The MAXWELL Lagrangian $\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$ has a CRUCIAL LOCAL SYMMETRY UNDER transformations of the form:

$$A_\mu \longrightarrow A_\mu' = A_\mu - \partial_\mu \theta(x) \qquad (5.6)$$

called GAUGE TRANSFORMATION (è locale perché $\theta(x)$ dipende dal punto su cui sto agendo la trasf., ogni punto viene trasformato diversamente).

I campi $\vec{E}, \vec{B}$ e dunque $F_{\mu\nu}$ e dunque $\mathcal{L}$ sono invarianti sotto tale trasf.

$$F_{\mu\nu}' = \partial_\mu A_\nu' - \partial_\nu A_\mu' = \partial_\mu A_\nu - \partial_\nu A_\mu - \partial_\mu \partial_\nu \theta + \partial_\nu \partial_\mu \theta = \partial_\mu A_\nu - \partial_\nu A_\mu = F_{\mu\nu}$$

so $F_{\mu\nu}' = F_{\mu\nu}$ c.v.d.!

so $\mathcal{L}' = \mathcal{L}$ c.v.d.!

Gauge invariance plays a NON TRIVIAL role in the quantization of field A^μ and in the definition of the Noether charges due to translational and Lorentz symmetries

When computing the Noether charges as $T^{\mu\nu}$ stress tensor or P^μ the momentum or $J^{\mu\nu}$ spin etc etc we must pay attention whether these quantities are or are not gauge invariant. A rule of thumb is:

All quantities which depend only by $\vec{E}, \vec{B}$ or $F_{\mu\nu}$ or $\mathcal{L}$ are gauge invariant. Quantities which depend by A_μ alone are NOT gauge invariant.

However, since all gauges must at the end give the same physics (because $\mathcal{L}$ is gauge invariant, la fisica cioè i campi sono gauge invariant) allora è possibile sempre definire anche le quantità "potenziali" in modo gauge invariant, cioè definirle in modo che siano gauge invariant anche a livello di potenziale e non solo di campo fisico effettivo.

↓

Ago di corrente

ES = $T^{\mu\nu}$ della def di carica di Noether non è gauge invariant ma il momento P^i sì. È comunque possibile def un nuovo $T^{\mu\nu}$ che dà lo stesso P^i, lo stesso fisica, e che sia anche invariante in ogni gauge

Coulomb or radiation Gauge

The coulomb or radiation gauge imposes:

$$A_0 \overset{!}{=} 0 \qquad \text{and} \qquad \vec{\nabla}\cdot\vec{A} \overset{!}{=} 0$$

In this gauge the Maxwell equations become:

$$\partial_\mu F^{\mu\nu} = 0$$
$$\partial_\mu \partial^\mu A^\nu - \partial_\mu \partial^\nu A^\mu = 0$$

but $\partial^\mu \partial_\mu A^\mu = \partial^\nu(\partial_0 A^0 + \partial_i A^i) = \partial^\nu(\partial_0 A^0 + \vec{\nabla}\cdot\vec{A}) = 0$ for conditions of Coulomb gauge, so:

$$\partial_\mu \partial^\mu A^\nu = 0$$

Hence the Maxwell equations in the Coulomb gauge are as Klein-Gordon equations for each component A^ν, but as mantles $m=0$ fields:

$$\partial_\mu F^{\mu\nu} = \boxed{\partial_\mu \partial^\mu} A^\nu = 0 \qquad \text{K-G equation for scalar } A^\nu \text{ with } m=0$$
$$= (\partial_0^2 - \vec{\nabla}^2) A^\nu = 0$$

So each component A^ν is a real scalar field following KG eq but with $m=0$

[Recall, K-G eq for scalar real ϕ is $\partial_\mu \partial^\mu \phi + m^2 \phi = 0$, now case now $A^\nu = \phi$ and $m=0$]

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Lorentz Gauge

the Lorentz Gauge is defined as by the condition:

$$\partial_\mu A^\mu = \partial_0 A^0 + \vec{\nabla}\cdot\vec{A} \overset{!}{=} 0$$

NOTE that the Coulomb Gauge implies also the Lorentz condition. However the Lorentz condition can be satisfied without the coulomb's (i.e $\partial_\mu A^\mu = 0$ can be zero without $A^0 = 0$ being zero.)

Also in the Lorentz Gauge we find the KG eq for A^ν as a massless field

$$\partial_\mu F^{\mu\nu} = \partial_\mu \partial^\mu A^\nu = 0 \qquad \text{KG eq of massless field } A^\nu$$

Translational symmetry — $T^{\mu\nu}$ & P^i

The energy momentum tensor as Noether current is:

$$T^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\rho)} \partial^\nu A_\rho - g^{\mu\nu}\mathcal{L}$$

which is not Gauge invariant 'cause it depends on A_ρ. Computing we find =

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\rho)} = \frac{\partial}{\partial(\partial_\mu A_\rho)}\left(-\tfrac{1}{2}\right)(\partial_\mu A_\rho \partial^\mu A^\rho - \partial_\mu A_\rho \partial^\rho A^\mu) = -(\partial^\mu A^\rho - \partial^\rho A^\mu) = -F^{\mu\rho}$$

hence

$$T^{\mu\nu} = -F^{\mu\rho}\partial^\nu A_\rho + \frac{1}{4} g^{\mu\nu} \overbrace{F^{\alpha\beta}F_{\alpha\beta}}^{F^2} \qquad \text{not (Gauge invariant)}$$

which gives the Hamiltonian:

$$H = \int d^3x\, T^{00} = \dots = \frac{1}{2}\int d^3x\, (\vec{E}^2 + \vec{B}^2) \qquad (\checkmark \text{ Gauge invariant})$$

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And a noether charge, momentum

$$P^i = \int d^3x\, T^{0i} = \dots = \int d^3x\, (\vec{E} \wedge \vec{B})^i$$ (also gauge invariant ✓)

So, even though $T^{\mu\nu}$ is not gauge invariant, its derived quantities H and P^i are gauge invariant. The reason is that under integration, imposing boundary conditions of fields vanishing to infinity, the terms not gauge invariant vanish, and only gauge invariant part is left. We can obtain an "improved" energy momentum tensor which is also gauge invariant by defining:

$$T^{\mu\nu} = F^{\mu\rho} F_\rho{}^\nu + \frac{1}{4} g^{\mu\nu} F^2$$

$$= -F^{\mu\rho} F^\nu{}_\rho + \frac{1}{4} g^{\mu\nu} F^2$$

Belinfante tensor
(✓ gauge invariant)

Lorentz symmetry

We want to compute the spin of the field A^μ. The spin is related to the Lorentz symmetry i.e the symmetry under spacetime rotations. We will consider just only the 3 space rotations in order to compute the spin, the time rotation symmetry gives another property.

a) coordinate transformation of Lorentz

$$x^\mu \longrightarrow x'^\mu = x^\mu + \delta x^\mu$$

Recalling the calculations at the end of the 1st notebook we got:

$$\delta x^\mu = \frac{1}{2} \omega^{\rho\sigma} a^\mu{}_{\rho\sigma}$$

where now I named $A^\mu{}_{\rho\sigma} \to a^\mu{}_{\rho\sigma}$ per non confondermi con il potenziale A^μ

$$a^\mu{}_{\rho\sigma} = \delta^\mu{}_\rho x_\sigma - \delta^\mu{}_\sigma x_\rho$$

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but now considering only spatial rotation $\rho\sigma \to jk$

$$a^{\mu(jk)} = \delta^{\mu j} x^k - \delta^{ek} x^j$$

so $\mu = 0 \quad a^{0jk} = \delta^{0j} x^k - \delta^{0k} x^j = 0$

so we remain with just

$$a^{i(jk)} = \delta^{ij} x^k - \delta^{ik} x^j$$

$$A'^i(x') - A^i(x) = \underbrace{\frac{1}{2}\omega^{jk}}_{\varepsilon^{(jk)}} \underbrace{(\delta^i_j A_k - \delta^i_k A_j)}_{F^{i(jk)}}$$

- Transformation of the field

Since the field now is a 4vector, it transforms exactly as x^μ so:

$$F^{i(jk)} = \delta^{ij} A^k - \delta^{ik} A^j$$

- Conserved current

We can now compute the conserved current $j^{0(jk)}$, the component we will insert in the calculation of the conserved charge, which, spoiler, would be the spin / angular momentum. Using 3.4 with $\phi^i \to A^i$ $A^\nu_a \to a^{\nu(jk)}$:

$$j^{0(jk)} = + \frac{\partial \mathcal{L}}{\partial(\partial_0 A_i)} \left(a^{\ell(jk)} \partial_\ell A_i - F^{A(jk)}_i \right) + \underbrace{a^{0(jk)}}_{=0} \mathcal{L} =$$

(sum only over spatial ℓ since $a^{0(jk)} = 0$)

(sum only over spatial A_i components because we are considering only space rotation)

$$= \partial^0 A^i \left((\delta^{\ell j} x^k - \delta^{\ell k} x^j) \partial_\ell A_i - (\delta^{kj}_i A^k - \delta^{jk}_i A^j) \right) =$$

$$j^{0(jk)} = + \partial^0 A^i \left(x^j \partial^k - x^k \partial^j \right) A_i + \left(A^k \partial_0 A^j - A^j \partial_0 A^k \right)$$

$(A_0 \neq 0)$ Coulomb gauge: $F^{0i} = \partial^0 A^i - \partial^i A^0 = \partial^0 A^i$

$\frac{\partial \mathcal{L}}{\partial(\partial_0 A_i)} = -F^{0i} = -\partial^0 A^i$

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o) Conserved charge

Hence the conserved charge associated to 3D spatial rotations:

$$J^{jk} = \int d^3x \; j^{0(jk)} = \int d^3x \; + \partial_0 A^i (x^j \partial^k - x^k \partial^j) A_i \; + \; + [A^k \partial_0 A^j - A^j \partial_0 A^k]$$

the 1st term is the matrix term of the orbital momentum L^{jk}

$$\langle A | L^{jk} | A^i \rangle = \frac{i}{2} \int d^3x \left[A^i \partial_0 (L^{jk} A_i) - (\partial_0 A^i) L^{jk} A_i \right] \quad \leftarrow (3.25)$$

from eq 3.25 (1st page of this notebook) but with a sum over i.
(because in eq 3.25 we were considering ϕ scalar fields, now $\phi_1 = A^i$ and $\phi_2 = L^{jk} A_i$ and we are summing over i index)

$$L^{jk} = i (x^j \partial^k - x^k \partial^j)$$

the 2nd term instead is new! It compares only for vector field A^μ, for scalar field is zero! We can identify it as a SPIN term

$$\langle S^{jk} \rangle = \int d^3x \left(-A^j \partial_0 A^k + A^k \partial_0 A^j \right)$$

↓

expectation value of the spin S^{ij} on vector field A^μ

(@) We don't impose the existence of spin, we find it as a natural consequence of symmetry

$$J^{jk} = \langle L^{jk} \rangle + \langle S^{jk} \rangle$$

(gauge invariant ✓)

(@ L^{ij} and S^{ij} alone are not gauge invariant, but J^{jk} TOT IS gauge invariant)

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$$S^i = \frac{1}{2}\varepsilon^{ijk} S^{jk} = \frac{1}{2}\varepsilon^{ijk}\int d^3x\,(-A^j\partial_0 A^k + A^k\partial_0 A^j) \quad \Rightarrow \quad \varepsilon^{ijk} = -\varepsilon^{ikj}$$

$$= -\frac{1}{2}\int d^3x\,(\varepsilon^{ijk} A^j\partial_0 A^k + \varepsilon^{ikj} A^k\partial_0 A^j) =$$

$$= -\int d^3x\ \varepsilon^{ijk} A^j\partial_0 A^k$$

in RADIATION GAUGE we have $A^0 = 0$ so $E^i = -F^{0i} = -\partial^0 A^i + \partial^i A^0$ ($\partial^i A^0 = 0$)

so $E^i = -\partial^0 A^i = -\partial_0 A^i$ so

$$= -\int d^3x\ \varepsilon^{ijk}(-A^j E^k)$$

hence

$$S^i = -\int d^3x\ \varepsilon^{ijk} E^j A^k \quad \rightarrow \quad \vec{S} = -\int d^3x\,(\vec{E}\times\vec{A})$$

in RADIATION GAUGE

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5.2 QUANTIZATION IN COULOMB GAUGE

Gauge invariance plays a nontrivial role in the quantization. We will address this coherence problem later, now we just pick a gauge and quantize in it. We pick the Coulomb gauge. As we saw, the coulomb gauge

$$A_0 \overset{!}{=} 0 \quad \& \quad \vec{\nabla}\cdot\vec{A} \overset{!}{=} 0$$

leads to the equations

$$\partial_\mu \partial^\mu A^\nu = 0$$

which are K-G equations for the scalar field A^ν with a mass $m=0$ zero

So each component A^ν is a scalar field with mass $=0$ zero, so the solution for each A^ν is the solution of a scalar field

However now we imposed $A_0 \overset{!}{=} 0$ so we need to look just for $\vec{A}$ spatial, 3 dimensions, 3 possible directions of polarization. However the 2nd equation $\vec{\nabla}\cdot\vec{A}=0$ constrains into just 2 degree of freedom instead of 3. We can then write

$$\hat{\vec{A}} = \int \frac{d^3p}{(2\pi)^3\sqrt{2\omega_p}} \sum_{\lambda=1}^{2} \left(\vec{\varepsilon}(\vec{p},\lambda)\, \hat{a}_{\vec{p},\lambda}\, e^{-ip\cdot x} + \vec{\varepsilon}^{\,*}(\vec{p},\lambda)\, \hat{a}^{+}_{\vec{p},\lambda}\, e^{ip\cdot x} \right)$$

$\omega_{\vec{p}} = p^0 = |\vec{p}|$ because $m=0$

λ index for polarization $\lambda = 1,2$ just two degrees of freedom

polarization vector, depends on $\vec{p}$ and on λ, with two possibilities.

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Commutation relations

ora seguiamo un procedimento logico un po diverso da quanto fatto per i campi scalari ϕ, il perché di questa scelta sarà chiaro più avanti ed è che fare con la libertà di gauge.

Ora IMPONIAMO che valgano le relazioni di commutazione tra $\hat{a}_p$ and $\hat{a}_p^\dagger$

$$[\hat{a}_{\vec{p},\lambda}, \hat{a}^\dagger_{\vec{p}',\lambda'}] \overset{!}{=} (2\pi)^3 \delta^{(3)}(\vec{p}-\vec{p}')\,\delta\lambda\lambda'$$

$$[\hat{a}_{\vec{p},\lambda}, \hat{a}_{\vec{p}'\lambda'}] = [\hat{a}^\dagger_{\vec{p}\lambda}, \hat{a}^\dagger_{\vec{p}'\lambda'}] \overset{!}{=} 0$$

Having imposed these commutations, recalling that for A^μ vector the conjugate momentum is

$$\Pi^0 = \phi \qquad \Pi^i = -F^{0i} = E^i$$

we can deduce the commutations:

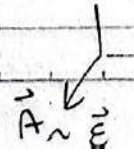
$$[A^i(t,\vec{x}), \Pi^j(t,\vec{x}')] = -i\int \frac{d^3\vec{p}}{(2\pi)^3} e^{i\vec{p}\cdot(\vec{x}-\vec{x}')} \frac{1}{2}\sum_{\lambda=1}^{2}\Big(\mathcal{E}^i_{(\vec{p},\lambda)}\mathcal{E}^{j*}_{(\vec{p},\lambda)} + \mathcal{E}^{i*}_{(-\vec{p},\lambda)}\mathcal{E}^j_{(-\vec{p},\lambda)}\Big)$$

Picking the frame $\vec{p} = (0\,0\,\overset{z}{p})$ (the result will be frame invariant) we can choose $\mathcal{E}(\vec{p},1) = (\overset{x}{1}\,0\,0)$ and $\vec{\mathcal{E}}(\vec{p},2) = (0\,\overset{y}{1}\,0)$ (because $\vec{A}$ is $\perp$ to the propagation direction) the linear polarization (its just one possibility, not the general one possibility), we get

$$\frac{1}{2}\sum_\lambda\left(\mathcal{E}^i_{(p)}\mathcal{E}^{j*}_{(p)} + \mathcal{E}^{i*}_{(-p)}\mathcal{E}^j_{(-p)}\right) = \delta^{ij} - \frac{p^i p^j}{|p|^2}$$

resulting in:

perché con Fotone allora $\vec{A}$ è $\perp$ alla direzione di prop, allora $\vec{\mathcal{E}}$ è $\perp$ alla direz di prop uscente da $\vec{p}$



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$$\left[A^i(t,\vec{x}),\, \Pi^j(t,\vec{x}')\right] = -i\int \frac{d^3p}{(2\pi)^3}\, e^{i\vec{p}\cdot(\vec{x}-\vec{x}')} \left(\delta^{ij} - \frac{p^i p^j}{|p|^2}\right) \qquad (5.9)$$

This term is specific for the Coulomb Gauge

If $p^i p^j/|p|^2$ were not there, integrating over $\vec{p}$ we would get

$$\int \frac{d^3p}{(2\pi)^3}\, e^{i\vec{p}\cdot(\vec{x}-\vec{x}')} = \delta^{(3)}(\vec{x}-\vec{x}')$$

and $\left[A^i(t,\vec{x}),\, \Pi^j(t,\vec{x}')\right] = -i\,\delta^{(3)}(x-x')\,\delta^{ij}$

thus recovering the commutation of the scalar field ϕ

Taking ∂_x or $\partial_{x'}$ and calculating $\partial \cdot \square = p^j(\delta^{ij} - \frac{p^i p^j}{|p|^2}) = 0$ so this would correspond to either

$$\left[\vec{\nabla}\cdot\vec{A}(t,\vec{x}),\, \vec{E}(t,\vec{x}')\right] = \left[\hat{A}(t,\vec{x}),\, \vec{\nabla}\cdot\vec{E}(t,\vec{x}')\right] = 0$$

This can be explained by the Gauge position $\vec{\nabla}\cdot\vec{A} \overset{!}{=} 0$

And this can be expected because $\vec{\nabla}\cdot\vec{E} = \rho$ is a complex number and not an operator so $[\hat{A}, \rho] = 0$ (ρ: number)

Vacuum state and Fock space

We impose the $\exists$ of $|0\rangle$ vacuum state imposing

$\hat{a}_{p\lambda}|0\rangle \overset{!}{=} \phi$ for all $\vec{p}$ and λ

and build the Fock spaces by $\sqrt{2\omega_p}\,\hat{a}^\dagger_{p\lambda}|0\rangle = |p_\lambda\rangle$

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Hamiltonian in normal ordering

the classical Hamiltonian was

$$H = \frac{1}{2}\int d^3x \ (\vec{E}^2 + \vec{B}^2)$$

hence we compute

$$\hat{H} = \frac{1}{2}\int d^3x : (\vec{E}^2 + \vec{B}^2) : = \dots = \int \frac{d^3p}{(2\pi)^3} \ \omega_{\vec{p}} \sum_{\lambda=1}^{2} \hat{a}^+_{p\lambda}\, \hat{a}_{p\lambda}$$

Traslational symmetry — momentum and mass

classical $P^i = \int d^3x \ T^{0i} = \dots = \int d^3x \ (\vec{E} \times \vec{B})^i$ which is the POYNTING VECTOR

$$\hat{P}^i = \int d^3x : (\vec{E}\times\vec{B})^i : = \dots = \int \frac{d^3p}{(2\pi)^3} \sum_{\lambda=1}^{2} p^i \, \hat{a}^+_{\vec{p}\lambda}\, \hat{a}_{\vec{p}\lambda}$$

↓ Noether operator charge of the whole field A^μ

↓ eigenvalue of each $|p\lambda\rangle$

that is very similar to the scalar field with just an additional sum over λ polarizations.

$\Rightarrow$ this tells us that $\sqrt{2\omega_p}\, a^+_{p\lambda}|0\rangle = |p\lambda\rangle$ is a state of momentum p and polarization λ, with 0 mass $m=0$ because of coulomb gauge $\Rightarrow$ this would be the photon!

Spatial rotations (Lorentz) symmetry — Spin

classical $S^{ij} = \int d^3x \ (A^i \partial_0 A^j - A^j \partial_0 A^i)$

quantization $\hat{S}^{ij} = \int d^3x : (A^i \partial_0 A^j - A^j \partial_0 A^i) : = \dots$ some algebra $\dots$

$$\hat{S}^{ij} = i \int \frac{d^3p}{(2\pi)^3} \sum_{\lambda,\lambda'} \left[\varepsilon^i(p,\lambda')\, \varepsilon^{j*}(p,\lambda) - \varepsilon^{i*}(\vec{p},\lambda)\, \varepsilon^j(p,\lambda') \right] \hat{a}^+_{p,\lambda}\, \hat{a}_{p,\lambda'}$$

$$\hat{S}^i = \frac{1}{2}\varepsilon^{ijk}\, S^{jk} = \int d^3x \ \varepsilon^{ijk} (-\partial_0 A^j) A^k = \int d^3x \ \varepsilon^{ijk}\, E^j A^k$$

$$\vec{S} = \int d^3x \ (\vec{E}\times\vec{A}) \quad \text{coulomb gauge} \quad -\partial_0 A^j = E^j \ \text{in coulomb gauge}$$

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While for $\hat{H}$ and $\hat{P}^i$ the particle states $a^+_{\vec{p}\lambda}|0\rangle$ are eigenstates of $\hat{H}$ and $\hat{P}^i$, for $\hat{S}^{ij}$ it is not so trivial. We want to compute the eigenstates of S^{ij}: let's start by computing the action of S^{ij} on $a_{p\lambda}|0\rangle$ generic particle state

$$\hat{S}^{ij}\underbrace{\hat{a}^+_{\vec{k}\lambda''}|0\rangle}_{\text{particle state of momentum } \vec{k} \text{ and polarization } \lambda''} =$$

$$= i\int \frac{d^3p}{(2\pi)^3} \sum_{\lambda\lambda'} [\dots] a^+_{p\lambda} a_{p\lambda'} a^+_{k\lambda''}|0\rangle =$$

$$\left([a_{p\lambda'}, a^+_{k\lambda''}] + a^+_{k\lambda''} a_{p\lambda'}\right)|0\rangle$$

$= 0$ because $a_{p\lambda}|0\rangle \equiv \emptyset$ annihilates

$$= i\int \frac{d^3p}{(2\pi)^3} \sum_{\lambda\lambda'} [\dots] a^+_{p\lambda}\left([a_{p\lambda'}, a^+_{k\lambda''}]\right)|0\rangle =$$

$$= i\int \frac{d^3p}{(2\pi)^3} \sum_{\lambda\lambda'} [\dots] (2\pi)^3 \delta^{(3)}(\vec{p}-\vec{k})\, \delta_{\lambda'\lambda''}\, a^+_{p\lambda} = \rightarrow$$

→ a) Svolgo la Sommatoria $\delta_{\lambda'\lambda''}$: $\lambda' \to \lambda''$

b) e ho $\int d^3p\, \delta^{(3)}(\vec{p}-\vec{k})$ delta di Dirac $\vec{p} \to \vec{k}$

$$= i\sum_{\lambda}\left[\varepsilon^i(\vec{k}\lambda'')\varepsilon^{j*}(\vec{k},\lambda) - \varepsilon^{j*}(\vec{k},\lambda)\varepsilon^i(k\lambda'')\right] a^+_{k\lambda}|0\rangle$$

LINEAR POLARIZATION STATES

a) Now we choose the particle with a momentum along the z axis
$\vec{k} \equiv (0\ 0\ k)$ (choice)

b) Then we choose to compute the spin along the same direction of $\vec{k}$ the momentum, which is the HELICITY h

$$h = \frac{\vec{k}\cdot\vec{s}}{|\vec{s}-\vec{k}|}$$

$\vec{s} = (s_1, s_2, s_3)$ spin vector
$\vec{k} = (k_1, k_2, k_3)$ momentum vector

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The tensor S^{ij} and vector of spin $\vec{S} = (S^1\ S^2\ S^3)$ are related by the relation

$$S^k = \frac{1}{2}\varepsilon^{ijk} S^{ij} \qquad S^{21} = -S^{12}$$

Hence $S^3 = \frac{1}{2}(S^{12} - S^{21}) = S^{12}$

So we want to compute S^{12} to compute the spin in the z direction which is proportional to the helicity.

$i = 1 \quad j = 2 \rightarrow S^3 = S^{12}$ (choice) lyo z

•) Then we decide to look first at the simplest polarization i.e the Linear polarization, with respect to transverse requirement $\vec{p}\cdot\vec{E} = 0$ given by the $\vec{\nabla}\cdot\vec{A} = 0$ of Coulomb gauge, the Linear polarization is only transverse in direction 1 and 2

$$\vec{\varepsilon}(\vec{k},1) = (1\ 0\ 0) \quad \text{lyo x}$$
$$\vec{\varepsilon}(k,2) = (0\ 1\ 0) \quad \text{lyo y}$$

$\hat{1}$, $\hat{2}$

$\mapsto \varepsilon^i(k\lambda) = \delta^i_\lambda$ with $i = 1 \text{ or } 2$, $\lambda = 1 \text{ or } 2$

Hence with this choices

$$\hat{S}^{12}\, a^\dagger_{k\lambda''}|0\rangle = i \sum_\lambda \left[\varepsilon^1(k\lambda'')\,\varepsilon^{2*}(k\lambda) - \varepsilon^{1*}(k\lambda)\,\varepsilon^2(k\lambda'')\right] a^\dagger_{k\lambda}|0\rangle =$$

$$= i \sum_\lambda (\delta^1_{\lambda''}\delta^2_\lambda - \delta^1_\lambda \delta^2_{\lambda''})\, a^\dagger_{k\lambda}|0\rangle$$

$S^3 = S^{12}$

So we see that

$$S^{12}\, a^\dagger_{k1}|0\rangle = i\, a^\dagger_{k2}|0\rangle$$
$$S^{12}\, a^\dagger_{k2}|0\rangle = -i\, a^\dagger_{k1}|0\rangle$$

Linear polarization states $a^\dagger_{k\lambda}|0\rangle$ are NOT eigenstate of the Helicity S^{12} !!

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CIRCULAR POLARIZATION STATES:

Defing the circular polarization states:

$$a^{+}_{\vec{k}\pm} = \frac{1}{\sqrt{2}} \left(a^{+}_{\vec{k}1} \pm i a^{+}_{\vec{k}2} \right)$$

We see that

$$S^{M} = \hat{S}^{12} \, a^{+}_{\vec{k}+} |0\rangle = a^{+}_{\vec{k}+} |0\rangle$$
$$\hat{S}^{12} \, a^{+}_{\vec{k}-} |0\rangle = - a^{+}_{\vec{k}-} |0\rangle$$

Circular polarization state $a^{+}_{\vec{k}+}|0\rangle$ & $a^{+}_{\vec{k}-}|0\rangle$ are eigenstates of helicity S^{12} with eigenvalues $+1$ and -1

=> conclusion: states $a^{+}_{\vec{k}\pm}|0\rangle$ are particles with momentum $\vec{k}$, energy $\omega_{\vec{k}} = |\vec{k}|$, mass $m = 0$, spin 1 (cause vector field A^{μ} is related to spin 1 and cause eigenvalues of S^{12} are ± 1) and helicity ± 1

=> these particles are PHOTONS

Final considerations

Even though we choose a specific gauge which was not Lorentz (Poincaré) invariant cause of the request $\vec{\nabla} \cdot \vec{A} = 0$ $A^{0} = 0$ in a specific frame, all the results we obtained are Lorentz (& Poincaré) invariant.

=> we do not demonstrate it but it's true => so all the results we obtain are valid in general and not only in our gauge in our specific frame, valid for every frame

=> we just need to pay attention that H, P^{i} and J^{ior} are also gauge invariant, but $T^{\mu\nu}$, S and L alone are not gauge invariant but specific for the transverse gauge.

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5.3 COVARIANT QUANTIZATION (M 4.3.2)

We now see another way to quantize A^μ, more general, supposing to quantize A_μ with no conditions, no gauge choice.

Trying straightforward quantization → 1° problem

Following the natural procedure of quantization we try to impose the commutations between A^μ and π^ν ($g^{ij} = -1$ se $i=j$ spaziale)

$$[A^\mu(t,\vec{x}), \pi^\nu(t,\vec{x}')] \stackrel{!}{=} i g^{\mu\nu} \delta^{(3)}(\vec{x}-\vec{x}') \qquad (5.12)$$

($g^{\mu\nu}$: new term just to account for $\mu\nu$ components)

But we see immediately a problem, an incoherence in this equation!

From one side $\pi^\nu = \frac{\partial \mathcal{L}}{\partial(\partial_0 A_\nu)} \rightarrow \pi^0 = -F^{00} = 0$ cause $\mathcal{L} = -\frac{1}{4} F^{\mu\nu} F_{\mu\nu}$

$\rightarrow$ so $[A^0, \pi^0] = [A^0, 0] = 0$

From other side $[A^0, \pi^0] \stackrel{!}{=} i g^{00} \delta^{(3)}(\vec{x}-\vec{x}') = i\delta^{(3)}(\vec{x}-\vec{x}') \neq 0$ not zero!

$\Rightarrow$ so this definition with this $\mathcal{L}$ is not a good choice!

1° solution = Gupta-Bleuler quantization $\mathcal{L}'$

To fix this problem we decide to change our $\mathcal{L} \rightarrow \mathcal{L}'$ into

$$\mathcal{L}' = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2}(\partial_\mu A^\mu)^2 \qquad (5.14)$$

which adds this new term which is NOT GAUGE INVARIANT, so $\mathcal{L}'$ is no more gauge invariant! However the additional term is a total divergence hence doesn't affect the Euler-Lagrange eq, so the physics it's the same

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$$\mathcal{L}_{GB} = -\frac{1}{2}(\partial_\alpha A^\alpha)^2 = -\frac{1}{2}(\partial_\alpha A_\beta g^{\alpha\beta})^2 = -\frac{1}{2}(X(\partial_\alpha A_\beta))^2 \quad | \quad X = \partial_\alpha A^\alpha = \partial_\alpha A_\beta g^{\alpha\beta}$$

$$\frac{\partial \mathcal{L}}{\partial(\partial_\alpha A_\mu)} = \frac{\partial \mathcal{L}}{\partial X}\frac{\partial X}{\partial(\partial_\alpha A_\mu)} = -X\frac{\partial X}{\partial(\partial_\alpha A_\mu)} = -\partial_\alpha A^\alpha g^{\alpha\mu}$$ CVD

With this new definition of $\mathcal{L}'$ the problem is fixed because

$$\pi'^\mu = \frac{\partial \mathcal{L}'}{\partial(\partial_0 A_\mu)} = +F^{0\mu} - g^{0\mu}\partial_\rho A^\rho$$

↓ new term

Hence
$$\begin{cases} \pi'^0 = -\partial_\rho A^\rho & \text{the time term is no more zero!} \\ \pi'^i = -F^{0i} = E^i & \text{the space term is the same as before} \end{cases}$$

$\Rightarrow$ the problem is now fixed so because $\pi'^0 \neq 0$ no more zero so

$$[A^\mu(t,\vec{x}), \pi'^\nu(t,\vec{x}')] \stackrel{!}{=} i g^{\mu\nu}\delta^{(3)}(\vec{x}-\vec{x}') \qquad (5.12)$$

is now a good imposition for quantization ✓

Quantization

The equations of motion derived from the new Lagrangian $\mathcal{L}'$ reduce one time more, to Klein-Gordon equation for massless field:

$$\partial_\mu\left(\frac{\partial \mathcal{L}'}{\partial(\partial_\mu A_\nu)}\right) - \frac{\partial \mathcal{L}'}{\partial A_\nu} = 0 \quad \Rightarrow 0$$

$$\partial_\mu(-\partial^\mu A^\nu + \partial^\nu A^\alpha) - \partial_\mu\partial^\nu A^\mu = 0$$ ← from new part

$$(\partial_\mu\partial^\mu)A^\nu = 0$$ K-G equation for scalar A^ν components with $m=0$ zero

So we can quantize A^ν each component as a scalar field. However, now, A^μ can take 4 DIFFERENT DIRECTION of polarization because $A^0 \neq 0$ is no more zero and because there are no gauge conditions to satisfy which would reduce the degrees of freedom of polarization

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$$\hat{A}_\mu(t,\vec{x}) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{p}}}} \sum_{\lambda=0}^{3} \left[\varepsilon_\mu(\vec{p},\lambda)\, \hat{a}_{\vec{p}\lambda}\, e^{-ip\cdot x} + \varepsilon^*_\mu(\vec{p},\lambda)\, \hat{a}^\dagger_{\vec{p}\lambda}\, e^{ip\cdot x} \right]$$

$\omega_{\vec{p}} = p^0 = |\vec{p}|^2$ as it is massless

$\lambda = 0\,1\,2\,3$ polarization can be in all 4 direction, time x y and z

polarization vector ε^μ is now a 4-vector with index $\mu = 0123$ and not just spatial $j = 123$

Commutations → 2° problem

Good! Let's now impose the commutation (5.12) between A^ν and Π^μ

$[A^\mu(t,\vec{x}), \Pi^\nu(t,\vec{x}')] \overset{!}{=} i g^{\mu\nu} \delta^{(3)}(\vec{x}-\vec{x}')$, these lead us to the commutation relations:

$$[\hat{a}_{\vec{p}\lambda}, \hat{a}^\dagger_{\vec{p}',\lambda'}] = -(2\pi)^3 \delta^{(3)}(\vec{p}-\vec{p}')\, g_{\lambda\lambda'} \qquad (5.15)$$

FOR $\lambda = 1,2,3$

For spatial polarization $\lambda = 1,2,3$ we get $g_{\lambda\lambda} = -1$ with an additional minus sign which makes our relation positive ($g_{\lambda\lambda'} = -\delta_{\lambda\lambda'}$ if $\lambda,\lambda' = 123$)

$$[\hat{a}_{p\lambda}, \hat{a}^\dagger_{p',\lambda'}] = (2\pi)^3 \delta^{(3)}(\vec{p}-\vec{p}')\, g_{\lambda\lambda'}$$ ok ✓

FOR $\lambda = 0$

For time polarized states we have a problematic − sign! $g_{00} = 1$ so:

$$[\hat{a}_{p0}, \hat{a}^\dagger_{p',0}] = -(2\pi)^3 \delta^{(3)}(\vec{p}-\vec{p}')$$ ⚠

This is a problem because leads to a norm of $|p,0\rangle$ state which is negative! → see dietro

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state $|p\lambda\rangle = \sqrt{2\omega_p}\, a^\dagger_{p\lambda}|0\rangle \xrightarrow{\lambda=0} |p0\rangle = \sqrt{2\omega_p}\, a^\dagger_{p0}|0\rangle$

norm $\langle p0|p0\rangle = 2\omega_p \langle 0| a_{p0} a^\dagger_{p0}|0\rangle = \langle 0|[a_{p0}, a^\dagger_{p0}]|0\rangle$ $\Rightarrow$ $a_{p0}a^\dagger_{p0} = [a_{p0}, a^\dagger_{p0}] + a^\dagger_{p0}a_{p0}$, a_{p0} zero when acting on $|0\rangle$, $=0$

$= 2\omega_p \langle 0|[a_{p0}, a^\dagger_{p0}]|0\rangle =$

$= 2\omega_p(-(2\pi)^3 \delta(0) g_{00})$

$= -2\omega_p (2\pi)^3 \delta(0) < 0$ IS NEGATIVE!

This is a problem because in Q. Mechanics the norm of states are probabilities hence they cannot be negative!

2nd solution = physical states

We solve this problem by defining which states we consider as physical, (= valid state) or not physical (= not valid state). In sostanza RESTRINGIAMO lo spazio degli stati su cui lavoriamo, non li consideriamo tutti, solo alcuni, e li chiamiamo "stati fisici". The condition which defines a physical state is:

$$\langle \text{phys}|\, \partial_\mu A^\mu\, |\text{phys}\rangle \overset{!}{=} 0 \qquad (5.13)$$

i.e. a physical state is a state where $\partial_\mu A^\mu = 0$, so when considering the subspace of $|\text{phys}\rangle$ states we are working in a space with a Lorentz gauge condition $\partial_\mu A^\mu = 0$ is satisfied.

We can say we IMPOSE the Lorentz gauge $\partial_\mu A^\mu \overset{!}{=} 0$ BUT only at OPERATOR LEVEL ACTING ON PHYSICAL STATES

$\Rightarrow$ this imposition fix the problem cause, as we will demonstrate in a minute, $|p0\rangle$ is not a physical state ✓

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- We define $\partial_\mu A^\mu := (\partial_\mu A^\mu)^{(+)} + (\partial_\mu A^\mu)^{(-)}$

$$(\partial_\mu A^\mu)^{(+)} := \int \frac{d^3p}{(2\pi)^3 \sqrt{2\omega_p}} \sum_{\lambda=0}^{3} \left(-ip_\mu \varepsilon^\mu(\vec{p},\lambda)\, e^{-ip\cdot x}\, \hat{a}_{\vec{p}\lambda} \right)$$

$$(\partial_\mu A^\mu)^{(-)} := \int \frac{d^3p}{(2\pi)^3 \sqrt{2\omega_p}} \sum_{\lambda=0}^{3} \left(ip_\mu \varepsilon^\mu(\vec{p},\lambda)\, e^{ip\cdot x}\, \hat{a}^\dagger_{\vec{p}\lambda} \right)$$

such that the requirements 5.13 of a physical state become:

$$5.13 \Rightarrow \langle phys | \partial_\mu A^\mu | phys \rangle = \langle phys | (\partial_\mu A^\mu)^{(-)} + (\partial_\mu A^\mu)^{(+)} | phys \rangle \overset{!}{=} 0$$

$$\Rightarrow (\partial_\mu A^\mu)^{(+)} | phys \rangle \overset{!}{=} 0$$

$$\Rightarrow \langle phys | (\partial_\mu A^\mu)^{(-)} \overset{!}{=} 0$$

Cause in fact $(\partial_\mu A^\mu)^{(-)} = (\partial_\mu A^\mu)^{(+)\dagger}$ (↑ adjoint sign; ↓ index, label of $(\partial_\mu A^\mu)^{(+)}$)

- Consider now a 1-particle states completely general as a superposition of all 4 polarizations

$$|\psi\rangle = \sum_{\lambda=0}^{3} C_\lambda\, \hat{a}^\dagger_{\vec{p}\lambda} |0\rangle \qquad \text{with } C_\lambda \in \mathbb{C}$$

choosing the momentum along z axis (with no losses of generality because we can always choose our axis in order to make $\vec{u}$ along $\vec{z}$) we have:

$$p^\mu = p(1, 0, 0, 1) \qquad (t, x, y, z)$$

→ This form for the time like component because $p^2 = p_\mu p^\mu \overset{!}{=} 0$ must be zero since A^μ is a light like vector of a particle with $m = 0$. So fixed 1 along z the choice of 1 along t is obliged.

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- now we impose $|\psi\rangle$ to be physical hence

$$(\partial_\mu A^\mu)^{(+)} |\psi\rangle \stackrel{!}{=} 0$$

that gives $C_0 + C_3 \stackrel{!}{=} 0 \rightarrow C_0 \stackrel{!}{=} -C_3$

So $|\psi\rangle = \sum_{\lambda=0}^{3} C_\lambda \, a^\dagger_{p\lambda} |0\rangle$ is physical only if $C_0 + C_{\hat{p}} = 0$ where $C_{\hat{p}}$ is the direction of the spatial $\vec{p}$ momentum

$\Rightarrow$ BUT this means that states $|\psi_0\rangle = a^\dagger_{p0}|0\rangle$ and $|\psi_3\rangle = a^\dagger_{p3}|0\rangle$ alone are not physical! because we must always have $C_0 = -C_3$ not just one of the two!

$\Rightarrow$ This fix the problem of negative sign of the norm of $|p_0\rangle \sim a^\dagger_{p0}|0\rangle$ because $|p_0\rangle$ is not a physical state! ✓

Equivalence class of physical states

The imposition of physical states is more interesting than just denying the existence of $|p_0\rangle$ or $|p_3\rangle$

$|p_1\rangle$ and $|p_2\rangle$ are physical states $\Rightarrow$ the TRANSVERSE states (transverse respect to the momentum $\vec{p}$) are physical state, as well as every combination of them $|\psi_T\rangle = (C_1 a^\dagger_{p1} + C_2 a^\dagger_{p2})|0\rangle$

But also every superposition of $|p_0\rangle$ and $|p_3\rangle$ with coeff $C_0 = -C_3$ is a physical states! the simplest with $C_0 = -C_3 = 1$ is

$$|\phi\rangle := |p_0\rangle - |p_3\rangle = (a^\dagger_{p0} - a^\dagger_{p3})|0\rangle$$

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- NORM OF $|\phi\rangle$ IS zero $\rightarrow \langle\phi|\phi\rangle = 0$

$$\langle\phi|\phi\rangle = \langle 0|(a_{p0} - a_{p3})(a^\dagger_{p0} - a^\dagger_{p3})|0\rangle =$$

since $a_3 a_0^\dagger$ and $a_3 a_0^\dagger$ are zero since generate $\langle P_0|P_3\rangle = 0$ $\langle P_3|P_0\rangle = 0$ otherwise

$$= \langle 0|\, a_{p0}a^\dagger_{p0} + a_{p3}a^\dagger_{p3}\,|0\rangle = \langle p_0|P_0\rangle + \langle P_3|P_3\rangle$$

$$= \langle 0|\,[a_{p0}, a^\dagger_{p0}] + [a_{p3}, a^\dagger_{p3}]\,|0\rangle =$$

$$= \langle 0|\, -(2\pi)^3\delta(0) + (2\pi)^3\delta(0)\,|0\rangle = 0$$

$\Rightarrow \langle\phi|\phi\rangle = 0$ norm is zero!

- $|\phi\rangle$ is orthogonal to $|\psi_T\rangle \rightarrow \langle\psi_T|\phi\rangle = 0$

For every transverse state $|\psi_T\rangle = (C_1 a^\dagger_{p\vec{1}} + C_2 a^\dagger_{p\vec{2}})|0\rangle$, $|\phi\rangle$ is orthogonal

i.e.:

$$\langle\phi|\psi_T\rangle = \langle 0|(a_{p0} - a_{p3})(C_1 a^\dagger_{k\vec{1}} + C_2 a^\dagger_{k\vec{2}})|0\rangle = \dots = 0$$

$\Rightarrow \langle\phi|\psi_T\rangle = \langle\psi_T|\phi\rangle = \emptyset$ — they commute $a_{p0}\, a^\dagger_{k\vec{2}} = a^\dagger_{k\vec{2}}\, a_{p0}$

- $|\phi\rangle$ is orthogonal to every $|phys\rangle$ state $\rightarrow \langle\phi|phys\rangle = 0$

$$\langle\phi|phys\rangle = \langle 0|(a_{p0} - a_{p3})|phys\rangle = \langle 0|(\hat{a}_{p0} - \hat{a}_{p3})\sum_{\lambda=0}^{3} C_\lambda a^\dagger_{k\lambda}|0\rangle =$$

not zero only when $\lambda = 0$ or $\lambda = 3$ because of commutation $a_{p0}\, a^\dagger_{k_{1/2}}$ and $a^\dagger_{k_{1/2}}\, a_{p0}|0\rangle = \emptyset$ because $a_{p0}|0\rangle = \emptyset$ annihilates

$$= \langle 0|(a_{p0}\, a^\dagger_{k0}\, C_0 - a_{p3}\, a^\dagger_{k3}\, C_3)|0\rangle =$$

as usual $aa^\dagger = [a\,a^\dagger] + a^\dagger a$, zero on $|0\rangle$

$$= \langle 0|\, C_0\,[a_{p0}, a^\dagger_{k0}] - C_3\,[a_{p3}, a^\dagger_{k3}])|0\rangle =$$

$$= -(2\pi)^3\,\delta^{(3)}(\vec{k} - \vec{p})\,(C_0 + C_3) = 0 \qquad \text{CVD}$$

$= 0$ on physical states!

$$\rightarrow (a_{p0} - a_{p3})|phys\rangle = \emptyset \qquad (5.17)$$

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$\Rightarrow$ This tells that all scalar products of $|\psi_T\rangle + C|\phi\rangle$ with any other physical state are the same as the scalar product of $|\psi_T\rangle$

$$(\langle\psi_T| + C\langle\phi|)|phys\rangle = \langle\psi_T|phys\rangle$$

- $\langle phys'|-a^\dagger_{p^0}a_{p^0}+a^\dagger_{p^3}a_{p^3}|phys\rangle = 0$ is zero on all phys states

$\langle phys'|-a^\dagger_{p^0}a_{p^0}+a^\dagger_{p^3}a_{p^3}|phys\rangle$ with $|phys'\rangle$ and $|phys\rangle$ two different physical states

$= \langle phy'|(-a_{p^0}{}^\dagger + a_{p^3}{}^\dagger)a_{p^3}|phys\rangle = \emptyset$

5.17: $a_{p^0}|phys\rangle = +a_{p^3}|phys\rangle$

$= 0$ because of 5.17 on the hermitian coniugate $\langle phys|(a^\dagger_{p^0} - a^\dagger_{p^3}) = \emptyset$ QED!

$\Rightarrow$ This tells that $\hat{H}$ energy and $\hat{P}^i$ momentum (which contain in their expression the therms $-a^\dagger_{p^0}a_p + a^\dagger_{p^3}a_{p^3}$) receive contributions from just the transverse modes $a^\dagger_{p^1}a_{p^1} + a^\dagger_{p^2}a_{p^2}$!!

In conclusion we can add a $|\phi\rangle$ term to any transverse state (physical) without changing its norm, scalar products and Energy an momentum

i.e.

There is an equivalent class of physical transverse states defined up to (a meno di) a factor $C|\phi\rangle$:

$$|\psi_T\rangle \leftrightarrow |\psi_T\rangle + C|\phi\rangle \qquad C\in\mathbb{C}$$

$\Downarrow$

So we identify the photons with these equivalence classes (un fotone è una classe di equivalenza di stati). Come rappresentate della classe possiamo prendere $|\psi_T\rangle$ PURO TRASVERSO che COINCIDE con quello studiato nel Gauge di Coulomb! $\Rightarrow$ Possiamo quindi costruire tutto la quantizzazione fatta nel Gauge di Coulomb (quando faccio interagire le particelle devono poter controllare che l'interazione non dia stati non fisici)

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6

SPINOR FIELDS

6.1 WEYL FIELDS (M. 2.5 3.4.1 - 3.4.3)

Recalling notes in 1st notebook, spinor fields are the basis of $(0, 1/2)$ or $(1/2, 0)$ representation of Lorentz transformations

Left Handed Weyl spinor $\rightarrow (1/2, 0) \rightarrow \psi_L = \begin{pmatrix} \psi_{1L} \\ \psi_{2L} \end{pmatrix}$

Right Handed Weyl Spinor $\rightarrow (0, 1/2) \rightarrow \psi_R = \begin{pmatrix} \psi_{1R} \\ \psi_{2R} \end{pmatrix}$

Lorentz tx on these fields are represented by 2x2 matrices Λ_L and Λ_R:

$$\psi_L \longrightarrow \psi_L' = \Lambda_L \psi_L = e^{(-i\vec{\theta} - \vec{n}) \cdot \frac{\vec{\sigma}}{2}} \psi_L$$

$$\psi_R \longrightarrow \psi_R' = \Lambda_R \psi_R = e^{(-i\vec{\theta} + \vec{n}) \cdot \frac{\vec{\sigma}}{2}} \psi_R$$

Where $\vec{\sigma} = (\sigma_1, \sigma_2, \sigma_3)$ Pauli matrices vector

Charge conjugation (M 2.5)

Before studying the classical field theory of spinors ie the Weyl lagrangian and equation of motion, let us analyze an interesting property of Weyl spinors = charge conjugation.

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•) first we observe that using the property of pauli matrices:

$\sigma^2 \sigma^i \sigma^2 = -(\sigma^i)^*$ (no dimostrazione) (where σ^2 is the 2 matrix)

we obtain the following relation:

$$\sigma^2 \Lambda_L^* \sigma^2 = \Lambda_R \qquad (V)$$

(2×2) (2×2) (2×2) (2×2)

DIM:

$$\sigma^2 \Lambda_L^* \sigma^2 = \sigma^2 e^{(i\theta - \vec{n})\cdot \frac{\vec{\sigma}^*}{2}} \sigma^2 = \sigma^2 \sum_{n=0}^{\infty} \frac{\left((i\vec{\theta} - \vec{n})\cdot \vec{\sigma}^*/2\right)^n}{n!} \sigma^2 =$$

$$= \sigma^2 \sum_{n=0}^{\infty} \underbrace{\sigma^2 \cdot \sigma^2}_{\mathbb{1}} \frac{\left((i\vec{\theta} - \vec{n})\cdot \vec{\sigma}^*/2\right)^n}{n!} \underbrace{\sigma^2 \cdot \sigma^2}_{\mathbb{1}} \cdot \sigma^2 =$$

↓ property $\sigma^2 \sigma^i \sigma^2 = -(\sigma^i)^*$

$$= \underbrace{\sigma^2 \cdot \sigma^2}_{\mathbb{1}} \sum \frac{\left[(i\vec{\theta} - \vec{n})\cdot(-\vec{\sigma})/2\right]^n}{n!} \underbrace{\sigma^2 \cdot \sigma^2}_{\mathbb{1}} =$$

$$= \sum \frac{\left[(-i\vec{\theta} + \vec{n})\cdot \vec{\sigma}/2\right]^n}{n!} = e^{(-i\vec{\theta} + \vec{n})\cdot \vec{\sigma}/2} = \Lambda_R \qquad \text{CVD!}$$

FASTER: $\sigma^2 e^{(i\theta - n)\frac{\vec{\sigma}^*}{2}} \sigma^2 = e^{(i\theta - n)(-\frac{\vec{\sigma}}{2})} = e^{(i\theta + n)\frac{\vec{\sigma}}{2}} = \Lambda_R$ CVD

•) using property above (V) we can demonstrate that the field

$\xi_R := \sigma^2 \varphi_L^* \in (0, 1/2)$ is a Right H Weyl spinor

DIM

Let's apply a Lorentz tx on $\sigma^2 \varphi_L^*$, only φ_L^* is transformed, σ^2 is constant, does not depend on coordinate x

$$\sigma^2 \varphi_L^* \longrightarrow (\sigma^2 \varphi_L^*)' = \sigma^2 (\Lambda_L^* \varphi_L^*) = \underbrace{\sigma^2 \Lambda_L^* \sigma^2}_{\Lambda_R \text{ for (V)}} \underbrace{\mathbb{1}}\, \sigma^2 \varphi_L^* \overset{(V)}{=}$$

$$= \Lambda_R (\sigma^2 \varphi_L^*)$$

→ Hence $\sigma^2 \varphi_L^*$ transform as a RH spinor! So $\sigma^2 \varphi_L^* \in (1/2, 0)$! CVD!

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- These allow us to define the CHARGE CONJUGATION OPERATION as an operation which transforms ψ_L into ψ_R spinors

$\psi_L^C := i\sigma^2 \psi_L^*$ $\in (1/2, 0)$ is a RH Weyl spinor

and its inverse $\psi_L = -i\sigma^2 (\psi_L^C)^*$ (note that $i\sigma^2$ is real because σ^2 is purely imaginary, hence $(i\sigma^2)^* = i\sigma^2$)

- Repeating the same process on Λ_R we can define a CHARGE CONJUGATION on RH spinor which transforms ψ_R into ψ_L spinors as

$\psi_R^C := -i\sigma^2 \psi_R^*$ $\in (0, 1/2)$ is a LH spinor

and its inverse $\psi_R = i\sigma^2 (\psi_R^C)^*$

- The coefficient i is so chosen so that iterating the conjugation twice we have the IDENTITY, i.e. Charge conjugation is IDEMPOTENT

$(\psi_L^C)^C = (i\sigma^2 \psi_L^*)^C = -i\sigma^2 i\sigma^2 \psi_L = \psi_L$ $\quad -i\sigma^2 (i\sigma^2 \psi_L^*)^* =$

$= -i\sigma^2 (-i\sigma^{*2}) \psi_L = \sigma^{*2} = -\sigma^2$

$= -\sigma^2 \sigma^{*2} \psi_L =$?

$= \sigma^2 \sigma^2 \psi_L = \mathbb{1} \psi_L = \psi_L$

$(\psi_R^C)^C = \ldots = \psi_R$

The physical meaning of such operation will become clear once we will quantize spinor fields and find their particles. For now we can anticipate that charge conjugation exchange the PARTICLE with the ANTIPARTICLE

NOTE

ψ_L^* is not a LH spinor! It transform with Λ_L^* not with Λ_L !!

$\psi_L \rightarrow$ transform with Λ_L $\longrightarrow$ LH

$\psi_L^* \rightarrow$ // Λ_L^* none

$\psi_L^C \rightarrow$ // Λ_R $\longrightarrow$ RH

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Weyl Lagrangian and equation (M 3.4)

Our aim now is to define the Lagrangian for the spinor fieed, for a free spinor $\mathcal{L}$ is just the kinetic term. We want our kinetic Lagrangian to be LORENTZ INVARIANT, to respect Poincare symmetry (we do not need to impose also traslational invariance, we can take it for granted because we IMPOSED that all our fields are scalar under traslations, hence the traslational symmetry is also verified. One may ask why Trasl symmetry is special? We want that Energy and momentum are conserved when the particle is a free particle with no interaction → that's why we work only with fields that are scalar under translation! So to assure the conservation of En and momentum!)

← COOL OBSERVATION!

So we want to define a Lorentz invariant Lagrangian.

- Define 4vector of Pauli matrices:

$$\sigma^\mu := (\mathbb{1}, \vec{\sigma}) \qquad \bar{\sigma}^\mu = (\mathbb{1}, -\vec{\sigma})$$

- It is possible to demonstrate that the following vector are 4 controvariant vectors:

$\psi_R^\dagger \sigma^\mu \psi_R$ — 1×2 (line) · 2×2 (matrix) · 2×1 (column)

$\psi_L^\dagger \bar{\sigma}^\mu \psi_L$ — 1×2 $(\psi_{1L}^*, \psi_{2L}^*)$ (line) · 2×2 $\begin{bmatrix} \cdot & \cdot \\ \cdot & \cdot \end{bmatrix}$ (matrix) · 2×1 $\begin{pmatrix} \psi_{1L} \\ \psi_{2L} \end{pmatrix}$ (column) → element (number) of the vector, with $\mu = 0,1,2,3$

where $\psi_R^\dagger := (\psi_{1R}^*, \psi_{2R}^*)$ is a line vector of complex cojugate

where $\psi_L := \begin{pmatrix} \psi_{1L} \\ \psi_{2L} \end{pmatrix}$ is a column vector

Hence $\psi^\dagger \sigma^\mu \psi$ is the $()^\mu$ component of a 4vector, is 1×1 number

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More in general, given the $(1/2\ 1/2)$ Dirac representation, given two general fields ψ_L and $\tilde{\psi}_R$ an element of $(1/2\ 1/2)$ can be written as the pair $(\psi_L, \tilde{\psi}_R)$

charge conjugation gives $(\psi_L, \tilde{\psi}_R) \xrightarrow{C} (\tilde{\psi}_L, \psi_R)$

and $\tilde{\psi}_R^+ \sigma^\mu \psi_R$ or $\tilde{\psi}_L^+ \bar{\sigma}^\mu \psi_L$ or $\psi_L^+ \bar{\sigma}^\mu \tilde{\psi}_L$ or $\psi_R \sigma^\mu \tilde{\psi}_R$ are always 4 contravariant vector. Also when ψ and ψ are two different fields

Hence the following $\mathcal{L}$ is Lorentz invariant because it is the contraction (scalar product) of 4 vectors:

$$\mathcal{L}_L = i \psi_L^+ \bar{\sigma}^\mu \partial_\mu \psi_L \qquad (6.3) \qquad \mathcal{L}_R = i \psi_R^+ \sigma^\mu \partial_\mu \psi_R$$

$1\times 2 \quad 2\times 2 \quad 2\times 1$ (for $\mathcal{L}_L$) and $1\times 2 \quad 2\times 2 \quad 2\times 1$ (for $\mathcal{L}_R$)

WEYL EQUATION

The E-L equation obtained by varying with respect to ψ_L^* / ψ_R^*, considering ψ_L and ψ_L^* two independent fields, is:

$$\frac{\partial \mathcal{L}_L}{\partial(\psi_L^+)} - \partial_\mu \frac{\partial \mathcal{L}_L}{\partial(\partial_\mu \psi_L^+)} = 0$$

$\frac{\partial \mathcal{L}_L}{\partial(\partial_\mu \psi_L^+)} = 0$ cause $\mathcal{L}_L$ does not depend on $\partial_\mu \psi_L^+$

$$\frac{\partial \mathcal{L}_L}{\partial(\psi_L^+)} = 0$$

$$\bar{\sigma}^\mu \partial_\mu \psi_L = 0 \quad \text{con } \sigma^0 = \mathbb{1} \qquad \sigma^\mu \partial_\mu \psi_R = 0$$

$$(\partial_0 - \sigma^i \partial_i)\psi_L = 0 \qquad (\partial_0 + \sigma^i \partial_i)\psi_R = 0$$

Weyl equation

1° order equation

$$\bar{\sigma}^\mu \partial_\mu \psi_L = 0 \qquad (6.4)$$

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It is nice to observe that Weyl equation gives K-G equation for massless particle!

$$(\partial_0 - \sigma^i \partial_i)\psi_L = 0 \qquad //$$

$$\partial_0 \psi_L = \sigma^i \partial_i \psi_L$$

$$\partial_0^2 \psi_L = (\sigma^i \partial_i)(\sigma^j \partial_j)\psi_L$$

$$= \sigma^i \sigma^j \partial_i \partial_j \psi_L$$

using $\sigma^i \sigma^j = \delta^{ij} + i\varepsilon^{ijk}\sigma^k$

$$(\delta^{ij} + i\varepsilon^{ijk}\sigma^k)\partial_i \partial_j \psi_L$$

antr sim

$$\partial_0^2 \psi_L = \partial_i^2 \psi_L$$

Massless K-G equation!

$$\Box \psi_L = \partial^\mu \partial_\mu \psi_L = 0 \qquad \Box \psi_R = \partial^\mu \partial_\mu \psi_R = 0$$

However the Weyl equation it's more informative and interesting because is a 1st order derivative equation, while KG is 2nd order

Hamiltonian and conjugate momentum

Treating ψ_L^+, ψ_R^+ and ψ_L, ψ_R as two independent fields, the conjugate momentum Π_L and Π_R and the Hamiltonian density are by definition

$$\Pi_{\psi_L} = \frac{\partial \mathcal{L}_L}{\partial(\partial_0 \psi_L)} = i\psi_L^+ \qquad \Pi_{\psi_R} = \frac{\partial \mathcal{L}_R}{\partial(\partial_0 \psi_R)} = i\psi_R^+$$

which are line vectors $(\cdot,\cdot)$ 2×1

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which has plane wave solutions

$$\psi_L(x) = u_L(p)\, e^{-ip\cdot x} \qquad \psi_R(x) = u_R\, e^{-ip\cdot x}$$

where u_L & u_R are constant spinor on x, but do depend on p

with $p^2 = p^\mu p_\mu = 0$ because massless

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And the Hamiltonian density:

$$\mathcal{H}_L = \Pi_L \partial_0 \psi_L - \mathcal{L}_L \qquad \mathcal{H}_R = \Pi_R \partial_0 \psi_R - \mathcal{L}_R$$

$$\mathcal{H}_L = -i \psi_L^\dagger \bar{\sigma}^j \partial_j \psi_L \qquad \mathcal{H}_R = -i \psi_R^\dagger \sigma^j \partial_j \psi_R$$

(∂_j: only spatial sum; ψ_R: 2×1)

Translational symmetry $= T^{\mu\nu}$ & p^i

COMMENT – UNIVERSALITY OF TRANSLATIONAL SYMMETRY

We never have to worry about the translational symmetry – we always have T symmetry because we IMPOSED that all our fields are scalar under T. Why? Because T symmetry implies conservation of Energy and momentum and we want to impose energy & momentum conservation. Ideally we are working with isolated systems

By definition: as Noether current.

$$T_L^{\mu\nu} = \frac{\partial \mathcal{L}_L}{\partial(\partial_\mu \psi_L)} \partial^\nu \psi_L + \underbrace{\frac{\partial \mathcal{L}_L}{\partial(\partial_\mu \psi_L^\dagger)}}_{=0} \partial^\nu \psi_L^\dagger - g^{\mu\nu} \mathcal{L}_L$$

$=0$ because $\mathcal{L}$ does not depend on $\partial\psi^\dagger$

for ψ_L satisfying eq of motion (6.a)(Weyl eqtn) we have that $\mathcal{L}_L = 0$ part $\bar{\sigma}^\mu \partial_\mu \psi_L = 0$

Hence

$$T_L^{\mu\nu} = i \psi_L^\dagger \bar{\sigma}^\mu \partial^\nu \psi_L$$

equivalently

$$T_R^{\mu\nu} = i \psi_R^T \sigma^\mu \partial^\nu \psi_L$$

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$$= -i\psi^{\dagger}\bar{\sigma}^{i}\partial_i\psi_L$$

Using Weyl equation $\partial_0\psi_L = \sigma^i\partial_i\psi_L = -i\bar{\sigma}^i\partial_i\psi_L$

Then Hamiltonian density can be denoted also by

$$\mathcal{H}_L = T^{00}_L = i\psi_L^{\dagger}\partial_0\psi_L \qquad \mathcal{H}_R = T^{00}_R = i\psi_R^{\dagger}\partial_0\psi_R$$

And the P^i momentum (Noether charge) is

$$P^i_L = \int d^3x\, T^{0i} = \int d^3x\, i\psi_L^{\dagger}\partial^i\psi_L \qquad P^i_R = \int d^3x\, i\psi_R^{\dagger}\partial^i\psi_R$$

Lorentz symmetry – helicity

$\mathcal{L}$ is also Lorentz invariant so the spin and angular momentum are conserved. We computed in chapter 2.4 that for spinors the angular momentum was:

$$\vec{J} = \vec{S} = \frac{\vec{\sigma}}{2} \quad \text{Left and Right}$$

Weyl equation is very informative, we can observe that substituting the solution of a plane wave $\psi_L = u_L e^{-ip\cdot x}$ inside Weyl equation 6.9

$$(\partial_0 - \sigma^i\partial_i)\psi_L = 0 \quad \rightarrow \psi_L = u_L e^{-ip\cdot x} = u_L e^{-iEt + i\vec{p}\cdot\vec{x}}$$

$$(-iE - i\vec{\sigma}\cdot\vec{p})\, u_L e^{-iEt + i\vec{p}\cdot\vec{x}} = 0$$

$$\frac{\vec{\sigma}\cdot\vec{p}}{E}\, u_L = -u_L$$

From KG eq we know $m=0$ hence $E = |\vec{p}|$ hence

$$\vec{\sigma}\cdot\frac{\vec{p}}{|p|}\, u_L = -u_L$$

$$\vec{\sigma}\cdot\hat{p}\, u_L = -u_L$$

$\hat{p}$ versor

$$\frac{\vec{\sigma}}{2}\cdot\hat{p}\, u_L = -\frac{1}{2}u_L$$

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But as we noticed before, the spin for Weyl spinor is $\frac{\vec{\sigma}}{2}$ hence $\frac{\hat{\sigma}}{2}\cdot\hat{p} = h$ is the helicity! (Projection of spin on momentum direction)

$$(\hat{p}\cdot\vec{s})\,\psi_L = -\frac{1}{2}\psi_L$$

equivalently

$$(\hat{p}\cdot\vec{s})\,\psi_R = \frac{1}{2}\psi_R$$

LH and RH Weyl spinors are massless particles with helicity $-\frac{1}{2}$ and $\frac{1}{2}$. Plane wave are helicity eigenstates with eigenvalues $\pm\frac{1}{2}$ → they are fermions

U(1) phase symmetry

Weyl $\mathcal{L}$ is invariant also under internal phase (global) symmetry

- Transformation on the field

$$\psi_{L,R} \to \psi'_{L,R} = e^{i\theta}\psi_{L,R} \underset{\theta\to 0}{=} (1+i\theta)\psi_{L,R} \quad \text{hence } F = i\psi_{L,R}$$

- Current

$$j^\mu_L = \frac{-\partial\mathcal{L}_L}{\partial(\partial_\mu\psi_L)} F = \psi_L^+ \bar{\sigma}^\mu \psi_L \qquad j^\mu_R = \psi_R^+ \sigma^\mu \psi_R \qquad (6.6)$$

- charge ↑ stretche $u(p)e^{-ipx}$ sono particelle con carica +1
  $v(p)e^{+ipx}$ sono antiparticelle con carica −1

$$Q_L = \int d^3x\, j^0 = \int d^3x\, \psi_L^+\psi_L \qquad Q_R = \int d^3x\, \psi_R^+\psi_R$$

(This is not an electric charge! Electric charge is related to U(1) phase symmetry $\theta(x)$, now we have global phase symmetry θ and the charge is a sort of number of particle conservation)

Parity = Weyl $\mathcal{L}$ is not invariant under parity

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6.2 DIRAC FIELDS

Dirac fields ψ are $(\frac{1}{2},0)\oplus(0,\frac{1}{2})$ Lorentz representation. Hence ψ is a 4 component vector, made of both ψ_L and ψ_R spinors. How to combine ψ_L and ψ_R is a free choice, the most natural is

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = \begin{pmatrix} \psi_{L1} \\ \psi_{L2} \\ \psi_{R1} \\ \psi_{R2} \end{pmatrix}$$

which is called the chiral representation.
We will come back to that in a moment.

Let us for the moment be very general and consider the theory that we can build having at our disposal BOTH ψ_L and ψ_R Weyl spinors. Now we can construct TWO NEW LORENTZ INVARIANT SCALAR!

a) $\psi_L^\dagger \psi_R$ and $\psi_R^\dagger \psi_L$ are Lorentz invariant

DLM noting that $\Lambda_L^\dagger \Lambda_R = \Lambda_R^\dagger \Lambda_L = \mathbb{1}$ we have

$$\psi_L^\dagger \psi_R \longrightarrow \psi_L'^\dagger \psi_R' = \psi_L^\dagger \Lambda_L^\dagger \Lambda_R \psi_R = \psi_L^\dagger \psi_R$$

NOTE: $\psi_L' = \Lambda_L \psi_L$ hence

$$\psi_L'^\dagger = (\Lambda_L \psi_L)^\dagger = \psi_L^\dagger \Lambda_L^\dagger$$

(2×2 $\quad$ 2×1 $\quad$ 1×2 $\quad$ 2×2)

From these two quantities we can build two Lorentz scalar which are Real scalars:

a) $\psi_L^\dagger \psi_R + \psi_R^\dagger \psi_L$ & $i(\psi_L^\dagger \psi_R - \psi_R^\dagger \psi_L)$

however the 2nd one is a pseudo vector under parity $\psi_L \leftrightarrow \psi_R$ (gains a minus sign) hence is not good to put in our Lagrangian, since we want also parity conserved for free $\mathcal{L}$.

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Now we are all set to build our DIRAC $\mathcal{L}_D$ which is Lorentz invariant and also PARITY INVARIANT!

$$\mathcal{L}_D = \underbrace{i\psi_L^\dagger \bar{\sigma}^\mu \partial_\mu \psi_L}_{\mathcal{L}_L} + \underbrace{i\psi_R^\dagger \sigma^\mu \partial_\mu \psi_R}_{\mathcal{L}_R} - \underbrace{m(\psi_L^\dagger \psi_R + \psi_R^\dagger \psi_L)}_{\text{new mass term}} \qquad (6.8)$$

! $\mathcal{L}_D$ IS PARITY INVARIANT ! PARITY: $\begin{pmatrix} t \to t \\ \vec{x} \to -\vec{x} \end{pmatrix}$

A differenza delle $\mathcal{L}_L$ and $\mathcal{L}_R$ da sole che NON sono invarianti sotto parità, $\mathcal{L}_D$ è invariante anche sotto parità e non solo sotto traslazione, Lorentz o U(1) global phase!

$$\psi_L \overset{P}{\longleftrightarrow} \psi_R \qquad \partial_i \overset{P}{\longleftrightarrow} -\partial_i \qquad \bar{\sigma}^\mu \partial_\mu \overset{P}{\longleftrightarrow} \sigma^\mu \partial_\mu$$

allora $\mathcal{L}_D \overset{P}{\longleftrightarrow} \mathcal{L}_D$ invariante!

DIRAC EQUATIONS

Writing the eq. of motions for ψ_L and ψ_D separately in the Lagrangian we obtain two different eq. of motion for ψ_L and ψ_D (now from $\mathcal{L}_D$ and not from $\mathcal{L}_L$ or $\mathcal{L}_R$ alone) with an additional mass term!

$$\partial_\mu \frac{\partial \mathcal{L}_D}{\partial(\partial_\mu \psi_L^\dagger)} - \frac{\partial \mathcal{L}_D}{\partial \psi_L^\dagger} = 0 \qquad \partial_\mu \frac{\partial \mathcal{L}_D}{\partial(\partial_\mu \psi_R^\dagger)} - \frac{\partial \mathcal{L}_D}{\partial \psi_R^\dagger} = 0$$

$\frac{\partial(\partial_\mu \psi_L^\dagger)}{\partial(\partial_\mu \psi_L^\dagger)} \geq 0$ ↓ $\qquad \frac{\partial(\partial_\mu \psi_R^\dagger)}{\partial(\partial_\mu \psi_R^\dagger)} = 0$ ↓

$$i\bar{\sigma}^\mu \partial_\mu \psi_L = m\psi_R \qquad i\sigma^\mu \partial_\mu \psi_R = m\psi_L \qquad (6.9)$$

$$i(\sigma^0 \partial_0 + \sigma^j \partial_j)\psi_R = \psi_L m$$

These are Dirac's eqtns, very similar to Weyl's but with an additional mass term which MIXES ψ_L and ψ_R!! (→ come vedremo ciò implica che pure onde ψ_L and ψ_R non sono più autovettori dell'elicità ... come ha senso visto che ho $m \neq 0$!)

↓

c'è il + perché $\partial_\mu = (\partial_0, +\vec{\nabla})$

$\sigma^\mu = (\sigma^0, +\vec{\sigma})$

le ∂_μ usano convenzione mostly + ! Hanno il − se ∂^μ hanno invece quello!

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Even though DKOC equations now mixes ψ_L and ψ_R, they still give the KG equation ... but with mass now!

- Applying $i\sigma^\nu \partial_\nu$ to both sides of:

$$i\bar{\sigma}^\mu \partial_\mu \psi_L = m\psi_R$$

- we get

$$-\sigma^\nu \bar{\sigma}^\mu \partial_\mu \partial_\nu \psi_L = \underbrace{i\sigma^\nu \partial_\nu m \psi_R}_{= m^2 \psi_L \text{ because is that equation}}$$

$$\sigma^\nu \bar{\sigma}^\mu \partial_\mu \partial_\nu = \frac{1}{2}(\sigma^\nu \bar{\sigma}^\mu)(\partial_\mu \partial_\nu + \partial_\nu \partial_\mu) = \frac{1}{2}(\sigma^\nu \bar{\sigma}^\mu + \sigma^\mu \bar{\sigma}^\nu)\partial_\mu \partial_\nu$$

- $-\frac{1}{2}(\sigma^\nu \bar{\sigma}^\mu + \sigma^\mu \bar{\sigma}^\nu)\partial_\mu \partial_\nu \psi_L = m^2 \psi_L$

- Applying the property:

$$\sigma^\mu \bar{\sigma}^\nu + \sigma^\mu \bar{\sigma}^\nu = 2g^{\mu\nu}$$

which is tedious but easy to prove, we get

$$-\partial^\mu \partial_\mu \psi_L = m^2 \psi_L$$

$$(\Box + m^2)\psi_L = 0$$

- Applying $i\bar{\sigma}^\nu \partial_\nu$ to:

$$i\sigma^\mu \partial_\mu \psi_R = m\psi_L$$

- we get

$$-\bar{\sigma}^\nu \sigma^\mu \partial_\mu \partial_\nu \psi_R = i\bar{\sigma}^\nu \partial_\nu m \psi_L$$

//

//

$$(\Box + m^2)\psi_R = 0$$

These are KG equations for particles with some mass $m \neq 0$!!

The general solutions are superposition of plane waves:

$$\psi_L(x) = u_L(p)\, e^{\pm ip\cdot x} \qquad \psi_R(x) = u_R(p)\, e^{\pm ip\cdot x}$$

with $p^2 = p^\mu p_\mu = m^2$ (before it was $p^\mu p_\mu = 0$ for Weyl spinors)

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Dirac vs Weyl spinors = mass or no mass?

ψ_L and ψ_R spinors in Dirac and Weyl representations are two different things, one as $m \neq 0$ and the other $m = 0$. Physically this means that the same mathematical object can be used to describe two very different particles! But all with spin = 1/2. An example of massive Dirac spinors? electron & positron ... An example of massless? Neutrinos ... but with some controversial.

Mathematically with ψ_L or ψ_R alone in Weyl Lagrangian I could not introduce a mass term otherwise $\mathcal{L}_L$ or $\mathcal{L}_R$ would not have been Lorentz invariant. In Dirac's $\mathcal{L}_D$ instead we could introduce a mass term maintaining a Lorentz invariance, so now to be completely general we consider the mass term also.

SYMMETRIES OF $\mathcal{L}_D$ = HELICITY IS MISSING

$\mathcal{L}_D$ is symmetric under Translations as $\mathcal{L}_L$ and $\mathcal{L}_R$ $\rightarrow T^{\mu\nu}$ p^ν
under Lorentz as $\mathcal{L}_L$ and $\mathcal{L}_R$ $\rightarrow J^{\mu\nu}$
under Phase U(1) as $\mathcal{L}_L$ and $\mathcal{L}_R$ $\rightarrow Q$

But is no more Helicity eigenstates! To explain better: Before for Weyl spinor from Weyl eq $(\partial_0 - \sigma^i \partial_i)\psi_L = 0$ applied to plane wave $\psi_L = u_L e^{-iPx}$ we found that $(\hat{p} \cdot \vec{J})\, u_L = -\frac{1}{2} u_L$ where helicity eigenstates

Now we have Dirac eq $i(\partial_0 - \sigma^i \partial_i)\psi_L = m\psi_R$ and, applying it to the plane wave $\psi_L = u_L e^{-iP\cdot x}$ we DO NOT END anymore with an eigenstate equation!

$\Rightarrow$ Introducing a mass term makes plane waves ψ_L and ψ_R NO MORE EIGENSTATES OF HELICITY $\Rightarrow$ Questo è in perfetto accordo con lo capper di lavorare in base di particelle in cui avevamo visto come oggetti massivi e non massivi sono tot diversi e solo oggetti senza massa sono eigenstate dell'elicità!

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Chiral representation (Weyl representation)

It is convenient to write everything in terms of the Dirac spinor:

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = \begin{pmatrix} \psi_{L1} \\ \psi_{L2} \\ \psi_{R1} \\ \psi_{R2} \end{pmatrix} \qquad \text{DIRAC SPINOR}$$

this defines the CHIRAL REPRESENTATION (which is not the only one! We may also take $\psi = \frac{1}{\sqrt{2}} \begin{pmatrix} \psi_R + \psi_L \\ \psi_R - \psi_L \end{pmatrix}$ or others...)

Defining the matrices: 4×4 matrices in the chiral repr.:

$$\gamma^0 := \begin{bmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{bmatrix} \qquad \gamma^i := \begin{bmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{bmatrix} \quad \Rightarrow \quad \gamma^\mu := \begin{bmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{bmatrix} \qquad (6.10)$$

We note that γ^μ satisfies a CLIFFORD ALGEBRA:

$$\{\gamma^\mu, \gamma^\nu\} = \gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu}\mathbb{1} \qquad \text{CLIFFORD ALGEBRA} \qquad (6.11)$$

($g^{\mu\nu}$: Minkowski)

Hence in terms of Dirac spinors the Dirac eq becomes:

$$(i\not{\partial} - m)\psi = 0 \qquad \text{Dirac equation} \qquad (6.12)$$

Where we used the Feynman slash notation:

$$\not{\partial} := \partial_\mu\gamma^\mu = \gamma^\mu\partial_\mu$$

that for a general vector a_μ is:

$$\not{a} := \gamma^\mu a_\mu = a_\mu\gamma^\mu$$

} Feynman slash notation

⚠ $\gamma^\mu\partial_\mu = \gamma^0\partial_0 + \gamma^1\partial_1 + \gamma^2\partial_2 + \gamma^3\partial_3 = \gamma^0\partial_0 + \vec{\gamma}\cdot\vec{\nabla}$

pure $\partial_\mu = (\partial_0, \vec{\nabla})$ $\gamma^\mu = (\gamma^0, \gamma^1\gamma^2\gamma^3)$ $\vec{\gamma} = (\gamma^1\gamma^2\gamma^3)$

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Noticing that, or well, defining the DIRAC ADJOINT AS:

$$\bar{\psi} := \psi^+ \gamma^0 \qquad \text{DIRAC ADJOINT} \qquad (6.113)$$

which in the chiral representation is

$$\bar{\psi} = \psi^+ \gamma^0 = (\psi_L^+ \ \psi_R^+) \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix} = (\psi_R^+ \ \psi_L^+)$$

the product $\bar{\psi}\psi$ is a Lorentz invariant:

$$\bar{\psi}\psi = (\psi_L^+ \psi_R^+) \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix} \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = \psi_R^+ \psi_L + \psi_L^+ \psi_R$$

Lorentz and parity invariant

Hence we can rewrite the $\mathcal{L}$ DIRAC as:

$$\mathcal{L}_D = \bar{\psi} (i\partial\!\!\!/ - m)\psi \qquad \text{DIRAC LAGRANGIAN} \qquad (6.114)$$

1×4 $\quad 4\times4$ $\quad 4\times1$

Defining $\gamma^5 := i\gamma^0\gamma^1\gamma^2\gamma^3$

in chiral repr is $\gamma^5 = \begin{pmatrix} -\mathbb{1} & 0 \\ 0 & \mathbb{1} \end{pmatrix}$

noticing that γ^5 has the following properties:

$$\{\gamma^5, \gamma^\mu\} = 0 \qquad \forall \mu \qquad (\gamma^5)^2 = \mathbb{1}$$

→ DIM pog di ello

$$\gamma^\mu \partial_\mu = \gamma_\mu \gamma^\mu = \gamma^0 \partial_0 + \vec{\gamma}\cdot\vec{\nabla}$$

$$\gamma_\mu = (\gamma_0, -\gamma^1, -\gamma^2, -\gamma^3) = (\gamma_0, \gamma_1, \gamma_2, \gamma_3)$$

$$\partial_\mu = (\partial_0, -\vec{\nabla})$$

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DIM

$$\{\gamma^5, \gamma^\mu\} = i\gamma^0\gamma^1\gamma^2\gamma^3\gamma^\mu + i\gamma^\mu\gamma^0\gamma^1\gamma^2\gamma^3$$

- Considering property $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}\mathbb{1}$ we have:

$$\gamma^\mu\gamma^\nu \begin{cases} -\gamma^\nu\gamma^\mu & \text{if } \mu \neq \nu \quad \text{ANTI} \\ +\gamma^\nu\gamma^\mu & \text{if } \mu = \nu \quad \text{COMM} \end{cases}$$

- Hence in the 2nd, volendo portare γ^μ tutto a dx, μ sarà uguale a uno solo tra γ^0, γ^1 γ^2 e γ^3 quindi ci farà anticommutare 3 VV e commutare 1V perciò guadagno un segno meno, per es nell'ipotesi $\mu = 3$ si ha

$$i\gamma^\mu\gamma^0\gamma^1\gamma^2\gamma^3 = -i\gamma^0\gamma^\mu\gamma^1\gamma^2\gamma^3 = +i\gamma^0\gamma^1\gamma^\mu\gamma^2\gamma^3 = -i\gamma^0\gamma^1\gamma^2\gamma^\mu\gamma^3 = -i\gamma^0\gamma^1\gamma^2\gamma^3\gamma^\mu$$

(ANTI, ANTI, ANTI, COMM!)

- Per cui

$$\{\gamma^5, \gamma^\mu\} = i\gamma^0\gamma^1\gamma^2\gamma^3\gamma^\mu - i\gamma^0\gamma^1\gamma^2\gamma^3\gamma^\mu = 0 \qquad \text{c.v.d.}$$

Hence $\frac{\mathbb{1} \pm \gamma^5}{2}$ are projectors that project a bi-spinor ψ in two single spinor ψ_L or ψ_R

$$P_L := \frac{\mathbb{1} - \gamma^5}{2} \;(4\times4,\ 1) \longrightarrow P_L\psi = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} \psi_L \\ \psi_R \end{bmatrix} = \begin{bmatrix} \psi_L \\ 0 \end{bmatrix}$$

$$P_R := \frac{\mathbb{1} + \gamma^5}{2} \;(4\times4,\ 1) \longrightarrow P_R\psi = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} \psi_L \\ \psi_R \end{bmatrix} = \begin{bmatrix} 0 \\ \psi_R \end{bmatrix}$$

and $P_{L,R}^2 = P_{L,R}$ idempotent, as $(\gamma^5)^2 = \mathbb{1}$

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OTHER REPRESENTATIONS = STANDARD REPRESENTATION

Any other repr of ψ which is a UNITARY TRANSF of ψ is EQUIVALENT, in the sense that all equations are expressed in the same form

$$\begin{cases} \psi' = U\psi & \text{with } U \text{ CONSTANT UNITARY MATRIX} \\ \psi = U^+\psi' & \text{action of } \psi \end{cases}$$

and defining the transf of γ^μ as:

$$\begin{cases} \gamma'^\mu = U\gamma^\mu U^+ & \\ \gamma^\mu = U^+\gamma'^\mu U & \text{action on } \gamma^\mu \end{cases}$$

Hence we have

$$\psi'^+ = \psi^+ U^+ \quad \& \quad \psi^+ = \psi'^+ U$$

$$\bar\psi' = \psi'^+\gamma'^0 = \psi^+U^+U\gamma^0U^+ = \psi^+\gamma^0U^+ = \bar\psi U^+ \quad \& \ \bar\psi = \bar\psi' U$$

Hence we can prove that

$$\mathcal{L} = \bar\psi(i\partial\!\!\!/ - m)\psi =$$

$$\bar\psi U \quad U^+\partial\!\!\!/' U \quad U^+\psi'$$

$$= \bar\psi' U U^+ (i\partial\!\!\!/' - m) U U^+ \psi' =$$

$$= \bar\psi'(i\partial\!\!\!/' - m)\psi' = \mathcal{L}'$$

$\Longrightarrow \mathcal{L} = \mathcal{L}'$ SAME EXPRESSION

In the new representation the $\mathcal{L}'$ has exactly the same form in terms of ψ' and γ' which the old $\mathcal{L}$ has in terms of ψ and γ ⇓

So all representations are EQUIVALENT, all quantities have the same definition in all representations, not only for chiral representation!

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An example is the STANDARD REPRESANTATION where:

$$\gamma^{0'} = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix} \qquad \gamma^{i'} = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix} \quad \text{and} \quad \gamma^{5'} = \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix}$$

acting on a basis of vectors.

$$\psi' = \frac{1}{\sqrt{2}} \begin{pmatrix} \psi_R + \psi_L \\ \psi_R - \psi_L \end{pmatrix}$$

HIGH AND LOW ENERGY REGIMES = CHIRAL VS STANDARD REPR.

Standard repr is commonly used when the particle is not relativistic hence the kinetic energy is low. Chiral repr instead is used when the particle is very energetic and relativistic (= HIGH ENERGY). That's because in high energy regimes we can approximate the particle to be massless

$$E = \sqrt{|\vec{p}|^2 + m} \simeq |p| \qquad \text{i.e} \quad \frac{m}{E} \to 0$$

if $|\vec{p}| \to \infty$

This allows us to treat high energy particles as massless, leading to way more easier calculations. If energy is high the experimental results obtained with $m \sim 0$ massless approx are in very good agreement with the the calculations. Accounting for $m \neq 0$ would not change significantly the values obtained (even though m and $m=0$ particles are theoretically very different objects

($m = 0 \rightarrow$ object moves at speed of light $v = c$ in every frame
$m \neq 0 \rightarrow$ object never reaches speed of light and $\exists$ exist a rest frame where $v = 0$)

Chiral repr is the best for high energy approximation because it separates, in the case $m=0$ massless, ψ_R and ψ_0 Weyl spinors which becomes eigenstates of helicity (if $m=0$)

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Hamiltonian and conjugate momentum

From Dirac Lagrangian in chiral repr.: $\mathcal{L}_D = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi$

we obtain the conjugate momentum:

$$\Pi_\psi = \frac{\partial \mathcal{L}_D}{\partial(\partial_0 \psi)} = i\bar{\psi}\gamma^0 = i\psi^\dagger$$

→ 1×4 vector

$\bar{\psi} = \psi^\dagger \gamma^0$ ↧

$\bar{\psi}\gamma^0 = \psi^\dagger \gamma^0 \gamma^0 = \psi^\dagger$

$= \mathbb{1}$

and the Hamiltonian density therefore is:

$$\mathcal{H}_D = \Pi_\psi \partial_0 \psi - \mathcal{L}_D = \dots = \bar{\psi}(-i\gamma^j \partial_j + m)\psi$$

$\gamma^0\gamma^0 = \begin{bmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{bmatrix}\begin{bmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{bmatrix} = \begin{bmatrix} \mathbb{1} & 0 \\ 0 & \mathbb{1} \end{bmatrix}$

and dirac Hamiltonian

$$H_D = \int d^3x\, \mathcal{H}_D = \int d^3x\, \bar{\psi}(-i\vec{\gamma}\cdot\vec{\nabla} + m)\psi$$

(just spatial)

Solutions of Dirac equation

Working in the chiral representation the Dirac equation is:

$$(i\partial\!\!\!/ - m)\psi = 0 \qquad \text{with} \rightarrow \psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}$$

We already noticed that both ψ_L ψ_R satisfy KG equation for massive particles hence they are superpositions of plane waves. The single plane wave is:

4 component spinor $U = \begin{pmatrix} U_L \\ U_R \end{pmatrix} = \begin{pmatrix} U_{L1} \\ U_{L2} \\ U_{R1} \\ U_{R2} \end{pmatrix}$

$$\psi_p(x) = \begin{pmatrix} \psi_L(x) \\ \psi_R(x) \end{pmatrix} = \begin{pmatrix} U_L(p)\, e^{-ip\cdot x} \\ U_R(p)\, e^{-ip\cdot x} \end{pmatrix} = U(p)\, e^{-ip\cdot x}$$

PARTICLE $+p$

$$\psi_{-p}(x) = \begin{pmatrix} V_L(p)\, e^{ip\cdot x} \\ V_R(p)\, e^{ip\cdot x} \end{pmatrix} = V(p)\, e^{+ip\cdot x}$$

Antiparticle $-p$

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The Dirac eq on plane waves becomes:

$$(i\not{\partial} - m)\psi = (i\gamma^\mu \partial_\mu - m)\psi = (i\gamma^\mu \partial_\mu - m)\left(u(p)\, e^{-iP\cdot x}\right) =$$

$$= (P_\mu \gamma^\mu - m)\, u(p)\, e^{-iP\cdot x} = (\not{P} - m)\, u(p)\, e^{-iP\cdot x} = 0$$

so we get, defining $\not{P} = P_\mu \gamma^\mu$:

$$(\not{P} - m)\, u(p) = 0 \qquad \text{Dirac eq. for PARTICLE}$$

$$(\not{P} + m)\, v(p) = 0 \qquad \text{Dirac eq. for antiparticle} \qquad (6.116)$$

PARTICLE SOLUTION

- We work in the chiral representation where $u(p) = \begin{pmatrix} u_L(p) \\ u_R(p) \end{pmatrix}$, we consider $p^2 = m^2 \neq 0$ non zero. Hence we can consider, without loss of generality, the rest frame $p^\mu = (m, 0\,0\,0)$

- under such hypothesys $p^\mu = (m\,000)$ with $p^0 = m$ $p^s = 0$ we have that Dirac eq is:

$$m(\gamma^0 - 1)\, u(p) = 0 \rightarrow (\gamma^0 - 1)\begin{pmatrix} u_L \\ u_R \end{pmatrix} = 0 \rightarrow \begin{pmatrix} u_R - u_L \\ u_L - u_R \end{pmatrix} = 0$$

$$\Rightarrow u_R = u_L$$

- A convenient choice of normalization is

$$u_R = u_L = \sqrt{m}\,\xi \qquad \text{with } \xi^+\xi = 1 \qquad \text{with } \xi \text{ a 2 component spinor as } u_L \text{ or } u_R$$

Hence

$$u(p) = \sqrt{m}\begin{pmatrix} \xi \\ \xi \end{pmatrix}$$

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- We consider $\vec{p} = (0\ 0\ p^3)$ a boost in the z direction. We do not have lon of generality because we can always orient axes such that $\vec{p} = (0\ 0\ p^3)$ is along the z axis . With some calculation it is possible to prove that 6.16 leads to two solutions:

$$u^s(p) = \begin{pmatrix} \left[\sqrt{E+p^3}\ \frac{1-\sigma^3}{2} + \sqrt{E-p^3}\ \frac{1+\sigma^3}{2}\right]\xi^s \\ \left[\sqrt{E+p^3}\ \frac{1+\sigma^3}{2} + \sqrt{E-p^3}\ \frac{1-\sigma^3}{2}\right]\xi^s \end{pmatrix} \qquad (6.17)$$

where $s = 1,2$ index for the two solutions , and $\xi^1 = \begin{pmatrix}1\\0\end{pmatrix}$ $\xi^2 = \begin{pmatrix}0\\1\end{pmatrix}$

It is also possible to demonstrate that a more compact way to write 6.17 is the following; for p along any direction :

$$u^s(p) = \begin{pmatrix} \sqrt{p\cdot\sigma}\ \xi^s \\ \sqrt{p\cdot\bar{\sigma}}\ \xi^s \end{pmatrix} \qquad (6.18)$$

↓

prodotto scalare tra 4 vettori: $p\cdot\sigma = p^\mu\sigma_\mu = p^0\,1 - \vec{p}\cdot\vec{\sigma}$; $p\cdot\bar{\sigma} = p^\mu\bar{\sigma}_\mu = p^0\,1 + \vec{p}\cdot\vec{\sigma}$

dove $\sqrt{}$ di una matrice significa prendere la radice positiva degli autovalori, della matrice diagonalizzata

dato $p^\mu = (p^0, p^1 p^2 p^3)$

$\sigma^\mu = (\sigma^0\ \sigma^1\ \sigma^2\ \sigma^3)$

$\sigma_\mu = (\sigma_0\ -\vec{\sigma})$

$\bar{\sigma}^\mu = (\sigma^0\ -\vec{\sigma})$

$\sigma_\mu = (\sigma_0\ +\vec{\sigma})$

allora

$p^\mu\sigma_\mu = p^0 - \vec{p}\cdot\vec{\sigma}$

$p^\mu\bar{\sigma}_\mu = p^0 + \vec{p}\cdot\vec{\sigma}$

- <u>ULTRA RELATIVISTIC LIMIT — $m \to 0$</u>

In the ultra relativistic limit where the mass is like zero mass hence $p \to E(1\ 0\ 0\ 1)$ hence $p^2 = p^\mu p_\mu = 0$ we get

$$u^1(p) = \sqrt{2E}\begin{pmatrix}0\\ \xi^1\end{pmatrix} \qquad u^2 = \sqrt{2E}\begin{pmatrix}\xi^2\\ 0\end{pmatrix}$$

u^1 is only RH spinor and u^2 is only a LH spinor

Compleo, ho solo le componenti $R\ \xi^1$ e solo $L\ \eta\xi^2$, ma per uno spinor TOT $\begin{pmatrix}0\\ \xi^1\end{pmatrix}_R^L$ $\begin{pmatrix}\xi^2\\ 0\end{pmatrix}_R^L$

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ANTIPARTICLE SOLUTION

Similarly for the antiparticle spinor we obtain:

$$V^s(p) = \begin{pmatrix} \sqrt{p\cdot\sigma}\,\eta^s \\ -\sqrt{p\cdot\bar{\sigma}}\,\eta^s \end{pmatrix} \qquad (6.19)$$

with $s = 1,2$ and $\eta^1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $\eta^2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

PROPERTIES

Given two indeces s and r running along $1,2$ it is possible to prove the following relations:

$$U^{r+}(p)\,U^s(p) = V^{r+}(p)\,V^s(p) = 2E_p\,\delta^{rs} \qquad (6.20)$$

Recalling the Dirac adjoint $\bar{\psi} = \psi^+\gamma^0$ we have:

$$\bar{U}^r(p)\,U^s(p) = 2m\,\delta^{rs} \qquad \bar{V}^r(p)\,V^s(p) = -2m\,\delta^{rs} \qquad (6.21)$$

$$\bar{U}^r(p)\,V^s(p) = \bar{V}^r(p)\,U^s(p) = 0 \qquad (6.22)$$

$$\sum_{s=1}^{2} U^s(p)\,\bar{U}^s(p) = \not{p} + m \qquad \sum_{s=1}^{2} V^s(p)\,\bar{V}^s = \not{p} - m \qquad (6.23)$$

(NOTE the $\bar{U}U$ is a number while $U\bar{U}$ is a 4×4 matrix)

DIRAC ADJOINT

$$\bar{U}(p)\,(\not{p} - m) = 0$$
$$\bar{V}(p)\,(\not{p} + m) = 0$$

Dirac eq for Dirac adjoint of $U(p)$ and $V(p)$ (6.24)

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we have two solutions $u^s(p)$ with $s = 1,2$ as before.
Inserting an u^s or v^s term in eq 6.24 we get

$$\bar{u}^r(p)(\not{p} - m)\, u^s(p) = 0$$
$$\bar{v}^r(p)(\not{p} + m)\, v^s(p) = 0$$

$$\bar{u}^r(p) \not{p}\, u^s(p) = m\, \bar{u}^r(p) u^s(p) = 2m^2 \delta^{rs} = 2 p^\mu p_\mu \delta^{rs} \quad (6.21)$$
$$\bar{v}^r(p) \not{p}\, v^s(p) = -m\, \bar{v}^r(p) v^s(p) = 2m^2 \delta^{rs} = 2 p^\mu p_\mu \delta^{rs}$$

$$\bar{u}^r(p) \gamma^\mu p_\mu u^s(p) = 2 p^\mu p_\mu \delta^{rs}$$
$$\bar{v}^r(p) \gamma^\mu p_\mu v^s(p) = 2 p^\mu p_\mu \delta^{rs}$$

$$\rightarrow \quad \bar{u}^r(p) \gamma^\mu u^s(p) = 2 p^\mu \delta^{rs} \qquad (6.25)$$
$$\rightarrow \quad \bar{v}^r(p) \gamma^\mu v^s(p) = 2 p^\mu \delta^{rs}$$

Bilinear combinations of fermion

it is interesting and useful to study some properties of the bilinear quantities one can compute from two fermions, hence

general bilinear combination is = $\bar{\psi} \Gamma \psi$

(ψ: 1×4 vector; Γ: 4×4 matrix; $\bar{\psi}$: 4×1 vector)

anticommute so there are $4! : 2! = 6$ matrices

1 1 4 4

We observe that the 16 matrices $\mathbb{1}$, γ^5, γ^μ, $\gamma^\mu \gamma^5$ and $\sigma^{\mu\nu}$ where $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$ are linear independent hence the most general 4×4 matrix Γ can be expressed as a combination of them

$$\Gamma = \left\{ \mathbb{1},\ \gamma^5,\ \gamma^\mu,\ \gamma^\mu \gamma^5,\ \sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu] \right\}$$

Hence the most general bilinear is a combination of $\bar{\psi}\psi$, $\bar{\psi}\gamma^5\psi$ ecc ecc cioè di $\bar{\psi}\Gamma\psi$ dove Γ è uno di base.

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Each particular $\bar{\psi}\Gamma\psi$ is a different object with particular properties under Lorentz and Parity transformations:

$\boxed{\Gamma = \mathbb{1}}$

$\bar{\psi}\psi = \psi_R^+ \psi_L + \psi_L^+ \psi_R$

is a Lorentz scalar and Parity even (perché sotto parità $\bar{\psi}\psi \xrightarrow{P} \bar{\psi}\psi$)

$\boxed{\Gamma = \gamma^5}$

$\bar{\psi}\gamma^5\psi = \psi_L^+ \psi_R - \psi_R^+ \psi_L$

is a Lorentz scalar and Parity odd (ritorno in se stesso sotto Lorentz proprio ma prende segno meno sotto parità) $\Rightarrow$ Lorentz PSEUDOSCALAR

$\boxed{\Gamma = \gamma^\mu}$

$j^\mu_V := \bar{\psi}\gamma^\mu\psi = \psi_L^+ \bar{\sigma}^\mu \psi_L + \psi_R^+ \sigma^\mu \psi_R$

is a Lorentz vector and Parity even

$\boxed{\Gamma = \gamma^\mu\gamma^5}$

$j^\mu_A := \bar{\psi}\gamma^\mu\gamma^5\psi = \psi_L^+ \bar{\sigma}^\mu \psi_L - \psi_R^+ \sigma^\mu \psi_R$

is a Lorentz 4vector and Parity odd $\rightarrow$ Lorentz PSEUDOVECTOR

$\boxed{\Gamma = \sigma^{\mu\nu}}$

$\bar{\psi}\sigma^{\mu\nu}\psi$ is a Lorentz true tensor

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LORENTZ TRANSFORMATION ON Ψ DIRAC FIELDS

In notebook 1 we had written the Lorentz tranformation on a Dirac field as with Λ_D matrix in chiral representation:

$$\psi' = \Lambda_D \psi \qquad \Lambda_D = \begin{pmatrix} \Lambda_L & 0 \\ 0 & \Lambda_R \end{pmatrix} \qquad \Lambda_{\frac{1}{2}} = e^{(-i\vec{\theta} \mp \vec{n})\frac{\vec{\sigma}}{2}}$$

it is straightforward to demonstrate that with $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$ we can express the Lorentz transf with:

$$\Lambda_D = e^{-\frac{i}{2}\omega_{\mu\nu}\frac{\sigma^{\mu\nu}}{2}} \longrightarrow \frac{1}{2}\sigma^{\mu\nu} = J^{\mu\nu} \text{ è il generatore di Lorentz per gli spinori di Dirac}$$

where $\omega_{\mu\nu}$ are parameters (coincident for $\vec{\theta}$ and $\vec{n}$)

(a) we did it in chiral representation but is valid in any representation, bastarà cambiare con $\tilde{\sigma}^{\mu\nu}$ nella nuova rappresentazione

$$\psi' = \Lambda_D \psi \qquad \Lambda_D = e^{-\frac{i}{4}\omega_{\mu\nu}\sigma^{\mu\nu}} \qquad \text{con } \sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$$

↓ nuova repr

$$\tilde{\psi}' = \tilde{\Lambda}_D \tilde{\psi} \qquad \tilde{\Lambda}_D = e^{-\frac{i}{4}\omega_{\mu\nu}\tilde{\sigma}^{\mu\nu}} \qquad \text{con } \tilde{\sigma}^{\mu\nu} = \frac{i}{2}[\tilde{\gamma}^\mu, \tilde{\gamma}^\nu]$$

con:

$$\tilde{\Lambda}_D = U \Lambda_D U^{-1} \qquad \tilde{\sigma}^{\mu\nu} = U \sigma^{\mu\nu} U^{-1} \qquad \tilde{\psi} = U\psi$$

$$[J^{\mu\nu}, J^{\rho\sigma}] = i(g^{\nu\rho} J^{\mu\sigma} - g^{\mu\rho} J^{\nu\rho} - g^{\nu\sigma} J^{\mu\rho} + g^{\mu\sigma} J^{\nu\rho})$$

↓ si può dimostrare che, se $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}\mathbb{1}$ allora $\frac{\sigma^{\mu\nu}}{2} = \frac{i}{4}[\gamma^\mu, \gamma^\nu]$ soddisfa

$$\left[\frac{\sigma^{\mu\nu}}{2}, \frac{\sigma^{\rho\sigma}}{2}\right] = i\left(g^{\mu\rho}\frac{\sigma^{\nu\sigma}}{2} - g^{\nu\rho}\frac{\sigma^{\mu\sigma}}{2} - g^{\mu\sigma}\frac{\sigma^{\nu\rho}}{2} + g^{\nu\sigma}\frac{\sigma^{\mu\rho}}{2}\right)$$

Soddisfa le stesse regole di commutazione di $J^{\mu\nu}$ del generatore di Lorentz, dunque $\frac{\sigma^{\mu\nu}}{2}$ sono i generatori di Lorentz in rappresentazione spinoriale!

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Phase and Chiral symmetry

Let us consider the following two transformations

PHASE TRANSFORMATION

$$\psi' = e^{i\theta}\psi$$

Hence:

$$\begin{cases} \psi_L' = e^{i\theta}\psi_L \\ \psi_R' = e^{i\theta}\psi_R \end{cases}$$

same angle θ for L and R

CHIRAL TRANSFORMATION

$$\psi' = e^{i\theta\gamma^5}\psi$$

Hence:

$$\begin{cases} \psi_L' = e^{-i\theta}\psi_L \\ \psi_R' = e^{i\theta}\psi_R \end{cases}$$

opposite angle $-\theta$ and θ for L R

MASSLESS DIRAC $\mathcal{L}_D$

The massless $m=0$ Dirac Lagrangian $\mathcal{L}_D = \bar{\psi} i\not{\partial}\psi$ is invariant under both transformations so it has both PHASE and CHIRAL SYMMETRY

Phase invariance:

$$\bar{\psi}' i\not{\partial}\psi = (e^{i\theta}\psi)^\dagger \gamma^0 i\not{\partial} e^{i\theta}\psi = \psi^\dagger e^{-i\theta}\gamma^0 i\not{\partial} e^{i\theta}\psi =$$
$$= \psi^\dagger \gamma^0 e^{-i\theta} e^{i\theta} i\not{\partial}\psi = \bar{\psi} i\not{\partial}\psi$$ ✓

Chiral invariance:

$$\bar{\psi}' i\not{\partial}\psi = (e^{i\theta\gamma^5}\psi)^\dagger \gamma^0 i\not{\partial} e^{i\theta\gamma^5}\psi = \psi^\dagger e^{-i\theta\gamma^5}\gamma^0 i\not{\partial} e^{i\theta\gamma^5}\psi =$$
$$= \psi^\dagger \gamma^0 e^{-i\theta\gamma^5} e^{i\theta\gamma^5} i\not{\partial}\psi = \bar{\psi} i\not{\partial}\psi$$ ✓

1, 2 : γ^μ and γ^5 anticommute hence each time I swap $e^{i\theta\gamma^5}$ with $\not{\partial}$ or γ^0 I gain a minus sign. I do that two times so at the end it doesn't change.

$$\gamma^0\gamma^\mu e^{i\theta\gamma^5} = \gamma^0 e^{-i\theta\gamma^5}\gamma^\mu = e^{i\theta\gamma^5}\gamma^0\gamma^\mu$$

from $\not{\partial} = \gamma^\mu\partial_\mu$

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The Noether current associated to phase symmetry $U(1)$ is the vector current j^μ_V, hence phase $U(1)$ symety is also called VECTOR $U(1)$

$$j^\mu_V = \bar{\psi}\gamma^\mu\psi$$ Vector

The Noether current associated with chiral symmetry is the pseudovector j^μ_A hence Chiral symmetry is also called AXIAL $U(1)$:

$$j^\mu_A = \bar{\psi}\gamma^\mu\gamma^5\psi$$ pseudovector

The Noether charges associated to vector $U(1)$ transformation or Axial $U(1)$ are the n° of particles/antiparticles for phase $U(1)$ and a sort of n° of Right fermions − n° of left fermions.

$$Q = \int d^3x\, j^0_V = \int d^3 \psi^\dagger\psi = \int d^3x(\psi_R^\dagger\psi_R + \psi_L^\dagger\psi_L) \rightarrow \text{n° particle} - \text{n° antiparticles}$$

$$Q_5 = \int d^3 j^0_A = \int d^3 \psi^\dagger\gamma^5\psi = \int d^3x(\psi_R^\dagger\psi_R - \psi_L^\dagger\psi_L) \rightarrow \text{n° of RH fermion} - \text{n° of LH fermions}$$

(When global $U(1)$ phase becomes local (i.e. $e^{i\theta(x)}$) the conserved charge is the electric charge)

(massless $\mathcal{L}_D$ is invariant under general transformation $U(1)\times U(1)$ Acting on RH or LH separately independently, with independent θ_R and θ_L angles

$$\psi_L' = e^{i\theta_L}\psi_L \quad \& \quad \psi_R' = e^{i\theta_R}\psi_R$$

MASSIVE $\mathcal{L}_D$ DIRAC

The massive $\mathcal{L}_D$ is invariant only under Phase $U(1)$, it is not invariant under chiral $U(1)$! the mass term is not invariant!

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$$\mathcal{L}_D = \bar{\psi}(i\gamma - m)\psi = \bar{\psi} i\gamma \psi - m\bar{\psi}\psi$$

MASS TERM

chiral transformation = not invariant:

$$\bar{\psi}\psi m \rightarrow m\bar{\psi}'\psi' = m\left(e^{i\theta\gamma^5}\psi\right)^+ \gamma^0 e^{i\theta\gamma^5}\psi = m\psi^+ e^{-i\theta\gamma^5} \gamma^0 e^{i\theta\gamma^5}\psi =$$

$$= m\psi^+\gamma^0 e^{i\theta\gamma^5} e^{i\theta\gamma^5}\psi = m\bar{\psi} e^{+2i\theta\gamma^5}\psi$$

$\Rightarrow$ NOT invariant! $m\bar{\psi}'\psi' = m\bar{\psi} e^{+2i\theta\gamma^5}\psi$!!

Contenta con il fatto che mass terms mix ψ_R and ψ_L in Dirac equations:

$$\begin{cases} i\bar{\sigma}^\mu \partial_\mu \psi_L = m\psi_R \\ i\sigma^\mu \partial_\mu \psi_R = m\psi_L \end{cases}$$

NOTHER current j^μ_V IS invariant

using $\mathcal{L}_D = \bar{\psi} i\gamma\psi - m\bar{\psi}\psi$ we obtain the eq of motion $(i\gamma - m)\psi = 0$ (V) which for $\bar{\psi}$ becomes $(\partial_\mu\bar{\psi})\gamma^\mu = i m\bar{\psi}$ (**) hence we can prove that:

$$\partial_\mu j^\mu_V = \partial_\mu(\bar{\psi}\gamma^\mu\psi) = (\partial_\mu\bar{\psi})\gamma^\mu\psi + \bar{\psi}\gamma^\mu(\partial_\mu\psi) \overset{(**)}{=} i m\bar{\psi}\psi + \bar{\psi}\gamma\psi \overset{(V)}{=}$$

$$= i m\bar{\psi}\psi - i m\bar{\psi}\psi = 0$$

$\rightarrow \partial_\mu j^\mu_V = 0$ is conserved with massive fermions ✓

NOETHER CURRENT j^μ_A IS NOT CONSERVED

instead, with similar passages, we can prove that j_A^μ axial current is not conserved!

$$\partial_\mu j^\mu_A = \dots = 2im\bar{\psi}\gamma^5\psi$$ ✗

$\Rightarrow$ MASSIVE DIRAC FERMIONS only have global U(1) symmetry and don't have chiral symmetry!

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TEORICA 1

n. 3

DOMITILLA ALOI

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6.3 QUANTIZATION OF DIRAC FIELD

→ normal ordering for spinor is different (!)

C.7 INTERACTING THEORIES

7.1 THE S-MATRIX OF TIME EVOLUTION

7.2 LSZ REDUCTION FORMULA

7.3 PERTURBATIVE EXPANSION OF GREEN FUNCTION

C.8 FEYMAN DIAGRAMS

8.1 FEYMAN PROPAGATOR FOR SCALAR FIELDS

8.2 WICK'S THEOREM (FOR SCALAR FIELDS)

8.3 DIAGRAMS BETWEEN TWO POINTS FOR ϕ^4 THEORY

8.4 DIAGRAMS BETWEEN FOUR POINTS FOR ϕ^4 THEORY

8.5 LSZ FORMULA WITH DIAGRAMS - part 1

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6.3 QUANTIZATION OF DIRAC FIELDS

Spin-statistic theorem

The spin-statistic theorem requires that fields with HALF-INTEGER SPINS are quantized composing ANTICOMMUTATION RELATIONS, and spin in INTEGER with COMMUTATION. (we do not discuss the theorem) (for more info see "How not to quantize a Dirac Field" - P&S 5.2-5.7)
The reason behind this choice is that, chosing commutation would lead to unbounded energy.

$$\{\psi_a(t,\vec{x}), \psi_b^+(t,\vec{y})\} \stackrel{!}{=} \delta^{(3)}(\vec{x}-\vec{y})\,\delta_{ab} \qquad (6.28)$$

where $a,b = 1,2,3,4$ are Dirac indeces (hence L1, L2, R1, R2)
where t time is the same for both fields

↓

Since Dirac fields satisfy K-G equation they can be expanded in plane waves so as:

$$\psi(x) = \int \frac{d^3\vec{p}}{(2\pi)^3}\frac{1}{\sqrt{2E_{\vec{p}}}} \sum_{s=1,2} \left(a_{\vec{p},s}\, v^s(p)\, e^{-ip\cdot x} + b^+_{p,s}\, v^s(p)\, e^{ip\cdot x} \right)$$

$$\bar{\psi}(x) = \int \frac{d^3\vec{p}}{(2\pi)^3}\frac{1}{\sqrt{2E_{\vec{p}}}} \sum_{s=1,2} \left(b_{p,s}\, \bar{v}^s(p)\, e^{-ip\cdot x} + a^+_{p,s}\, \bar{u}^s(p)\, e^{ip\cdot x} \right)$$

(p → 4 vector p^μ)

- where $u^s(p)$ is the solution for a $+p$ plane wave (see eq 6.17) and $v^s(p)$ is the one for $-p$ plane wave (eq 6.19) antiparticle
- where ψ is the whole Dirac spinor (4 components), where $\bar{\psi}$ is the Dirac adjoint $\bar{\psi} = \psi^+\gamma^0$
- where s runs over the two spin possibility of K-G equation, as showed before

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using imposing anticommutators (6.28) we obtain:

$$\{a_{\vec{p}r}, a^{+}_{\vec{q}s}\} = \{b_{\vec{p}r}, b^{+}_{\vec{q}s}\} = (2\pi)^3 \delta^{(3)}(\vec{p}-\vec{q})\, \delta_{rs} \qquad (6.29)$$

r, s indeces of spins

$\vec{p}, \vec{q}$ momentum index

$$\{a_{pr}, a_{qs}\} = \{b_{pr}, b_{qs}\} = \{a^{+}_{pr}, a^{+}_{qs}\} = \{b^{+}_{pr}, b^{+}_{qs}\} = 0$$

$$\{a_{pr}, b_{qs}\} = \{a^{+}_{pr}, b^{+}_{qs}\} = \{a^{+}_{pr}, b_{qs}\} = \{a_{pr}, b^{+}_{qs}\} = 0$$

Vacuum state and Fock space

We impose the $\exists$ of a vacuum state $|0\rangle$ defined by the annihilation:

$$a_{ps}\,|0\rangle \stackrel{!}{=} \emptyset \qquad \text{same } |0\rangle \text{ for particle and antiparticle}$$
$$b_{ps}\,|0\rangle \stackrel{!}{=} \emptyset \qquad (6.30)$$

We generate two copies of Fock space acting with creator operators of a_{ps}^{+} particles and b_{ps}^{+} antiparticles (Fermions and antifermions)

$$|\vec{p}, s\rangle_a := \sqrt{2E_{\vec{p}}}\; a^{+}_{ps}\,|0\rangle$$
$$|\vec{p}\, s\rangle_b := \sqrt{2\epsilon_p}\; b^{+}_{ps}\,|0\rangle$$

With orthogonality relations. (following from 6.29):

$${}_a\langle q r | p s\rangle_a = (2\pi)^3\, 2E_{\vec{p}}\, \delta_{rs}\, \delta^{(3)}(\vec{p}-\vec{q})$$

$${}_b\langle q r | p s\rangle_b = (2\pi)^3\, 2\epsilon_p\, \delta_{rs}\, \delta^{(3)}(\vec{p}-\vec{q})$$

$${}_a\langle q r | p s\rangle_b = \emptyset$$

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Hamiltonian in normal ordering and momentum

the classical density of Hamiltonian and hamiltonian were,

$$\mathcal{H} = \bar{\psi}(-i\partial_j\gamma^j + m)\psi \qquad H = \int d^3x\ \bar{\psi}(-i\gamma^j\partial_j + m)\psi$$

so imposing the mode expansion of fields ψ and $\bar{\psi}$ and the normal ordering we get

$$\hat{H} = \int d^3x :\bar{\psi}(-i\gamma^j\partial_j + m)\psi: \; =$$

NB = ogni swap $\hat{a^\dagger a}$ nel caso dei fermioni, guadagno segno meno! $: a_r a_s^\dagger : = -a_s^\dagger a_r$

$$\hat{H} = \int \frac{d^3p}{(2\pi)^3} \sum_{s=1,2} E_{\vec{p}}\,(a_{ps}^\dagger a_{ps} + b_{ps}^\dagger b_{ps}) \qquad (6.30)$$

If we used commutation relation instead of anticommutation, in front of $b^\dagger b$ term we would have had a MINUS SIGN! So adding antiparticles would have diminished the energy! Avremmo potuto costruire ad esempio antiparticelle (con energia negativa) diminuendo sempre più l'energia in un sistema "stabile" → unbounded energy in ogni caso limit → NOT PHYSICAL !!
(⇒ MARE DI DIRAC)

Imposing correct anticommutations we obtain that antiparticles created by $b_{ps}^\dagger$ (particle with $e^{i\vec{p}\cdot\vec{x}}$ i.e negative wave momentum $-p$) create antiparticles of POSITIVE energy ϵ_p, as well as normal particles created by $a_{ps}^\dagger$!

The momentum of the whole field (conserved charge of translational symmetry) is therefore:

$$P^j = -\int d^3x\ \Pi\,\partial^j\psi = \dots = \int \frac{d^3p}{(2\pi)^3}\ p^j \sum_{s=1,2} (a_{ps}^\dagger a_{ps} + b_{ps}^\dagger b_{ps})$$

so P^j of the whole field is positive ($+b_p^\dagger b_p$) also for antiparticles, even though the wave momentum of antiparticle is apparently negative

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Lorentz symmetry - spin and angular momentum

From notebook 1, section 2.4.1 of fields representation, the Lorentz transformation on Weyl and Dirac fields could be written as:

$$\Lambda = e^{-\frac{i}{2}\omega_{\alpha\beta} J^{\alpha\beta}} \qquad J^{\alpha\beta} = \underbrace{S^{\alpha\beta}}_{\text{spin}} + \underbrace{L^{\alpha\beta}}_{\text{angular}}$$

Where we could define the

$$S^i = \frac{i}{2}\varepsilon^{ijk} S^{jk} = \frac{\sigma^i}{2} \qquad \text{for both } \psi_L \text{ and } \psi_R$$

As shown for vector field, the Noether charge derived from Lorentz symmetry is the expectation value of orbital and spin angular momentum generators L^i and S^i

Therefore the Noether current (CHARGE?) associated with the spin is:

$$\hat{S}^i = \int dx^3 :\psi^+ \frac{\Sigma^i}{2} \psi: \qquad \text{where } \Sigma^i = \begin{bmatrix} \sigma^i & 0 \\ 0 & \sigma^i \end{bmatrix} \text{ for } \psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} \text{ Dirac}$$

Let's find its expression in terms of creator and annihilator operators of the field ψ. Expanding $\psi^+ \to \int dp \ldots \psi \to \int dq$ in normal modes I get:

$$S^i = \int d^3x \int \frac{d^3p}{(2\pi)^3\sqrt{2E_p}} \int \frac{d^3q}{(2\pi)^3\sqrt{2E_q}} \sum_{r,r'} :\left(a^+_{pr} U^{+r}_p e^{ipx} + b_{pr} V^{+r}_p e^{-ipx}\right) \frac{\Sigma^i}{2} \left(\ldots\right.$$

$$\left. a_{qr'} U^{r'}_q e^{-iqx} + b^+_{qr'} V^{r'}_q e^{iqx}\right): =$$

$$= \int d^3x \int \frac{d^3p}{(2\pi)^3\sqrt{2E_p}} \int \frac{d^3q}{(2\pi)^3\sqrt{2E_q}} \sum_{rr'} :\Big(a^+_{pr} a_{qr'} U^{+r}_p \otimes U^{r'}_q e^{-i(q-p)x} + a^+_{pr} b^+_{qr'} U^{+r}_p \otimes V^{r'}_q e^{i(p+q)x} + b_{pr} a_{qr'} V^{+r}_p \otimes U^{r'}_q e^{-i(p+q)x} + b_{pr} b^+_{qr'} V^{+r}_p \otimes V^{r'}_q e^{-i(p-q)x} \Big):$$

in mezzo ci sono U^+U c'è la matrice Σ^i che devo diagonalizzare

$$\otimes = \frac{\Sigma^i}{2}$$

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so I have four components

$(a^\dagger_{pr} a_{qr'})$ (1) $(a^\dagger_{pr} b^\dagger_{qr'})$ (2) $(b_{pr} a_{qr'})$ (3) $(b_{pr} b^\dagger_{qr'})$ (4)

Alla fine a noi interessa quale è il valore dello spin delle particelle cioè degli stati $a^\dagger_{us}|0\rangle \sim |ks\rangle_a$ o delle antiparticelle $b^\dagger_{us}|0\rangle \sim |ks\rangle_b$

IMPONGO che sul vuoto lo spin sia nullo

$S^i|0\rangle \stackrel{!}{=} \phi$

AZIONE SULLE PARTICELLE $a^\dagger_{us}|0\rangle$

Quando faccio agire S^i su uno stato di particella $\sim a^\dagger_{us}|0\rangle$ gli ultimi 3 termini si annullano infatti:

$S^i a^\dagger_{ks}|0\rangle = [S^i, a^\dagger_{us}]|0\rangle$ perché $S^i|0\rangle = 0$

allora

(1) anticommutano $\{a^\dagger_{us} a^\dagger_{pr}\} = 0$

(2) $\rightarrow [a^\dagger_{pr} b^\dagger_{qr'}, a^\dagger_{us}]|0\rangle = (a^\dagger_{pr} b^\dagger_{qr'} a^\dagger_{us} - a^\dagger_{us} a^\dagger_{pr} b^\dagger_{qr'})|0\rangle =$

(2) anticommutano

$= (a^\dagger_{pr} b^\dagger_{qr'} a^\dagger_{us} - a^\dagger_{pr} b^\dagger_{qr'} a^\dagger_{us})|0\rangle = \phi|0\rangle = \phi$

uguali ora! si annullano!

(3) $\rightarrow$ stesso discorso, si annullano!

anticommutano

(4) $\rightarrow [b_{pr} b^\dagger_{qr'}, a^\dagger_{us}]|0\rangle = (b_{pr} b^\dagger_{qr'} a^\dagger_{us} - a^\dagger_{us} b_{pr} b^\dagger_{qr'})|0\rangle =$

$= (b_{pr} b^\dagger_{qr'} a^\dagger_{us} - b_{pr} b^\dagger_{qr'} a^\dagger_{us})|0\rangle = \phi$

anticommutano

$b_{pr} a_{qr'} a^\dagger_{us} - a^\dagger_{us} b_{pr} a_{qr'} = b_{pr}(a_{qr'} a^\dagger_{us} + a^\dagger_{us} a_{qr'})|0\rangle =$

$= b_{pr}\{a_{qr'}, a^\dagger_{us}\}|0\rangle = (\text{number}) b_{pr}|0\rangle = \phi$

number

annihilatore

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- solo il primo termine è non nullo! Infatti:

$$[a^\dagger_{pr}\, a_{qr'},\, a^\dagger_{us}]|0\rangle = (a^\dagger_{pr}\, a_{qr'}\, a^\dagger_{us} - a^\dagger_{us}\, a^\dagger_{pr}\, a_{qr'})|0\rangle =$$

($a^\dagger_{us}$ anticommutano)

$$= a^\dagger_{pr}\,(a_{qr'}\, a^\dagger_{us} + a^\dagger_{us}\, a_{qr'})|0\rangle = a^\dagger_{pr}\,\{a_{qr'}, a^\dagger_{us}\}|0\rangle =$$

(loro qua non anticommutano poiché sono un a con un $a^\dagger$)

$$= (2\pi)^3 \delta_{r's}\, \delta(\vec{u}-\vec{q})\, a^\dagger_{pr}|0\rangle \qquad \text{non zero!}$$

- → so I keep just the first term in S^i

$$S^i = \int dx^3 \int \frac{d^3p}{(2\pi)^3\sqrt{2\epsilon_p}} \int \frac{d^3q}{(2\pi)^3\sqrt{2\epsilon_q}} \sum_{rr'} :a^\dagger_{pr}\, a_{qr'}: U^{\dagger r}_p \frac{\sigma^i}{2} U^{r'}_q\, e^{-i(p-q)x} =$$

$$= \int \frac{d^3p}{\sqrt{2\epsilon_p}} \int \frac{d^3q}{(2\pi)^3\sqrt{2\epsilon_q}} \sum_{rr'} a^\dagger_{pr}\, a_{qr'}\, U^{\dagger r}_p \frac{\sigma^i}{2} U^{r'}_q\, \delta(\vec{p}-\vec{q}) =$$

$$= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\epsilon_p} \sum_{rr'} a^\dagger_{pr}\, a_{pr'}\, U^{\dagger r}_p \frac{\sigma^i}{2} U^{r'}_p$$

spin acting on particle fermions

- ricordando l'azione di S^i sulla particella

$$S^i\, a^\dagger_{us}|0\rangle = [S^i, a^\dagger_{us}]|0\rangle = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\epsilon_p} \sum_{rr'} [a^\dagger_{pr}\, a_{pr'}, a^\dagger_{us}]\, U^{\dagger r}_p \frac{\sigma^i}{2} U^{r'}_p|0\rangle =$$

$$= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\epsilon_p} \sum_{rr'} (2\pi)^3 \delta_{r's}\, \delta^{(3)}(\vec{p}-\vec{u})\, a^\dagger_{pr}\, U^{\dagger r}_p \frac{\sigma^i}{2} U^{r'}_p|0\rangle =$$

$$= \frac{1}{2\epsilon_u} \sum_{rr'} \delta_{r's}\, a^\dagger_{ur}\, U^{\dagger r}_u \frac{\sigma^i}{2} U^{r'}_u|0\rangle =$$

(all terms with momentum $\vec{u}$ of the particle we are hitting on)

$$= \frac{1}{2\epsilon_u} \sum_r a^\dagger_{ur}\, U^{\dagger r}_u \frac{\sigma^i}{2} U^{s}_u|0\rangle$$

(U^s_u → dello portello; $a^\dagger_{ur}$ → dello portello)

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- choosing a particle in the rest frame $\vec{u} = 0$ doria $E_u = m$ e $v^s(\vec{u}=0) = \sqrt{m}\begin{pmatrix}\xi^s\\ \xi^s\end{pmatrix}$ Hence

$$S^i a_{0s}^+ |0\rangle = \frac{1}{2m}\sum_r (\xi^{r+}\ \xi^{r+})\frac{\sigma^i}{2}\begin{pmatrix}\xi^s\\ \xi^s\end{pmatrix} a_{0i}^+ |0\rangle = \frac{1}{2}\sum_r \xi^{r+}\sigma^i\xi^s a_{0r}^+|0\rangle$$

- using $\xi^{1=r} = \begin{pmatrix}1\\0\end{pmatrix}$ $\xi^{2=r} = \begin{pmatrix}0\\1\end{pmatrix}$ ottengo per S^3

$$S^3 a_{0\,s=1}^+ |0\rangle = +\frac{1}{2} a_{0\,s=1}^+ |0\rangle$$
$$S^3 a_{0\,s=2}^+ |0\rangle = -\frac{1}{2} a_{0\,s=2}^+ |0\rangle$$

} autostati di $S^3 = S^z$

If I choose $\vec{u} = (0\ 0\ u^i)$ along the same direction of S^i (cioè se S^z scelgo $\vec{u} = u\hat{z} = (0\,0\,u)$ se S^y uso $(0\,u\,0)$ ecc...) computation lead to the same result so

$$S_z\, a_{\vec{u}s}^+ |0\rangle = \pm\frac{1}{2} a_{\vec{u}s}^+ |0\rangle$$

per $\vec{u} = u\hat{z}$
+ per $s=1$
− per $s=2$

Particle con momento lungo z sono autostati di spin z con autovalore $+\frac{1}{2}$ o $-\frac{1}{2}$ a seconda dello spin S

AZIONE SULLE ANTIPARTICELLE

Se agisco i conti $S^i b_{\vec{u}s}^+ |0\rangle$ per l'azione su uno antiparticella fermionica $b_{\vec{u}s}^+ |0\rangle$ trovo che l'unico termine non nullo è il $C(\xi)$ $(b_{pr}\, b_{pr'}^+)$, ottengo una espressione analoga per S^i con $b_{pr'}^+ b_{pr}$ al posto di $a_{pi}^+ a_{pr'}$ e alla fine lo stesso risultato con solo segno opposto per $s=1$ e $s=2$

$$S_z\, b_{\vec{u}s}^+ |0\rangle = \mp\frac{1}{2} b_{\vec{u}s}^+ |0\rangle$$

$\vec{u} = u\hat{z}$
+ per $s=2$
− per $s=1$

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Phase symmetry - N° of particles & antiparticles

Dirac Lagrangian is invariant also under phase U(1) symmetry

$$\psi \rightarrow e^{i\theta}\psi$$

The charge associated to such symmetry is:

$$j^{\mu} = \bar{\psi}\gamma^{\mu}\psi \qquad \text{conserved current}$$

$$Q = \int dx^3 \, :\psi^{+}\psi: \qquad \text{conserved charge (quantized)}$$

expanding that in normal modes & obtain after some computations:

$$Q = \int \frac{d^3p}{(2\pi)^3} \sum_{s=1,2} \left(a_{ps}^{+} a_{ps} - b_{ps}^{+} b_{ps}\right)$$

particles have count +1, antiparticles count -1 - this is the # of particle minus # antiparticle

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7

INTERACTING THEORIES

So far we have considered FREE fields, ie particles alone isolated. The $\mathcal{L}$ is linear in ϕ hence the equation of motion can be solved by superposition of plane waves.

An interaction introduces a non linear term in $\mathcal{L}$ and the solutions of eq. of motion are no more plane waves

At a fixed time (initial time) it is true I can always expand the field in Fourier transform and plane waves modes, but then when the field evolves in time it evolves with an Hamiltonian which is not linear so it generates non linear terms in the fields, no more plane waves.

For example $a_p^\dagger |0\rangle$ can evolve in $(a_p^\dagger)^2|0\rangle$ or $(a_p^\dagger)^3|0\rangle$ ecc ecc

Such example show that with interactions it's possible to CREATE new particles! This is good, since special relativity allows creation or destruction of particles → we need however a theory which can describe that coherently with also QM formalism.

IMPORTANT NOTE

Hilbert space of interacting theories is DIFFERENT from the one of FREE theory.

$|\Omega\rangle$ vacuum state of interacting space / $|0\rangle$ vacuum state of free space

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In particle physics we are interested in scattering amplitude because, once integrated over all possible phase angles, gives the cross section, a sort of probability of interaction / collision between particles / events.

EXAMPLES OF INTERACTIONS

ϕ^4 theory: $\mathcal{L} = \underbrace{\frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2} m^2 \phi^2}_{\mathcal{L}_0 \text{ (free)}} - \underbrace{\frac{\lambda}{4!}\phi^4}_{\mathcal{L}_{int}}$ self interaction of scalar fields

QED: $\mathcal{L}_{qed} = \bar{\psi}(i\gamma - m)\psi - \frac{1}{4}(F_{\mu\nu})^2 - \underbrace{e\,\bar{\psi}\gamma^\mu A_\mu \psi}_{\mathcal{L}_{int}}$ electron interaction with A_μ fields

Yukawa: $\mathcal{L}_{yuk} = \underbrace{\bar{\psi}(i\gamma - m)\psi}_{\text{Dirac field}} + \underbrace{\frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2} m^2 \phi^2}_{\text{scalar field}} - \underbrace{g\bar{\psi}\psi\phi^2}_{\mathcal{L}_{int}}$ interaction between Dirac and scalar fields

ASSUMPTIONS

(1) **Local interactions** = i.e. connected by time-like case (Entanglement? Open point ...) → important for Locality in QFT

(2) **Interaction terms containing only fields and not their derivatives** = this could be relaxed, we impose it to have easier calculations but there is no real physical need

(3) **Coupling parameters** (λ, e, g ..) **sufficiently SMALL** to interpret the interaction as a perturbation → PERTURBATIVE THEORY

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because $\mathcal{H} = \pi \partial_0\phi - \mathcal{L}$ with $\pi = \frac{\partial \mathcal{L}}{\partial(\partial_0\phi)}$, if $\mathcal{L}_{int}$ not contains $\partial_0\phi$

then it gives $\pi_{int} = \frac{\partial \mathcal{L}_{int}}{\partial(\partial_0\phi)} = 0$ so $\mathcal{H}_{int} = -\mathcal{L}_{int}$

With such assumptions I can write general system as:

$$\mathcal{L} = \mathcal{L}_{free} + \mathcal{L}_{int}$$

And the Hamiltonian density

$$\mathcal{H} = \mathcal{H}_{free} + \underbrace{\mathcal{H}_{int}}_{\text{small, to be a perturbation}}$$

and Hamiltonian

$$H = \int d^3x\, \mathcal{H} = \int d^3x\, \mathcal{H}_{free} + \int d^3x\, \mathcal{H}_{int} = H_{free} + H_{int}$$

and if $\mathcal{L}_{int}$ does not contain derivatives of the fields we can write:

$$H_{int} = \int d^3x\, \mathcal{H}_{int} = -\int d^3x\, \mathcal{L}_{int} \qquad \mathcal{H}_{int} = -\mathcal{L}_{int}$$

7.1 S-MATRIX OF TIME EVOLUTION (M 5.1)

So far we have been working almost exclusively in Heisenberg picture, because it treats operators as a function of time also. Hence an operator (hence a field) is a function of both space $\vec{x}$ and time t and is better when for Lorentz invariance. However now, introducing the time evolution of the perturbation, is more intuitive to introduce the evolution on the states, so to work in Schroedinger picture. After having understood the meaning of such evolution on the states, we will come back to Heisenberg picture.

SCHRÖDINGER PICTURE

In the schrodinger picture consider a state $|\alpha(t)\rangle$ function of time. At time $t = T_i$ (initial time) $|\alpha(T_i)\rangle \equiv |\alpha\rangle$ is a state which is an

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eigenstate of some commuting operators. I label the eigenstates by the parameters "a" (sarà $\vec{p}$ momento ecc ecc). Similarly I consider a state $|b(t)\rangle$ which, at another time $t = T_F$ (final time) will be an eigenstate $|b\rangle$ of other commuting operators. For now $|a(t)\rangle$ and $|b(t)\rangle$ are completely general different states.

$$|a(T_i)\rangle = |a\rangle \qquad |b(T_F)\rangle = |b\rangle$$

such states evolve under H hamiltonian as:

$$|a(t)\rangle = e^{-iH(t-T_i)}\,|a(T_i)\rangle$$

At time $t = T_F$ we have:

$$|a(T_F)\rangle = e^{-iH(T_F-T_i)}\,|a(T_i)\rangle$$
$$|a(T_i)\rangle = |a\rangle$$

The probability amplitude for an initial state $|a\rangle = |a(T_i)\rangle$ to evolve at time T_F into $|b\rangle = |b(T_F)\rangle$ is therefore:

$$\langle b|a(T_F)\rangle = \langle b|e^{-iH(T_F-T_i)}|a\rangle$$

(Q such definition is the same in Shrod and Heisenberg cause scalar products are the same in the two pictures)

S-MATRIX DEFINITION

We define the S matrix as:

$$\langle b|S|a\rangle := \lim_{(T_F-T_i)\to\infty} \langle b|e^{-iH(T_F-T_i)}|a\rangle \qquad (7.1)$$

Schrod.

S is an operator which maps an initial state $|a\rangle$ into a final state $S|a\rangle$ and the scattering amplitudes are given by its matrix elements $\langle b|S|a\rangle$

Shrodinger picture operator: $S_{\text{shrodyer}} = \lim_{(T_F-T_i)\to\infty} e^{-iH(T_F-T_i)}$

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Why infinite time gap $T_F - T_i \to \infty$? Because we want to describe initial and final states as FREE FIELDS and so, to be sure that they don't interact, we separate them by an infinite time gap.

PROPERTIES OF S

(1) $S S^\dagger = S^\dagger S = \mathbb{1}$ unitary

Operator is a unitary operator. We can see that by IMPOSING the conservation of probability: given a basis set $|n\rangle$ that $\sum |n\rangle\langle n| = \mathbb{1}$, the prob conservation imposes that the evolving from $|\alpha\rangle$ to $\to S|\alpha\rangle$ also $S|\alpha\rangle$ is normalized in probabilities ie:

$$\sum_n |\langle n|S|\alpha\rangle|^2 \overset{!}{=} 1$$

$$\sum_n \langle\alpha|S^\dagger|n\rangle\langle n|S|\alpha\rangle = \langle\alpha|S^\dagger S|\alpha\rangle \overset{!}{=} 1$$

($\sum |n\rangle\langle n| = \mathbb{1}$)

since $\langle\alpha|\alpha\rangle = 1$ the only way such that $\langle\alpha|S^\dagger S|\alpha\rangle = 1$ is that $S^\dagger S = \mathbb{1}$ hence that S is unitary. (cvd)

(2) $S := \mathbb{1} + iT$ (7.2) → $2\,\mathrm{Im}\, T_{aa} = \sum_n |T_{an}|^2$ (7.9)

T: interaction term → $-i(T - T^\dagger) = TT^\dagger$ (7.13)

- Defining S with an interaction term T and a non interaction $\mathbb{1}$ term, $S = \mathbb{1} + iT$,
- Imposing the unitarity:
  $S^\dagger S = \mathbb{1} \to -i(T - T^\dagger) = TT^\dagger$
- Writing $T_{ab} = \langle b|T|a\rangle$ we have
  $-i(T - T^\dagger) = TT^\dagger \to -i(T_{ab} - T_{ba}^*) = \sum_n T_{bn} T_{an}^*$
- setting $a = b$ we get
  $2\,\mathrm{Im}\, T_{aa} = \sum_n |T_{an}|^2$ (cvd)

which is a very usefue expression in calculations.

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HEISEMBERG PICTURE

Now we want to represent all in Heisemberg picture because it's more useful in QFT. In Heisemberg picture states does not depend on time but operators do.

General

$$|a(t)\rangle_{Schrod} = e^{-iHt}\,|a\rangle_{Heisenb} \longrightarrow |a\rangle_{Heis} = e^{iHt}\,|a(t)\rangle_{Schrod}$$

where in this def if $t=0$ $|a(0)\rangle = |a\rangle$ Heisbg.

Fixing $t=T_i$

If I choose $t=T_i$ I obtain

$$|a,T_i\rangle_H = e^{iHT_i}\,|a(T_i)\rangle_{Schrod}$$

$$|a,T_i\rangle_{Heis} = e^{iHT_i}\,|a\rangle_{Schrod}$$

↓

label to remember that when $t=T_i$ the schrodger state in $|a\rangle = |a(T_i)\rangle$.

Fixing $t=T_f$

Equivalently =

$$|b,T_f\rangle_{Heis} = e^{iHT_f}\,|b\rangle_{Schrod}$$

Se il tempo fosse positivo $\langle a|T|b\rangle_H = e^{-iH(T-t)}\,\langle a|b\rangle_S = \langle a|b\rangle_S$

Calcolare tra stato schrodger e stato heisbg evoluto dopo lo stesso tempo si ricava, è ovvio! → Noi vivremo sempre una cosa diversa = guardiamo all'evoluzione dello stato di S in a tempi diversi, procedendo come occupandoci a due studi di S in a tempi $T_F - T_i$

FINIRE nello IALO dopo un tempo $T_F - T_i$ consideriamo il tempo finale come il tempo iniziale

operators transforms as:

General

$$S_{Heis}(t) = e^{iHt}\,S_{Schr}\,e^{-iHt}$$

Fixing times T_i, T_f

$$S(T_f,T_i) = e^{iHT_f}\,S_S\,e^{-iHT_i} = \mathbb{1}$$

$$S_S = \lim_{|T_f - T_i|\to\infty} e^{-iH(T_f-T_i)}$$

Matrix elements are the same (IF TIME IS THE SAME!)

General

$$\langle b|_S\,S_S\,|a\rangle_S = \underbrace{\langle b|_S e^{-iHt}}_{\langle b|_H}\;\underbrace{e^{iHt}\,S_S\,e^{-iHt}}_{S_H}\;\underbrace{e^{iHt}\,|a\rangle_S}_{|a\rangle_H} = \langle b|_H\,S_H\,|a\rangle_H$$

se il tempo è lo stesso ad Quello che in Schrod è il valor di aspettazione di S tra $\langle a|$ e $|b\rangle$, in Heisbg non è più il valor di aspettazione di S_I è solo l'overlap tra due stati

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We can write:

Fixing T_F, T_i:

$$\langle b|S|a\rangle = \lim_{T_F - T_i \to \infty} \langle b|e^{-iH(T_F - T_i)}|a\rangle = \lim_{T_F - T_i \to \infty} \langle b, T_F | a, T_i\rangle \qquad (7.5)$$

$\Downarrow$ Schrodyer

$\Downarrow$ "Heisemberg"

Meaning of $\langle b, T_F | a, T_i\rangle$ in Heisemberg picture:

When I make the overlap of $|a\,T_i\rangle_H$ and $|b\,T_F\rangle_H$ in Heisemberg picture I'm overlapping TWO states which are TIME INDEPENDENT, H picture is time independent on states.

The time dependence of the overlap is HIDDEN in the operator which connects the states

It's like = I have two people, they are Fixed, what changes is the external operator $O(t)$. Same results if I see instead the operator fixed O and who change are the two people

$$\underbrace{\langle b|}_{H}\ \underbrace{O(t)}_{H}\ \underbrace{|a\rangle}_{H} = \underbrace{\langle b(t)|}_{S}\ \underbrace{O}_{S}\ \underbrace{|a(t)\rangle}_{S}$$

persone fisse, cambio ed esce quantità: O

quantità fisse, cambiano le persone

[ES = Due persone si amano a e b, O è il loro amore. Nel tempo, se fino a e b, il loro amore cambia. Oppure l'amore è l'amore, poco cambiano le persone e la quantità di amore tra essi cambia per questo.]

For eq 7.5 it's the same logic, with only note that we have chosen TWO DIFFERENT TIMES!

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SHRODINGER

$|a(t)\rangle = |a(T_i)\rangle =: |a\rangle$ ($t = T_i$)

$|b(t)\rangle = |b(T_f)\rangle =: |b\rangle$ ($t = T_f$)

Time dependent — Fixed Times

$S_S = \lim_{T_F - T_i \to \infty} e^{-iH(T_F - T_i)}$

time independent always

$\langle b(t)| S_S |a(t)\rangle$

T_f — Fixing Times — T_i

$\langle b(T_f)| S_S |a(T_i)\rangle$

$\langle b| S |a\rangle$

FIXED TIME — FIXED TIME

HEISEMBERG

$S(t,t) = S(T_f,T_i) = e^{+iHT_f} S_S e^{-iHT_i} = \mathbb{1}$ ($t = T_f$, $t = T_i$)

Time dependent — Fixed Times

$|a\rangle_H =: |aT_i\rangle$

$|b\rangle_H =: |bT_f\rangle$

always time independent (T_i, T_F are just labels)

$\langle bT_f|_H \; S_H(t,t) \; |aT_i\rangle_H$

T_f — Fixing Times — T_i

$\langle bT_f|_H \; S_H(T_f,T_i) \; |a,T_i\rangle_H$

$\langle b,T_f| \; \mathbb{1} \; |a,T_i\rangle$

FIXED TIMES

DIVERSI

Nello vuoto beg ⇒ pos gli

⇒ Fisso l'operatore ma prendendo due tempi diversi e ottengo l'identità per quei tempi scelti.

cos tempi ... dal tempo, così ... con ... S

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SHRODINGER

$|a(t)\rangle = |a(T_i)\rangle =: |a\rangle$ ($t \to T_i$)

$|b(t)\rangle = |b(T_f)\rangle =: |b\rangle$ ($t \to T_f$)

Time dependent — Fixed Times

$S_S = \lim_{T_f - T_i \to \infty} e^{-iH(T_f - T_i)}$

Time independent always

HEISEMBERG

$S(t,t) = S(T_f, T_i) = e^{+iHT_f} S_S e^{-iHT_i} = \mathbb{1}$ ($t \to T_f$, $t \to T_i$)

Time dependent — Fixed Times

$|a\rangle_H =: |a T_i\rangle$

$|b\rangle_H =: |b T_f\rangle$

always time independent (T_i, T_f are just labels)

⇓

È un po' strano quello che stiamo facendo! In realtà dovremo considerare un unico tempo iniziale per entrambi, invece stiamo considerando due tempi diversi.

Nella visione di Schrödinger ⇒ posso cambiare gli stati ⇒ Fisso i due stati iniziali e finali a DUE TEMPI DIVERSI

Nella visione di Heisberg ⇒ possono cambiare gli operatori ⇒ Fisso l'operatore Ma prendendo due tempi diversi e ottengo l'identità per quei tempi scelti.

In entrambi i casi ho FISSATO i tempi sulle quantità che dip dal tempo, così ho ottenuto un oggetto che NON DIPENDE dal tempo

$$\langle b | S | a \rangle = \langle b, T_f | a, T_i \rangle$$

per $T_f - T_i \to \infty$

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7.2 LEHMANN-SYMANZIK-ZIMERMAN (LSZ) REDUCTION FORMULA (M.5.2)

Consider S-matrix of n initial states $|\vec{k}_1, \vec{k}_2 \dots \vec{k}_n, T_i\rangle$ labeled by $\vec{k}_i$ and m final states $\langle \vec{p}_1, \vec{p}_2 \dots \vec{p}_n, T_f|$ labeled by $\vec{p}_j$ =

$$S_{pk} = \langle \vec{p}_1, \vec{p}_2 \dots \vec{p}_n, T_f | \vec{k}_1, \vec{k}_2 \dots \vec{k}_m, T_i \rangle = \langle p_1 \dots p_n | S | k_1 \dots k_m \rangle$$

Heisenberg — Schrodyer

We want to compute such overlap, so the scattering amplitude of S matrix evolving $\vec{k}$ into $\vec{p}$

AIM = COMPUTE THE S-MATRIX S_{pk} ELEMENTS

Remember = The formula above is in Heisenberg notation! The two states written as above are time independent ... it accounts for every time = I can look at the interaction at any intermediate time

Assumption = $\vec{p}_i \neq \vec{k}_j \quad \forall i,j$ initial and final states are DIFFERENT

Hence we are evaluating only the iT term in $S = \mathbb{1} + iT$, cause identity $\mathbb{1}$ would lead to equal initial and final state. $\langle p_1 \dots p_n T_f | k_n \dots k_n T_i \rangle = \langle p \dots | iT | k \dots \rangle$

FREE FIELDS FOR INITIAL AND FINAL TIME

Consider a FREE scalar field $\phi_{free}(x)$. It expands as depicted in chapter 4.1 in plane waves =

$$\hat{\phi}_{free}(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} \left(\hat{a}_{\vec{p}}\, e^{-ip\cdot x} + \hat{a}^{+}_{\vec{p}}\, e^{ip\cdot x} \right)$$

such eq can be inverted by:

$$\sqrt{2E_k}\, \hat{a}_k = i \int d^3x\, e^{ikx} \overleftrightarrow{\partial_0} \hat{\phi}_{free} \qquad (7.6)$$

$$\sqrt{2E_k}\, \hat{a}^{+}_k = -i \int d^3x\, e^{-ikx} \overleftrightarrow{\partial_0} \hat{\phi}_{free}$$

with $i A \overleftrightarrow{\partial_0} B = +i (A \partial_0 B - B \partial_0 A)$

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One can verify such invesion substituting the expression of the ϕ_{free} in plane waves and performing the integration over x. (inverting eq of ϕ_{free} directly is not so easy).

As we can already notice, $\hat{U}_H$ does not depend on time

INTERACTION

At time $t \to +\infty$ and $t \to -\infty$ such that $(T_F - T_i) \to +\infty$ particles are very well separated and does not interact = for $t \to \pm\infty$ the interacting theory ($\phi(x)$ interacting) collapses into free theory (ϕ_{in} & ϕ_{out})

So we call $\phi(x)$ our field of the interaction theory, of the system governed by $H = H_{free} + H_{int}$ in the world of interacting space with $|\Omega\rangle$ vacuum state

ϕ_{in} and ϕ_{out} are instead two DIFFERENT FREE fields of two Different Hilbert spaces, their vacuum state is labeled $|0\rangle$ as for free theories.

We consider $\phi(x)$ collapsing into free initial and final fields for $t \to \pm\infty$

$$\phi(x) \text{ interacting} \longrightarrow \sqrt{Z}\,\phi_{in}(x) \quad \text{free} \quad \text{for } t \to -\infty \qquad (7.7)$$
$$\phi(x) \text{ interacting} \longrightarrow \sqrt{Z}\,\phi_{out}(x) \quad \text{free} \quad \text{for } t \to +\infty$$

Wher $\sqrt{Z}$ is a normalization constant.

$$\lim_{t \to -\infty} \phi(x) = \sqrt{Z}\,\phi_{in}(x)$$

$$\lim_{t \to +\infty} \phi(x) = \sqrt{Z}\,\phi_{out}(x)$$

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we can then rewrite eq (7.6) of the free fields as the limits of interacting $\phi(x)$ field:

$$\sqrt{2E_k}\,\hat{a}_k^{+\,(in)} = -i\int d^3x\, e^{-ik\cdot x}\overleftrightarrow{\partial_0}\hat{\phi}_{in} = -i\,z^{-1/2}\lim_{t\to-\infty}\left(\int d^3x\, e^{-ik\cdot x}\overleftrightarrow{\partial_0}\hat{\phi}\right)$$

($\hat{\phi}_{in}$: FREE; $\hat{\phi}$: interacting)

$$\sqrt{2E_p}\,\hat{a}_p^{+\,(out)} = -i\int d^3x\, e^{-ip\cdot x}\overleftrightarrow{\partial_0}\hat{\phi}_{out} = -i\,z^{-1/2}\lim_{t\to+\infty}\left(\int d^3x\, e^{-ip\cdot x}\overleftrightarrow{\partial_0}\hat{\phi}\right)$$

($\hat{\phi}_{out}$: free; $\hat{\phi}$: interacting)

notice that $\hat{a}_k^{+\,(in)}$ and $\hat{a}_k^{+\,(out)}$ don't depend on time but they refers and act in DIFFERENT SPACES, those of INITIAL and FINAL states, which are DIFFERENT states

Removing one particle from the initial state

Given the matrix element =

$$\langle \vec{p}_1 \cdots \vec{p}_n, \text{IF} \,|\, \vec{k}_1 \cdots \vec{k}_m, \text{Ii}\rangle$$

our aim is to relate it to the expectation value of some operator on the vacuum state $|\Omega\rangle$. To do that we will eliminate (remove) one particle at a time from initial and final state, ending with the vacuum $|\Omega\rangle$ on both sides.

Let's start by removing one particle to the initial state. The state $|\vec{k}_1, \vec{k}_2 \cdots \vec{k}_n, \text{Ti}\rangle$ is written in Heisenberg picture and corresponds to the "initial" state (even though in this picture states do not depend on time, are always the same). Operator $\hat{a}_{\vec{k}_1}^{+\,(in)}$ act on the space of initial state ie act on $|\vec{k}_2 \cdots \text{Ti}\rangle$

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we can see $\vec{k}_1$ particle as if it was created by $\hat{a}^{+}_{k_1}$ acting on the vacuum

$$|\vec{k}_1, \vec{k}_2 \dots \vec{k}_n, T_i\rangle = \sqrt{2E_{k_1}}\ \hat{a}^{+(in)}_{k_1} |0, \vec{k}_2, \dots \vec{k}_m, T_i\rangle \qquad (\star)$$

Observations

a) $\hat{a}^{+}_{k_1}$ is an operator in Heisenberg picture so it acts on Heisenberg states $|k_1 \dots T_i\rangle$

b) usually op in Heis. picture depend on time, $\hat{a}_{k_1}$ however does not depend on time

c) Neither $|k_1 \dots T_i\rangle$ depends on the time by definition of Heis state.

$\Rightarrow$ we can perform such creation because, thinking of initial time, the field is a FREE FIELD ϕ_{in}

d) we could do the same with $\hat{a}^{+(out)}_{p}$ BUT on $|\vec{p}_1 \dots \vec{p}_n T_F\rangle$ which is a DIFFERENT STATE. Always time independent but acting on different states / spaces!

$\hat{a}^{+(out)}_{p}$ $\rightarrow$ Act only on $|\vec{p}_n - T_F\rangle$ $\qquad$ $\hat{a}^{+(in)}_{k_1}$ $\rightarrow$ acts only on $|\vec{k}_n - T_i\rangle$

$\Rightarrow$ Nella visione di Heisenberg posso vedere $|k\rangle$ and $|p\dots\rangle$ come CAMPI LIBERI, ma ognuno DIVERSO con il suo personale op di creazione. SPAZIO iniziale e spazio finale sono diversi

⊙ Given $(\star)$ we can write: remove one particle from the initial state:

$$\langle \vec{p}_1 \dots \vec{p}_n) T_F | \vec{k}_1 \dots \vec{k}_n, T_i\rangle \overset{(\star)}{=} \sqrt{2E_{k_1}}\ \langle p_1, \dots p_n) T_F |\ \hat{a}^{+(in)}_{k_1} |0, k_2 \dots T_i\rangle =$$

$$= -i\, z^{-1/2} \lim_{t \to -\infty} \int d^3x\ e^{-ik_1 x} \langle \vec{p}_1 \dots \vec{p}_n, T_F |\ \overleftrightarrow{\partial_0} \phi\ | 0, \vec{k}_2 \dots \vec{k}_m, T_i\rangle$$

interacting !!

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we had removed one particle from the initial state free state but at the cost of introducing an interaction field acting on the vacuum. We will come again to this point.

First we want to notice that such equation is not evidently Lorentz covariant = It is Lorentz invariant but is not so clear = terms of $\int d^3x$ and ∂_0 look as if they are treating space and time differently. It's not the case, but we want a to write such eq in a nice clear covariant expression

$$\langle p_1, p_2 \dots p_n, T_F | k_1, k_2 \dots k_m, T_i \rangle = \frac{-i}{\sqrt{2}} \lim_{t \to -\infty} \int d^3x \, e^{-ik_1 \cdot x} \langle p_1, p_2 \dots p_n T_F | \overleftrightarrow{\partial_0} \phi | 0, k_2 \dots T_i \rangle$$

$\int d^3x$: apparently not covariant

$\overleftrightarrow{\partial_0}$

⊙ First we start noticing that:

$$\sqrt{2E_{k_1}} \langle \vec{p}_1 \dots T_F | a^{+(in)}_{k_1} - a^{+(out)}_{k_1} | 0 \vec{k}_2 \dots T_i \rangle = \sqrt{2E_{k_1}} \langle p_1 \dots T_F | a^{+(in)}_{k_1} | 0 \, k_2 \dots T_i \rangle$$

because $a^{+(out)}_{k_1}$ acts on $\langle p_1 \dots p_n T_F |$ annihilating it,

$\langle \vec{p}_1 \dots \vec{p}_n T_F | a^{+(out)}_{k_1} = \emptyset$ → because $a^{(out)}_{k_1} | p_1 \dots p_n T_F \rangle = \emptyset$ if $\vec{k}_1 \neq \vec{p}_j \ \forall j$ as we assumed

⊙ Good! Let's now study better the term in $a^{(in)}_k - a^{+(out)}_{k_1}$

$$\sqrt{2E_k} \left(a^{+(in)}_k - a^{+(out)}_k \right) \overset{*}{=} \overset{*}{=} +i \bar{z}^{1/2} \int dx^3 \left(\lim_{t \to +\infty} - \lim_{t \to -\infty} \right) \left(e^{-ik \cdot x} \overleftrightarrow{\partial_0} \phi \right) =$$

f

Fundamental theorem of calculus:

$$\left(\lim_{t \to \infty} - \lim_{t \to -\infty} \right) f(t) = \int_{-\infty}^{+\infty} dt \, \frac{\partial f}{\partial t}$$

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$$= i\tilde{z}^{-1/2} \underbrace{\int d^4x}_{\int dt \int d^3x} \partial_0 \underbrace{\left(e^{-ik\cdot x} \overleftrightarrow{\partial_0} \phi \right)}_{f} = i\tilde{z}^{-1/2} \int d^4x\, \partial_0 \left(e^{-ik\cdot x} \overleftrightarrow{\partial_0} \phi \right) =$$

$$= i\tilde{z}^{-1/2} \int d^4x\, \partial_0 \left(e^{-ik\cdot x} \partial_0 \phi - \phi\, \partial_0 e^{-ik\cdot x} \right) =$$

$$= i\tilde{z}^{-1/2} \int d^4x \left(\partial_0 \partial_0 \phi + \partial_0^2 \phi - \partial_0 \partial_0 \phi - \phi\, \partial_0^2 \right) =$$

$$= i\tilde{z}^{-1/2} \int d^4x \left(e^{-ik\cdot x} \partial_0^2 \phi - \phi\, \partial_0^2 e^{-ik\cdot x} \right) = (*)$$

① the k^μ in the exponential $e^{-ik\cdot x} = e^{-ik^\mu x_\mu}$ is the mass-shell momentum coming from plane wave expansion of free field hence

$$\partial_0^2 e^{-ik\cdot x} = -k_0^2 e^{-ik\cdot x} =$$

but $k^\mu k_\mu = k^2 = (k_0)^2 - |\vec{k}|^2 = m^2 \rightarrow k_0^2 = |\vec{k}|^2 + m^2$

$$= (-|\vec{k}|^2 - m^2)\, e^{-ik\cdot x} = (\vec{\nabla}^2 - m^2)\, e^{-ik\cdot x}$$

hence

$$(*) = i\tilde{z}^{-1/2} \int d^4x \left(e^{-ik\cdot x} \partial_0^2 \phi - \phi\, (\vec{\nabla}^2 - m^2) e^{-ik\cdot x} \right) =$$

② integrating by parts the $\vec{\nabla}^2$ term, imposing boundary conditions in the space integral (i.e. = suppongo che per $\vec{x} \to \pm\infty$ il campo si annulli - ciò ha senso, ed posso spiegare come se stessi sovrapponendo il campo iniziale e finale in un pacchetto di onde che quindi è localizzato nello spazio e all'infinito si annulla) (NB = NON sto localizzando anche nel tempo! È un pacchetto localizzato in spazio ma no tempo! Nel tempo ci scelgo $t \to \pm\infty$ per def Free Fields at initial and final states) we obtain:

$$= i\tilde{z}^{-1/2} \int d^4x\, e^{-ik\cdot x} (\Box + m^2)\, \phi(x)$$

$$\Rightarrow \sqrt{2E_k} \left(a_k^{+(in)} - a_k^{+(out)} \right) = \frac{i}{\sqrt{z}} \int d^4x\, e^{-ik\cdot x} (\Box + m^2)\, \phi(x) \qquad (7.9)$$

now clearly Lorentz invariant!

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⊙ Recovering the overlap $\langle p_1 .. p_n T_f | .. | 0 k_2 . T_i \rangle$ we finally obtain our Lorentz invariant formula:

$$\langle p_1 .. T_f | k_1 .. T_i \rangle = \boxed{\sqrt{2 E_{k_1}} \langle \vec{p}_n .. T_f | a^{+(in)}_{k_1} - a^{+(out)}_{k} | 0 \vec{k}_2 . T_i \rangle} =$$

$$= i Z^{-1/2} \int d^4x \; e^{-i k_1 x} (\Box + m^2) \langle \vec{p}_1 . T_f | \hat{\phi} | 0 \vec{k}_2 . T_i \rangle$$

$$\Longrightarrow \langle \vec{p}_1 .. T_f | \vec{k}_1 .. T_i \rangle = \frac{i}{\sqrt{Z}} \int d^4x \; e^{-i k_1 \cdot x} (\Box + m^2) \langle p_1 .. T_f | \hat{\phi}(x) | 0 \vec{k}_2 .. T_i \rangle \qquad (7.10)$$

overlap of two free fields, of initial and final state, with n particles

interacting field acting on the vacuum of $\vec{k}_1$ initial particle

This tells us that we can look at the overlap of two free fields of initial and finalstate of n particles. Looking at it as we had one less particle at initial state (starts from $|0 k_2 ..\rangle$ instead of $|k_1 k_2 -\rangle$) is possible BUT we need an interacting field $\phi(x)$ acting on the vacuum left by $\vec{k}_1$ particle

$\langle p_1 .. p_n T_f | k_1 k_2 T_i \rangle$ is written in the Heisenberg picture so I can see it in many perspectives = the formula accounts for every time!

- as two free fields at time T_i e T_f
- as interacting particles for intermediate times

Removing one particle from final state

we removed one particle from initialstate and obtained 7.10

$$\langle \vec{p}_2 .. | \vec{k}_1 .. \rangle = \langle p_1 .. | a^{+(in)}_{k_1} | 0 .. \rangle \longrightarrow (7.10)$$

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Now we want to remove also one particle from final state

$$\langle p_1 \dots | k_1 \dots \rangle = \langle 0\, \vec{p}_2 \dots | \, a^{(out)}_{p_1}\, a^{+(in)}_{k_1} \, | 0\, \vec{k}_2 \dots \rangle = \tag{7.110}$$

creating p_1 — creating k_1

$$= i\tilde{z}^{1/2} \int d^4x\, e^{-ik_1\cdot x} (\Box_{tom})^2 \langle 0\, p_2 \dots | \, a^{(out)}_{p_1}\, \phi(x) \, | 0\, k_2 \dots \rangle \sqrt{2\epsilon_{p_1}}$$

↓ esplicitando l'azione di $a^{+(in)}_{k_1}$

↓ a meno di calcolare ciò, esplicitare l'azione di $a^{out}_{p_1}$

TIME ORDERED PRODUCT

Define the time ordered product as: (7.111)

$$T\{\phi(x)\phi(y)\} = \theta(y^0-x^0)\phi(y)\phi(x) + \theta(x^0-y^0)\phi(x)\phi(y) = \begin{cases} \phi(y)\phi(x) & x^0<y^0 \\ \phi(x)\phi(y) & y^0<x^0 \end{cases}$$

↓ Heaviside function

since:

$a^{(in)}_p$ is constructed from $\phi(y)$ interacting as $y^0 \to -\infty$

$a^{(out)}_p$ is constructed from $\phi(y)$ interacting as $y^0 \to +\infty$

we have:

$$T\{a^{(in)}_p \phi(x)\} = \phi(x)\, a^{(in)}_p \quad \text{as } y^0 \to -\infty, \text{ so } y^0 < x^0 \tag{7.112}$$
$$T\{a^{(out)}_p \phi(x)\} = a^{(out)}_p \phi(x) \quad \text{as } y^0 \to +\infty \text{ so } x^0 < y^0$$

⊙ We note that we can rewrite $\Box$ as follows:

$$\sqrt{2E_{p_1}} \langle 0\, \vec{p}_2 \dots | \, T\{a^{out}_{p_1} - a^{in}_{p_1}\}\hat{\phi}(x)\} | 0\, k_2 \dots \rangle = \; \Box \tag{7.112}$$

$$= \sqrt{2\epsilon_{p_1}} \langle 0\, \vec{p}_2 \dots | \, a^{out}_{p_1}\phi(x) - \hat{\phi}(x) a^{(in)}_{p_1} | 0\, k_2 \dots \rangle =$$

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$$= \sqrt{2\epsilon_{p_1}}\, \langle 0\, p_2 \cdots | \, a^{out}_{p_1} \phi(x) - \phi(x) a_{p_1} \, | 0\, k_2 \cdots \rangle =$$

a_{p_1} annihilates because $\vec{p}_1$ does not coincide with any $\vec{k}_i$, so it's like $a_{p_1} |0 k_2 \cdots\rangle = \emptyset$ else $a_p |0\rangle = \emptyset$

$$= \sqrt{2\epsilon_{p_1}}\, \langle 0 \vec{p}_2 \cdots | \, a^{out}_{p_1} \phi(x) \, | 0 \vec{k}_2 \cdots \rangle$$ obtained the same term.

⊙ We note also that the conjugate of eq(7.9) is:

$$\sqrt{2\epsilon_{p_1}} \left(a^{out}_{p_1} - a^{in}_{p_1} \right) = +i \bar{z}^{-1/2} \int d^4y \, e^{+ip_1 \cdot y} \left(\Box_y + m^2 \right) \phi(y)$$ conjugate (7.9)

Hence our term is: (conjugate (7.9) ↓)

$$\sqrt{2\epsilon_{p_1}}\, \langle 0 \vec{p}_2 \cdots | \, T\{ a^{out}_{p_1} - a^{in}_{p_2} \} \phi(x) \, | 0 \vec{k}_2 \cdots \rangle =$$

$$= +i \bar{z}^{1/2} \int d^4y \, e^{+i p_1 \cdot y} \left(\Box_y + m^2 \right) \langle 0 \vec{p}_2 \cdots | \, T\{ \phi(y) \phi(x) \} | 0 \vec{k}_2 \cdots \rangle$$

⊙ So substituting in our expression with both one k_1 particle and one p_2 particle removed from initial and final state we have

$$\langle \vec{p}_1 \vec{p}_2 \cdots T F | \vec{k}_1 \vec{k}_2 \cdots T_i \rangle = \langle 0 p_2 \cdots T F | \, a^{out}_{p_1} \, a^{+in}_{k_1} \, | 0 k_2 \cdots T_i \rangle = \quad (7.13)$$

$$= \left(i \bar{z}^{-1/2} \right)^2 \int d^4y \, e^{i p_1 \cdot y} \left(\Box_y + m^2 \right) \int d^4x \, e^{-i k_1 \cdot x} \left(\Box_x + m^2 \right) \langle 0 p_2 \cdots T F | \, T\{ \phi(x) \phi(y) \} \, | 0 k_2 \cdots T_i \rangle$$

↓ TIME ORDERED

We have rewritten the overlap of initial and final state as the overlap of two states in an intermediate time with one less particle at both initial final state, with some interaction.

What is important here is the TIME ORDERED PRODUCT OF interacting fields! $\langle OUT | T\{ \phi(x) \phi(y) \} | IN \rangle$ → We want initial ϕ acting $|IN\rangle$ and final ϕ acting $\langle OUT |$!!

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Removing all particles from initial and final state

I can iterate removing particles from initial and final state - removing all n final particles and all m initial particles we obtain:

$$\langle \vec{p}_1 \dots \vec{p}_n ; \text{out} | \vec{k}_1 \dots \vec{k}_m \text{ in} \rangle = (iZ^{-1/2})^{n+m} \int \prod_{i=1}^{m} d^4x_i \int \prod_{j=1}^{m} d^4y_j \; e^{i\sum_{j=1}^{m} p_j y_j - i\sum_{i=1}^{m} k_i x_i} \cdot$$

(n final particles, m initial particles; product from 1)

$$\cdot (\Box_{x_1} + m^2)(\Box_{x_2} + m^2) \dots (\Box_{x_m} + m^2)(\Box_{y_1} + m^2) \dots (\Box_{y_m} + m^2) \cdot$$

$$\cdot \underbrace{\langle \Omega | T\{\phi(x_1) \dots \phi(x_n)\phi(y_1) \dots \phi(y_m)\} | \Omega \rangle}_{G(x_1 \dots y_m)} \qquad (7.12)$$

$|\Omega\rangle$: interacting vacuum state

LSZ reduction formula

Our AIM is to compute the element of the S matrix. I recall that $\langle p_1 \dots \text{out} | k_1 \dots \text{in} \rangle$ in Heisenberg picture is equivalent to the element of the S matrix in schrodyer picture

$$\underbrace{\langle \vec{p}_1 \dots \vec{p}_n | S | \vec{k}_1 \dots \vec{k}_m \rangle}_{\text{Schrodyer}} = \underbrace{\langle \vec{p}_1 \dots \vec{p}_n \text{ out} | \vec{k}_1 \dots \vec{k}_m \text{ in} \rangle}_{\text{Heisenberg}}$$

Recalling $S = \mathbb{1} + iT$

Recalling that we ASSUMED $\vec{p}_i \neq \vec{k}_j \;\; \forall i,j$ initial and final states completely different

We conclude that under such assumption we are computing only iT elements cause $\mathbb{1}$ term would lead to same initial and final state or at least partially same

$$\underbrace{\langle \vec{p}_1 \dots \vec{p}_n | i\hat{T} | \vec{k}_1 \dots \vec{k}_m \rangle}_{\text{Schrodyer}} = \underbrace{\langle \vec{p}_1 \dots \vec{p}_n \text{ out} | \vec{k}_1 \dots \vec{k}_m \text{ in} \rangle}_{\text{Heisenberg}} \qquad \text{for } \vec{p}_i \neq \vec{k}_j \;\; \forall i,j$$

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⚠ questo passaggio non è proprio corretto --- è vero solo se considero $\int dx (\Box + m^2) G(x)$

$= \int dx (-k^2 + m^2) G(x) e^{-ikx}$, dentro l'integrale in $\int dx\, e^{-ikx}$ ... altrimenti non è vero!

GREEN FUNCTION

We define/call the $\langle \Omega | \dots | \Omega \rangle$ a GREEN FUNCTION of x_i and y_j

$$G(x_1 \dots y_m) := \langle \Omega | T\{\phi(x_1) \dots \phi(y_m)\} | \Omega \rangle$$

the reason why G is a green function will be clear later.

we can always expand a function of coordinates x into a Fourier transform of momentum k

$$G(x) = \int \frac{d^4k}{(2\pi)^4} \cdot e^{-ik\cdot x} \tilde{G}(k) \qquad \tilde{G}(k) = \int d^4x\, e^{ik\cdot x} G(x)$$

considering more $x_1 \dots x_n$, $y_1 \dots y_m \Longrightarrow k_1 \dots k_n$, $p_1 \dots p_m$ the Fourier transform is

$$G(x_1 \dots x_n) \dots = \int \prod_{i=1}^{n} \frac{d^4k_i}{(2\pi)^4} e^{-i\sum k_i x_i} \tilde{G}(k_1 \dots k_n)$$

acting with a KG operator $(\Box_{x_j} + m^2)$ on $G(x_1 \dots x_n)$ as in eq 7.12 we obtain, acting on the Fourier transform of G:

$$(\Box_{x_j} + m^2) G(x_1 \dots x_n) = -\int \prod_{i=1}^{n} \frac{d^4k_i}{(2\pi)^4} e^{-i\sum k_i x_i} (k_j^2 - m^2) \tilde{G}(k_1 \dots k_n) \overset{⚠}{=}$$

$$\overset{⚠}{=} -(k_j^2 - m^2) G(x_1 \dots x_n)$$

For y_j coordinates I can chose to transform with $-p_j$ momenta that the resulting expression with KG operator is the same

$$(\Box_{y_j} + m^2) G(y_1 \dots y_m) = -\int \prod_{j=1}^{m} \frac{d^4p}{(2\pi)^4} e^{i\sum p_j \cdot y_j} (p_j^2 - m^2) \tilde{G}(p_1 \dots p_m) =$$

$$\overset{⚠}{=} -(p_j^2 - m^2) G(y_1 \dots y_m)$$

(NOTE = I can bring $(p_j^2 - m^2)$ outside the integral cause $p_j^2 = p_j^\mu p_{j\mu}$ = constant now is a constant number)

$$\Box e^{-ik\cdot x} = (\partial\partial^\mu - \nabla^2) e^{-ik^0 t + i\vec{k}\cdot\vec{x}} = (-ik^0)^2 - (i\vec{k})^2 = -k^{0\,2} + \vec{k}^2 = -k^2$$

con $k^2 = k^\mu k_\mu$

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LSZ REDUCTION FORMULA

Hence acting with multiple KG operators on $G(x_1 \dots x_n\, y_1 \dots y_m)$ we have

$$\prod_r^n (\Box_{x_r} + m^2) \prod_s^m (\Box_{y_s} + m^2)\, G(x_1 \dots y_m) = \prod_r^n (k_r^2 - m^2) \prod_s^m (p_s^2 - m^2)\, G(x_1 \dots y_m)$$

Substituting in eq 7.12 we have =

$$(i\sqrt{z}^{-1})^{n+m} \int \prod_i^m d^4x_i\, e^{-i k_i \cdot x_i} \int \prod_s^m d^4y_j\, e^{i p_s y_s} \prod_r^m (-k_r^2 + m^2) \prod_s^n (-p_s^2 + m^2)\, G(x_1 \dots y_m) =$$

$$= \langle p_1 \dots p_n | i\hat{T} | k_1 \dots k_m \rangle$$

Hence:

$$\int \prod_{i=1}^m d^4x_i\, e^{-i x_i \cdot k_i} \int \prod_{j=1}^n d^4y_j\, e^{i p_j \cdot y_j} \langle \Omega | T\{\phi(x_1) \dots \phi(y_m)\} | \Omega \rangle =$$

$$= \prod_{r=1}^m \frac{(i\sqrt{z})}{(k_r^2 - m^2)} \prod_{s=1}^n (i\sqrt{z})\, (p_s^2 - m^2)\, \langle p_1 \dots p_n | i\hat{T} | k_1 \dots k_m \rangle \qquad (7.14)$$

LSZ REDUCTION FORMULA

① This formula connects a green function on the left side with the $\hat{T}$ part of S matrix on the right

② In the Right part we have same poles for $k_r^2 = m^2$ and $p_s^2 = m^2$ hence when momenta are on shell . Since right part has poles also green function on the left must contain same poles . We can evaluate the integral of the green function with the Residue theorem thanks to the fact that the G function has poles . Computing the left part , since its equal to the S matrix (Right part) we computed the S matrix

↓

Now still completely general non-perturbative

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Recipe to compute the S-matrix

(1) $S = \mathbb{1} + iT$

we want a recipe to compute $\hat{T}$

(2) we now from LSZ that $\langle \vec{p}_n \cdots \vec{p}_1 | iT | \vec{k}_1 \cdots \vec{k}_m \rangle = i\, T_{fi}$ matrix element is related to the green function on the vacuum $\langle \Omega | T\{ \cdots \} | \Omega \rangle$

(3) FIRST we EVALUATE $\langle \Omega | T\{ \phi(x_1) \cdots \phi(y_m) \} | \Omega \rangle$ (green function)

(4) Then we compute the Fourier-transform of such green function thanks to the residue theorem in the limit of sending the momenta on "mass shell" i.e. $k^2 \to m^2$ $p^2 \to m^2$, the poles $\longrightarrow$ left part of LSZ formula

$$\int \prod d^4x_i \; e^{-ik_i x_i} \int \prod d^4y_j \; e^{ip_j y_j} \langle \Omega | T\{ \cdots \} | \Omega \rangle$$

(5) Taking the limit of mass shell ($p^2 = m^2$ $k^2 = m^2$) terms with $\frac{1}{k^2 - m^2}$ & $\frac{1}{p^2 - m^2}$ cancel out on the right side of LSZ formula

$\rightarrow$ we evaluated the T matrix scattering amplitude!

So now we can do everything except that we don't know how $\langle \Omega | T\{ \phi(x_1) \cdots \phi(y_m) \} | \Omega \rangle$ is made $\longrightarrow$ now all the problem has shifted in computing the $\langle \Omega | \cdots | \Omega \rangle$ function (green function)

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7.3 PERTURBATIVE EXPANSION OF GREEN FUNCTION

I Recall that we assumed a PERTURBATIVE REGIME i.e. the interaction is a small perturbation from the free case.

AIM = to compute $\langle \Omega | T\{\phi(x_1) \dots \phi(y_m)\} | \Omega \rangle$

PROBLEM = $|\Omega\rangle$ and ϕ fields are the interacting vacuum and fields, so I cannot expand them in the Fock space, there is no Fock space for the interacting fields and vacuum

SOLUTION = we will correlate $\langle \Omega | T\{\phi \dots\} | \Omega \rangle$ to free $|0\rangle$ vacuum and free fields ϕ_I (in "interaction" picture)

INTERACTION PICTURE

To describe perturbations which depend on time we introduce a picture midway from Heisemberg and Schrodinger's = the interaction picture.

For a system with: $H = H_0 + H_{int}(t)$

H_0 ↓ Free H_{int} ↓ interacting

The interaction picture treats:

-) Operators as Heisemberg operators evolving under H_0 only
-) States evolving a Schrodinger as under H_{int} only

Recall that:

SCHRODINGER: A_S $|\psi(t)\rangle_S = e^{-iH(t-t_0)} |\psi(t_0)\rangle_S$

Heisemberg: $A_H(t) = e^{iH(t-t_0)} A(t_0)_S e^{-iH(t-t_0)}$ $|\psi\rangle_H = |\psi(t_0)\rangle_S$

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Interaction picture:

$$A_I = e^{iH_0(t-t_0)} A_S e^{-iH_0(t-t_0)}$$

l'operatore evolve nel tempo per via solo di H_0 Free (dove $A_S = A_I(t_0)$)

$$|\psi,t\rangle_I = e^{iH_0(t-t_0)} |\psi,t\rangle_S = e^{iH_0(t-t_0)} e^{-iH(t-t_0)} |\psi,t_0\rangle_S$$

$e^{iH_0(t-t_0)}$: come evolverebbe con H_0; $e^{-iH(t-t_0)}$: evoluzione completa

evoluzione solo di $H_{int}(t)$

lo stato evolve solo due fase per effetto dello $H_{int}(t)$

NB = H_0 e $H_{int}(t)$ non commutano!

Allora non è possibile $e^{iH_0(t-t_0)} e^{-iH(t-t_0)} \neq e^{-i(\overbrace{H-H_0}^{H_{int}})(t-t_0)}$

Noo!! Ciò si può fare solo se commutano!

$e^A e^B = e^{AB}$ sse $[A,B] = 0$

$e^A e^B \neq e^{AB}$ se $[AB] \neq 0$

Ampiezze = prossimi

$$\underset{S}{\langle \psi_t | A_S | \psi_t \rangle_S} = \underset{S}{\langle \psi_t |} e^{-iH_0(t)+iH_0(t)} A_S e^{-iH_0(t)+iH_0(t)} |\psi_t\rangle_S = \langle \psi_i | A_I | \psi_I \rangle$$

(gli esponenziali si annullano)

Le ampiezze di probabilità si conservano anche in regime di interazione.

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Interaction picture applied to fields

A field in the interaction picture is a FREE FIELD cause it evolves only with H_0 (a field is an operator)

$$\hat{\phi}_I(t,\vec{x}) = e^{iH_0(t-t_0)}\,\phi_I(t_0,\vec{x})\,e^{-iH_0(t-t_0)}$$

Since is a free field we can expand it in normal modes

$$\phi_I(t,\vec{x}) = \int \frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2\epsilon_p}}\left(a_p e^{-ip\cdot x} + a_p^\dagger e^{ip\cdot x}\right)$$

Whole (Heisemberg) time evolution

In the Heisemberg picture whole time evolution is destined to operators hence to fields

$$A_H(t) = e^{iH(t-t_0)}\,A(t_0)\,e^{-iH(t-t_0)} \equiv A_I(t_0)$$

for fields:

$$\phi(t,\vec{x}) = e^{iH(t-t_0)}\,\phi_I(t_0,\vec{x})\,e^{-iH(t-t_0)}$$

The equation above shows how we can link our interacting (Heisenb.) field ϕ to a free field ϕ_I (interaction picture). FOR A SPECIFIC TIME t_0 HEISEMBERG AND INTERACTION (and also Shrodyer) PICTURES COINCIDE = I define ϕ_I free field such that it coincides with our interacting field ϕ exactly for time t_0

$$\phi(t_0,\vec{x}) = \phi_I(t_0,\vec{x})$$

NB = ciò è sempre possibile! Fissato un tempo t_0 è sempre possibile espandere ϕ interacting in normal modes come un free field ... poi quando lo evolvo non sarà più in normal modes ma se rimango in t_0, istantaneamente è un free field

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so I can link explicitly ϕ to ϕ_I :

$$\phi(t,\vec{x}) = e^{iH(t-t_0)}\,\phi_I(t_0,x)\,e^{-iH(t-t_0)} =$$

$$= e^{iH(t-t_0)}\,e^{-iH_0(t-t_0)}\,e^{iH_0(t-t_0)}\,\phi_I(t_0,x)\,e^{-iH_0(\cdot)}\,e^{iH_0(\cdot)}\,e^{-iH(\cdot)}$$

$$= e^{iH(t-t_0)}\,e^{-iH_0(t-t_0)}\,\phi_I(t,\vec{x})\,e^{iH_0(t-t_0)}\,e^{-iH(t-t_0)}$$

Hence

$$\phi(t,\vec{x}) = U^{\dagger}_{int}(t,t_0)\,\phi_I(t,x)\,U_{int}(t,t_0) \qquad (7.15)$$

dove $U_{int}(t,t_0) = e^{iH_0(t-t_0)}\,e^{-iH(t-t_0)}$

l'eq 7.15 mi dice che l'evoluzione totale temporale (Heisenberg) del campo $\phi(t)$ può essere ottenuto dal $\phi_I(t)$ evolvendolo con un op di evoluzione temporale che dipende solo da Hinterazione

infatti $\phi_I(t)$ è già calcolato nello stato dell con H_0 free, per ripristinare l'evoluzione totale alla Heisenberg devo reinserire solo Hint.

ATTENZIONE = siccome H_0 e H_{int} non commutano, sebbene è vero che alla fine dalla fisica $\exp(iH_0-)\exp(-iH-)$ sia una evoluzione che dip solo da H_{int} perché il termine di $+iH_0$ cancella l'evoluzione free destra $-iH$, TUTTAVIA H_0 e H_{int} NON COMMUTANO perciò non è immediato da vedere come tale U_{int} sia fatto !!

$$U_{int}(t,t_0) = e^{iH_0(\cdot)}\,e^{-iH} \neq e^{-i(H-H_0)(\cdot)} = e^{-iH_{int}(\cdot)} \quad !!$$

↓ Sarebbero solo se commutassero

TIME EVOLUTION OPERATOR $U_{int}(t,t_0)$

Il nostro op di evoluzione temporale U_{int} dip solo da H_{int} allo stesso modo di un po' particolare. Di fatto vedremo che è un operatore in cui evoluzione è data da H_{int} fatta evolvere con H_0.

Ora vogliamo studiare come U_{int} è fatto esplicitamente per potere dimostrare che valgono le seguenti proprietà (che ora non sono scontate!):

$$U_{int}^{+}(t,t_0) = U_{int}(t_0,t) \qquad (7.21)$$

$$U(t_2,t_0)U(t_1,t_0) = U(t_2,t_0) \qquad (7.21.b)$$

Eq di Schrodinger per U_{int}

$$i\frac{\partial U_{int}}{\partial t} = i\left(e^{iH_0(t-t_0)} \; iH_0 \; e^{-iH(t-t_0)}\right) + i\left(e^{iH_0(t-t_0)} \; (-i)H \; e^{-iH(t-t_0)}\right) =$$

$$= e^{iH_0(t-t_0)} \; H_{int} \; e^{-iH_0(t-t_0)} \; e^{iH_0(t-t_0)} \; e^{-iH(t-t_0)}$$

($e^{-iH_0(t-t_0)} \; e^{iH_0(t-t_0)} \; e^{-iH(t-t_0)}$ = U_{int})

$$\Downarrow$$

$$i\frac{\partial U_{int}}{\partial t} = H_I(t)\, U_{int}(t,t_0) \qquad 7.17$$

$$H_I(t) = e^{iH_0(t-t_0)} \; H_{int}(t) \; e^{-iH_0(t-t_0)} \qquad 7.18$$

$$\downarrow$$

Hamiltoniano H_{int} evoluto con lo H_0 free

Rispetto alla normale evoluzione di Schrodinger per gli operatori temporali cioè $i\frac{\partial U}{\partial t} = H(t)\,U$ con H_{TOT}
ora la mia U_{int} evolve con H_I che è lo H_{int} calcolato con H_0.

La soluzione dell'eq 7.17 differential equation è la SERIE DI DYSON (vedi quad 3 modulo 1 fisica quantistica)

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Solution = DYSON SERIES

$$U_I(t,t_0) = T\left\{ e^{-i\int_{t_0}^{t} dt' H_I(t')} \right\} \qquad (7.20)$$

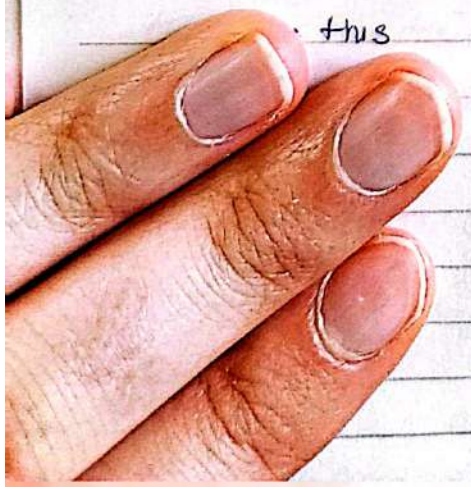
↓ prodotto cronologico

dove

$$T\left\{ e^{-i\int_{t_0}^{t} dt' H_I(t')} \right\} = T\left\{ \sum_{n=0}^{\infty} (-i)^n \frac{\left(\int d(H_1(t'))^n}{n!} \right\} =$$

$$= \sum_{n=0}^{\infty} \frac{1}{n!} T\left\{ \int \cdots \int \cdots \int \right\} = \sum_{n=0}^{\infty} (-i)^n \int_{t_0}^{t} dt_n \int_{t_0}^{t_n} dt_2 \int_{t_0}^{t_2} dt_3 \cdots \int_{t_0}^{t_{n-1}} dt_n \, H_1(t_n) \cdots H_1(t_n)$$

↓ n integrali



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Demonstration of 7.21

The proof of $U^+(t,t_0) = U(t_0,t)$ is direct from the definition of the Dyson serie.

Demonstration of 7.21 b – composition

$$i\frac{\partial}{\partial t}\left[U(t_2,t_1),U(t_1,t_0)\right] = i\frac{\partial}{\partial t}\left(U(t_2,t_1)U(t_1,t_0) - U(t_1,t_0)U(t_2,t_1)\right) =$$

$$= i\frac{\partial}{\partial t}U_{21}U_{10} - i\frac{\partial}{\partial t}U_{10}U_{21} \overset{7.17}{=} H_I\left[U(t_2,t_1),U(t_1,t_0)\right]$$

↓ ↓

each term satisfies same eq. differential

7.17 Hence

the initial condition $t_2 = t_1$ tells $U(t_1,t_1)U(t_1,t_0) = U(t_1,t_0)$

" $t_0 = t_1$ " $U(t_2,t_1)U(t_1,t_1) = U(t_2,t_1)$ $\quad (=\mathbb{1})$

Hence

$$U(t_2,t_1)U(t_1,t_0) = U(t_2,t_0) \qquad \text{QED}$$

1) Relating Green function to free fields ϕ_I

- Consider an already time ordered green function:

$\langle\Omega|\phi(x_1)\phi(x_2)\cdots\phi(x_n)|\Omega\rangle$ with $t_1>t_2>t_3\cdots>t_n$

- use eq 7.15 for every $\phi(x_j) = U^+(t_j,t_0)\,\phi_I(x_j)\,U(t_j,t_0)$ so:

$$\langle\Omega|\,U^+(t_1,t_0)\phi_I(x_1)U(t_1,t_0)U^+(t_2,t_0)\phi_I(x_2)U(t_2,t_0)\cdots U^+(t_n,t_0)\phi_I(x_n)U(t_n,t_0)\,|\Omega\rangle$$

- us 7.21 and 7.21 b to se each intermediate $U(t_i,t_0)U^+(t_{i+1},t_0) = U(t_i,t_0)U(t_0,t_{i+1}) = U(t_i,t_{i+1})$

(7.21 ↓ ; 7.21.b ↓)

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$$\langle \Omega | U^+(t_1,t_0)\,\phi_I(x_1)\,U(t_1,t_2)\,\phi_I(x_2)\,U(t_2,t_3)\cdots U(t_{n-1},t_n)\,\phi_I(x_n)\,U(t_n,t_0)\,|\Omega\rangle$$

- Now consider a large time t such that $t \gg t_1 > t_2 \cdots t_n \gg -t$

we can write:

$$U^+(t_1,t_0) = U(t_0,t_1) = U(t_0,t)\,U(t,t_1) = U^+(t,t_0)\,U(t,t_1)$$

$$U(t_n,t_0) = U(t_n,-t)\,U(-t,t_0)$$

so that in the green function

$$\langle \Omega | U^+(t,t_0)\,U(t,t_1)\,\phi_I(x_1)\cdots\phi(x_n)\,U(t_n,-t)\,U(-t,t_0)\,|\Omega\rangle$$

ϕ and U don't commute but I can switch because I'm inside the Time ordering $T\{\cdots\}$

this part is already time ordered

$$\Big\|$$

$$\langle \Omega | U^+(t,t_0)\,T\{U(t,t_1)\,\phi_I(x_1)\cdots\phi(x_n)\,U(t_n,-t)\}\,U(-t,t_0)\,|\Omega\rangle$$

$$\Big\|$$

$$T\{\phi_I(x_1)\cdots\phi_I(x_n)\;\underbrace{U(t,t_1)\,U(t_1,t_2)\cdots U(t_{n-1},t_n)\,U(t_n,-t)}_{U(t,-t)}\}$$

$$\Big\|$$

$$\langle \Omega | U^+(t,t_0)\,T\{\phi_I(x_1)\cdots\phi_I(x_n)\,U(t,-t)\}\,U(-t,t_0)\,|\Omega\rangle$$

Hence

interacting Fields ↑ — Free Fields ↑

$$\langle \Omega | \phi(x_1)\cdots\phi(x_n)|\Omega\rangle = \langle \Omega | U^+(t,t_0)\,T\{\phi_I(x_1)\cdots\phi_I(x_n)\,U(t,-t)\}\,U(-t,t_0)\,|\Omega\rangle \quad \text{(Free)}$$

with: $$U(t,-t) = T\left\{\exp\left(-i\int_{-t}^{t} dt'\,H_I(t')\right)\right\}$$

We have connected ϕ to ϕ_I free fields. Now we want to connect $|\Omega\rangle$ to $|0\rangle$.

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2) Relating $|\Omega\rangle$ to Free $|0\rangle$

Spoiler result: $U^+(t_0,-t)|\Omega\rangle = \lim_{t\to\infty(1-i\alpha)} \left(e^{-iE_0(t+t_0)}\langle\Omega|0\rangle\right)^{-1}|0\rangle$

- To prove that start by consider the free vacuum $|0\rangle$ and Act the total evolution given by H on it → let's compute the evolution of free vacuum in the interacting system: (suppose $t_0 = 0$ per facilità)

$$e^{-iHt}|0\rangle = \sum_n e^{-iE_n t}|n\rangle\langle n|0\rangle$$

↓ autostati di H

- Tra gli autostati di H quello con $n=0$ a energia minima è $|\Omega\rangle$ il vuoto interacting hence

$$e^{-iHt}|0\rangle = e^{-iE_0 t}\langle\Omega|0\rangle|\Omega\rangle + \sum_{n\neq\Omega} e^{-iE_n t}\langle n|0\rangle|n\rangle$$

where I can suppose $\langle\Omega|0\rangle \neq 0$ because we supposed PERTURBATIVE REGIME hence the change is small. If the vacuum from virtual free field $|0\rangle$ is totally different from $|\Omega\rangle$ the interaction would not be small but Big!

FOCUS = ZERO POINT ENERGY $E_0 = \langle\Omega|H|\Omega\rangle$

In the interacting system the Energy of the vacuum $|\Omega\rangle$ is not zero!

$E_0 = \langle\Omega|H|\Omega\rangle \quad (\neq 0)$

While instead in the free theory H_0 on $|0\rangle$ has zero energy

$0 = \langle 0|H_0|0\rangle$

⇒ possiamo vedere tale energia come un'energia POTENZIALE che sta per trasformarsi in particella e antiparticella ⇒ il vuoto ha un'energia solo quando in realtà non è vuoto... l'energia dell'interazione riempie il vuoto, è una massa in potenza.

TRUE EMPTY VACUUM $|0\rangle$ — $|\Omega\rangle$ Energy pure → ●↔● giving particles

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- now take the limit $t \to \infty(1-i\alpha)$ where α consider a phase $-i\alpha$ in order to send imaginary exponential to zero.

$$e^{-iE_n t} \xrightarrow[t\to\infty(1-i\alpha)]{} e^{-iE_n(1-i\alpha)\infty} = e^{-iE_n\infty} e^{-E_n\alpha\infty} \to 0$$

$e^{-iE_n\infty}$: oscillating; $e^{-E_n\alpha\infty}$: real term going to zero

Since E_0 is the smallest, only $|\Omega\rangle$ remains

$$e^{-iHt}|0\rangle \xrightarrow[\lim t\to\infty(1+i\alpha)]{} e^{-iE_0 t}\langle\Omega|0\rangle|\Omega\rangle$$

Hence

$$|\Omega\rangle = \lim_{t\to\infty(1-i\alpha)} \frac{1}{(\langle\Omega|0\rangle e^{-iE_0 t})} e^{-iHt}|0\rangle$$

- Now restore t_0-shifting $t \to into \to t+t_0$ (since $t-(-t_0)$).

$$|\Omega\rangle = \lim_{t\to\infty(1-i\alpha)} \frac{1}{(\langle\Omega|0\rangle e^{-iE_0(t+t_0)})} e^{-iH(t+t_0)}|0\rangle$$

- since $e^{iH_0(t+t_0)}|0\rangle = e^{i\phi}|0\rangle = |0\rangle$ since $|0\rangle$ is eigenstate of H_0 with zero ϕ eigenvalue ($e^{i\phi}=1$) we can add:

$$|\Omega\rangle = \lim_{t\to\infty(1-i\alpha)} \frac{1}{(\cdot)} e^{-iH(t+t_0)} e^{iH_0(t+t_0)}|0\rangle$$

$$U(t_0,-t)$$

$$|\Omega\rangle = \lim_{t\to\infty(1-i\alpha)} \frac{1}{(\cdot)} U(t_0,-t)|0\rangle$$

NB = intuitively should be =

$$U(t_0,-t) = e^{iH_0(t+t_0)} e^{-iH(t+t_0)} \neq e^{-iH(\cdot)} e^{iH_0 t}$$

è generale non è detto che si possano scambiare gli esponenti. Tuttavia nel nostro caso so che $U^\dagger(t,t_0) = U(t_0,t)$ cioè che

$$U^\dagger(t,t_0) = \left(e^{iH_0(t-t_0)} e^{-iH(t-t_0)}\right)^\dagger = e^{-iH(t_0-t)} e^{iH_0(t_0-t)}$$

$$U(t_0,t) = e^{iH_0(t_0-t)} e^{-iH(t_0-t)}$$

!) ovvero è possibile scambiare gli esponenti!

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c) now apply $U^\dagger(t_0,-t)$ to $|R\rangle$:

$$U^\dagger(t_0,-t)|R\rangle = U(-t,t_0)|R\rangle = \lim_{t\to\infty(1-i\alpha)} \frac{1}{\langle\Omega|0\rangle} U(-t,t_0)U(t_0,-t)|0\rangle$$

we get

$$U^\dagger(-t,t_0)|R\rangle = \lim_{t\to(1-i\alpha)\infty} \left(\frac{1}{(e^{-i\epsilon_0(t+t_0)}\langle\Omega|0\rangle)} |0\rangle \right) \quad *$$

equivalently, taking the adjoint of previous equation and switching $t\to -t$, I have:

$$\langle R|U^\dagger(t,t_0) = \lim_{t\to(1-i\alpha)\infty} \left(\langle 0| \frac{1}{(e^{-i\epsilon_0(t-t_0)}\langle 0|\Omega\rangle)} \right) \quad *$$

3) Relating green function to free $|0\rangle$

a) substituting the two relation above in the green function 7.22

$$\langle\Omega|\phi(x_1)\dots\phi(x_n)|\Omega\rangle = \langle R|U^\dagger(t,t_0)\,T\{\dots\}\,U(-t,t_0)|R\rangle = *$$

$$= \boxed{\lim_{t\to\infty(1-i\alpha)} \frac{1}{|\langle\Omega|0\rangle|^2 e^{-2i\epsilon_0 t}}} \langle 0|T\{\dots\}|0\rangle \qquad (7.23)$$

$$=: A$$

b) note that the limit is applied only to the $e^{-i\epsilon_0(t+t_0)}$ because no other terms depend on t, we want to calculate such A term. Note that

$$1 = \langle R|R\rangle = \langle\Omega|U^\dagger(t,t_0)\,U(t,t_0)\,U^\dagger(-t,t_0)\,U(-t,t_0)|\Omega\rangle = *$$

$$= A\,\langle 0|U(t,t_0)U^\dagger(-t,t_0)|0\rangle$$

$$\to A = \frac{1}{\langle 0|U(t,t_0)U^\dagger(-t,t_0)|0\rangle} \cdot \frac{1}{\langle 0|U(t,t_0)U(t_0,-t)|0\rangle} \cdot \frac{1}{\langle 0|U(t,-t)|0\rangle}$$

(a the limit $t\to\infty$)

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hence:

$$A = \frac{1}{\langle 0|\, T\{ \exp(-i\int_{-\infty}^{+\infty} dt' H_I(t')) \}\, |0\rangle}$$

•) so, Restoring also the Time ordering in the green function we finally obtain:

$$\langle \Omega|\, T\{\phi(x_1)\cdots\phi(x_n)\}\, |\Omega\rangle = \frac{\langle 0|\, T\{\phi_I(x_1)\cdots\phi_I(x_n)\, e^{-i\int_{-\infty}^{\infty} dt' H(t')}\}\, |0\rangle}{\langle 0|\, T\{\exp -i\int_{-\infty}^{\infty} dt' H_I(t')\}\, |0\rangle} \quad (7.24)$$

(interacting vacuum; interacting field; free fields; free vacuum)

we have related the green function to FREE FIELDS AND FREE VACUUM. The price we had to pay is to introduce a boson source in the evolution time $\exp(-i\int dt' H_I(t'))$

However I KNOW HOW TO CALCULATE $H_I(t')$ and its exponential!

$$\exp\left(-i\int_{-\infty}^{\infty} dt' H_I(t')\right) = \exp\left(-i\int_{-\infty}^{\infty} d^4x' \mathcal{H}_I(x')\right)$$

where $\mathcal{H}_I = e^{iH_0(t-t_0)}\, \mathcal{H}_{int}\, e^{-iH_0(t-t_0)}$

Now I can compute my Smatrix with the Recipe given before!

However

NO ONE USE THIS METHOD! IS TOO LONG AND PAINFUL!
NOWADAYS WE USE FEYNMAN DIAGRAMS.

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8

FEYNMAN DIAGRAMS

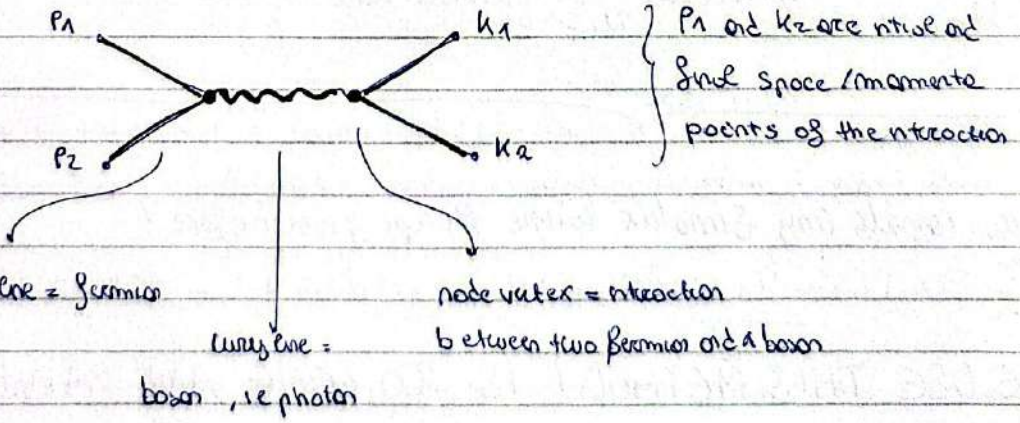
Feynman realized there was a much quicker way to compute the time ordered correlator (Green function) $\langle \Omega | T\{\phi(x_1)\dots\}|\Omega\rangle$ needed in the LSZ formula.

Feynman diagrams are a set of rules which allows to compute a SPECIFIC ORDER of the perturbative expansion (approximation) of the green function of the S-matrix (= scattering amplitude)

Feynman diagrams model the particle interactions with a pictorial representation that makes use of a bunch of:

- PROPAGATORS = lines which represent the particle propagating
- INTERACTION VERTEX = any vertex between lines/particle ns an interaction.

EX:



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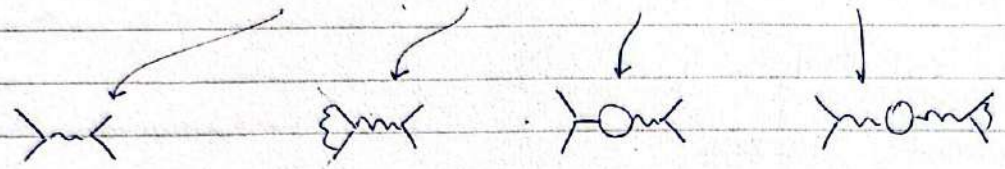
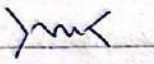
Each DIAGRAM corresponds to one term of the perturbative expansion

$$\langle \Omega | T\{\cdots\} | \Omega \rangle = \#_0 \lambda_0 + \#_1 \lambda_1 + \#_2 \lambda_2 + \#_3 \lambda_3 + \cdots$$



for each diagram:

 → sum → $\#_1$ number

ASSUMPTION: PERTURBATIVE REGIME

Feynman diagrams work only in the assumption of SMALL interactions i.e. perturbations, otherwise F. diagrams make no sense.

The perturbative expansion of $\langle \Omega | T\{\cdots\} | \Omega \rangle$ can be expanded in the POSITION space d^4x $(x^4 = x^0 x^1 x^2 x^3)$ or in the dual MOMENTA SPACE d^4p $(p^0 p^1 p^2 p^3)$ → usually momenta space is preferred, since also we labeled our initial and final state of the S matrix already with momenta (→ $\langle \vec{k}_1 \cdots \vec{k}_n | S | \vec{p}_1 \cdots \vec{p}_m \rangle$)

NOTATION

FROM NOW ON INITIAL STATES ARE $\vec{p}_i$ AND FINAL $\vec{k}_j$

$$\langle \vec{k}_1 \cdots \vec{k}_n | iT | \vec{p}_1 \cdots \vec{p}_m \rangle$$

$$\text{LSZ} = \prod_{j=1}^{n} \left(\frac{i\sqrt{z}}{k_j^2 - m_j^2} \right) \prod_{i=1}^{m} \left(\frac{i\sqrt{z}}{(p_i^2 - m_i^2)} \right) \langle \vec{k}_1 \cdots | iT | \vec{p}_1 \cdots \rangle = \int \prod_j d^4y_j \, e^{i k_j y_j} \int \prod_i d^4x_i \, e^{-i p_i x_i} \langle \Omega | \cdots | \Omega \rangle$$

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8.1 FEYNMAN PROPAGATOR (between two points) FOR REAL SCALAR FIELDS

The feynman propagator between two POINTS of spacetime is defined as:

$$D(x_1 - x_2) = \langle 0 | T\{\phi_I(x_1)\phi_I(x_2)\} | 0 \rangle$$

(free vacuum; free fields)

It expresses the AMPLITUDE for a particle to propagate from x_1^μ to x_2^μ without any interaction (a meno di correzioni di ordini superiori → vedi dopo) (infatti è il 1° ordine dell'espansione di $\langle \Omega | T\{\phi_1(x_1)\phi_2(x_2)\} | \Omega \rangle$ che è l'ampiezza di propagazione)

Rewriting $T\{\phi_1(x_1)\phi(x_2)\}$

- I define:

$$\phi_I = \phi_I^+ + \phi_I^-$$

annihilator $\phi_I^+ |0\rangle = 0$ — creator $\langle 0 | \phi_I^- = 0$

$$\phi_1^+ = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\epsilon_p}} a_p e^{-ip\cdot x} \qquad \phi_1^- = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\epsilon_p}} a_p^\dagger e^{ip\cdot x}$$

- assume $x_1^0 > x_2^0$

$$T\{\phi(x_1)\phi(x_2)\} = \phi(x_1)\phi(x_2) = \phi^+(x_1)\phi^+(x_2) + \phi^+(1)\phi^-(2) + \phi^-(1)\phi^+(2) + \phi^-(1)\phi^-(2) =$$

$$= \phi^+(1)\phi^+(2) + \phi^-(2)\phi^+(1) + [\phi^+(1), \phi^-(2)] + \phi^-(1)\phi^+(2) + \phi^-(1)\phi^-(2) =$$

except for $[\cdots]$ all other terms are NORMAL ORDERED (ie $a^\dagger a$, first creator then annihilator, here $\phi^-\phi^+$)

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$$x_1^0 > x_2^0 \rightarrow \quad :\phi_I(x_1)\phi_I(x_2): + [\phi_I^+(x_1), \phi_I^-(x_2)] = T\{\phi_I(x_1), \phi_I(x_2)\} \qquad 8.1(a)$$

Similarly

$$x_2^0 > x_1^0 \rightarrow \quad :\phi_I(x_1)\phi_I(x_2): + [\phi^+(x_2), \phi^-(x_1)] = T\{\phi_I(x_1)\phi_I(x_2)\} \qquad 8.1(b)$$

General for every time

$$T\{\phi_I(x_1)\phi_I(x_2)\} = :\phi_I(x_1)\phi_I(x_2): + \theta(x_1^0 - x_2^0)[\phi_I^+(x_1), \phi_I^-(x_2)] + \theta(x_2^0 - x_1^0)[\phi_I^+(x_2), \phi_I^-(x_1)] =$$

$$= :\phi_I(x_1)\phi_I(x_2): + D(x_1 - x_2) \qquad (8.2)$$

$D(x_1 - x_2)$ is what we will discover being the GREEN FONCTION. For now we note that $D(x-y)$ is no longer an operator but just a complex NUMBER (is a commutator between creator and annihilator operators $a, a^\dagger$)

When I apply the vacuum $\langle 0| T\{\ \}|0\rangle$ the $:\phi\phi:$ part gives zero cause normal order put $a^\dagger$ close to bra $\langle 0|$ and annihilator $a|0\rangle = \phi$ hence it annihilates. Just the D number remains:

$$\langle 0| T\{\phi_I(x_1)\phi_I(x_2)\}|0\rangle = D(x_1 - x_2) = \theta(x_1^0 - x_2^0)[\phi_I^+(x_1), \phi_I^-(x_2)] + \theta(x_2^0 - x_1^0)[\phi_I^+(x_2), \phi_I^-(x_1)]$$

- Now I want to express $D(x_1 - x_2)$ in Fourier transform

$$D(x_1 - x_2) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} \left(\theta(x_1^0 - x_2^0)\, e^{-ip\cdot(x_1 - x_2)} + \theta(x_2^0 - x_1^0)\, e^{ip\cdot(x_1 - x_2)} \right) \qquad 8.4$$

We will demonstrate that the Fourier transform of $D(x_1 - x_2)$ is:

$$D(x_1 - x_2) = \int \frac{d^4p}{(2\pi)^4} \tilde{D}(p)\, e^{-ip\cdot(x_1 - x_2)} \qquad \text{for scalar fields} \qquad 8.5$$

with $$\tilde{D}(p) = \frac{i}{p^2 - m^2 + i\varepsilon} \qquad \text{for scalar fields}$$

$$\tilde{D}(p) = \int d^4y\, D(y)\, e^{ip\cdot y}$$

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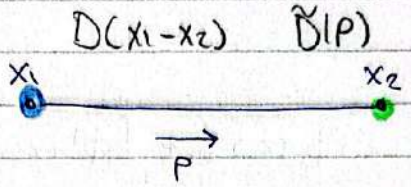
For a couple of point $x_1 - x_2$, a line, we associate one value of momentum p. This reflects how a line is a particle, existing between to spatial time point with a specific personal p



DIMOSTRAZIONE

$$\int \frac{d^4p}{(2\pi)^4} e^{-ip\cdot(x_1-x_2)} \frac{i}{(p^2-m^2+i\varepsilon)} = \int \frac{d^3p}{(2\pi)^3} e^{i\vec{p}\cdot(\vec{x}_1-\vec{x}_2)} \int \frac{dp_0}{2\pi} e^{-ip_0(x_0-x_{20})} \frac{i}{(p^2-m^2+i\varepsilon)} =$$

- Since $E_p^2 = |\vec{p}|^2 + m^2$ allora $p^2 - m^2 = p_0^2 - |\vec{p}|^2 - m^2 = p_0^2 - E_p^2$

$$= \int \frac{d^3p}{(2\pi)^3} e^{i\vec{p}\cdot(\vec{x}_1-\vec{x}_2)} \left[\int \frac{dp_0}{2\pi} e^{-ip_0(x_1^0-x_2^0)} \frac{i}{(p_0)^2 - E_p^2 + i\varepsilon} \right] = (*)$$

- I compute the integral in dp_0 with residue theorem expanding p_0 to be complex

$$p_0 = \mathrm{Re}p_0 + i\,\mathrm{Im}p_0$$

since:

$$e^{-ip_0(x_1^0-x_2^0)} = e^{-i\mathrm{Re}p_0(x_1^0-x_2^0)} \; e^{\mathrm{Im}p_0(x_1^0-x_2^0)}$$

$e^{-i\mathrm{Re}p_0(x_1^0-x_2^0)}$: oscillating as $p_0 \to \infty$

$e^{\mathrm{Im}p_0(x_1^0-x_2^0)}$: as $p_0 \to \infty$ we have want it to go to zero so we want to have a negative exponent

for $\mathrm{Im}p_0 \to \infty$ $\quad x_2^0 > x_1^0$
UPPER HALF PLANE

for $\mathrm{Im}p_0 \to -\infty$ $\quad x_1^0 > x_2^0$
LOWER HALF PLANE

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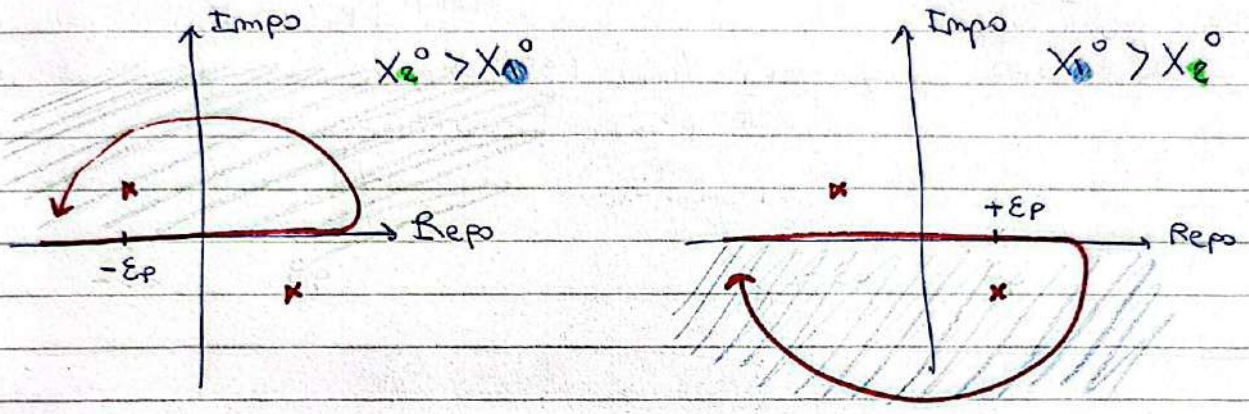
Hence if $x_2^0 > x_1^0$ ⟶ close the integral path in UPPER HALF PLANE

if $x_1^0 > x_2^0$ ⟶ close the path in LOWER HALF PLANE



- The integrand has poles in $p_0^2 - E_p^2 + i\varepsilon = 0$ so in

$p_0 = +E_p - i\varepsilon$ LOWER PLANE

$p_0 = -E_p + i\varepsilon$ UPPER PLANE

- Compute the integral with residue theorem for 1 degree pole order:

$x_2^0 > x_1^0$ | $\theta(x_2^0 - x_1^0)$

$$\int = \int - \cancel{\int} \to \int = \int = 2\pi i \, \text{Res}\left(f(p_0), -\varepsilon_p\right) =$$

(pole in the upper half plane; $\cancel{\int} = 0$ goes to zero)

$$= 2\pi i \lim_{p^0 \to -\varepsilon_p} (p_0 + \varepsilon_p) f(p_0) =$$

$$= 2\pi i \, i \lim_{p^0 \to -\varepsilon_p} \frac{(\cancel{p^0 + \varepsilon_p}) \, e^{-i p_0 (x_1^0 - x_2^0)}}{2\pi \, (p_0 - \varepsilon_p)(\cancel{p_0 + \varepsilon_p})} =$$

$$= -\frac{e^{i\varepsilon_p (x_1^0 - x_2^0)}}{-2E_p} = \frac{e^{i\varepsilon_p (x_1^0 - x_2^0)}}{2E_p}$$

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$x_1^0 > x_2^0$ | $\theta(x_1^0 - x_2^0)$

$-2\pi i$ perché giro senso orario

polo in the lower half plane

$$\int = \int + \oint \rightarrow \int dp_0 = \int = -2\pi i \, \mathrm{Res}\left(f(p), \varepsilon_p\right) =$$

$= 0$ goes to zero as $p_0 \to \infty$

$$= -2\pi i \frac{i}{2\pi} \lim_{p^0 \to \varepsilon_p} (p^0 - \varepsilon_p) \frac{e^{-i p_0 (x_1^0 - x_2^0)}}{(p^0 - \varepsilon_p)(p^0 + \varepsilon_p)} =$$

$$= \frac{e^{-i \varepsilon_p (x_1^0 - x_2^0)}}{2\varepsilon_p}$$

Quindi

$$\textcircled{X} = \int \frac{d_p^3}{(2\pi)^3} \cdot e^{i \vec{p}\cdot(\vec{x}_1 - \vec{x}_2)} \left(\theta(x_1^0 - x_2^0) \frac{e^{i \varepsilon_p (x_1^0 - x_2^0)}}{2\varepsilon_p} + \theta(x_2^0 - x_1^0) \frac{e^{-i E_p (x_1^0 - x_2^0)}}{2\varepsilon_p} \right) =$$

Here changing $\vec{p} \to -\vec{p}$ tanto $\varepsilon_p = \varepsilon_{-p}$ e l'integrale non cambia nulla

in the spatial part dp^3 va da $-\infty$ a $+\infty$

$$= \int \frac{d_p^3}{(2\pi)^3} \frac{1}{2E_p} \left(\theta(x_2^0 - x_1^0) e^{i p (x_1 - x_2)} + \theta(x_1^0 - x_2^0) e^{-i p (x_1 - x_2)} \right)$$

$= D(x_1 - x_2)$ è esattamente l'eq 8.9 !! Voilà! CVD

$\Rightarrow$ Hence I demonstrated $D(x_1 - x_2) = \int \frac{d_p^4}{(2\pi)^4} \frac{i}{(p^2 - m^2 + i\varepsilon)} \left(e^{-ip(x_1 - x_2)}\right)$ CVD ■

PROOF OF GREEN FUNCTION BEHAVIOUR

$$D(x_1 - x_2) = \int \frac{d_p^4}{(2\pi)^4} \tilde{D}(p) \, e^{-ip(x_1 - x_2)} \qquad \tilde{D}(p) = \frac{i}{p^2 - m^2}$$

is a green function, precisely is the sum of ADVANCED green function (for $x_1^0 > x_2^0$) and RETARDED (for $x_2^0 > x_1^0$). How can we see it?

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The definition of a Green function is a function which gives a δ di Dirac under the action of some linear operator L

$$L\, G(x) \propto \delta(x)$$

L: linear operator; G: Green function

our function $D(x_1 - x_2)$ is a green function for $(\Box + m^2)$ operator! Infatti:

$$(\Box + m^2) D(x_1 - x_2) = \int \frac{d^4p}{(2\pi)^4} \frac{i}{(p^2 - m^2)} (\Box + m^2)\, e^{-ip(x_1 - x_2)} =$$

$$= \int \frac{d^4p}{(2\pi)^4} \frac{i}{(p^2 - m^2)} - (p^2 - m^2)\, e^{-ip(x_1 - x_2)} =$$

$$= -i \int \frac{d^4p}{(2\pi)^4} e^{-ip(x_1 - x_2)} = -i\, \delta(x_1 - x_2) \quad \checkmark$$

$$\Rightarrow (\Box + m^2) D(x_1 - x_2) = -i\, \delta(x_1 - x_2)$$

8.2 WICK'S THEOREM (for Bosons in general) (for scalar fields)

Il teorema di Wick's allows to write the expectation value on the vacuum $|0\rangle$ of $T\{\phi_1(x_1) \dots \phi_1(x_n)\}$ (hence $\langle 0|T\{\dots\}|0\rangle$) as a combination of Feynman propagators $D(x_i - x_j)$.

We define a CONTRACTION of two fields ϕ_1 and ϕ_2 (NOTA: notation $\phi_i := \phi_i(x_i)$) as the Feynmann propagator between them:

$$\overbrace{\phi_1\, \phi_2} := D(x_1 - x_2) =: D_{12} \quad \leftarrow \text{it's a number!} \quad (8.9)$$

Contraction — Feynman propagator

PROPERTIES:

$$\overbrace{\phi_1 \phi_2} = \overbrace{\phi_2 \phi_1}$$

$$\overbrace{\phi_1 \phi_2}\, \overbrace{\phi_3 \phi_4} = \overbrace{\phi_3 \phi_4}\, \overbrace{\phi_1 \phi_2}$$

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Therefore from our definition of $T\{\phi_1\phi_2\}$ we have: from 8.2

$$T\{\phi_1\phi_2\} = :\phi_1\phi_2: + \overbrace{\phi_1\phi_2}$$

And since $\overbrace{\phi_1\phi_2}$ it's a number we can bring it inside the : :

$$T\{\phi_1\phi_2\} = :\phi_1\phi_2 + \overbrace{\phi_1\phi_2}:$$

WICK'S THEOREM

$$T\{\phi_1\phi_2\ldots\phi_n\} = :\phi_1\phi_2\ldots\phi_n: + :\{\text{sum of all possible normal ordery and contractions}\}:$$

EXAMPLE FOR 4 FIELDS

$$T\{\phi_1\phi_2\phi_3\phi_4\} = :\phi_1\phi_2\phi_3\phi_4: + \quad \rightarrow \text{pure normal ordery}$$

$$+ \overbrace{\phi_1\phi_2}:\phi_3\phi_4: + \overbrace{\phi_2\phi_3}:\phi_1\phi_4: + \overbrace{\phi_1\phi_3}:\phi_2\phi_4: + \overbrace{\phi_2\phi_4}:\phi_1\phi_3: + \overbrace{\phi_1\phi_4}:\phi_3\phi_2: + \overbrace{\phi_3\phi_4}:\phi_1\phi_2: + \quad \rightarrow \text{Mix of contractions and normal ord.}$$

$$+ \overbrace{\phi_1\phi_2}\overbrace{\phi_3\phi_4} + \overbrace{\phi_1\phi_3}\overbrace{\phi_2\phi_4} + \overbrace{\phi_1\phi_4}\overbrace{\phi_3\phi_2} \quad \rightarrow \text{pure contractions}$$

When we take the expection value on $|0\rangle$ vacuum all terms with : : give zero and only terms where all fields have been contracted survive

$$\langle 0|T\{\phi_1\phi_2\phi_3\phi_4\}|0\rangle = \overbrace{\phi_1\phi_2}\overbrace{\phi_3\phi_4} + \overbrace{\phi_1\phi_3}\overbrace{\phi_2\phi_4} + \overbrace{\phi_1\phi_4}\overbrace{\phi_3\phi_2} =$$

$$= D_{12}D_{34} + D_{13}D_{24} + D_{14}D_{32}$$

This is exactly a combination of Feynman propagators D_{ij} //

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PHYSICAL INTERPRETATION AND FEYNMAN DIAGRAMS

the previous equation has a vivid physical interpretation = we interpret

2 FIELDS POINTS PROPAGATOR

- $D_{12} = D(x_1 - x_2) = \langle 0|T\{\phi_1 \phi_2\}|0\rangle = \overbrace{\phi_1 \phi_2}$ is the AMPLITUDE for a particle to PROPAGATE from x_1^μ to x_2^μ without any interaction (ie = of zero order of the expansion)

⇓

I assign to each propagator D_{12} a DIAGRAM which is a simple line

$D(x_1 - x_2) \longrightarrow$

N POINTS PROPAGATOR

- $\langle 0|T\{\phi_1 \phi_2 \cdots \phi_n\}|0\rangle$ = all combinations of D_{ij} propagator from Wicks contractions, hence is the SUM of all combination of AMPLITUDE between 2 points

⇓

Each term in the sum is a Diagram:

$\langle 0|T\{\phi_1 \phi_2 \phi_3 \phi_4\}|0\rangle = D_{12} D_{34} + \Rightarrow$

(1)

$+ D_{31} D_{24} + \Rightarrow$

(2)

$+ D_{14} D_{23} \Rightarrow$

(3)

(1) ~~FIRST DIAGRAM~~: represent two particles, the first going from 1 to 2, the second from 2 to 4, without any interaction

(2) second diagram represent a particle from 1 to 3 and another from 2 to 4

(3) third diagram represent a particle from 1 to 4 and another from 2 to 3, with NO interaction

ALL THIS DIAGRAMS ARE AT ZERO ORDER, HENCE WITH NO INTERACTION (which will be pictured by a VERTEX ✕)

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È ovvio che per ora i diagrammi disegnati siano banali, solo propagatori, perché stò prendendendo $\langle 0|T\{\phi_1\phi_2\}|0\rangle$ che è l'ORDINE ZERO nell'espansione dell esponenziale dell'ampiezza di scattering completa, cioè:

$$\langle \Omega|T\{\phi(x_1)\cdots\phi(x_n)\}|\Omega\rangle = \langle 0|T\{\phi_1(x_1)\cdots\phi_1(x_n)\, e^{-i\int d^4\xi H_{int}}\}|0\rangle$$
$$\langle 0|T\{e^{-i\int d\xi H_I}\}|0\rangle$$

hamiltoniano di interazione, che dà i VERTICI nei diagrammi. All'ordine zero tale esponenziale è $\exp(\cdots) = 1 + (\cdots) + \cdots$ è proprio 1 perciò trovo che $\langle\Omega|T\{\cdots\}|\Omega\rangle$ all'ordine zero è proprio $\langle 0|T\{\cdots\}|0\rangle$ del teo di Wicks!

Per creare diagrammi più complessi che considerano anche le interazioni fra campi devono considerare ordine superiore nello sviluppo — ed è quello che facciamo nella pagina a fianco. Prima però vediamo la dimostrazione del teo di Wicks.

PROOF OF WICK'S THEOREM (con p.d.s)

We already proved the theorem to be right for 2 points $\langle 0|T\{\phi_1\phi_2\}|0\rangle$. Here we present a proof by INDUCTION. I will demonstrate that if the theorem is right for $n-1$ point then it has to be right also for n points. Since we start from 2 points being already true, I can then by induction affirm it is right for 3 then 4 then 5 ecc ecc

AIM = $T\{\phi_1\phi_2\cdots\phi_n\} = :\phi_1\phi_2\cdots\phi_n: + :\{\text{all contractions}\}:$ for n

IPOTESI = $T\{\phi_2\cdots\phi_n\} = :\phi_2\cdots\phi_n: + :\{\text{all contractions}\}:$ for $n-1$

NOTA INIZIALE

$$\overbrace{\phi_1\phi_2} = \langle 0|T\{\phi_1(x_1)\phi_1(x_2)\}|0\rangle = \begin{cases} [\phi_1^+(x_1), \phi_1^-(x_2)] & \text{if } x_1^0 > x_2^0 \\ [\phi_1^+(x_2), \phi_1^-(x_1)] & \text{if } x_2^0 > x_1^0 \end{cases}$$

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PROCEDIMENTO

- with no loss of generality I can assume $x_1^0 > x_2^0 > \dots x_n^0$ (if not I can always re-label the x_j) hence

$$T\{\phi_1 \phi_2 \cdots \phi_n\} = \phi_1 \phi_2 \cdots \phi_n = \phi_1 T\{\phi_2 \cdots \phi_n\} \overset{\text{IPOTESI WICK'S THEOREM FOR N-1 points}}{=}$$

$$= \phi_1 \left(:\phi_2 \cdots \phi_n: + :\{\text{all contractions never } \phi_1\}: \right) =$$

$$= (\phi_1^+ + \phi_1^-)\left(:\phi_2 \cdots \phi_n: + :\{\text{all never } \phi_1\}: \right) =$$

$$= \boxed{\phi_1^+ :\phi_2 \cdots \phi_n: + \phi_1^- :\phi_2 \cdots \phi_n: + (\phi_1^+ + \phi_1^-) :\{\text{never } \phi_1\}:}$$

$\square = \phi_1^- :\phi_2 \cdots \phi_n: = :\phi_1^- \phi_2 \cdots \phi_n:$ already normal ordered because ϕ^- is creator and ϕ^+ annihilator

- $\square = \phi_1^+ :\phi_2 \cdots \phi_n: = \phi_1^+ :\cdots: - :\cdots: \phi_1^+ + :\cdots: \phi_1^+ =$

$$= [\phi_1^+, :\phi_2 \cdots \phi_n:] + :\phi_2 \cdots \phi_n: \phi_1^+ =$$

now this term is normal ordered because ϕ^+ is annihilator goes to the end

$$= [\phi_1^+, :\phi_2 \cdots \phi_n:] + :\phi_1^+ \phi_2 \cdots \phi_n: =$$

$$= \boxed{[\phi_1^+, \phi_2^- \phi_3^- \cdots \phi_2^+ \phi_3^+ \cdots]} + :\phi_1^+ \phi_2 \cdots \phi_n: =$$

now use $[A, BCD\cdots] = [A,B]CD + B[A,C]D \cdots + BC[A,D]E \cdots + \cdots$

we obtain terms:

$[\phi_1^+, \phi_j^+] =$ but they are zero ϕ because the annihilators commute

$[\phi_1^+, \phi_j^-] = \phi_1 \phi_j$ since $x_1^0 > x_j^0 \quad \forall j \neq 1$

allora

$$= :\overbrace{\phi_1 \phi_2} \phi_3 \phi_4 \cdots \phi_n: + :\phi_2 \overbrace{\phi_1 \phi_3} \phi_4 \cdots: + :\phi_2 \phi_3 \overbrace{\phi_1 \phi_4} \cdots: \cdots + :\phi_1^+ \cdots \phi_n: =$$

$$= \boxed{:\phi_1^+ \phi_2 \cdots \phi_n: + :\{\text{all contractions with } \phi_1 \text{ or ONE OTHER FIELD}\}:}$$

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- so

$$\boxed{\quad} = \square + \square = :\phi_1 \overbrace{\phi_2 \cdots \phi_n}: + :\phi_1^{\dagger} \phi_2 \cdots \phi_n: + :\{\phi_s \text{ contracted with one other } \phi_j\}: =$$

- then

$$T\{\phi_1 \phi_2 \cdots \phi_n\} = \square + :\phi_1 \{\text{contractions without } \phi_1\}: =$$

$$= :\phi_1 \phi_2 \cdots \phi_n: + :\{\text{contraction } \phi_s \overbrace{\phi_j} \text{ in so far as contracted with } \phi_1 \text{ and with another } \phi_j\}: + :\phi_1 \{\text{all contractions without } \phi_1\}:$$

- Repeating this procedure for all ϕ_j and not only ϕ_1 would extract all contractions of ϕ_j with one other fields

- Then Repeating this procedure for a term inside $\{\text{single contraction } \phi_1 \overbrace{\phi_s}\}$ would give terms with more contractions. For example term $:\phi_2 \overbrace{\phi_s \phi_3} \phi_4:$ may give $:\overbrace{\phi_1 \phi_3} \overbrace{\phi_2 \phi_4}:$

$\Rightarrow$ This completes the proof

CVD

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8.3 FEYMAN DIAGRAMS BETWEEN TWO POINTS FOR ϕ^4-THEORY INTERACTION

Let's now compute $\langle \Omega | T\{\phi(x_1)\phi(x_2)\} | \Omega \rangle$ with Feyman Diagrams.
We expand $\langle \Omega | \dots | \Omega \rangle$ in the position space expanding the exponential.
Each term of the expansion correspond to a specific Diagram:

$$\langle \Omega | T\{\phi(x_1)\phi(x_2)\} | \Omega \rangle = \frac{\langle 0 | T\{\phi_I(x_1)\phi_I(x_2) e^{-i\int d^4z H_I}\} | 0 \rangle}{\{\langle 0 | T\{e^{-i\int d^4z H_I}\} | 0 \rangle\}}$$

Numerator / Denominator

- Consider ϕ^4-theory: $H_I = \frac{\lambda}{4!}\phi_I^4(z) = \frac{\lambda}{4!}\phi_I(z)\phi_I(z)\phi_I(z)\phi_I(z)$

NUMERATOR EXSPANSION

$$\langle 0 | T\{\phi_I(x_1)\phi_I(x_2) e^{-i\int d^4z H_I(z)}\} | 0 \rangle \xrightarrow{\text{expand the exp}} e^n = 1 + n + \frac{n^2}{2!} + \dots$$

$$= \langle 0 | T\{\phi_I(x_1)\phi_I(x_2)\} | 0 \rangle + \langle 0 | T\{\phi_I(x_1)\phi_I(x_2)(-i\int d^4z H_I)\} | 0 \rangle +$$

$$+ \langle 0 | T\{\phi_I(x_1)\phi_I(x_2)\frac{(-i\int d^4z H_I)^2}{2!} | 0 \rangle + \langle 0 | T\{\phi_I(x_1)\phi_I(x_2)\frac{(\ )^3}{3!}\} | 0 \rangle + \dots =$$

Zero order

It's exactly the Feyman propogator! It correspond to the particle propogating to x_1 to x_2 without any interaction!

$$\langle 0 | T\{\phi_I(x_1)\phi_I(x_2)\} | 0 \rangle = D(x_1 - x_2)$$

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1st order

$$= \langle 0|T\{\phi_I(x_1)\phi_I(x_2)\frac{(-i\lambda)}{4!}\int d^4z\,\phi_I(z)\phi_I(z)\phi_I(z)\phi_I(z)\}|0\rangle =$$

apply Wicks theorem for contractions . How many possibilities do I have?

1) contract $\phi(x_1)\phi(x_2)$ then I have 3 different way to contract the two di fhe $\phi(z)$ fieed internal

$\phi(1)\phi(2)\ \phi(z)\phi(z)\phi(z)\phi(z)$
$\phi(1)\phi(2)\ \phi(z)\phi(z)\phi(z)\phi(z)$
$\phi(1)\phi(2)\ \phi(z)\phi(z)\phi(z)\phi(z)$

} 3 equivalent contractions

2) contract x_1 to z and x_2 to z, contracting ext point with internal z points leaves 12 different pombilities

$\phi(x_1)\phi(z)\ \phi(y)\phi(z)\ \phi(z)\phi(z)$
$\phi(y)\phi(z)\ \phi(z)\phi(z)$
$\phi(y)\phi(z)\ \phi(z)\phi(z)$

} 3 equivalent $\times$ 4 choices of $\phi_0(z)$ either ϕ_b, ϕ_c, ϕ_d $= 12$

1) → corresponds to $3 \times D(x_1-x_2)\,D(z-z)\,D(z-z)$

2) → // $12 \times D(x_1-z)\,D(x_2-z)\,D(z-z)$

Hence

$$= \underbrace{\frac{3\,(-i\lambda)}{4!}\,D(x_1-x_2)\cdot\int d^4z\,D^2(z-z)}_{(1)} + \underbrace{12\,\frac{(-i\lambda)}{4!}\int d^4z\,D(x_1-z)\,D(x_2-z)\,D(z-z)}_{(2)}$$

(1): double loop on z

(2): single loop on z

it's possible to represent each term with a Diagram in the following way

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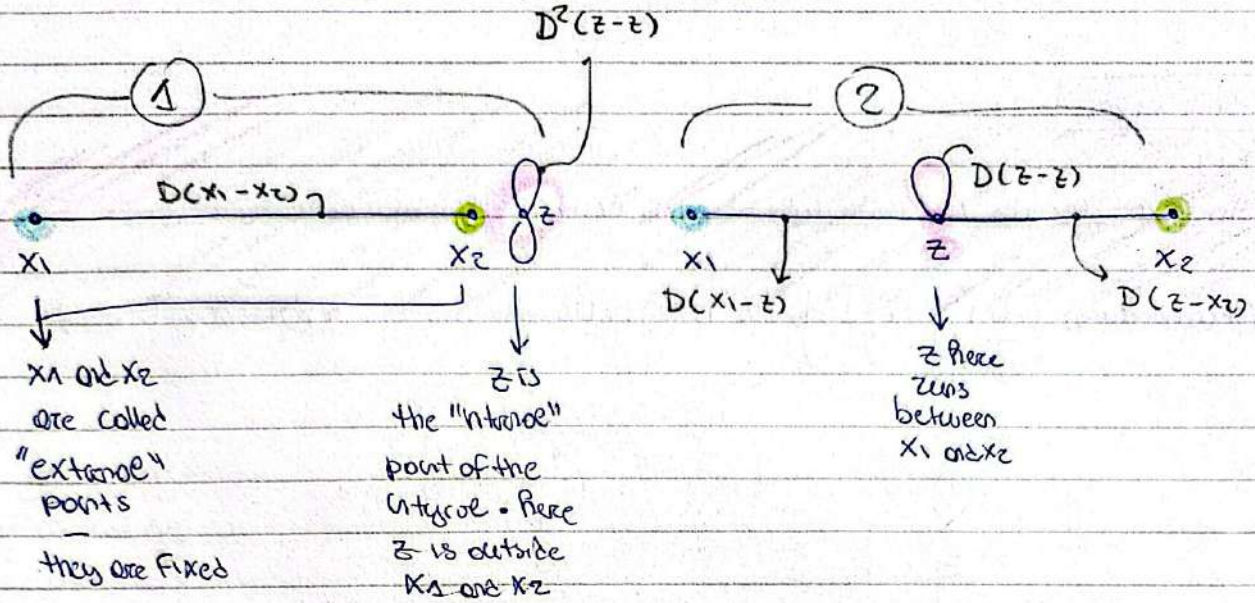


Small Spoiler = the Diagram where the z external points are not connected to any real x_1 x_2 ext points are called Disconnected diagrams and they will be cancelled by the denominator (which accounts gives only disconnected part since it is only the exponential with no external point field $\phi(x_1)\phi(x_2)$

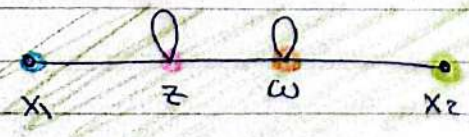
2 order

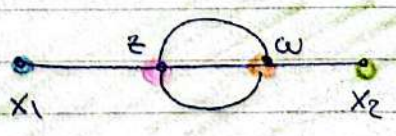
$$= \langle 0|T\{\phi(x_1)\phi(x_2)\ \frac{(-i\lambda)^2}{2!(4!)^2}\int d^4z\int d^4\omega\ \phi^4(z)\,\phi^4(\omega)\}|0\rangle$$

There are many possibilities of combination of contractions. Here we have two ext points (x_1, x_2) and 2 internal points (z, ω). An example of one of the Diagram which compares in this 2nd term is:

$$\frac{(-i\lambda)^2}{2!(4!)^2}\ D(x_1-x_2)\int dz\ D^2(z-z)\int d\omega\ D(\omega-\omega)$$

Or another possibility (just diagrams):





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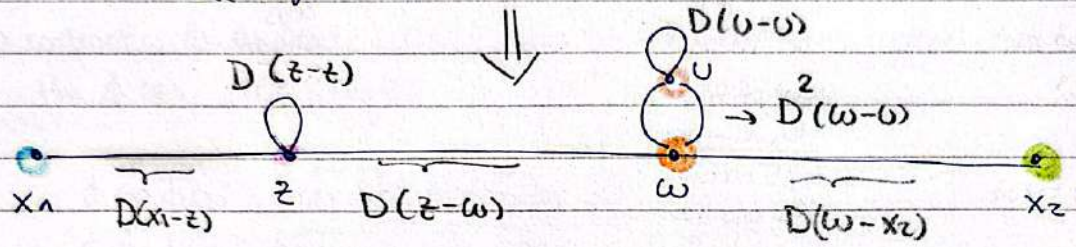
3rd order

An example of one diagram which compares in the 3d order is

$$(\#\text{combinations})\ \frac{(-i\lambda)^3}{3!(4!)^3}\int d^4z \int d^4v \int^4 dw\ D(x_1-z)\,D(z-v)\,D(z-w)\,D(w-x_2)\,D^2(w-v)\,D(v-v)$$



(To see what the # combination is (symmetry factor) see P&S pag 92-93)

Finally, the numerator is the sum over all possible diagrams with fixed external points x_1, x_2. Incrementing the order means incrementing the number of vertexes (i.e. $\times$ from lines)

$$\langle 0|T\{\phi_1(x_1)\phi_1(x_2)\, e^{-i\int d^4z\, \mathcal{H}_1}\}|0\rangle = \text{Sum of all Diagrams of fixed external points } x_1, x_2$$

0 zero order diagrams = contains zero vertex, just propagators (lines)
1 order diagrams = contains 1 vertex
2 order diagrams = contain 2 vertexes
....

vertexes = Order of the diagram

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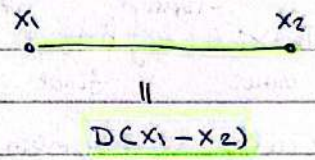
FEYNMAN RULES FOR DIAGRAMS IN POSITION SPACE FOR REAL SCALAR FIELD IN ϕ^4-THEORY

The following rules explain how to evaluate the analitic expression to associate in the expansion of $\langle \Omega | T\{\phi(x_1)\cdots\phi(x_n)\}|\Omega\rangle$ that corresponds to the diagram

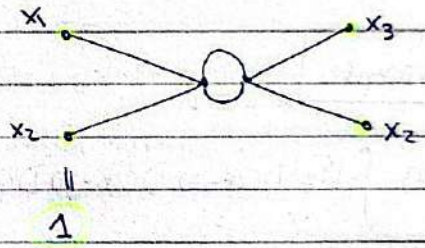
- **PROPAGATORS**

  All the lines comparing in the diagram. each line is labeled by its initial and final position space point and correspond to a green function



- **EXTERNAL POINTS / LEG**

  are the points which compare also in the first $\langle \Omega | T\{\phi(x_1)\cdots\phi(x_n)\}|\Omega\rangle$. They are fixed and contribute with 1 in the position space expansion

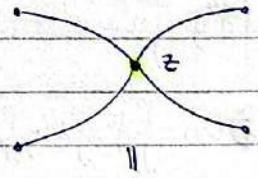


- **VERTICES (ϕ^4 - THEORY)**

  Vertices are cross section of 4 lines, they are the INTERNAL POINTS, refers to an integration over such internal point

  ↓

  in ϕ^4 theory where $H_I = \frac{\lambda}{4!}\phi^4$ they are the crossy of minimum 4 lines X and bring also a factor λ (For ϕ^3 theory they can be 3 lines Y)

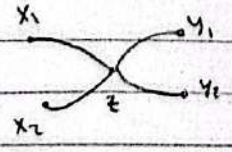


$$\frac{(-i\lambda)}{4!}\int d^4 z$$

- **SYMMETRY FACTOR**

  At final step adjust for the symmetry factor = count how many combinations of contractions lead to such diagram and multiply by it.

EX:



$$(-i\lambda)\int d^4 z\; D(x_1 - z)\, D(x_2 - z)\, D(z - y_1)\, D(z - y_2) \cdot \#$$

Symmetry factor

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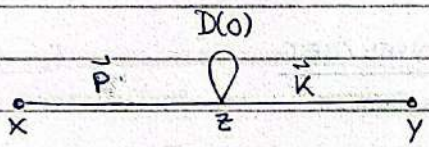
FEYNMANN RULES FOR DIAGRAMS IN MOMENTA SPACE FOR REAL SCALAR FIELD IN ϕ^4-THEORY

Since our aim is to compute the S matrix it is useful to take the Fourier transform of the diagrams as in the LSZ formula we take the Fourier transform of the correlation function $\langle \Omega | T\{\dots\} | \Omega \rangle$

LSZ: $\int_{\text{initial}} \prod dx_i \, e^{-ip_i x_i} \int_{\text{final}} \prod dy_i \, e^{ik_j y_j} \langle \Omega | T\{\phi(x_1)\dots\} | \Omega \rangle = \sim$ motrix element

How the rules of Feynman diagrams transform? Let's look at some examples

1st example



$$\frac{12}{4!}(-i\lambda)\int d^4z\, D(x-z)D(z-y)D(z-z)$$

$$\tilde f(k) = \int dy\, e^{ik\cdot y} f(y) \quad \text{final}$$

$$\tilde f(p) = \int dx\, f(x) e^{-ip\cdot x} \quad \text{initial}$$

↓ FOURIER TRANSFORM INTO MOMENTA SPACE

$$\frac{12}{4!}(-i\lambda)\int d^4x \int d^4y \int d^4z\, e^{-ip\cdot x} e^{ik\cdot y} \underbrace{D(0)}\, D(x-z)D(z-y)D(z-z) =$$

$$= \frac{12}{4!}(-i\lambda)\, D(0) \int d^4x \int d^4y \int d^4z\, e^{-ipx+iky} D(x-z)D(z-y) =$$

Shift $x = s+z$ $y = r+z$ hence

$$= \frac{12}{4!}(-i\lambda)\, D(0) \int d^4s \int d^4r \int d^4z\, e^{(-ip+ik)z} e^{-ip\cdot s} e^{+ikr} D(s) D(r) =$$

by def we had of green functions, $\tilde D(-k) = \tilde D(k)$ $\tilde D(p) = \tilde D(-p)$ for scalar field:

$$= \frac{12}{4!}(-i\lambda)\, D(0)\, \tilde D(k)\, \tilde D(p) \int d^4z\, e^{+i(k-p)\cdot z} =$$

$$= \frac{12}{4!}(-i\lambda)\, D(0)\, \tilde D(k)\, \tilde D(p)\, (2\pi)^4\, \delta^{(4)}(p-k)$$

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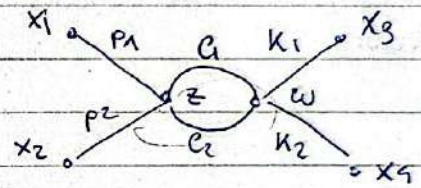
$$\frac{12}{4!}(-i\lambda)(2\pi)^4\delta^{(4)}(p-k)\,\tilde{D}(k)\tilde{D}(p)\,D(0)$$

symmetry factor — vertex in z — external legs — tadpole (legs closing loop in z point)

2nd example — Check signs

$$\frac{1}{2}(-i\lambda)^2\int d^4z\int d^4\omega\; D(x_1-z)D(x_2-z)D^2(z-\omega)D(\omega-x_3)D(\omega-x_4)$$

↓ Fourier transform on external points



$$\frac{1}{2}(-i\lambda)^2\int d^4x_1\int d^4x_2\int d^4x_3\int d^4x_4\, e^{i(p_1x_1+p_2x_2-k_1x_3-k_2x_4)}\cdot$$

$$\cdot\int d^4z\int d^4\omega\; D(x_1-z)D(x_2-z)D^2(z-\omega)D(\omega-x_3)D(\omega-x_4) = \quad \text{shift } x_i-z=s_i,\ x_3-\omega=s_3$$

$$= \frac{1}{2}(-i\lambda)^2\tilde{D}(p_1)\tilde{D}(p_2)\tilde{D}(k_1)\tilde{D}(k_2)\int d^4z\int d^4\omega\, e^{i(p_1+p_2)z}e^{-i(k_1+k_2)\omega}D^2(z-\omega) =$$

I remember $D(z-\omega) = \int\frac{d^4\ell}{(2\pi)^4}e^{-i\ell(z-\omega)}\tilde{D}(\ell)$ per cui

$$D^2(z-\omega) = \int\frac{d^4\ell_1}{(2\pi)^4}\int\frac{d^4\ell_2}{(2\pi)^4}e^{-i\ell_1(z-\omega)}\tilde{D}(\ell_1)\,e^{-i\ell_2(z-\omega)}\tilde{D}(\ell_2)$$

quindi

$$= \frac{1}{2}(-i\lambda)^2\tilde{D}(p_1)\tilde{D}(p_2)\tilde{D}(k_1)\tilde{D}(k_2)\int d^4z\int d^4\omega\int\frac{d^4\ell_1}{(2\pi)^4}\int\frac{d^4\ell_2}{(2\pi)^4}e^{-i(p_1+p_2+\ell_1+\ell_2)z}\,e^{-i(k_1+k_2+\ell_1+\ell_2)\omega}\tilde{D}(\ell_1)\tilde{D}(\ell_2) =$$

svolgo integrale su z e ω, ottengo le δ

$$= \frac{1}{2}(-i\lambda)^2\tilde{D}(p_1)\tilde{D}(p_2)\tilde{D}(k_1)\tilde{D}(k_2)\int\frac{d^4\ell_1}{(2\pi)^4}\int\frac{d^4\ell_2}{(2\pi)^4}(2\pi)^4\delta^4(p_1+p_2-\ell_1-\ell_2)\,\delta^4(k_1+k_2-\ell_1-\ell_2)(2\pi)^4\tilde{D}(\ell_1)\tilde{D}(\ell_2) =$$

$$\delta^4(p_1+p_2-k_1-k_2)\,\delta^4(k_1+k_2-\ell_1-\ell_2)$$

PROPRIETÀ':

$$\delta(A-C)\delta(B-C) = \delta(A-B)\delta(B-C)$$

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$$= \frac{1}{2}(-i\lambda)^2 \tilde{D}(\cdot)\cdots\tilde{D}(\cdot)\int \frac{dq_1}{(2\pi)^4}\int \frac{dq_2}{(2\pi)^4}\,(2\pi)^4(2\pi)^4\,\hat{\delta}(p_1+p_2-k_1-k_2)\,\delta^4(k_1+k_2-q_1-q_2)\,\tilde{D}(q_1)\,\tilde{D}(q_2)\,>$$

svolgo integrale su q_2

$$= \frac{1}{2}(-i\lambda)^2 \tilde{D}(\cdot)\cdots\tilde{D}(\cdot)\int \frac{dq_1}{(2\pi)^4}\,(2\pi)^4\,\delta^4(p_1+p_2-k_1-k_2)\,\tilde{D}(q_1)\,\tilde{D}(q_1-k_1-k_2) =$$

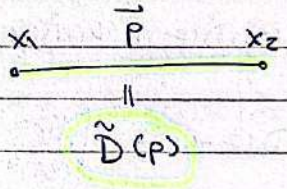
$$= \frac{1}{2}(-i\lambda)^2 \tilde{D}(p_1)\tilde{D}(p_2)\tilde{D}(k_1)\tilde{D}(k_2)\,(2\pi)^4\,\delta^4(p_1+p_2-k_1-k_2)\int \frac{d^4q_1}{(2\pi)^4}\,\tilde{D}(q_1)\,\tilde{D}(q_1-k_1-k_2)$$

- symmetry factor
- external points
- overall momenta conservation
- internal momentum conservation with integration over non fixed internal momentum

RULES I CAN EXTRACT

- **PROPAGATORS**

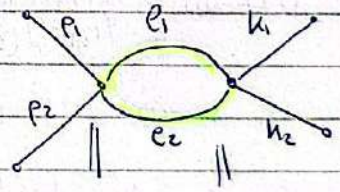
  Corresponds to Fourier green function of one momentum



- **VERTICES / INTERNAL LINES**

  At each vertex I impose the integration over one internal momentum as $\int \frac{dq}{(2\pi)^4}$ and I multiply per $(-i\lambda)$

  and I impose a delta of momenta conservation at that vertex per $(2\pi)^4$



$$(-i\lambda)\int \frac{dq_1}{(2\pi)^4}\,(2\pi)^4\,\delta^4(p_1+p_2-q_1-q_2)$$

$$(-i\lambda)\int \frac{dq_2}{(2\pi)^4}\,(2\pi)^4\,\delta^4(k_1+k_2-q_1-q_2)$$

- **SYMMETRY FACTOR**

  Accounts for n° of contractions and factor $\frac{1}{4!}$ in the ϕ^4 hamiltonian and $\frac{1}{n!}$ of expansion order.

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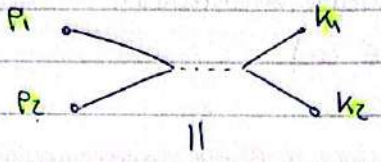
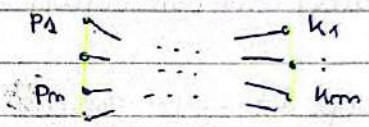
However at this stage I should continue with computation of the external loops integrations. At the end I remain with a formula which follows the following rules

- **EXTERNAL LEGS (PROPAGATORS)**
  Each external legs accounts for one $\tilde{D}(p_i)$



$$\tilde{D}(p_1)\tilde{D}(p_2)\tilde{D}(k_1)\tilde{D}(k_2)$$

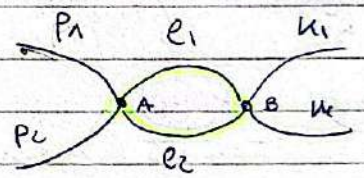
- **OVERALL MOMENTA CONSERVATION**
  It comes from momenta conservation imposition at each vertices



$$(2\pi)^4 \delta^4(p_1+p_2+\dots - k_1 - k_2 \dots - k_m)$$

- **INTERNAL PROPAGATORS / VERTICES**

(1) Impose momenta conservation at each vertex. Relate all momenta to a minimum number of "free" internal momenta and write all others internal momenta in terms of external p_i & k_i and such "free" internal ℓ_i momenta.



(1) A: $p_1+p_2=\ell_1+\ell_2$
B: $k_1+k_2=\ell_1+\ell_2$
$\longmapsto$ $\ell_2 = k_1+k_2-\ell_1$ (NON FREE ↑, FREE ↑)

(2) Write each propagator $\tilde{D}(\ell_i)$ eventually substituting the expression of non-free ℓ_i in terms of free ℓ_j.

(2) $\tilde{D}(\ell_1)\tilde{D}(\ell_2) \longrightarrow \tilde{D}(\ell_1)\tilde{D}(k_1+k_2-\ell_1)$

(3) Multiply per $(-i\lambda)^m$ with $n = \#$ vertices.

(4) Integrate over $\int \frac{d^4\ell_i}{(2\pi)^4}$ all free internal momenta

(3)(4) $(-i\lambda)^2 \int \frac{d^4\ell_1}{(2\pi)^4} \tilde{D}(\ell_1)\tilde{D}(k_1+k_2-\ell_1)$

- **SYMMETRY FACTOR**
  # contractions, factor $\frac{1}{4!}$ from ϕ^4 theory interaction and order of expansion n give $\frac{1}{n!}$ $\longrightarrow$ $\frac{\#}{(4!)^m\, m!}$

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Loops and tadpole may give divergent terms = RENORMALIZATION

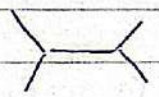
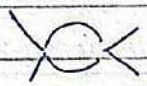
Ogni volta che ho un anello, come un tadpole tipo chiuso su se stesso ( ) o un anello fra due punti (op) posso ottenere termini che sotto integrazione DIVERGONO! ES una tadpole da sempre D(0) che DIVERGE! È ∞!

↓

Per ovviare a questo prob c'è un processo che noi chiamo RENORMALIZATION che sottrae alcuni termini (omettendo appunto diagrammi che annullano tale divergenza e rendono FINITI i contributi di loop and tadpole

↓

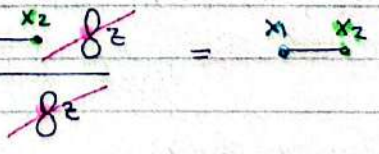
Osservo che loop & tadpole esistono solo in ϕ^4-theory, in ϕ^3 theory ammette anche Diagrammi ad "Albero" senza loop, perché presente vertici a 3 e non a 4!

ϕ^3 tree Diagram:  | ϕ^4 loop diagram 

See M. 5.6

DENOMINATOR ESXPANSION

Shortly, the denominator contains only disconnected graphs i.e vacuum to vacuum bubble. Hence it cancels all terms that are not connected to any of external x_i point

EX: $\frac{\text{Numerator}}{\text{denominator}} = \frac{\langle 0|T\{\phi(x_1)\phi(x_2)e^{-\int dz}\}|0\rangle}{\langle 0|T\{e^{\int dz}\}|0\rangle}$ = 

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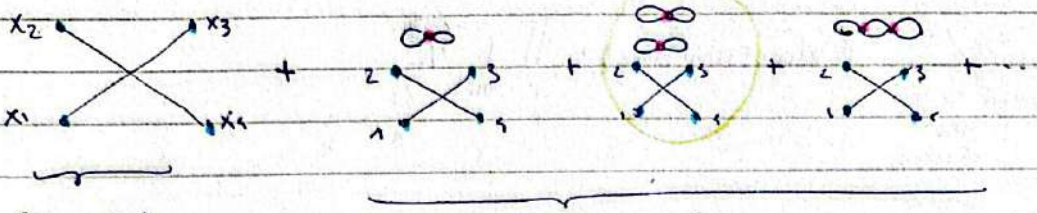
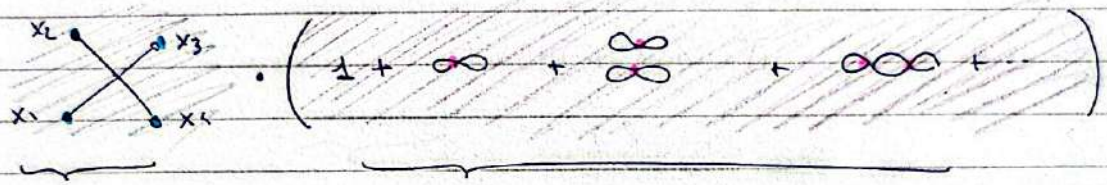
Take the following graph as example



Connected

Disconnected

It is possible to factorize

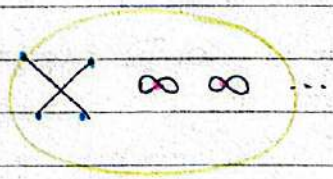


Value of connected piece

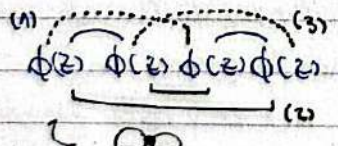
disconnected pieces

The disconnected pieces in the parenthesis are canceled by the denominator

a) Now let's consider a generic graph, one single diagram (not a generic term of the sum above). It's the product of connected part and many disconnected



Assign a value of each disconnected piece as V_i. Be n_i the number of possible equivalent contractions leading to the i-esim graph

Es: 

→ $n_i = 3$, 3 possibili combinazioni di ∞^2_2 cioè $D^2(z-z)$

This is only this part, just one V_i

Hence the value of ONE diagram is

$$(\text{Connected Value}) \cdot \left(\prod_i \frac{1}{m_i!} V_i^{m_i} \right)$$

m_i volte V_i pari ai n_i possibili contrazioni

where $\frac{1}{m_i!}$ accounts for the symmetry factor from interchanging m_i copies of V_i

$n_1 = 3$ $n_2 = 3$ $n_3 = 9$ (credo)

indirecte scorie di voi parti disconnected

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WHOLE

- Now consider the graph hence SUM over all possible diagrams

$$\sum_{\substack{\text{Connected}\\ \text{pieces}\\ \text{of each}\\ \text{diagram}}} \sum_{\substack{\text{sets of}\\ \{m_i\}\\ \text{of each}\\ \text{diagram}}} (\text{Value of connpiece}) \cdot \prod_i \frac{1}{m_i!} V_i^{m_i}$$

$$\sum_{\text{Connected}} (\text{Connected}) \prod_i \sum_{\{m_i\}} \frac{1}{m_i!} V_i^{m_i}$$

$$\sum_{\text{Con}} (\text{Connected}) \prod_i \exp(V_i)$$

$$\sum_{\text{Connected}} (\text{Connected}) \exp(\Sigma_i V_i)$$

The disconnected diagrams exponentiate!

- We demonstrated that the numerator can be separated into this two pieces

$$\underset{\text{NUMERATOR}}{\langle 0|T\{\phi(1)\phi(2)\exp^{-S}\}|0\rangle} = \left(\sum_{\text{Con}} \text{Con}_2\right)\left(\exp(\Sigma_i V_i)\right)$$

- The denominator contains only the exp $\langle 0|T\{\exp\{-i\int d\tau H_I\}\}|0\rangle$ Hence only the disconnected vacuum bubbles!

$$\underset{\text{Denominator}}{\langle 0|T\{\exp\cdots\}|0\rangle} = \left(\exp(\Sigma_i V_i)\right)$$

- The complete theorem is then only made of connected diagrams

$$\langle R|T\{\cdots\}|R\rangle = \frac{\text{NUMER}}{\text{DENOM}} = \frac{(\cdots)(\cdots)}{(\cdots)} = \sum_{\text{Connected}}$$

Only pure connected terms contribute to the correlation formula $\langle R|T\{\cdots\}|R\rangle$. Vacuum bubbles eliminate with denominator

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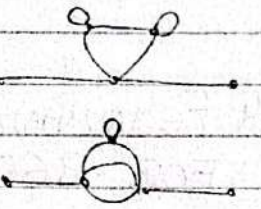
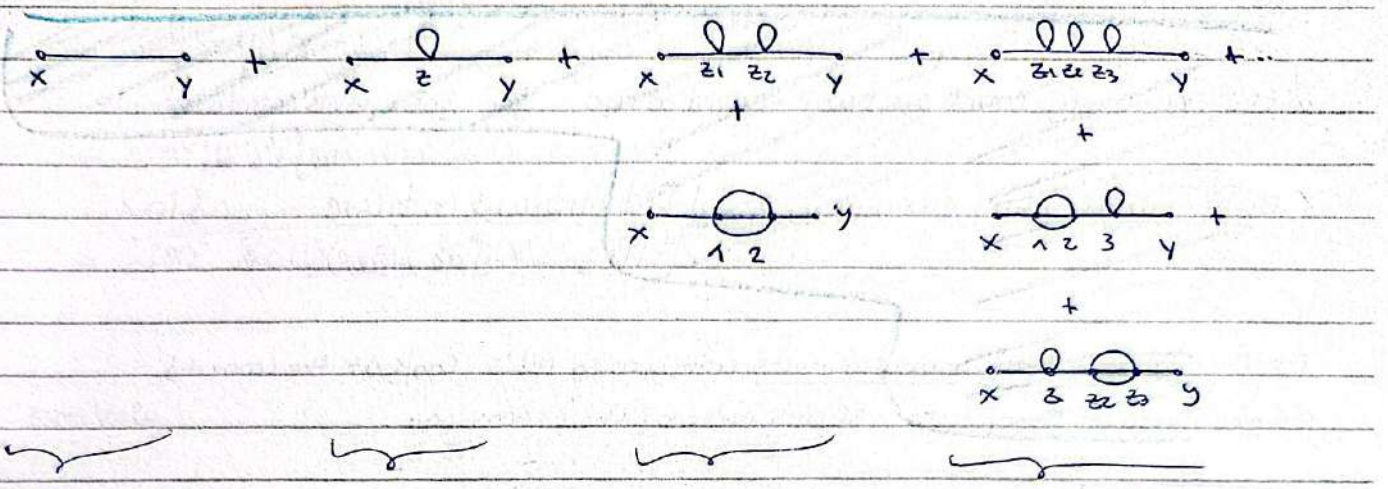
- and others -



Ex for tree parts:



$O(1)$	$O(\lambda)$	$O(\lambda^2)$	$O(\lambda^3)$
zero order	one order	2-order	3rd order
	(one node)	(2 nodes)	(3 nodes)

Disconnected parts = vacuum energy density

Physically the disconnected parts can be viewed as vacuum energy density, relative to the free vacuum $|0\rangle$.

- Example: $\int_z$ (x, y) $= D(x-y)\int d^4z\, D^2(z-z) = D(x-y)\, D^2(0)\int dz \sim V2T$

this is proportional to V = spatial volume, T = time volume which is $\int_{-T}^{T} dt$ for $T\to\infty(1-i\varepsilon)$. This term diverges! Ultra violet divergence!

That example shows us that each V_i value of each diagram $\sim 2VT$

- Then eq (7.83) shows $\lim_{t\to\infty(1-i\varepsilon)} \langle 0|T\{\phi(x)\cdots \exp^{-\int dt}\}|0\rangle = \lim_{\epsilon\to(1-i\varepsilon)\infty} |\langle 0|e\rangle|^2 e^{-i2E_0\epsilon}\langle e|T\{\cdots\}|e\rangle$

NUMERATOR

Hence the terms with t dependence are

$$e^{-i\int_{-t'}^{t'} H_1(t')\,dt'} \sim e^{-i2E_0t'}$$

but we just proved that this is the part containing the disconnected terms $\exp(\Sigma_i V_i)$ Hence

$\exp(\Sigma_i V_i) \sim e^{-i2E_0t} \Rightarrow$ Hence disconnected value V_i is proportional to vacuum energy E_0

Disconnected — vacuum energy E_0

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8.4 FEYNMAN DIAGRAMS FOR 4-POINTS CORRELATOR FOR REAL SCALAR FIELD FOR ϕ^4-INTERACTION

Let's now apply what we have learnt so far for four points correlator:

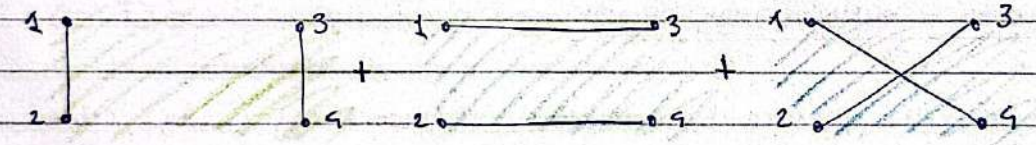
$$\langle \Omega | T\{\phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4)\} | \Omega \rangle = \frac{\langle 0 | T\{\phi_I(x_1)\phi_I(x_2)\phi_I(x_3)\phi_I(x_4) e^{-i\int d^4z\, \mathcal{H}_I}\} | 0 \rangle}{\langle 0 | T\{ e^{-i\int d^4z\, \mathcal{H}_I(z)}\} | 0 \rangle}$$

we learnt that only connected parts contribute so let's look at the connected graphs between four points at each order of the expansion:

zero order

$$\langle 0 | T\{\phi_I(x_1)\phi_I(x_2)\phi_I(x_3)\phi_I(x_4)\} | 0 \rangle \overset{\text{only connected parts}}{=}$$

$$= D(x_1-x_2)D(x_3-x_4) + D(x_1-x_3)D(x_2-x_4) + D(x_1-x_4)D(x_2-x_3)$$



FOURIER TRANSFORM Check signs

- Let's go in momenta space and take the fourier transform of each eterm. Chose to name x_3 x_4 as fixed points hence the momenta as $\vec{k}$

$$= \int dx_1 \int dx_2 \int dx_3 \int dx_4\, e^{i(p_1x_1 + p_2x_2 - k_1x_3 - k_2x_4)} D(x_1-x_2)D(x_3-x_4)$$

- Shift $X = x_1 - x_2$ $\qquad \mathcal{X} = \frac{x_1+x_2}{2}$

$x = x_3 - x_4$ $\qquad \mathcal{K} = \frac{x_3+x_4}{2}$

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$$= \int d^4x \int d^4X \; e^{i(p_1+p_2)X} \; e^{i(p_1-p_2)\frac{x}{2}} \; D(x) \cdot$$

$$\cdot \int d^4z \int d^4Z \; e^{-i(k_1+k_2)Z} \; e^{-i(k_1-k_2)\frac{z}{2}} \; D(z) =$$

$$= (2\pi)^4 \, \delta^{(4)}(p_1+p_2) \, \delta^{(4)}(k_1+k_2)(2\pi)^4 \; \tilde{D}(p_1) \, \tilde{D}(k_1)$$

$$= (2\pi)^4 \, \delta^{(4)}(p_1+p_2) \, \delta^{(4)}(k_1+k_2) \, (2\pi)^4 \; \frac{i}{(p_1^2 - m^2 + i\varepsilon)} \; \frac{i}{(k_1^2 - m^2 + i\varepsilon)} \qquad (8.16)$$

similarly:

$$= (2\pi)^4 (2\pi)^4 \delta^{(4)}(p_1 - k_1) \delta^{(4)}(p_2 - k_2) \; \tilde{D}(p_1) \; \tilde{D}(p_2)$$

$$= (2\pi)^4 (2\pi)^4 \delta^{(4)}(p_1 - k_2) \delta^{(4)}(p_2 - k_1) \; \tilde{D}(p_1) \; \tilde{D}(p_2)$$

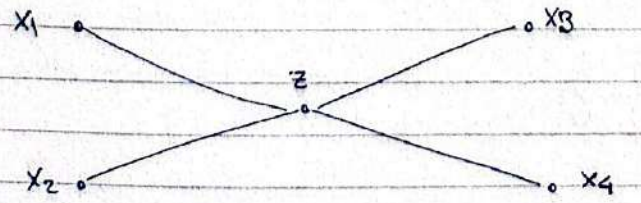
(all this zero order terms have just two out of four poles $\tilde{D}(\cdot)$ / 😊)

1st order

$$\langle 0|T\{\phi_1(x_1)\phi_1(x_2)\phi_1(x_3)\phi_1(x_4)\frac{(-i\lambda)}{4!}\int d^4z \, \phi_1^4(z)\}|0\rangle =$$

Connected diagrams come from contractions with at least $\overbrace{\phi(x_i)\phi(z)}$ connected, one x_i with one z. Contracting two $\overbrace{\phi(z)\phi(z)}$ lead to disconnected parts. There are $4!$ possible combinations that lead to:

$$-\frac{4!}{4!}(-i\lambda)\int d^4z \; D(x_1-z) \, D(x_2-z) \, D(x_3-z) \, D(x_4-z)$$



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FOURIER TRANSFORM

- $-i\lambda \int d^4z \int dx_1 \int dx_2 \int dx_3 \int dx_4 \; e^{i(p_1x_1+p_2x_2-k_1x_3-k_2x_4)} \; D(x_1-z)D(x_2-z)D(x_3-z)D(x_4-z) =$

- shift $y_i = x_i - z$ $\qquad \tilde{D}(p_1) \qquad \tilde{D}(p_2)$

$$= -i\lambda \int d^4z \int dy_1 \int dy_2 \int dy_3 \int dy_4 \; e^{i(p_1+p_2-k_1-k_2)z} \; e^{ip_1y_1} D(y_1) \; e^{ip_2x_2} D(y_2) \cdot$$

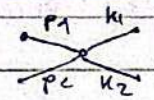
$$e^{-ik_1x_3} D(y_3) \; e^{-ik_2x_4} D(y_4) =$$

$\tilde{D}(-k_1) \qquad \tilde{D}(k_2)$

$$= -i\lambda \, \tilde{D}(k_1)\tilde{D}(k_2)\tilde{D}(p_1)\tilde{D}(p_2) \int d^4z \; e^{i(p_1+p_2-k_1-k_2)z} =$$

$$= -i\lambda \, \tilde{D}(k_1)\tilde{D}(k_2)\tilde{D}(p_1)\tilde{D}(p_2) \, (2\pi)^4 \, \delta^{(4)}(p_1+p_2-k_1-k_2) \qquad (8.17)$$

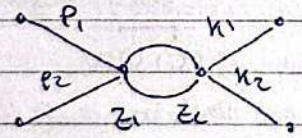
poles of all external points/legs

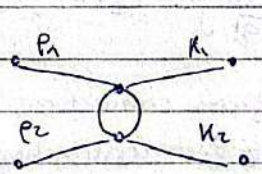


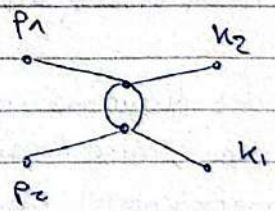
SECOND ORDER check signs

Going directly diagrammatically we observe diagrams like

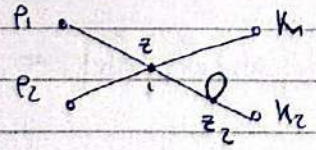
A)





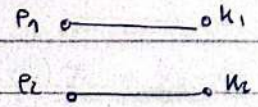


B)



$\ldots$

C)



$\int z_1 \int z_2 \ldots$

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A) pure connected with Fourier transform with all external poles $\tilde{D}(p_1)\tilde{D}(p_2)\tilde{D}(k_1)\tilde{D}(k_2)$ isolated

B) connected BUT not all poles of external points isolate when going to the Fourier transformation ⟶ Need AMPUTATION (vedi dopo)
(NO Meglio = non sono 1PI irriducibile e si tolgono sotto Renormalizzazione!)

C) Disconnected, do not contribute, eliminated by denominator

---

In conclusion we can deduce that in general, EXCEPT FOR ZERO ORDER (& DIAGRAMS WHICH WILL NEED AMPUTATION) all connected Diagrams in the MOMENTA (Fourier transf) space look like:

$$\underbrace{\tilde{D}(p_a)\tilde{D}(p_2)\cdots\tilde{D}(k_1)\tilde{D}(k_2)\cdots}_{\text{external legs}} \cdot \left(\text{connection core (internal) contributions}\right)$$

8.5 LSZ FORMULA WITH DIAGRAMS

The LSZ formula connects iT Matrix with correlation $\langle \Omega | T\Sigma \cdots \Sigma | \Omega \rangle$ function's Fourier transform:

$$\text{LSZ} \Rightarrow \underbrace{\left(\prod_i \frac{i\sqrt{Z}}{(p_i^2 - m^2)}\right)\left(\prod_j^m \frac{i\sqrt{Z}}{(k_j^2 - m^2)}\right)\langle k_1 \cdots k_m | i\hat{T} | p_1 \cdots p_m \rangle}_{\text{$i\hat{T}$ Matrix element multiplied by on-shell poles of external points } p_i\, k_j} = \underbrace{\int dx_1 \cdots \int dy_n \, e^{-ip_1 x_1 \cdots + ik_1 x \cdots} \langle \Omega | T\Sigma \phi(x_1) \cdots \phi(y_n) \Sigma | \Omega \rangle}_{\text{Fourier transform of } \langle \Omega | T\Sigma \phi(x_1) \cdots \phi(y_n) \Sigma | \Omega \rangle}$$

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The left side of LSZ formula contains poles of ALL external LEGS $p_1 \dots p_n$ $k_1 \dots k_m$, initial and final states. Infatti mandando idoneamente alle particelle incoming e final on shell cioè $p_i^2 \to m^2$ $k_j^2 \to m^2$ otteggo dei punti di discontinuità.

Di conseguenza, per il teorema dei residui, solo nel termine di DESTRA (Right hand side), cioè nello sviluppo di Fourier di $\langle e_1 \dots e_2 \rangle$, solo i termini con ALL external poles danno contributo alla matrice iT, perché si semplificano con i poli di SX!

$$\left(\prod \frac{i\sqrt{z}}{p_i^2 - m^2}\right)\left(\prod \frac{i\sqrt{z}}{p_j - m^2}\right)\langle k_1 \dots | iT | p_1 \dots \rangle = \tilde{D}(p_1) \dots \tilde{D}(k_m) \ (\text{core})$$

$$\left(\prod \frac{i\sqrt{z}}{(p_i^2 - m^2)}\right)\left(\prod \frac{i\sqrt{z}}{k_j^2 - m^2}\right)\langle k_1 \dots | iT | p_1 \dots \rangle = \frac{1}{p_1^2 - m^2} \dots \frac{1}{k_1 - m^2} \ (\text{core})$$

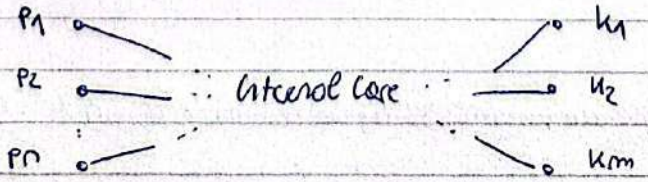
Se ho meno di tutti i poli, come il termine zero dell'espansione di $\langle e_1 \dots e_2 \rangle$, tale termine do contributo nullo alla matrice iT!

I termini che danno contributo nullo sono il termine zero nell'espansione e i diagrammi che non sono ancora stati AMPUTATI (vedi dopo). Dopo la PROCEDURA DI AMPUTAZIONE, TUTTI I DIAGRAMMI CONNESSI contribuiscono alla matrice (eccetto l'ordine zero) perché sono sempre formati da un coeff $\tilde{D}(p_1) \dots \tilde{D}(k_m) \dots$ di tutti i poli esterni. In più mostrano sempre anche la conservazione globale del momento $(2\pi)^4 \delta^{(4)}(p_1 + p_2 + \dots - k_{m-1} - k_m)$

GENERIC AMPUTED CONNECTED DIAGRAM (except zero order) (MOMENTA SPACE) $= \tilde{D}(p_1) \dots \tilde{D}(k_m)\ (2\pi)^4 \delta^{(4)}(p_1 + \dots - k_m)$ (internal core)

$\tilde{D}(p_1) \dots \tilde{D}(k_m)$: external poles

$(2\pi)^4 \delta^{(4)}(p_1 + \dots - k_m)$: global momenta conservation



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TEORICA 1

n. 4

DOMIFILLA ALOI

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8.5 LSZ FORMULA WITH DIAGRAMS – part 2

→ Matrix element full and summary of Feynman rules for ϕ^4 theory

8.6 FEYMAN RULES FOR FERMIONS

→ Time ordering and normal ordering for fermions

8.7 FEYMAN RULES FOR QED (BOSONS)

C.9 EXAMPLE OF CROSS SECTION COMPUTATION

9.1 S-MATRIX

→ γ^μ rules & Mandelstam invariants

9.2 CROSS SECTION

9.3 CROSSING SYMMETRY

C.10 PATH INTEGRAL FORMALISM

10.1 PATH INTEGRAL FORMALISM IN QM

10.1.1 evolution amplitude A

10.1.2 matrix element of $\hat{q}_H(t)$

10.2 PATH INTEGRAL FOR SCALAR FIELDS

10.3 FEYMAN PROPAGATOR OF SCALAR FIELDS FOM PATH INTEGRAL

10.4 GENERATING FUNCTIONAL (FOR SCALAR FIELDS)

10.5 FUNCTIONAL QUANTIZATION OF FERMIONS SPINOR FIELDS

- Grassmann numbers
- Dirac propagator
- generating functional

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Per cui:

$$\left(\prod_i^n \frac{i\sqrt{Z}}{(p_i^2-m^2)}\right)\left(\prod_j^m \frac{i\sqrt{Z}}{k_j^2-m^2}\right)\langle k_n \dots k_0|\, i\hat{T}\,|p_1 \dots p_m\rangle = \int dx_1 e^{-ip_1x_1} \dots \int dy_m e^{ik_m y_m} \langle \Omega|T\{\dots\}|\Omega\rangle$$

Taking on shell limit $p_i^2 \to m^2$, $k_j^2 \to m^2$ only ↓ AMPUTATED DIAGRAMS contribute

$$\left(\prod_i^n \frac{i\sqrt{Z}}{(p_i^2-m^2)}\right)\prod_j^m \left(\frac{i\sqrt{Z}}{k_j^2-m^2}\right)\langle k_n \dots k_0|\, iT\,|p_1 \dots p_m\rangle = \left(\prod_i \tilde{D}(p_i)\right)\left(\prod_j \tilde{D}(k_j)\right)(2\pi)^4 \delta^{(4)}(p_1 - \dots - k_n)\,(\text{sum of internal cores of all amputated diagrams})$$

$$\langle k_n \dots k_0|\, iT\,|p_1 \dots p_m\rangle = (2\pi)^4 \delta^{(4)}(p_1 + \dots - k_n)\,(\text{sum} \dots)$$

Per cui ottengo:

$$\langle k_n \dots k_0|\, i\hat{T}\,|p_1 \dots p_m\rangle = (2\pi)^4 \delta^{(4)}(p_1 + p_2 + \dots - k_n)\; i\mathcal{M}$$

dove $\mathcal{M}$ = sum of all internal cores contribution of all AMPUTATED DIAGRAMS (in Fourier transform)

$\mathcal{M}$ è l'elemento di matrice di scattering T che deriva dai diagrammi di Feynman nel mondo dei momenti dove ho AMPUTATO le leggs $\tilde{D}(p_1) \dots \tilde{D}(k_j)$ esterne ed Estratto la conservazione globale del momento $(2\pi)^4 \delta^{(4)}(p_1 \dots - k_n)$

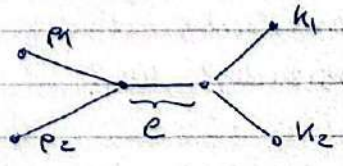
SHORT CUTS

a) For ϕ^4 theory At 1st order $O(\lambda)$ for 4ports:

$$i\mathcal{M}(p_1 p_2 \to k_1 k_2) = -i\lambda + O(\lambda^2)$$

b) For ϕ^3 theory At 2nd order $O(\lambda^2)$ for 4ports:

$$i\mathcal{M}(p_1 p_2 \to k_1 k_2) = (-i\lambda)^2 \underbrace{\tilde{D}(p_1+p_2)}_{e} + O(\lambda^3)$$



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EXPLICT CALCULATION FOR ZERO ORDER CANCELATION

Consider the example of ϕ^4 theory for $p_1 p_2 \to k_1 k_2$ four particles. The zero order term of the Fourier expansion was proportional to

$$(9.16) \to \quad N \frac{1}{(p_1^2 - m^2)} \frac{1}{(k_1^2 - m^2)}$$

only two poles, while LSZ has FOUR ($p_1, k_1, p_2 k_2$) at $p_1^2 = m^2$ $k_1^2 = m^2$ $p_2^2 = m^2$ $k_2^2 = m^2$

Taking the on shell limit $p^2 \to m$ $k^2 \to m^2$ for real virtual and final (external) particles leads to contribute zero for $O(1)$ order term:

1) $$\frac{1}{(p_1^2 - m^2)} \frac{1}{(p_2^2 - m^2)} \frac{1}{(k_1^2 - m^2)} \frac{1}{(k_2^2 - m^2)} \langle k_1 k_2 | iT | p_1 p_2 \rangle = -(2\pi)^8 \delta^{(4)}(p_1 + p_2) \delta^{(4)}(k_1 + k_2) \frac{1}{(p_1^2 - m^2)} \frac{1}{(k_1^2 - m^2)}$$

2) multiply per $(p_1^2 - m^2)(p_2^2 - m^2)(k_1^2 - m^2)(k_2^2 - m^2)$ both sides:

$$\langle k_1 k_2 | iT | p_1 p_2 \rangle = -(2\pi)^8 \delta^{(4)}(p_1 + p_2) \delta^{(4)}(k_1 + k_2) (k_2^2 - m^2)(p_2^2 - m^2)$$

3) Taking on shell limit $p^2 \to m^2$ $k^2 \to m^2$

$$\lim_{\substack{p^2 \to m^2 \\ k^2 \to m^2}} \langle k_1 k_2 | iT | p_1 p_2 \rangle = 0$$

No contribution from zero order!

AMPUTATION PROCEDURE

Amputation = removing the external legs propagators to extract the T-matrix element contributions

In every diagram in momento space $\underbrace{\tilde{\delta}(p_1) \cdots \tilde{\delta}(k_m)}_{\text{external leg}} \times (\text{internal core})$

and we already motivated why matematically the external legs simplify. But physically? why?

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S describe propagation between two spacetime points. However for final and initial states, incoming and outgoing particles (external legs) they do not propagate because

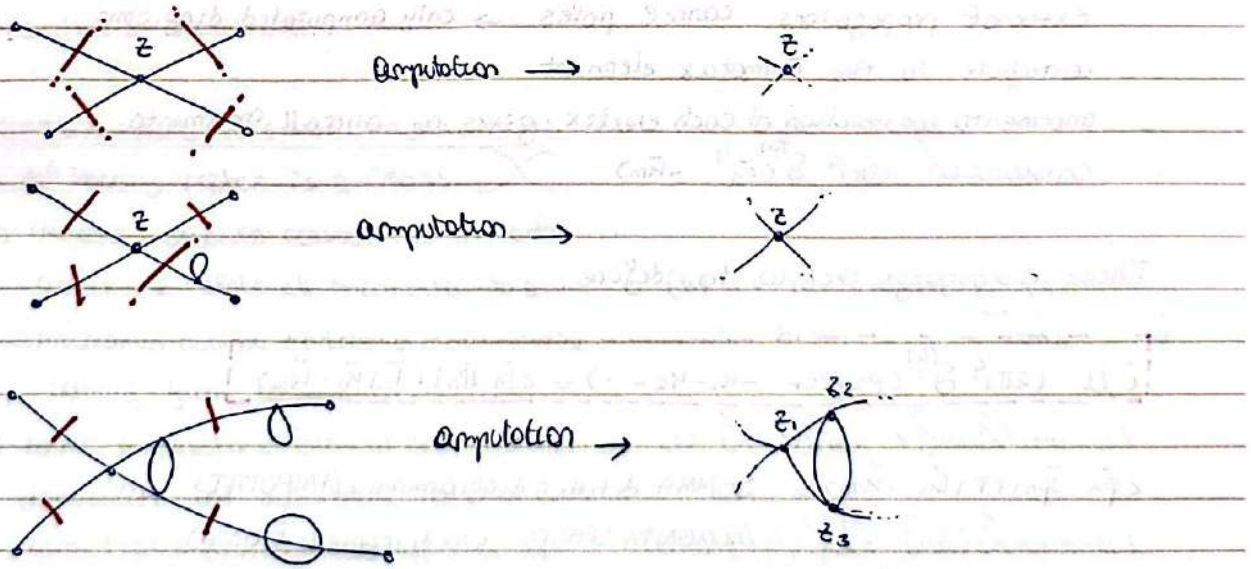
- initial → already prepared for $T \to -\infty(1-i\varepsilon)$
- final → already prepared for $T \to +\infty(1+i\varepsilon)$

Hence their propagation should not be part of the interaction amplitude. This is what happens mathematically in the LSZ formula when taking the on shell limit

the on shell limit is taken because external particles are REAL PARTICLES with mass m ($p_i^2 = m^2$ $k_y \geq mc$). (intermediate particles of the diagram may not be on shell instead).

AMPUTATION DIAGRAMATICALLY

Start at the end of each external leg = find the last part at which the diagram can be cut by cutting one single line (propagator) and such that the cut separates the leg from the rest of the diagram



(?)

Perché Amputiamo ste parti? Danno contributo! Però ipotizzo di aver già considerato e tolto le figure?

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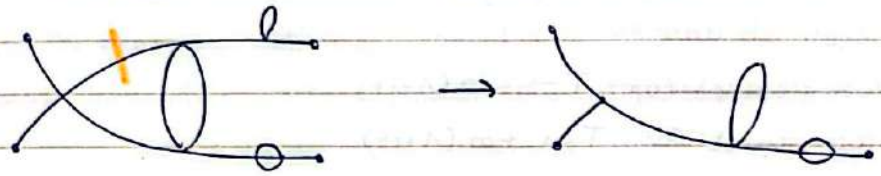
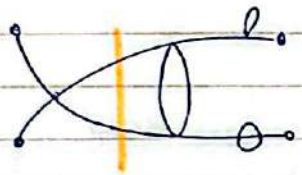
ERRORS



not a good cut cause the peg doesn't separate!



not a good cut because I'm cutting two lines! And not just one!

SUMMARY OF FEYNMANN RULES FOR REAL SCALAR FIELDS

So far we can resume what we have learnt in the list:

- Only connected diagrams contribute to $\langle \Omega | T\{\phi_1 \cdots \phi_n\} | \Omega \rangle$
- In the LSZ formula, taking the Fourier transform of $\langle \Omega | \phi_1 \cdots \phi_n | \Omega \rangle$ external propagators cancel poles → only amputated diagrams contribute to the T-matrix element
- momenta conservation at each vertex gives an overall momenta conservation $(2\pi)^4 \delta^{(4)}(p_1 \cdots - k_m)$

These discoveries lead us to define:

$$i\mathcal{M}\,(2\pi)^4 \delta^{(4)}(p_1+p_2+\cdots-k_1-k_2-\cdots) = \langle \vec{k}_1 \cdots \vec{k}_n | i\hat{T} | \vec{p}_1 \cdots \vec{p}_m \rangle$$

$\langle \vec{k}_1 \cdots \vec{k}_n | iT | \vec{p}_1 \cdots \vec{p}_m \rangle$ = SOMMA di tutti i diagrammi AMPUTATI in MOMENTA SPACE (vedi regole a fianco)

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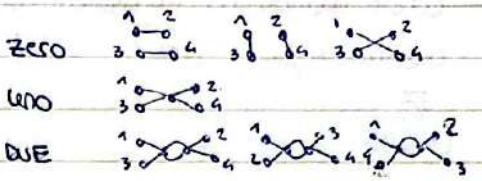
(1) Disegna punti iniziali e punti finali

(2) Disegno tutti i diagrammi CONNESSI di interesse ⇒ Ordine del diagramma = = n° Vertici interni



(3) AMPUTA TUTTI i diagrammi

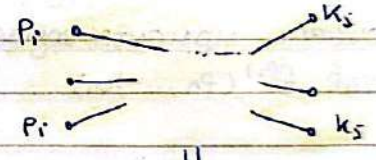
(4) CALCOLA IL VALORE DI OGNI DIAGRAMMA E SOMMA TUTTO $\Sigma(\ )$

(5) Uguaglia la somma $Z(\ ) = \langle k_1 \dots | T | p_1 \dots \rangle = (2\pi)^4 \delta^{(4)}(p_1 \dots - k_m)\ i\mathcal{M}$

VALORE DEL DIAGRAMMA IN ϕ^4 THEORY $\quad H_I = \frac{\lambda}{4!}\phi^4(x)$

- EXTERNAL PROPAGATORS

  Since they have been amputated, count as 1



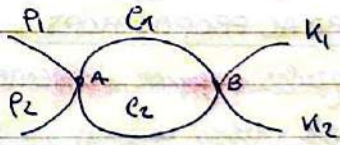
- OVERALL MOMENTA CONSERVATION

  It comes from momenta conservation imposition at each vertex

$$(2\pi)^4 \delta^{(4)}(p_1 + p_2 \dots - k_m)$$

- VERTEX AND INTERNAL PROPAGATORS

  In ϕ^4 theory vertex is a CROSS



(a) IMPOSE MOMENTA conservation at each vertex = relate all momenta to a minimum number of free internal momenta coming from conservation relations

(a) A: $p_1 + p_2 = q_1 + l_2$ | $l_2 = k_1 + k_2 - l_1$ (↓ no free, ↓ free)

B: $k_1 + k_2 = l_1 + l_2$

(b) write propagators $\tilde{D}(l_i)$ of each internal propagators and substitute non free l_i with their expression in terms of free l_j

(b) $\tilde{D}(l_1)\tilde{D}(l_2) \to \tilde{D}(l_1)\tilde{D}(k_1 + k_2 - l_1)$

(c) multiply per $\frac{(-i\lambda)^n}{n!}$ n = n° vertices

(d) integrate $\int \frac{dl_i}{(2\pi)^4}$ over all free internal l_i

(c)(d) $\frac{(-i\lambda)^2}{2}\int \frac{dl_1}{(2\pi)^4}\tilde{D}(l_1)\tilde{D}(k_1 + k_2 - l_1)$

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- **SYMMETRY FACTOR**

Count number of equivalent contractions # which lead to such diagram and multiply per:

$$\frac{\#}{(4!)^m m!}$$

= n° of contractions → for $\times \phi^4$ vertex is 4! contractions so it cancels with 4! denominator of $\frac{\lambda}{4!}\phi^4$

4! = comes from ϕ^4 theory

m! = degree of diagram

hence

$\times \rightarrow -i\lambda$ 1° order

so for each order is $\frac{(-i\lambda)^n}{m!}$ without 4! terms

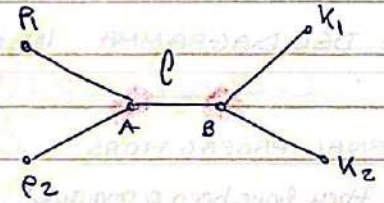
VALORE DEL DIAGRAMMA IN ϕ^3 THEORY $H_1 = \frac{\lambda}{3!}\phi^3(x)$

- **EXTERNAL PROPAGATORS**

—— = 1

- **OVERALL MOMENTA CONSERVATION**

$(2\pi)^4 \delta^{(4)}(p_1 \dots - k_m)$



- **VERTEX**

is a intersection of 3 lines

(a) A: $p_1+p_2=\ell$ | B: $k_1+k_2=\ell$ → $\ell = k_1+k_2$ (Free / non Free)

- **INTERNAL PROPAGATORS**

Same rules = impose MOMENTA conservation at each vertex ecc ecc

(b) $\tilde{D}(\ell) \rightarrow \tilde{D}(k_1+k_2) = \tilde{D}(p_1+p_2)$

(c)(d) $\frac{(-i\lambda)^2}{2!}\tilde{D}(k_1+k_2)$

- **SYMMETRY FACTOR**

$$\frac{\#(-i\lambda)^n}{(3!)^m m!}$$

- n° of contractions
- (coupling constant)^n
- $\frac{1}{n!}$

$$\Downarrow$$

$$\# \frac{(2\pi)^4 \delta^{(4)}(p_1+p_2-k_1-k_2)(-i\lambda)^2 \tilde{D}(k_1+k_2)}{(3!)^2 2!}$$

(Internal particles are not necessarily on shell! $\ell^2 \neq m^2$, there is no condition imposing that! Anzi nell'esempio $\ell = p_1+p_2$ e quindi $\neq m$)

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8.6 FEYNMANN RULES FOR FERMIONS

→ see P&S 4.7 Feynman Rules For Fermions

We now want to understand the Feynman rules for more interesting interaction theories like QED which treats also fermions (electrons) and gauge bosons (photons). Start whith fermion's rules.

Wick's theorem for fermions

The Feynmann propogator between two points is defined as: for fermions:

$$S(x-y) = \langle 0| T\{\psi(x)\bar{\psi}(y)\}|0\rangle$$

→ ↳ Free vacuum & free fields

In order to apply Wick's theorem, however, we need to generalize the definition of time-ordering and normal ordering for fermions fields

TIME ORDERING FOR FERMIONS

$$T\{\psi(x)\bar{\psi}(y)\} = \begin{cases} \psi(x)\bar{\psi}(y) & \text{if } x^0 > y^0 \quad \text{no exchange needed} \\ -\bar{\psi}(y)\psi(x) & \text{if } y^0 > x^0 \quad \underline{\text{one}} \text{ exchange needed} \end{cases} \tag{8.118}$$

Rule = to generale for time ordering of n fields, when, in the process of re-ordering the fields in order to achieve the final time ordered order, each time I EXCHANGE TWO FIELDS I GET A MINUS SIGN

$$T\{\psi(x_1)\psi(x_2)\psi(x_3)\} \begin{cases} \psi(x_1)\psi(x_2)\psi(x_3) & \text{if } 1>2>3 \\ \psi(2)\psi(3)\psi(1) & 2>3>1 \\ \psi(3)\psi(1)\psi(2) & 3>1>2 \\ -\psi(1)\psi(3)\psi(2) & 1>3>2 \\ -\psi(2)\psi(1)\psi(3) & 2>1>3 \\ -\psi(3)\psi(2)\psi(1) & 3>2>1 \end{cases}$$

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EXAMPLE

START WITH = $T\{\psi_{(4)}\psi_{(2)}\psi_{(3)}\psi_{(1)}\}$

SUPPOSE THAT = $x^0_4 > x^0_3 > x^0_2 > x^0_1$

$$T\{\psi_1\psi_2\psi_3\psi_4\} = -T\{\psi_1\psi_3\psi_2\psi_4\} = T\{\psi_1\psi_3\psi_4\psi_2\} = \psi_1\psi_3\psi_4\psi_2$$

(− sign; − sign; that's the correct order!)

SUPPOSE THAT = $x^0_3 > x^0_1 > x^0_4 > x^0_2$

$$T\{\psi_1\psi_2\psi_3\psi_4\} = -T\{\psi_1\psi_3\psi_2\psi_4\} = T\{\psi_3\psi_1\psi_2\psi_4\} = -T\{\psi_3\psi_1\psi_4\psi_2\} = -\psi_3\psi_1\psi_4\psi_2$$

(− sign; − sign; − sign; correct order)

NORMAL ORDERING FOR FERMIONS

normal ordering a bunch of operators means bringing all the creation operators $a^\dagger$ to the left ← and all annihilation a to the right →. For fermions each time I exchange two operators I gain a minus sign, exactly as for time ordering

$$:a_p a_q a_r^\dagger: = -:a_p a_r^\dagger a_q: = :a_r^\dagger a_p a_q: = a_r^\dagger a_p a_q = -a_r^\dagger a_q a_p$$

(− sign; − sign; correct order!; − sign because a_p and a_q anticommute!)

I define, as done for scalar fields, two parts of a Dirac field as:

$$\psi(x) = \psi^+(x) + \psi^-(x)$$

annihilator part — creator part

$$\psi^+(x) = \int \frac{d^3p}{(2\pi)^3\sqrt{2\epsilon_p}} \sum_{s=1,2} a_{p,s} v^s(p) e^{-ip\cdot x}$$

$$\psi^-(x) = \int \frac{d^3p}{(2\pi)^3\sqrt{2\epsilon_p}} \sum_{s=1}^{2} b^\dagger_{p,s} v^s(p) e^{ip\cdot x}$$

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$$\bar{\psi}(x) = \bar{\psi}^+(x) + \bar{\psi}^-(x)$$

annihilator — creator part

$$\bar{\psi}^+(x) = \int \frac{d^3p}{(2\pi)^3\sqrt{2\epsilon_p}} \sum_{s=1,2} b_{ps}\, \bar{v}^s(p)\, e^{-ip\cdot x}$$

$$\bar{\psi}^-(x) = \int \frac{d^3p}{(2\pi)^3\sqrt{2\epsilon_p}} \sum_{s=1,2} a^+_{ps}\, \bar{u}^s(p)\, e^{ip\cdot x}$$

Hence the normal ordering for fermions applied to $\bar{\psi}^+$ and ψ^- is as example, for field at same spacetime point x:

$$:\psi^+(x)\,\bar{\psi}^-(x): \;=\; -\bar{\psi}^-(x)\,\psi^+(x)$$

$$:\psi^-\,\psi^+: \;=\; \psi^-\,\psi^+$$

$$:\psi^+\,\bar{\psi}^-: \;=\; -\bar{\psi}^-\,\psi^+$$

ec ec and all combinations

WICK'S THEOREM FOR FERMIONS

Such generalization of time and normal ordering is needed for fermions because in Wick's theorem formula we make use of commutators ... but for FERMIONS we know the ANTICOMMUTATORS RELATIONS And NOT THE COMMUTATORS! So in order to define Wick's theorem with ANTICOMMUTATORS we needed to change also definition of time and normal ordering. With all that said it is possible to prove that Wick's theorem generalizes for fermions as:

$$T\{\psi(x)\,\bar{\psi}(y)\} = :\psi(x)\,\bar{\psi}(y): + \overbrace{\psi(x)\,\bar{\psi}(y)} \qquad (8.119)$$

where

$$\overbrace{\psi(x)\,\bar{\psi}(y)} := \begin{cases} \{\psi^+(x), \bar{\psi}^-(y)\} & \text{if } x^0 > y^0 \\ -\{\bar{\psi}^+(y), \psi^-(x)\} & \text{if } y^0 > x^0 \end{cases}$$

Anticommutators

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With contraction rules:

$\overbrace{\psi\bar{\psi}}$ = is a matrix since $\psi\ \bar{\psi} = \overbrace{\psi\bar{\psi}}$

$\begin{pmatrix} \vdots \\ \end{pmatrix} \begin{pmatrix} \cdots \end{pmatrix} = \begin{bmatrix} \ddots \\ \end{bmatrix}$

$\overbrace{\psi\psi} = \overbrace{\bar{\psi}\bar{\psi}} \overset{!}{=} 0$ Have no sense so we imposed it to be zero

$\overbrace{\bar{\psi}\psi} = \overset{?}{0}$ weird number

Then in order to define Wicks theorem for more than two fields let me explain one last procedure = contraction under normal ordering.

A contraction under normal ordering gains a minus sign each time I need to exchange to fields to express the contraction

Recall = $:\overbrace{\psi\bar{\psi}}: = \overbrace{\psi\bar{\psi}}$ since $\overbrace{\psi\bar{\psi}}$ is no more an operator, normal ordering does not affect the contraction (see also under scalar fields)

EXAMPLE =

$:\overbrace{\psi_1\psi_2}\bar{\psi}_3\bar{\psi}_4: = -:\overbrace{\psi_1\bar{\psi}_3}\psi_2\bar{\psi}_4: = -\overbrace{\psi_1\bar{\psi}_3}:\psi_2\bar{\psi}_4:$

$:\overbrace{\psi_1\psi_2\bar{\psi}_3\bar{\psi}_4}: = -:\overbrace{\psi_1\psi_2\bar{\psi}_4}\bar{\psi}_3: = :\overbrace{\psi_1\bar{\psi}_4}\overbrace{\psi_2\bar{\psi}_3}: = \overbrace{\psi_1\bar{\psi}_4}\ \overbrace{\psi_2\bar{\psi}_3}$

Aim = contract always a ψ to the left with a $\bar{\psi}$ to the right

⇓

Hence general Wick's theorem for fermions is:

$$T\{\psi_1\bar{\psi}_2\psi_3\dots\} = :\psi_1\bar{\psi}_2\psi_3\dots + \text{all possible contractions}: \qquad (8.20)$$

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Propagators

Since $S(x-y) = \langle 0|T\{\phi(x)\bar{\psi}(y)\}|0\rangle = \langle 0|:\psi\bar{\psi}:|0\rangle + \langle 0|\overbrace{\psi\bar{\psi}}|0\rangle$

annihilates because normal ordering

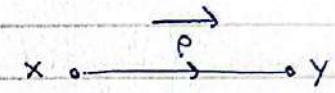
$\overbrace{\psi\bar{\psi}}\langle 0|0\rangle = 1$

Hence

$$S(x-y) = \overbrace{\psi(x)\bar{\psi}(y)}$$ Feynmann propagator of fermions

It can be computed explicitly and the result is

$$S(x-y) = \int \frac{d^4p}{(2\pi)^4}\, \tilde{S}(p)\, e^{-ip\cdot(x-y)}$$



with $$\tilde{S}(p) = \frac{i(\not{p}+m)}{p^2-m^2+i\varepsilon}$$ (8.21)

where $\not{p} = p_\mu \gamma^\mu$ $\quad p^2 = p_\mu p^\mu$

PRACTICAL EXPRESSION OF $\tilde{S}(p)$

note that: $(\not{p}+m)(\not{p}-m) = (p_\mu\gamma^\mu + m)(p_\nu\gamma^\nu - m) = p_\mu p_\nu \gamma^\mu\gamma^\nu - m^2 =$

$= \frac{1}{2}(p_\mu p_\nu \gamma^\mu\gamma^\nu + p_\nu p_\mu \gamma^\nu\gamma^\mu) - m^2 =$

$= \frac{1}{2} p_\mu p_\nu \underbrace{(\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu)}_{2g^{\mu\nu}} - m^2 = p^2 - m^2$

general statement $\not{a}\cdot\not{a} = a^2 = a_\mu a^\mu$ but $\not{a}\not{b} \neq a\cdot b$

we can use $(\not{p}+m)(\not{p}-m) = p^2-m^2$ to write $\tilde{S}(p)$ as:

$$\tilde{S}(p) = \frac{i(\not{p}+m)(\not{p}-m)}{(p^2-m^2+i\varepsilon)(\not{p}-m)} = \frac{i(p^2-m^2)}{(p^2-m^2+i\varepsilon)(\not{p}-m)} = \frac{i}{(\not{p}-m)}$$

if pole prescription understood

$$\tilde{S}(p) \simeq \frac{i}{(\not{p}-m)}$$ with pole prescription understood → not analytically correct but it will be useful in computations

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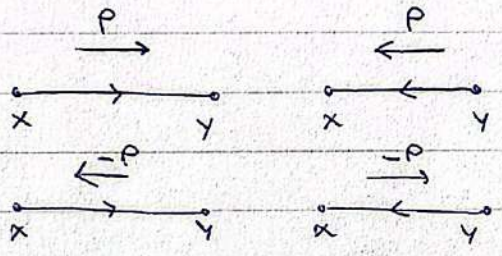
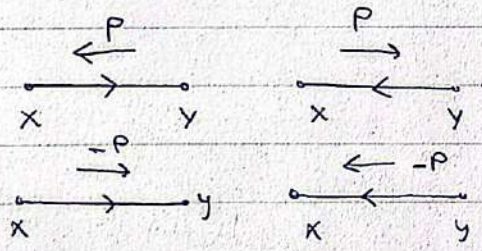
$S(x-y)$ and $\tilde{S}(p)$ for



$S(y-x)$ and $\tilde{S}(-p)$ for



FOR SCALAR FIELDS : $\tilde{D}(p) = \tilde{D}(-p)$ and $D(x-y) = D(y-x)$

FOR FERMIONIC FIELDS $S(x-y)$ & $\tilde{S}(p)$ or $S(y-x)$ & $\tilde{S}(-p)$

BUT

$$\frac{i(\not{p}+m)}{p^2-m^2+i\varepsilon} = \tilde{S}(p) \neq \tilde{S}(-p) = \frac{i(-\not{p}+m)}{p^2-m^2+i\varepsilon}$$

AND ALSO

$$\frac{i(\not{p}+m)}{p^2-m^2+i\varepsilon} = \tilde{S}(p) \neq -\tilde{S}(-p) = \frac{-i(-\not{p}+m)}{p^2-m^2+i\varepsilon}$$

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$$\not{a}\not{a} = \gamma^\mu a_\mu \gamma^\nu a_\nu = \gamma^\mu\gamma^\nu a_\mu a_\nu = (2g^{\mu\nu} - \gamma^\nu\gamma^\mu) a_\mu a_\nu =$$

(numeri)

$$= 2a_\mu a^\mu - \gamma^\nu a_\nu \gamma^\mu a_\mu = 2(a\cdot a) - \not{a}\not{a}$$

$$\Rightarrow \not{a}\not{a} = 2a^2 - \not{a}\not{a} \Longrightarrow 2\not{a}\not{a} = 2a^2 \Longrightarrow \not{a}\not{a} = a^2$$

---

se invece $\not{a}$ e $\not{b}$ sono due versi non volte per perché

$$\not{a}\not{b} = \gamma^\mu\gamma^\nu a_\mu b_\nu = (2g^{\mu\nu} - \gamma^\nu\gamma^\mu) a_\mu b_\nu = 2a\cdot b - \gamma^\nu b_\nu \gamma^\mu a_\mu =$$

posso spostare il numero b_ν ma con γ^ν verso b_ν !!
γ^ν è MATRICE!

$$= 2a\cdot b - \not{b}\not{a}$$

$$\Rightarrow \not{a}\not{b} = 2a\cdot b - \not{b}\not{a} \rightarrow \not{a}\not{b} + \not{b}\not{a} = 2a\cdot b$$ ma $\not{a}\not{b} \neq \not{b}\not{a}$!

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A POSTERIORI MOTIVATION OF $\tilde{S}(p)$ = GREEN FUNCTION

The expression $\tilde{S}(p) = \frac{i(\not{p}+m)}{(p^2-m^2+i\epsilon)}$ is a good expression because with such definition $S(x-y)$ acts like a GREEN FUNCTION under DIRAC LINEAR OPERATOR

$$(i\not{\partial}-m)\,S(x-y) = i\,\delta^{(4)}(x-y)$$

DIM

$$(i\not{\partial}-m)\,S(x-y) = \int \frac{d^4p}{(2\pi)^4}\,(i\not{\partial}-m)\,e^{-ip\cdot(x-y)}\,\tilde{S}(p) =$$

$$= \int \frac{d^4p}{(2\pi)^4}\,\underbrace{(\not{p}-m)\,\tilde{S}(p)}_{\sim i}\,e^{-ip(x-y)} = i\int \frac{d^4p}{(2\pi)^4}\,e^{-ip\cdot(x-y)} = i\,\delta^{(4)}(x-y)$$

c.v.d.

External legs

In the case of scalar fields, acting with the field operator $\hat{\phi}$ on the vacuum $|0\rangle$ to create a particle brings a factor e^{ipx} while destroying a particle brings e^{-ipx} :

$$\phi = \int \frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2\epsilon_p}}\left(a_p^+ e^{ip\cdot x} + a_p e^{-ip\cdot x}\right)$$

(a_p^+: create; a_p: destroy)

This is the origin of the factor $e^{ip_i x_i}$ for each final particle and $e^{-ip_i x_i}$ for each initial particle in the LSZ Formula for scalar fields :

$$(\cdot\cdot)(\cdot\cdot)\,\langle \vec{k}_1 \cdots \vec{k}_n\,|\,iT\,|\vec{p}_1 \cdots \vec{p}_m\rangle = \int dx_i \int dy_j\; e^{-ip_i\cdot x_i} \cdots e^{ik_j y_j}\,\langle \Omega|T\{\phi(\cdot)\cdots\}|\Omega\rangle$$

factors $e^{-ip_i x_i}$ and $e^{+ik_j y_j}$

These leads to the factor 1 for external legs in momenta space Feynman rules for scalar fields.

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From expression of ψ fields we see that for fermions together with the exponential factor we have additional factor spinor u or v when creating or destroying a particle

$$\psi = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} \sum_{s=1,2} \left(a_{ps}\, u^s(p)\, e^{-ip\cdot x} + b^{+}_{ps}\, v^s(p)\, e^{ip\cdot x} \right)$$

$a_{ps}\, u^s(p)\, e^{-ip\cdot x}$: destroying a particle fermion (initial)

$b^{+}_{ps}\, v^s(p)\, e^{ip\cdot x}$: creating a fermion particle (final)

$$\bar{\psi} = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_p}} \sum_{s=1,2} \left(b_{ps}\, \bar{v}^s(p)\, e^{-ip\cdot x} + a^{+}_{ps}\, \bar{u}^s(p)\, e^{ip\cdot x} \right)$$

$b_{ps}\, \bar{v}^s(p)\, e^{-ip\cdot x}$: destroying an (initial) Antifermion

$a^{+}_{ps}\, \bar{u}^s(p)\, e^{ip\cdot x}$: creating an antifermion particle (final)

Hence in the LSZ formula we get:

- for each initial fermion = get a factor $u^s(p)\, e^{-ip_i\cdot x}$ from $\psi\,|p\rangle$
- " initial Antifermion = " $\bar{v}^s(p)\, e^{-ip_i\cdot x}$ $\bar{\psi}\,|p\rangle$
- " final fermion = " $\bar{u}^s(p)\, e^{ip_f\cdot x}$ $\langle p|\,\bar{\psi}$
- final Antifermion = " $v^s(p)\, e^{ip_f\cdot x}$ $\langle p|\,\psi$

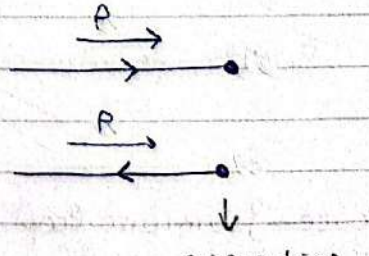
the exponential factor eliminates the poles in the LSZ formula hence give a factor one $\rightarrow$ we remain with the spinors $u^s, v^s, \bar{u}^s, \bar{v}^s$ factors !

FEYMAN RULES FOR FERMIONS EXT LEGS (8.22)

In the rules for the value of the Diagram in momenta space fo fermions, therefore, external legs doesn't count anymore as only one 1 but gain spinors

a) initial fermion $\overrightarrow{\psi\,|p,s\rangle} \Rightarrow u^s(p)$

b) initial antifermion $\overrightarrow{\bar{\psi}\,|p,s\rangle} \Rightarrow \bar{v}^s(p)$



$\rightarrow$ see quad of Lecture 1 for derivation

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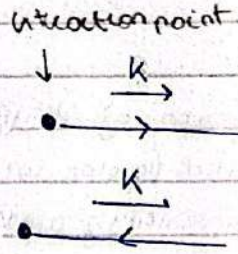


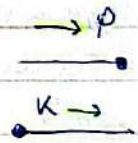
-) final fermion $\langle ks| \bar{\psi} \Rightarrow \bar{v}^s(k)$
-) final antifermion $\langle ks| \psi \Rightarrow v^s(k)$

[NOTE = a fermion is $\psi|ps\rangle$ hence is BRA is $\langle ps|\bar{\psi}$; that is why the final fermion is depicted described by $\bar{\psi}$ and not ψ. Aguelmente a final antifermion is $\langle ps|\psi$, described by fermionic ψ]

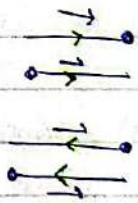
ARROWS RULES

For fermions is important to understand also from the Diagram if the particle is a fermion or antifermion , initial or final , cause it changes the value assigned. People created the arrows notation =

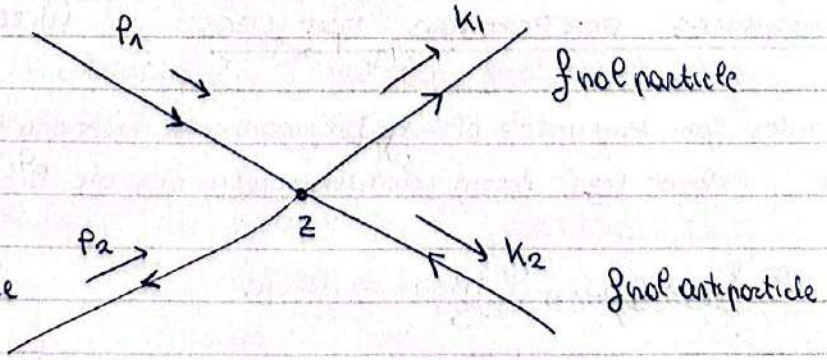
-) $\vec{p}$ of initial state is an arrow INCOMING in the point
-) $\vec{k}$ OF FINAL state is an arrow OUTGOING from the point



-) FOR FERMIONS the arrows on the line have same direction as the one of the momentum
-) FOR ANTIFERMIONS // opposite //

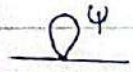


EXAMPLE =



closed fermionic loop

for any closed fermionic loop we gain a $-$ sign in Feynman rules



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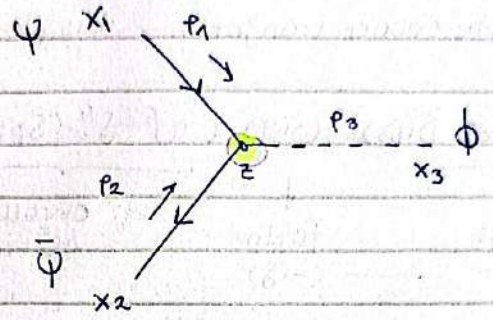
Vertex for Yukawa interaction $H_I = g\,\psi\bar{\psi}\phi$

Vertices for Yukawa interaction are 3 lines vertices with interaction of a fermion ψ, an anti fermion $\bar{\psi}$ and a scalar ϕ



$$H_I = g\,\bar{\psi}\psi\phi$$

coupling constant

At each Yukawa vertex in Feynman rules at momenta space is associated the value

$-ig$ & momentum conservation $p_1 + p_2 = p_3$ (8.23)

Quick Demonstration

a) let's compute $\langle 0|\, T\{\psi(x_1)\bar{\psi}(x_2)\phi(x_3)\, e^{-i\int d^4z\, g\,\bar{\psi}(z)\psi(z)\phi(z)}\}\,|0\rangle$ correlation function. The zero order is trivial and doesn't contribute to the M matrix in the momenta space. Let's consider directly the 1 order

a) $\langle 0|\, T\{\psi_1\bar{\psi}_2\phi_3(-ig)\int d^4z\,\psi(z)\bar{\psi}(z)\phi(z)\}\,|0\rangle$

I can contract only ψ with $\bar{\psi}$ and ϕ with ϕ so there is just one possible combination of contractions

$$\langle 0|(-ig)\,T\Big\{\int d^4z\,\psi_1\psi_2\phi_3\bar{\psi}_z\psi_z\phi_z\Big\}|0\rangle =$$

NOW with FERMIONS I NEED TO PUT ATTENTION WHEN SWAPPING FIELDS I GAIN A MINUS SIGN!

$$= \langle 0|(-ig)\,T\Big\{\int d^4z\,\psi_1\bar{\psi}_2\bar{\psi}_z\psi_z\phi_3\phi_z\Big\}|0\rangle - \#\,\psi_1\bar{\psi}_z\bar{\psi}_2\psi_z\phi_3\phi_z\,\# =$$

$-$ sign

$-$ sign

need to swap to get correct contraction $\bar{\psi}\psi$

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$$= (-ig) \int d^4z \; S(x_1 - z)\, S(z - x_2)\, D(x_3 - z)$$

where $\overbrace{\psi(x_1)\bar{\psi}(z)} = S(x_1 - z)$, $\overbrace{\psi(z)\bar{\psi}(x_2)} = S(z - x_2)$, $\overbrace{\phi(x_3)\phi(z)} = D(x_3 - z)$

NOTE = while for scalar $\overbrace{\phi_3\phi_z} = \overbrace{\phi_z\phi_3} = D(x_3 - z) = D(z - x_3)$

for fermions no! $\overbrace{\psi_z\bar{\psi}_2} = S(z - x_2) \neq S(x_2 - z) = \overbrace{\psi_2\bar{\psi}_z}$

- Now taking the Fourier transform it is possible to show that I get:

$$\tilde{S}(p_1)\,\tilde{S}(p_2)\,\tilde{D}(p_3)\,(-ig)\,(2\pi)^4\,\delta^4(p_1 + p_2 - p_3)$$

external legs — factor $(-ig)$ — overall momento conservation of the vertex

- comparing with LSZ formula the ext legs got canceled and we obtain exactly

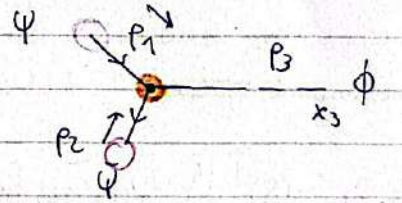
$$\langle p_3 | iT | p_1 p_2 \rangle = (-ig)\,(2\pi)^4\,\delta^4(p_1 + p_2 - p_3)\; u^s(p_1)\,\bar{v}^s(p_2)$$

vertex — momento conservation — external spinor legs

comparing to our definition of $\langle p_3 | iT | p_1 p_2 \rangle = i\mathcal{M}\,(2\pi)^4\,\delta^{(4)}(p_1 + p_2 - p_3)$ we get:

$i\mathcal{M} = (-ig)\,u^s(p_1)\,\bar{v}^s(p_2)$

vertex — ext legs

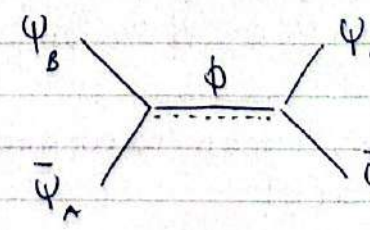


CVD!

- Usual diagram of Yukawa interaction

Usually the scalar particle is only an intermediate product

$\bar{\psi}_A \psi_B \longrightarrow \psi_C \bar{\psi}_D$



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8.7 FEYNMANN RULES FOR QED (M 5.5.4, P&S 4.8)

Photon propagator

For Spin 1 fields, they are vectors $A_\mu(x)$. The photon propagator is defined as

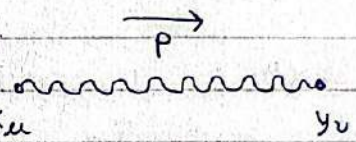
$$D_{\mu\nu}(x-y) = \langle 0|T\{A_\mu(x)A_\nu(y)\}|0\rangle = \overbrace{A_\mu(x)\,A_\nu(y)} \qquad (8.24)$$

$$D_{\mu\nu}(x-y) = \int \frac{d^4p}{(2\pi)^4} e^{-ip(x-y)} \tilde{D}_{\mu\nu}(p)$$

$$\tilde{D}_{\mu\nu}(p) = \int d^4y\, e^{ip\cdot y} D_{\mu\nu}(y)$$

$$\tilde{D}_{\mu\nu}(p) = \frac{-i\, g_{\mu\nu}}{p^2 + i\varepsilon} \qquad (8.25)$$

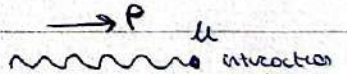
(Minkowski space)



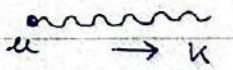
[As for scalar field $\tilde{D}_{\mu\nu}(p) = \tilde{D}_{\mu\nu}(-p)$ and $D_{\mu\nu}(x-y) = D_{\mu\nu}(y-x)$
and also, since $g_{\mu\nu} = g_{\nu\mu}$ in Minkowski space $D_{\mu\nu} = D_{\nu\mu}$]

External legs of photon

a) Initial photon $\overbrace{A_\mu}\,|\vec{p},\lambda\rangle = \epsilon_\mu(p)\lambda$



b) Final photon $\langle \vec{k},\lambda|\,\overbrace{A_\mu} = \epsilon^*_\mu(k)\lambda$



Since:

$$A_\mu(x) = \int \frac{d^3p}{(2\pi)^3\sqrt{2\omega_p}} \sum_{\lambda=0}^{3} \left(\epsilon_\mu(p,\lambda)\, a_{p,\lambda}\, e^{-ip\cdot x} + \epsilon^*_\mu(p,\lambda)\, a^\dagger_{p\lambda}\, e^{ip\cdot x}\right)$$

$a_{p,\lambda}$: destroying; $a^\dagger_{p\lambda}$: creating

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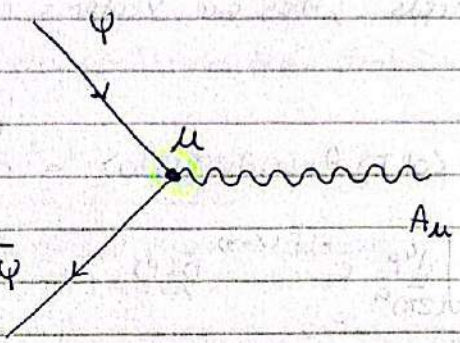
Vertex for QED interaction $H_I = \int e \bar{\psi} \gamma^\mu \psi A_\mu$

Vertices of QED interaction are 3 lines vertices with a photon and a pair of fermions

$H_I = \int e \bar{\psi} \gamma^\mu \psi A_\mu$

coupling constant



+ gives the factor:

$-i\,|Q|\,e\,\gamma^\mu$ & momenta conservation

where Q is the charge of the ψ fermionic particle ψ ($Q_{e^-} = -1$, $Q_{e^+} = +1$, $Q_{\mu^-} = -1$, $Q_{\mu^+} = +1$, $Q_U = \frac{2}{3}$ up quark...)

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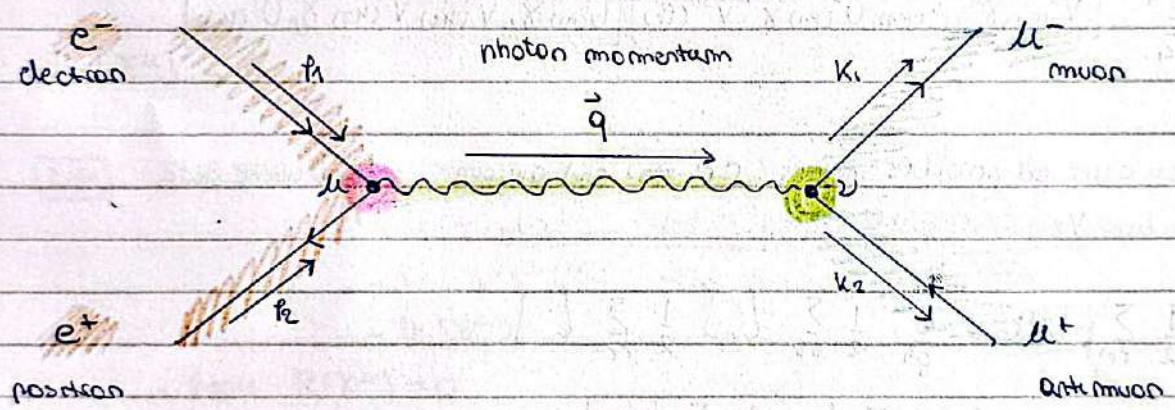
9

CROSS SECTION COMPUTATION (P.S 5.1)

9.1 S-MATRIX

Let's now compute an example explicitly: we will compute first the S matrix (via M matrix) according to the rules presented in chapter 8. Then we will present a recipe to compute the cross section.

$e^-(p_1)\ e^+(p_2) \longrightarrow \mu^-(k_1)\ \mu^+(k_2)$ electrons into muons



The rules for fermions and QED give:

$$i\mathcal{M} = \bar{v}^{s'}(p_2)(-ie\gamma^\mu)u^s(p_1)\left(\frac{-i\delta_{\mu\nu}}{q^2}\right)\bar{u}^r(k_1)(-ie\gamma^\nu)v^{r'}(k_2) \qquad (9.1)$$

$\bar{v}^{s'}(p_2)$: initial particles or antip; $(-ie\gamma^\mu)$: vertex μ; $u^s(p_1)$: initial particles or antip; $\left(\frac{-i\delta_{\mu\nu}}{q^2}\right)$: photon propagator; $\bar{u}^r(k_1)$: final particles; $(-ie\gamma^\nu)$: vertex ν; $v^{r'}(k_2)$: final particles

$$= \frac{ie^2}{q^2}\left(\bar{v}^{s'}(p_2)\gamma^\mu u^s(p_1)\right)\left(\bar{u}^r(k_1)\gamma_\mu v^{r'}(k_2)\right)$$

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- $|\mathcal{M}|^2 = \frac{e^4}{q^4} (\bar{v}^{s'} \gamma^\mu u^s)(\bar{v}^{s'} \gamma^\beta u^s)^\dagger (\bar{u}^r \gamma_\mu v^{r'})(\bar{u}^r \gamma_\beta v^{r'})^\dagger$

  ↓ new dummy index β for adjoint parts

- now use $(\bar{v} \gamma^\mu u)^\dagger = \bar{u} \gamma^\mu v$

  $(\gamma^\mu)^\dagger = \gamma^0 \gamma^\mu \gamma^0$!

  Because $(\bar{v} \gamma^\mu u)^\dagger = u^\dagger (\gamma^\mu)^\dagger \bar{v}^\dagger = u^\dagger (\gamma^\mu)^\dagger (\gamma^0)^\dagger v =$

  $= u^\dagger \gamma^0 \gamma^\mu \gamma^0 (\gamma^0)^\dagger v = u^\dagger \gamma^0 \gamma^\mu \gamma^0 \gamma^0 v =$

  $(\gamma^0)^\dagger = (\gamma^0)^3 = \gamma^0$ (since $(\gamma^0)^2 = \mathbb{1}$!) $\gamma^0 \gamma^0 = \mathbb{1}$

  $= u^r \gamma^0 \gamma^\mu v = \bar{u} \gamma^\mu v$ cvd

then

$$|\mathcal{M}|^2 = \frac{e^4}{q^4} \left[\bar{v}^s(p_2) \gamma^\mu u^{s'}(p_1) \bar{u}^{s'}(p_1) \gamma^\beta v^s(p_2) \right] \left[\bar{u}^r(k_1) \gamma_\mu v^{r'}(k_2) \bar{v}^{r'}(k_2) \gamma_\beta u^r(k_1) \right]$$

- now sum over all possible spinors $r r' s s'$ and average over all initial states

  hence introduce $1/2$ for each initial state s s'

$$\frac{1}{2}\frac{1}{2} \sum_{ss'} \sum_{rr'} |\mathcal{M}|^2 = \frac{e^4}{q^4} \frac{1}{2} \sum_{ss'} [\,\bigcirc\,] \frac{1}{2} \sum_{rr'} [\,\square\,]$$

$$\frac{1}{4} \sum_{\text{spins}} |\mathcal{M}|^2 = \frac{e^4}{q^4} \frac{1}{4} \sum_{ss'} [\,\bigcirc\,] \sum_{rr'} [\,\square\,]$$

initial spins – average final spins – just sum

- consider 1st factor ◯ and make explicit the spinor indices a, b that run over 4 spinor indices $\left(\psi = (\psi_a) = (\psi_1 \psi_2 \psi_3 \psi_4) = (\psi_{L1} \psi_{L2} \psi_{R1} \psi_{R2})\right)$

$$\sum_{s,s'} \bar{v}^{s'}_a(p_2) (\gamma^\mu)_{ab} u^s_b(p_1) \bar{u}^s_c(p_1) (\gamma^\beta)_{cd} v^{s'}_d(p_2) =$$

$$= \sum_{s'} \bar{v}^{s'}_a(p_2) v^{s'}_a(p_2) \sum_s u^s_b(p_1) \bar{u}^s_c(p_1) (\gamma^\mu)_{ab} (\gamma^\beta)_{cd} =$$

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$=$ from eq 6.23 I recall that $\sum_s U^s(p)\bar{U}^s(p) = \not{p} + m$, $\sum_s V^s(p)\bar{V}^s(p) = \not{p} - m$!

$= (\not{p}_2 - m)_{dc}\,(\not{p}_1 + m)_{ba}\,(\gamma^\mu)_{ab}\,(\gamma^\beta)_{cd} =$

$= \mathrm{Tr}\left[\gamma^\mu(\not{p}_1 + m)\gamma^\beta(\not{p}_2 - m)\right]$

- Repeating the same for 2. factor we obtain

$$\frac{1}{4}\sum_{\text{spins}}|\mathcal{M}|^2 = \frac{e^4}{4q^4}\,\mathrm{Tr}\left[\gamma^\mu(\not{p}_1 + m_e)\gamma^\nu(\not{p}_1' - m_e)\right]\mathrm{Tr}\left[\gamma_\mu(\not{p}_2' - m_\mu)\gamma_\nu(\not{p}_2 + m_\mu)\right]$$

so now we want to compute these traces

ASIDE = TRACES OF γ-MATRICES

- $\mathrm{Tr}[\gamma^\mu] = 0$ — one γ^μ matrix

DIM $\mathrm{Tr}(\gamma^\mu) = \mathrm{Tr}(\gamma^5\gamma^5\gamma^\mu)$ as $(\gamma^5)^2 = \mathbb{1}$

$= -\mathrm{Tr}(\gamma^5\gamma^\mu\gamma^5)$ as $\{\gamma^\mu, \gamma^5\} = 0$ anticommute

$= -\mathrm{Tr}(\gamma^5\gamma^5\gamma^\mu)$ for ciclicity of the trace

$= -\mathrm{Tr}(\gamma^\mu)$

→ hence $\mathrm{Tr}(\gamma^\mu) = 0$ CVD!

Easy generalization for an odd number of γ matrices leads to

- $\mathrm{Tr}[\gamma^{\mu_1}\gamma^{\mu_2}\dots\gamma^{\mu_{2n+1}}] = 0$ — odd γ^μ matrices

- $\mathrm{Tr}[\gamma^\mu\gamma^\nu] = 4g^{\mu\nu}$ — two $\gamma^\mu\gamma^\nu$ matrices

DIM $\mathrm{Tr}[\gamma^\mu\gamma^\nu] = \mathrm{Tr}[2g^{\mu\nu}\mathbb{1} - \gamma^\nu\gamma^\mu] =$ as $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}\mathbb{1}$

$= 8g^{\mu\nu} - \mathrm{Tr}[\gamma^\nu\gamma^\mu] = 8g^{\mu\nu} - \mathrm{Tr}[\gamma^\mu\gamma^\nu]$

→ $2\,\mathrm{Tr}[\gamma^\mu\gamma^\nu] = 8g^{\mu\nu}$ ⟶ $\mathrm{Tr}[\gamma^\mu\gamma^\nu] = 4g^{\mu\nu}$ CVD

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- For an EVEN number n of γ matrices, the trace reduces to the sum of the traces of n-2 matrices by anticommuting 1st γ matrix to last place & using ciclicity of the trace.

For 4 γ matrices:

$$Tr[\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma] = 4[g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho}]$$

FOUR γ MATRICES

- $g^{\mu\nu}g^{\rho\sigma}$: initial order
- $-g^{\mu\rho}g^{\nu\sigma}$: minus sign swapping $\nu \to \rho$
- $+g^{\mu\sigma}g^{\nu\rho}$: double minus hence plus swapping $\sigma \leftrightarrow \nu$, $\nu \leftrightarrow \sigma$

TRACES WITH γ^5

- $Tr[\gamma^5] = 0$ with $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$

DIM $Tr(\gamma^5) = Tr[\gamma^0\gamma^0\gamma^5]$ as $\gamma^0\gamma^0 = \mathbb{1}$
$= -Tr[\gamma^0\gamma^5\gamma^0]$ as $\{\gamma^5, \gamma^0\} = \emptyset$ anticomute
$= -Tr[\gamma^0\gamma^0\gamma^5]$ ciclicity
$= -Tr(\gamma^5)$
$\to Tr(\gamma^5) = 0$ QED

- $Tr[(\text{odd n° of } \gamma \text{ matrices})\gamma^5] = 0$

DIM simply because γ^5 is a product of four γ-matrices hence 4 + odd is still an odd number and $Tr[\text{ODD n° of } \gamma] = 0$

- $Tr[\gamma^\mu\gamma^\nu\gamma^5] = 0$

DIM $Tr[\gamma^\mu\gamma^\nu\gamma^5] = Tr[\gamma^\mu\gamma^\nu(\gamma^\sigma)^2\gamma^5] =$ since $(\gamma^\sigma)^2 = \mathbb{1}$
$= -Tr[\gamma^\mu\gamma^\nu\gamma^\sigma\gamma^5\gamma^\sigma] =$ since $\{\gamma^5, \gamma^\sigma\} = 0$
$= -Tr[\gamma^\sigma\gamma^\mu\gamma^\nu\gamma^\sigma\gamma^5] =$ ciclicity

now I impose that $\sigma \neq \mu$ and $\sigma \neq \nu$ in order to use anti-comutation $\{\gamma^\mu, \gamma^\sigma\} = 0$ otherwise if $\mu = \sigma$ $\{\gamma^\mu\gamma^\mu\} = 2\mathbb{1}$

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$= Tr[\gamma^\mu \gamma^\sigma \gamma^\nu \gamma^\sigma \gamma^5]$ = since $\{\gamma^\sigma \gamma^\mu\} = 0$

$= -Tr[\gamma^\mu \gamma^\nu (\gamma^\sigma)^2 \gamma^5]$ since $\{\gamma^\sigma, \gamma^\nu\} = 0$

$= -Tr[\gamma^\mu \gamma^\nu \gamma^5]$

$\longrightarrow Tr[\gamma^\mu \gamma^\nu \gamma^5] = 0$ cvd

c) $Tr[\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma \gamma^5] = \begin{cases} 0 & \text{if two indices are the same (ie } \mu=\nu \text{ or } \mu=\sigma \text{ or } \rho=\sigma \ldots) \\ \neq \phi & \text{if } \mu \neq \nu \neq \rho \neq \sigma \end{cases}$

DIM if two indices are the same ie $\mu = \nu$ we obtain $Tr[(\gamma^\mu)^2 \gamma^\rho \gamma^\sigma \gamma^5]$
$= Tr[\gamma^\rho \gamma^\sigma \gamma^5] = 0$ from previous relation

if $\mu \neq \nu \neq \rho \neq \sigma$ we cannot insert another $(\gamma^\alpha)^2$ with α different from all indeces μ, ν, ρ, σ cause we have only four possibilities 0, 1, 2, 3 and all already taken in $\mu \neq \nu \neq \rho \neq \sigma$.
Then following the passages in the previous demonstration we cannot prove anymore that such trace $Tr[\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma \gamma^5]$ is zero always!

CONTRACTIONS OF γ-MATRICES

$\gamma \cdot \gamma = \gamma^\mu \gamma_\mu = 4\mathbb{1}$

DIM $\gamma^\mu \gamma_\mu = g_{\mu\nu} \gamma^\mu \gamma^\nu = g_{\mu\nu} \frac{1}{2} \{\gamma^\mu \gamma^\nu\} = g_{\mu\nu} g^{\mu\nu} \mathbb{1} = 4\mathbb{1}$

$\gamma^\mu \gamma^\alpha \gamma_\mu = -2\gamma^\alpha$

DIM $\gamma^\mu \gamma^\alpha \gamma_\mu = \gamma^\mu (2g^\alpha{}_\mu - \gamma_\mu \gamma^\alpha) = 2\gamma^\alpha - 4\gamma^\alpha = -2\gamma^\alpha$

$\gamma^\mu \gamma^\alpha \gamma^\beta \gamma_\mu = 4 g^{\alpha\beta} \mathbb{1}$

DIM $\gamma^\mu \gamma^\alpha \gamma^\beta \gamma_\mu = \gamma^\mu \gamma^\alpha (2g^\beta{}_\mu - \gamma_\mu \gamma^\beta) = 2\gamma^\beta \gamma^\alpha - \gamma^\mu (2g^\alpha{}_\mu) \gamma^\beta + \underbrace{\gamma^\mu \gamma_\mu}_{4} \gamma^\alpha \gamma^\beta =$
$= 2\gamma^\beta \gamma^\alpha - 2\gamma^\alpha \gamma^\beta + 4\gamma^{\alpha\beta} = 2(\gamma^\alpha \gamma^\beta + \gamma^\beta \gamma^\alpha) = 2\{\gamma^\alpha, \gamma^\beta\} = 2g^{\alpha\beta}\mathbb{1}$

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- With all that showed for traces of γ matrices we can now compute

$$\sum_{\text{spin}} |\mathcal{M}|^2 = \frac{e^4}{4q^4} \mathrm{Tr}\left[\gamma^\mu (\not{p}_1 + m_e)\gamma^\nu (\not{p}_2 - m_e)\right] \mathrm{Tr}\left[\gamma_\mu (\not{k}_2 - m_\mu)\gamma_\nu (\not{k}_1 + m_\mu)\right]$$

$$= \frac{e^4}{4q^4} \mathrm{Tr}\Big[\gamma^\mu \not{p}_1 \gamma^\nu \not{p}_2 + \gamma^\mu m_e \gamma^\nu \not{p}_2 - \gamma^\mu \not{p}_1 \gamma^\nu m_e - m_e^2 \gamma^\mu \gamma^\nu \Big] \cdot$$

($\gamma^\mu m_e \gamma^\nu \not{p}_2 = 0$, $\gamma^\mu \not{p}_1 \gamma^\nu m_e = 0$)

$$\cdot \mathrm{Tr}\left[\gamma_\mu \not{k}_2 \gamma_\nu \not{k}_1 - \gamma_\mu m_\mu \gamma_\nu \not{k}_1 + \gamma_\mu \not{k}_2 \gamma_\nu m_\mu - m_\mu^2 \gamma_\mu \gamma_\nu\right]$$

($\gamma_\mu m_\mu \gamma_\nu \not{k}_1 = 0$, $\gamma_\mu \not{k}_2 \gamma_\nu m_\mu = 0$)

- terms proportional to m_μ or m_e are zero cause $\mathrm{Tr}[\not{a}\gamma^\mu\gamma^\nu] = \emptyset$ hence cause is an odd n° of γ inside the trace since $\not{p} = p_\alpha \gamma^\alpha$ with p_α a number and γ^α a matrix

$$\cdot \sum_{\text{spins}} |\mathcal{M}|^2 = \frac{e^4}{4q^4} \left(\mathrm{Tr}\left[\gamma^\mu \not{p}_1 \gamma^\nu \not{p}_2\right] - m_e^2 \mathrm{Tr}\left[\gamma^\mu \gamma^\nu\right]\right) \cdot$$

$$\cdot \left(\mathrm{Tr}\left[\gamma_\mu \not{k}_2 \gamma_\nu \not{k}_1\right] - m_\mu^2 \mathrm{Tr}\left[\gamma_\mu \gamma_\nu\right]\right) =$$

- using $\mathrm{Tr}[\gamma_\mu \gamma_\nu] = 4 g_{\mu\nu} \mathbb{1}$

$$= \frac{e^4}{4q^4} \left(\mathrm{Tr}\left[\gamma^\mu \not{p}_1 \gamma^\nu \not{p}_2\right] - m_e^2 4 g^{\mu\nu}\right) \cdot$$

$$\cdot \left(\mathrm{Tr}\left[\gamma_\mu \not{k}_2 \gamma_\nu \not{k}_1\right] - m_\mu^2 4 g_{\mu\nu}\right)$$

(labels for muons! not index of spacetime)

- using $\not{p} = \gamma^\alpha p_\alpha$ where p_α is a number and γ^α is a matrix, we can then bring p_α out of the trace obtaining

$$= \frac{e^4}{4q^4} \left(\mathrm{Tr}\left[\gamma^\mu \gamma^\alpha \gamma^\nu \gamma^\beta\right] p_{1\alpha} p_{2\beta} - m_e^2 4 g^{\mu\nu}\right) \cdot$$

$$\cdot \left(\mathrm{Tr}\left[\gamma_\mu \gamma_\alpha \gamma_\nu \gamma_\beta\right] k_2^\alpha k_1^\beta - m_\mu^2 4 g_{\mu\nu}\right)$$

- using $\mathrm{Tr}[\gamma^\mu\gamma^\alpha\gamma^\nu\gamma^\beta] = 4 g^{\mu\alpha} g^{\nu\beta} - g^{\mu\nu} g^{\alpha\beta} + g^{\mu\beta} g^{\alpha\nu}$ we get

$$= \frac{e^4}{4q^4} \left(4\left(g^{\mu\alpha} g^{\nu\beta} - g^{\mu\nu} g^{\alpha\beta} + g^{\mu\beta} g^{\alpha\nu}\right) p_{1\alpha} p_{2\beta} - m_e^2 4 g^{\mu\nu}\right) \cdot$$

$$\cdot \left(4\left(g_{\mu\alpha} g_{\nu\beta} - g_{\mu\nu} g_{\alpha\beta} + g_{\mu\beta} g_{\alpha\nu}\right) k_2^\alpha k_1^\beta - m_\mu^2 4 g_{\mu\nu}\right) =$$

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$$= \frac{e^4}{4q^2}\left[4(p_1^\mu p_e^\nu + p_1^\nu p_e^\mu) - 4g^{\mu\nu}(p_1\cdot p_2 + m_e^2)\right]\cdot$$
$$\cdot\left[4(k_{2\mu}k_{2\nu} + k_{2\nu}k_{2\nu}) - 4g_{\mu\nu}(k_1\cdot k_2 + m_\mu^2)\right]$$

• HIGH ENERGY APPROXIMATION $m_e \doteq 0$

The energy has to be larger than m_μ of the muon in order to create a muon! The interaction $e^+e^- \to \mu^+\mu^-$ occurs only if the interaction energy is larger than m_μ ... Hence since $m_e \ll m_\mu \ll$ Energy we can approximate $m_e \sim 0$ (Because in the final formula we will obtain terms as $\frac{m_e}{E} \to 0$ going to zero so we can simply put $m_e = 0$ from now)

$$\frac{1}{4}\sum_{spins}|\mathcal{M}|^2 = \frac{8e^4}{q^4}\left[(p_1\cdot k_1)(p_2\cdot k_2) + (p_2\cdot k_1)(p_1\cdot k_2) + m_\mu^2(p_1\cdot p_2)\right] \qquad (9.2)$$

$|\mathcal{M}|^2$ WITH MANDELSTAM INVARIANTS

- Define mandelstam invariants (Frame invariant):

$$s := (p_1+p_2)^2 = 2p_1\cdot p_2 = (k_1+k_2)^2 = 2k_1\cdot k_2 + 2m_\mu^2$$
$$t = (p_1-k_1)^2 = -2p_1\cdot k_1 + m_\mu^2 = (p_2-k_2)^2 = -2p_2\cdot k_2 + m_\mu^2$$
$$u = (p_1-k_2)^2 = -2p_1\cdot k_2 + m_\mu^2 = (p_2-k_1)^2 = -2p_2\cdot k_1 + m_\mu^2$$

noting that

$$s + t + u = \underbrace{2p_1\cdot(p_2-k_1-k_2)}_{p_2 \text{ for momento conservation}} + 2m_\mu^2 = \overbrace{2m_\mu^2}^{\text{as } p_1^2 = m_e^2 \doteq 0 \text{ high energy}}$$

in general $s+t+u = \sum_j m_j^2$ for external masses

- Hence in the limit $m_\mu \to 0$ we obtain

$$\frac{1}{4}\sum_{spins}|\mathcal{M}|^2 = \frac{8e^4}{s^2}\left[\left(\frac{t}{2}\right)^2 + \left(\frac{u}{2}\right)^2\right] \qquad \text{with } s = q^2 = (p_1+p_2)^2$$

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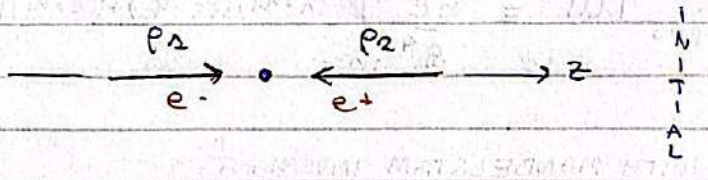
CENTRE OF MASS FRAME

Since $|\mathcal{M}|^2$ is frame independent we can choose to work in one frame of choice that the value of $|\mathcal{M}|^2$ is always the same. We choose to work in centre of mass frame

> ! NOTE that the expression of $|\mathcal{M}|^2$ CHANGES from frame to frame! the value is the same but the expression in terms of energies and momenta changes because energies and momenta change from frame to frame!

- In the centre of mass frame, in High energy approximation $m_e \overset{!}{=} 0$ the momenta are for initial massless electrons:

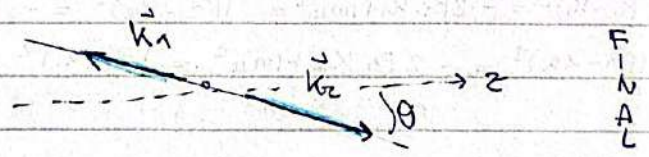
$p_1 := E(1\,0\,0\,1)$
$p_2 := E(1\,0\,0\,-1)$



nel centro di massa sist di rif, le due particelle incidenti hanno stesso impulso ma verso opposto, decido di orientare l'asse z come asse di incidenza

- using generic momenta for muons (massive particles $m_\mu \neq 0$)

$k_1 = (E_{k_1}, \vec{k}_1)$
$k_2 = (E_{k_2}, \vec{k}_2)$



Imposing conservation of energy and 3-momenta from final to initial states:

$\vec{p}_1 + \vec{p}_2 \overset{!}{=} \vec{k}_1 + \vec{k}_2$

$0 = E - E \overset{!}{=} \vec{k}_1 + \vec{k}_2 \longrightarrow \vec{k}_1 = -\vec{k}_2 = -\vec{k}$

$2E \overset{!}{=} E_{k_1} + E_{k_2}$ → since $E_{k_1} = E_{k_2}$ since $|k_1| = |k_2| \longrightarrow E_{k_1} = E_{k_2} = E$

hence

$k_1 = (E, \vec{k})$
$k_2 = (E, -\vec{k})$

note that the direction of spatial final $\vec{k}$ may not be the same as of initial particles!

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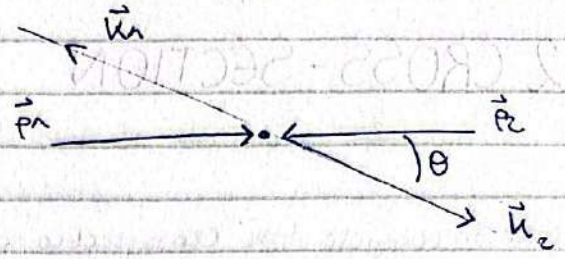
- Hence



$$p_1 \cdot p_2 = 2E^2$$

$$p_1 \cdot k_1 = p_2 \cdot k_2 = E^2 - E|\vec{k}|\cos\theta$$

$$p_1 \cdot k_2 = p_2 \cdot k_1 = E^2 + E|\vec{k}|\cos\theta$$

Putting all these relations in (9.2) we obtain our final operative formula for matrix element $|M|^2$ in the centre of mass frame

$$\frac{1}{4}\sum_{\text{spins}} |M|^2 = e^4 \left[\left(1 + \frac{m_\mu^2}{E^2}\right) + \left(1 - \frac{m_\mu^2}{E^2}\right)\cos^2\theta \right] \qquad (9.3)$$

with $q^2 = (p_1 + p_2)^2 = p_1 \cdot p_2$ (= 5 mandelstam invariant)

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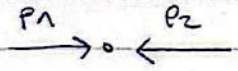
9.2 CROSS-SECTION

We want to compute the cross-section of a reaction $p_1 p_2 \dots p_n \to k_1 k_2 \dots k_m$ ie the "probability" of such reaction to occur

DIFFERENTIAL CROSS SECTION

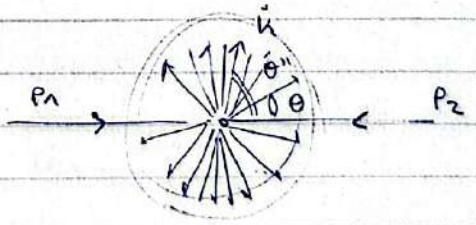
$$\frac{d\sigma}{d\Omega} = (\text{FLUX FACTOR})\ (\text{PHASE FACTOR})\ \sum_{s}\overline{|\mathcal{M}|^2}$$

Accounts for initial kinematics which is FIXED since we control the initial particles we prepare to collide



Initial = FIXED

ACCOUNTS for FINAL PHASE DISTRIBUTION: the matrix $|\mathcal{M}|^2$ depends on a fixed specific $\vec{k}$ and $\vec{p}$ with an angle θ specific. However the MEASURABLE cross section is a MEAN over all phase θ distribution of the outgoing final $\vec{k}$ particles = we DON'T KNOW with which θ phase the final $\vec{k}$ particles will spawn, however we know some θ are more likely to occur than other = taking a mean over this phase distribution is what the phase factor does



Final = MEAN over all phases θ

WHOLE CROSS SECTION

$$\sigma = \int d\Omega \frac{d\sigma}{d\Omega}$$ integration over all solid angle $d\Omega$

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FLUX FACTOR

Flux factor depends only on the initial particle kinematics (hence it depends on which FRAME we are considering)

FOR 2 initial and 2 final states in the CENTRE OF MASS FRAME

$$\text{Flux factor} = \frac{1}{2E_{CM}^2}$$

↳ Energy of centre of mass

see PdS pag 102

We will use only that in the exam, it is not required to compute flux factor

PHASE FACTOR

To compute phase factor we compute as follows:

$$PS^{(n)} = \prod_{j=1}^{n} \int \frac{d^4k_j}{(2\pi)^4} \theta(E_{kj}) (2\pi)\delta(k_j^2 - m_j^2) (2\pi)^4 \delta^{(4)}(P - k_1 - k_2 \dots - k_n)$$

- $PS^{(n)}$: # of final particles
- $\frac{d^4k_j}{(2\pi)^4}$: integral of all final momenta
- $\theta(E_{kj})$: Heaviside function emphasys to be positive orthochronus Lorentz transformations
- $(2\pi)\delta(k_j^2 - m_j^2)$: request for final particle to be on shell - means final i particles are real
- $(2\pi)^4 \delta^{(4)}(P - k_1 - k_2 \dots - k_n)$: overall 4 momenta conservation
  - $P = p_1 + p_2 + \dots$ sum of initial momenta

Going through calculations we rewrite the PS as

$$PS^{(n)} = \int d\mathcal{R} \quad (\text{PHASE FACTOR})$$

↓

this is the phase factor to insert in the cross section

PS is the integration over all Lorentz invariant phase space (LIPS) for each final state particle, ensuring conservation of 4 momentum

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PHASE FACTOR FOR TWO FINAL PARTICLES $e^-e^+ \to \mu^-\mu^+$

- Let's compute all passages for a $PS^{(2)}$, that is what we will mainly use.

$$PS^{(2)} = \int \frac{d^4k_1}{(2\pi)^4} \int \frac{d^4k_2}{(2\pi)^4} \theta(E_{k1})\theta(E_{k2})\, \delta(k_1^2 - m_1^2)(2\pi)\, \delta(k_2^2 - m_2^2)(2\pi)\, \delta^{(4)}(p - k_1 - k_2)(2\pi)^4$$

where $p = p_1 + p_2$

- Let's consider the centre of mass frame for the example we had before Hence for two initial massless particles.

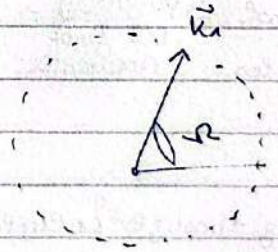
$p_1 = E\,(1\,0\,0\,1)$
$p_2 = E\,(1\,0\,0\,-1)$
$\Big\} \to p = p_1 + p_2 = 2E(1000) = E_{cm}(1000)$ with $E_{cm} = 2E$

then resolving the $\delta^4(p - k_1 - k_2)$ I put $E_{k2} = E_{cm} - E_{k1}$ and $k_2 = p - k_1$ (since $\delta^{(4)}(p - k_1 - k_2) = \delta(E_{cm} - E_{k1} - E_{k2})\,\delta^{(3)}(\vec{p} - \vec{k_1} - \vec{k_2})$ Since $\vec{p_1} + \vec{p_2} = 0$ in the spatial $\vec{p}$)

k_2: total 4 vector! 1 with also E_{k2}, not only spatial!

$$PS^{(2)} = \int \frac{d^4k_1}{(2\pi)^4} \theta(E_{k1})\,\theta(E_{cm} - E_{k1})\,\delta(k_1^2 - m_1^2)(2\pi)\,\delta((p - k_1)^2 - m_2^2)(2\pi)$$

- $\int d^4k_1 = \int dE_{k1} \int d^3\vec{k}_1 = \int dE_{k1} \int d\Omega \int d|\vec{k}_1|\,|\vec{k}_1|^2$



as $\int d^3\vec{x} = \int d\Omega \int dr\, r^2$ with r = radius

Hence

$$PS^{(2)} = \int \frac{dE_{k1}\, d\Omega\, d|\vec{k}_1|\,|\vec{k}_1|^2}{(2\pi)^2}\, \theta(E_{k1})\,\theta(E_{cm} - E_{k1})\,\delta(E_{k1}^2 - |\vec{k}_1|^2 - m_1^2)\,\delta((E_{cm} - E_{k1})^2 - |\vec{k}_1|^2 - m_2^2)$$

$k_1^2 = k_1^\mu k_{1\mu} = E_{k1}^2 - |\vec{k}_1|^2$

$(p - k_1)^2 = (E_{cm} - E_{k1})^2 - |\vec{k}_1|^2$

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- use $\delta(x^2-a^2) = \frac{1}{2|a|}\left(\delta(x-a)+\delta(x+a)\right)$

but with $a^2 = E_{ui}^2 - m_i^2$ and $x^2 = |\vec{k}_i|^2$. Hence

$$\delta(|\vec{k}_i|^2 - (E_{ui}^2 - m_i^2)) = \frac{1}{2\sqrt{E_{ui}^2 - m_i^2}}\left[\delta(|\vec{k}_i| - \sqrt{E_{ui}^2 - m_i^2}) + \delta(|\vec{k}_i| + \sqrt{E_{ui}^2 - m_i^2})\right]$$

Since $|\vec{k}_i|$ always positive cannot be equal to $-\sqrt{\dots}$

then

$$|\vec{k}_i| = \sqrt{E_i^2 - m_i^2}$$

$$PS^{(2)} = \int d|\vec{k}_i|\, d\Omega\, dE_{ui}\, \theta(E_{ui})\theta(E_{cm}-E_{ui})\, |\vec{k}_i|^2\, \delta(|\vec{k}_i| - \sqrt{\ }) \delta(\ ) =$$

$$= \int \frac{d\Omega}{(2\pi)^2}\, dE_{ui}\, \theta(E_{ui})\theta(E_{cm}-E_{ui})\, \frac{\sqrt{E_{ui}^2 - m_i^2}}{2}\, \delta(E_{cm}^2 - 2E_{ui}E_{cm} + m_1^2 - m_2^2) =$$

- $\frac{\delta(x-a)}{a} = \delta(xa)$ Hence

$$= \int \frac{d\Omega}{(2\pi)^2}\, dE_{ui}\, \theta(E_{ui})\theta(E_{cm}-E_{ui})\, \frac{\sqrt{E_{ui}^2 - m_i^2}}{2}\, \frac{\delta\left(E_{ui} - \frac{E_{cm}}{2} - \frac{(m_1^2-m_2^2)}{2E_{cm}}\right)}{2E_{cm}} =$$

using $\theta(E_{ui})\,\theta(E_{cm}-E_{ui})$:

$$= \int \frac{d\Omega}{(2\pi)^2} \int_0^{E_{cm}} dE_{ui}\, \frac{1}{2E_{cm}}\, \delta\left(E_{ui} - \frac{E_{cm}}{2} - \frac{(m_1^2-m_2^2)}{2E_{cm}}\right) =$$

$$= \int \frac{d\Omega}{(2\pi)^2}\, \frac{1}{4E_{cm}} \left[\left(\frac{E_{cm}}{2} + \frac{(m_1^2-m_2^2)}{2E_{cm}}\right)^2 - m_1\right]^{1/2} =$$

- $\square =: \frac{1}{4E_{cm}^2}\lambda(E_{cm}^2, m_1^2, m_2^2)$ where $\lambda(a,b,c) := (a-b-c)^2 - 4bc$

$$\square^{1/2} = \frac{1}{2E_{cm}}\lambda^{1/2}(E_{cm}^2, m_1^2 m_2^2) \quad \text{Hence}$$

$$= \int \frac{d\Omega}{(2\pi)^2}\, \frac{1}{8E_{cm}^2}\, \lambda^{1/2}(E_{cm}^2, m_1^2, m_2^2)$$

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- In our example $m_1 = m_2 = m_\mu$ $E_{CM} = 2E$ hence

$$\lambda^{1/2}(E_{CM}^2, m_\mu^2, m_\mu^2) = E_{CM}^2 \sqrt{\left(1 - \frac{4m_\mu^2}{E_{CM}^2}\right)}$$

$$\frac{\lambda^{1/2}(\dots)}{8E_{CM}^2} = \frac{1}{8}\sqrt{\left(1 - \frac{4m_\mu^2}{E_{CM}^2}\right)} = \frac{1}{8}\sqrt{\left(1 - \frac{m_\mu^2}{E^2}\right)}$$

then

$$PS^{(2)} = \int \frac{d\Omega}{32\pi^2}\sqrt{1 - \frac{m_\mu^2}{E^2}}$$

$\Rightarrow$ FOR $e\bar{e}^+ \to \mu\bar{\mu}\mu^+$ phase factor is $= \frac{1}{32\pi^2}\sqrt{1 - \frac{m_\mu}{E^2}}$
for E in the centre of mass, massless electrons

CROSS SECTION FOR $e^-e^+ \to \mu^-\mu^+$

Hence we can write the differential cross section for $e\bar{e}^- \to \mu\bar{\mu}^+$ reaction

$$\frac{d\sigma}{d\Omega} = \frac{1}{2E_{CM}^2}\frac{1}{32\pi^2}\sqrt{1 - \frac{m_\mu^2}{E^2}}\sum_{spins}\frac{|\mathcal{M}|^2}{4} \qquad \text{with } E_{CM} = 2E$$

introducing eq (9.13) for $\sum \frac{|\mathcal{M}|^2}{4}$ we get:

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4E_{CM}^2}\sqrt{1 - \frac{m_\mu^2}{E^2}}\left[\left(1 + \frac{m_\mu^2}{E^2}\right) + \left(1 - \frac{m_\mu^2}{E^2}\right)\cos^2\theta\right] \qquad \alpha = \frac{e^2}{4\pi}$$

integrating over all sphere $d\Omega$ → CROSS SECTION OF $ee \to \mu\mu$ in the centre of mass and for $m_e \neq 0$

$$\sigma = \frac{4\pi\alpha^2}{3E_{CM}^2}\sqrt{1 - \frac{m_\mu^2}{E^2}}\left(1 + \frac{1}{2}\frac{m_\mu^2}{E^2}\right) \longrightarrow$$

$$\int d\Omega = \int_0^\pi d\theta \sin\theta \int_0^{2\pi} d\varphi = \int_{-1}^{1} d\cos\theta \int_0^{2\pi} d\varphi = 4\pi$$

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HIGH ENERGY LIMIT

When $E \gg m_\mu$ hence

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4E_{CM}^2}(1+\cos^2\theta) \longrightarrow$$ CROSS SECTION OF $e e \to \mu\mu$ in high energy limit $E \gg m_\mu \gg m_e$ in the centre of mass frame

$$\sigma = \frac{4\pi\alpha^2}{3E_{CM}^2} \longrightarrow$$ with $\alpha = \frac{e^2}{4\pi}$

9.3 CROSSING-SYMMETRY (P&S 5.4)

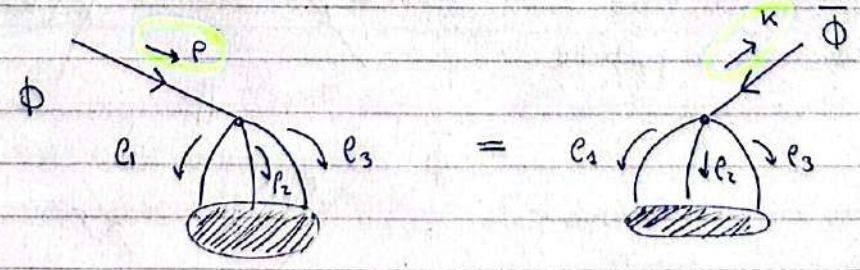
S matrix for process with initial particle of momentum p ($\phi(p)$) is equal to the S-Matrix for same process but for a final antiparticle of momentum $k = -p$ ($\bar{\phi}(k)$) (up to a phase) :

$$|\mathcal{M}(\phi(p) + \dots \to \dots)| = |\mathcal{M}(\dots \to \dots + \bar{\phi}(k=-p))|\,(-1)$$ ← $e e \to \mu\mu$ in fermion

IF $\psi(p)$ is a fermion, we gain a -1 sign when summing over all spin polarization :

$$\sum_{\text{spins}} |\mathcal{M}(\psi(p) + \dots \to \dots)|^2 = \sum_{\text{spins}} |\mathcal{M}(\dots \to \dots + \bar{\psi}(k=-p))|^2 (-1)^2$$

DIAGRAMATICALLY



$$p = e_1 + e_2 + e_3 \qquad -k = e_1 + e_2 + e_3$$

$$|\mathcal{M}|_p = (-1)|\mathcal{M}_k|$$

$$|\mathcal{M}_p|^2 = |\mathcal{M}_k|^2$$

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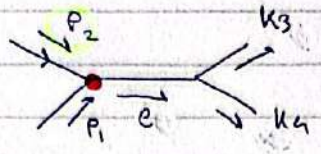
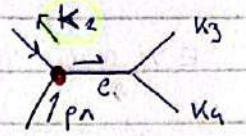
MOMENTA CONSERVATION RULES

Left to right connection



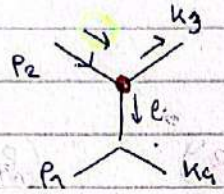
$p_2 + p_1 = q$



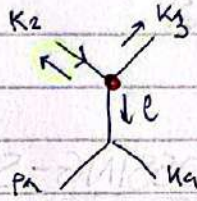
$p_1 - k_2 = q$

assign + sign to momenta flowing to the right and − to momenta flowing backwards to the left

Up to down connection



$p_2 - k_3 = q$

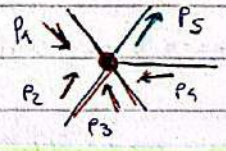


$-k_2 - k_3 = q$

assign + sign to momenta flowing down and − to momenta flowing up

More concisely = look at a node, assign + to entering momenta and − to exiting momenta and equal the sum to zero

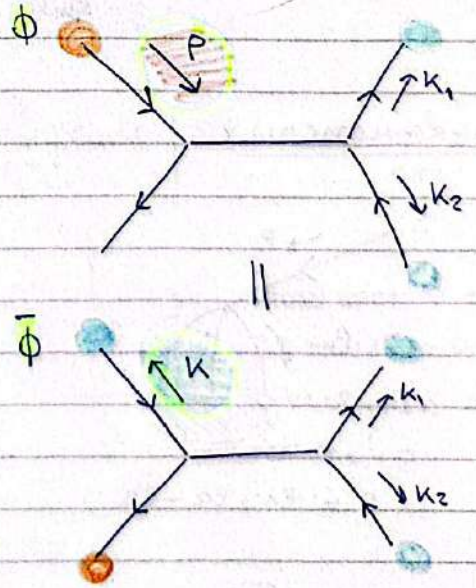
$$p_1 + p_2 + p_3 + p_4 - p_5 = 0$$



DIAGRAMMATICALLY

Mantieni tutto il diagramma uguale, solamente scambi $p \rightarrow$ con $k \leftarrow$ nella direzione opposta. Così trasformi automaticamente la particella in una antiparticella e viceversa.

Nel conto di $|M|$ basta considerare il campo scambiato come finale e non più iniziale (come con k appunto e non con p) con l'eventuale -1 dei fermioni già a posto!



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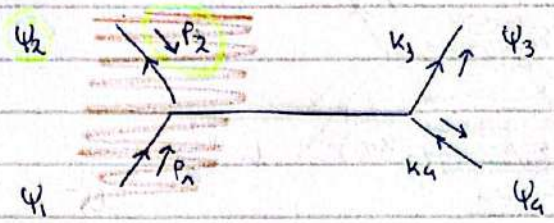
EXAMPLE

ψ_2

$\mu + \bar{\nu}_\mu \rightarrow e^- + \bar{\nu}_e$

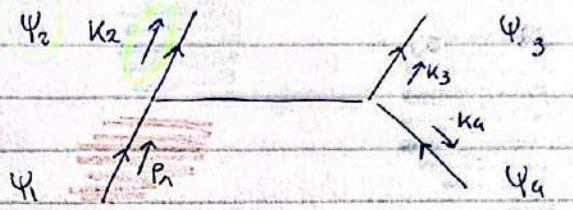


has same matrix element of:

$p_1 + p_2 = k_3 + k_4$

$\bar{\psi}_2$

$\mu \rightarrow e^- + \bar{\nu}_e + \nu_\mu$



if $p_2 = -k_2$

if $p_{\bar{\nu}_\mu} = -k_{\nu_\mu}$

$p_1 - k_2 = k_3 + k_4$

REASON OF (-1) FOR FERMIONS

For scalar the propagators $\tilde{D}(p) = \tilde{D}(-p)$ hence if $k = -p$ $\tilde{D}(p) = \tilde{D}(k)$

For fermions no! $\tilde{S}(p) \neq \tilde{S}(-p)$ and neither $\tilde{S}(p) = -\tilde{S}(-p)$. The reason of the (-1) occurring in the formula is that for external leg factor:

$$\sum_{spins} u^s(p)\bar{u}^s(p) = \not{p} + m = -\not{k} + m = -(\not{k} - m) = -\sum_{spins} \bar{v}^s(k) v^s(k)$$

$p = -k$

u for Fermion — v for antifermion

EXAMPLE - FROM $e^- e^+ \rightarrow \mu^- \mu^+$ TO $e^- \mu^- \rightarrow \mu^- e^-$

We already computed $\Sigma |\mathcal{M}|^2$ for $e^-(p_1)\, e^+(p_2) \rightarrow \mu^-(k_1)\, \mu^+(k_2)$

Note that I can swap initial

$e^+(p_2)$ into final $\rightarrow e^-(k_2 = -p_2)$ new

and final

$\mu^+(k_2)$ into initial $\rightarrow \mu^-(p_2 = -k_2)$ new

$e^-(p_1)\, \mu^-(p_2) \rightarrow \mu^-(k_1)\, e^-(k_2)$ new

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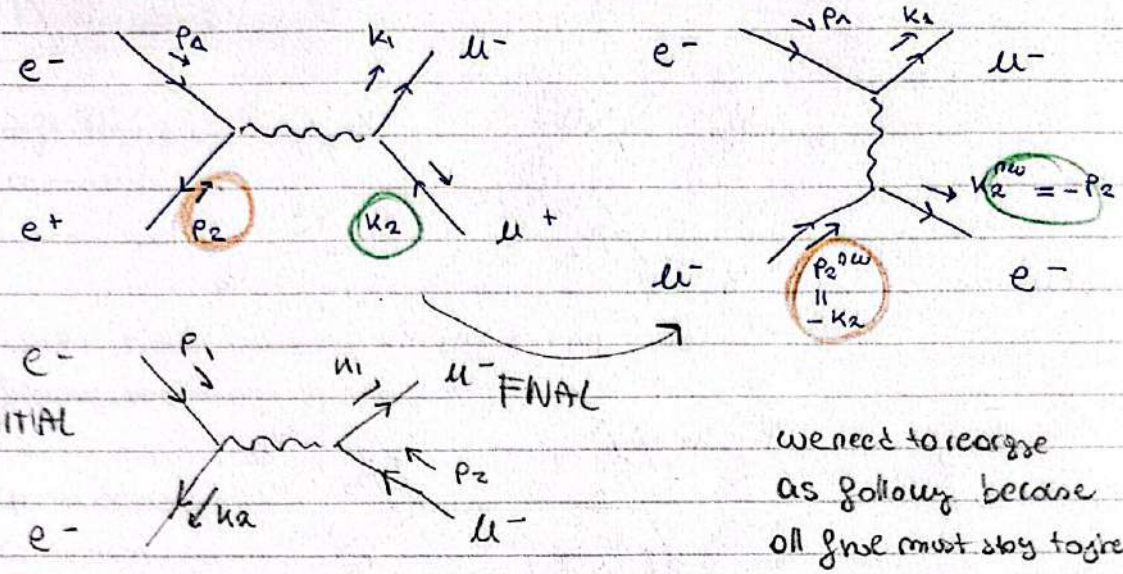
$e^- e^+ \longrightarrow \mu^- \mu^+$ $\qquad$ $e^- \mu^- \longrightarrow \mu^- e^-$



we need to reorder as follows because all that must stay together in the same node and all written together

So we note that the $\sum |M|^2$ matrix is the same but just swapping what we called $p_2 \to -k_2^{new}$ and $k_2 \to -p_2^{new}$ (two swaps on the same name but different things). Therefore simply

$$p_2 \leftrightarrow -k_2$$

Eq (9.2) swapping $p_2 \leftrightarrow -k_2$ (−) swaps for two fermions swapping

$$\frac{1}{4}\sum_s |M(e^-\mu^- \to \mu^- e^-)|^2 = (-)^2(-)^2 \frac{8e^4}{q^4}\left[(p_1\cdot k_1)(k_2\cdot p_2) + (k_2\cdot k_1)(p_1\cdot p_2) - m_\mu^2 (p_1\cdot k_2)\right] =$$

this exchange $s \leftrightarrow u$ in Mandelstam invariant formula

$$\frac{1}{4}\sum_s |M|^2 = \frac{8e^4}{u^2}\left[\left(\frac{t}{2}\right)^2 + \left(\frac{s}{2}\right)^2\right] \qquad \text{with } m_e = m_\mu = 0$$

in the centre of mass frame with $p_1 = k(1,0,0,1)$ $p_2 = (Ek, 0, 0, -k)$

$k_1 = (E, \vec{u})$ $k_2 = (k, -\vec{u})$ where $E = \sqrt{\vec{u}^2 + m_\mu^2}$ taking $m_\mu \neq 0$

$$\frac{1}{4}\sum_s |M|^2 = \frac{2e^4}{k^2(1-\cos\theta)^2}\left[(E+k)^2 + (E+k\cos\theta)^2 - m_\mu^2(1-\cos\theta)\right]$$

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10

PATH INTEGRAL FORMALISM

(M 9.1 9.2 9.3)

Now we will present a new way of quantization = until now we have used canonical quantization which promotes classical fields into operator. In the '40s Feynmann developed a new way to quantize, the path integral formalism, in which fields and position trajectories remain FUNCTION of x and t and are NOT OPERATOR.

CANONICAL QUANTIZATION

In the canonical formulation of quantum mechanics we have operators which, in Heisemberg picture, are functions of time. In QFT we want Lorentz invariance hence we want to treat position and time equally. We have two options: promoting also time t as operator as the position $\hat{q}$ already is OR DEGRADING position to be a parameter as t. Canonical quantization of QFT chooses the 2nd way = we substitute $\hat{q}$ with a Field $\phi(x,t)$ depending both on t and space x

Canonical Quantization			Canonical FIELD QUANTIZATION
$q\ p$	$\rightarrow$ $\hat{q}\ \hat{p}$	promoting position and momenta to operators	$\phi\ \pi \rightarrow \hat{\phi}\ \hat{\pi}$
classical	operators		
$[\hat{q}, \hat{p}] = i\hbar$		imposing commutators	$[\phi(x,t), \pi(x',t)] = i\hbar\delta^{(3)}(\vec{x}-\vec{x}')$
			↓ ↓ Same TIME

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In this formulation however Lorentz invariance is not so evident and time is still treated in a preferencial way. The fundamental equation is Schrödinger equation which treats time specially and makes use of the Hamiltonian, an operator:

$$i\hbar \frac{\partial}{\partial t}\psi(x,t) = \hat{H}\,\psi(x,t)$$

PATH INTEGRAL FORMALISM

Path integral in quantum mechanics is a formalism which preserves the classical feature of trajectories $q(t)$ and the quantum behaviour is seen as the SUM of all possible trajectories (paths) between two points, and this sum is weighted by a phase which introduces interference between paths. So that way $q(t)$ (and fields) remains a function.

PATH INTEGRAL

$$A = \int_{q(t)=q_i}^{q(t)=q_f} [dq]\; e^{\frac{iS}{\hbar}}$$

amplitude to go from q_i to q_f

$\Leftarrow$

PATH INTEGRAL OF FIELDS

$$A = \int_{\phi_i}^{\phi_f} [d\phi]\; e^{\frac{iS}{\hbar}}$$

For QFT in the path integral formalism we still promote q into a field ... or better = substitute the ϕ field to the position, and the field depend on both space and time. Here Lorentz invariance is more evident 'cause the fundamental equation is based on the Action S which is intrinsically Lorentz invariant and treat x and t equally:

$$S = \int d^4x\, \mathcal{L}(\phi, \partial_\mu \phi)$$

Nel limite classico solo un cammino contribuisce alla vera traiettoria della particella = quello che minimizza l'azione. Nel path integral in generale tutti i cammini contribuiscono.

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KEY DIFFERENCES

CANONICAL QUANTIZATION	PATH INTEGRAL
• Based on Schrodger equation - t is peculiar - $\hat{H}$ is operator	• Path integral • X^μ Minkoski 4 vector • S Action – Lagrangian
• $\hat{q}$ and $\hat{p}$ (or $\hat{\phi}$ and $\hat{\pi}$) are operators defined instante per instante da commutatori a tempo fissato $[\hat{q}(t), \hat{p}(t)] = i\hbar \cdots$	• $q(t)$ (e $\phi(x_t)$) sono funzioni classiche - la quantistica emerge dalla somma di tutte le possibili cammini che interferiscono

REASONS TO PREFER PATH INTEGRAL FOR QFT

1) Path integral works for every interaction - non perturbative also

Until now we have treated only perturbative interactions, hence small interactions. Often there are case where interactions are big, as in QCD quantum chromo dyanics. For strongly interacting process Path integral formalism is used

2) Lorentz invariance is more evident and simplier . Path formalism is more natural to face quantization in Gauge theories

3) Particle's statistics (Fermi or Bose) appears very natural

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10.1 PATH INTEGRAL FORMALISM IN QM

10.1.1 Amplitude of evolution from state q_i to q_f

SHRODINGER PICTURE	HEISEMBERG PICTURE
$\hat{A}_S$ Operators fixed	$A_H(t) = e^{iHt} A_S e^{-iHt}$ time dep
$\lvert\psi(t)\rangle_S = e^{-iHt} \lvert\psi(0)\rangle_S$ time dependent	$\lvert\psi\rangle_H = \lvert\psi(0)\rangle_S$ fixed

amplitudes are the same

$$\langle\psi_{tf}\rvert_S A_S \lvert\psi_{ti}\rangle_S = A = \langle\psi_H\rvert A_H(t) \lvert\psi_{ti}\rangle$$

Eigenstates are FIXED:	Eigenstate have a time LABEL:
$A_S \lvert n\rangle_S = \alpha_n \lvert n\rangle_S$	$A_H(t) \lvert n,t\rangle_H = \alpha_n \lvert n,t\rangle_H$

Because at each time instant the Heisemberg op $A_H(t)$ is a different operator with different state, different set of eigenstates

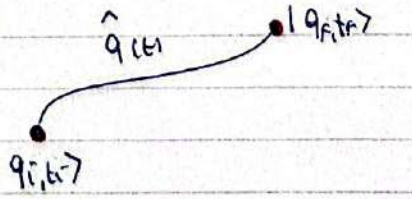
---

Let's consider POSITION OPERATOR $\hat{q}_H(t)$ at two different times t_i initial and t_f final and denote eigenstate of initial and final time

$$\hat{q}_H(t_i) \lvert q_i, t_i\rangle_H = q_i \lvert q_i, t_i\rangle_H \quad (10.2)$$

$$\hat{q}_H(t_f) \lvert q_f, t_f\rangle_H = q_f \lvert q_i, t_f\rangle_H$$

these are two different operators!



These are two different states that in Schrodinger picture are referred to the eigenstate $\lvert q\rangle_S$ of position $\hat{q}_S$ but evolved at two different times.

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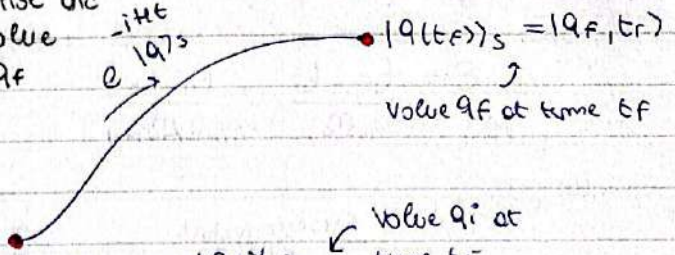
$\hat{q}_S |q_i\rangle_S = q_i |q_i\rangle_S$
$\hat{q}_S |q_f\rangle_S = q_f |q_f\rangle_S$
} two different eigenstate for unitive one final value $q_i \neq q_f$

$\hat{q}_S |q\rangle = q |q\rangle_S$

then $|q(t_i)\rangle_S = e^{-iHt} |q\rangle_S$

evolves in time



each $|q_F\rangle_S$ and $|q_i\rangle_S$ evolve in time $|q_i(t_i)\rangle_S = e^{-iHt}|q_i\rangle_S$ & $|q_f(t_f)\rangle = e^{-iHt}|q_F\rangle_S$

The amplitude to evolve from state $|q,t_i\rangle_H$ to $|q\,t_F\rangle_H$ is given by:

$$\langle q_F, t_F | q_i, t_i\rangle_{H\,H} = \langle q_F | e^{-iH(t_F - t_i)} | q_i\rangle_S =: A \qquad (10.1)$$

since $|q,t\rangle_H = e^{iHt} |q\rangle_S$ (inverse time evolution)

Our aim now is to express this amplitude overlap, to find a formula to compute such amplitude of transition-evolution

---

a) At some time $|q,t\rangle_H$ is a basis ie $\int dq\, |q,t\rangle_H \langle q,t|_H = \mathbb{1}$ Hence inserting N discrete step of times such that:

$$t_i < t_n < t_2 < \dots t_{n-1} < t_F$$
$$\| \qquad\qquad\qquad\qquad \| \qquad\qquad\qquad\qquad \|$$
$$t_0 \qquad\qquad\qquad\qquad t_n$$

(we count such order in time for simplify calculations after) we obtain:

$$A = \langle q_F t_F | q_i t_i\rangle = \int dq_n \langle q_F t_F | q_n t_n\rangle \langle q_n t_n | q_i t_i\rangle = \quad \forall\, n \text{ step}$$

$$= \int dq_1\, dq_2 \dots dq_{n-1} \langle q_F t_F | q_{n-1}, t_{n-1}\rangle \cdots \langle q_n t_n | q_i t_i\rangle$$

calling $q_0 = q_i$ $t_0 = t_i$, $q_F = q_n$ $t_F = t_n$ we have

$$A = \int dq_1\, dq_2 \dots dq_{n-1} \prod_{k=0}^{n-1} \langle q_{k+1}\, t_{k+1} | q_k\, t_k\rangle \qquad (10.3)$$

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THIS IS EXACTLY THE SAME AMPLITUDE OF THE S-MATRIX

Stradger picture

$$|Q(t)\rangle_S = e^{-iH(t-t_i)} \underbrace{|Q(t_i)\rangle_S}_{=:|Q\rangle_S}$$

$$|Q(t_f)\rangle_S = e^{-iH(t_f-t_i)} |Q\rangle$$

$$|b(t)\rangle_S = e^{-iH(t-t_i)} |b(t_i)\rangle_S$$

$$\underbrace{|b(t_f)\rangle}_{=:|b\rangle} = e^{-iH(t_f-t_i)} |b(t_i)\rangle_S$$

Amplitude to propagate from $|Q(t_i)\rangle$ to $|b(t_f)\rangle$ in $(t_f-t_i)=\Delta t$ is the overlap of a and b at time t_f

$$\langle b(t_f)|Q(t_f)\rangle = \langle b(t_f)| e^{-iH(t_f-t_i)} |Q(t_i)\rangle = \langle b| e^{-iH(t_f-t_i)} |Q\rangle$$

this is the S-MATRIX //

In Heisemberg picture

$$|Q\rangle_H = e^{iH(t-t_0)} |Q(t_0)\rangle$$

choose $t_0=0$ (time at which Heisebg and stradger coincec $|Q\rangle_H = |Q(0)\rangle_S$ I can define

$$|Q,t_i\rangle_H = e^{iHt_i} |Q(t_i)\rangle_S \qquad |b,T_f\rangle_H = e^{iHt_f} |b(t_f)\rangle_S$$

$|Q\rangle$ (label, NO dependence!)

$|b\rangle$

$$\langle b|_S e^{-iH(t_f-t_i)} |Q\rangle_S = \langle b, T_f|_H |Q, T_i\rangle_H$$

NOW WE CALL $Q=q_i$ $b=q_f$ TWO DIFFERENT EIGENSTATES OF POSITION OPERATOR !! TWO DIFFERENT POSITIONS! EACH $|q_i(t)\rangle_S$ $|q_f(t)\rangle_S$ evolves in time (in Stradger picture)

$$\langle q_f| e^{-iH(t_f-t_i)} |q_i\rangle = \langle q_f|T_f|_H |q_i, T_i\rangle_H =: A$$

So now A is exactly the S-matrix BUT EVALUETED Between two eigenstate of POSITION! No more between $|p\rangle$ and $|k\rangle$ momenta space ... A is S matrix in position space !!

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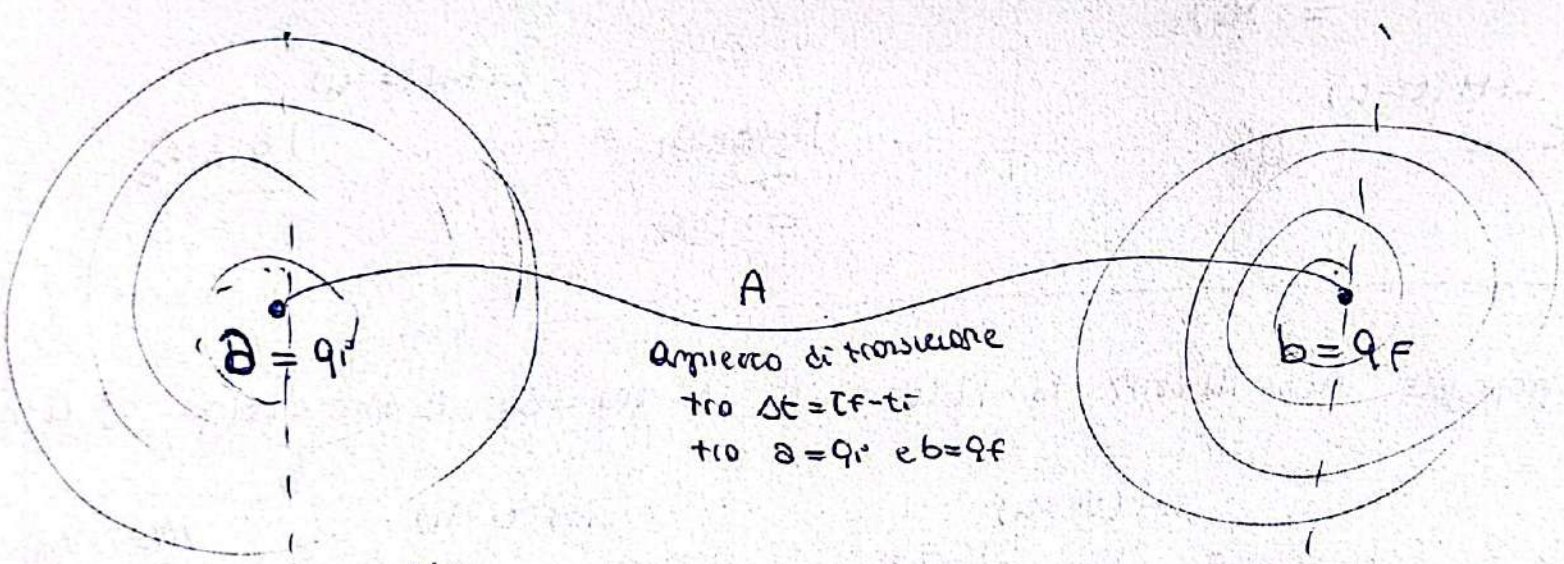
⚠ q_f NON è l'evoluto temporale di q_i nel tempo!! sono due posizioni diverse!! $q_f = b$ $q_i = a$, e ciascun $b(t)$ e $a(t)$ evolve nel tempo!



$|q_i(t)\rangle = e^{-iHt} |q_i(0)\rangle$

evolve nel tempo

$|q_f(t)\rangle = e^{-iHt} |q_f(0)\rangle$

evolve nel tempo

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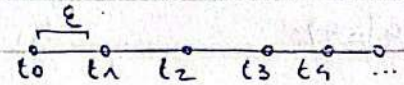
a) choosing time points to be evenly spaced as $t_{k+1} - t_k = \varepsilon$ with

$$\varepsilon := \frac{t_F - t_i}{m} = \frac{t_n - t_0}{m}$$



we get

$$\langle q_{k+1}, t_{k+1} | q_k, t_k \rangle = \langle q_{k+1} |_S \, e^{-i\hat{H}(t_{k+1} - t_k)} \, | q_k \rangle =$$

(Schrödinger picture)

$$= \langle q_{k+1} | e^{-iH\varepsilon} | q_k \rangle = \int dp_{k+1} \langle q_{k+1} | p_{k+1} \rangle \langle p_{k+1} | e^{-i\hat{H}\varepsilon} | q_k \rangle = *$$

b) now, normalizing

- $\langle q | p \rangle = e^{ipq}$

- and writing $\hat{H} = \hat{H}(\hat{p}, \hat{q})$ as a function of operators $\hat{q}$ and $\hat{p}$ such that all $\hat{p}$ are left to $\hat{q}$

EX: $\hat{H} = \hat{p}^2 \hat{q}$ ok

$\hat{H} = +\hat{p}\hat{q}\hat{p}$ no good but I can adjust knowing that

$[q\,p] = i\hbar$ $qp - pq = i\hbar$ so

$\hat{H} = +\hat{p}\hat{q}\hat{p} = \hat{p}^2\hat{q} + i\hbar\hat{p}$

such that if $\hat{H} = \hat{p}^2 \hat{q}$

$$\langle p_k | \hat{H}(\hat{p}\hat{q}) | q_k \rangle = \langle p_k | \hat{p}^2 \hat{q} | q_k \rangle = p_k^2 q_k \langle p_k | q_k \rangle = H(pq) \langle p_k | q_k \rangle$$

all $\hat{p}$ but B.c.a $\langle p_k |$ all $\hat{q}$ but Het $| q_k \rangle$ — exactly the Hamiltonian but with eigenvalue and no more operators

Then

$$* = \int dp_k \langle q_{k+1} | p_k \rangle \langle p_k | e^{-i\hat{H}(\hat{p}\hat{q})\varepsilon} | q_k \rangle =$$

$$= \int dp_k \, e^{i q_{k+1} p_k} \langle p_k | 1 - i\varepsilon \hat{H}(\hat{p}\hat{q}) + \dots | q_k \rangle =$$

(operators)

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$$= \int dp_n \; e^{i q_{n+1} p_n} \; (1 - i\varepsilon H(p_n, q_n) + \cdot) \langle p_n | q_n \rangle =$$

(eigenvalues)

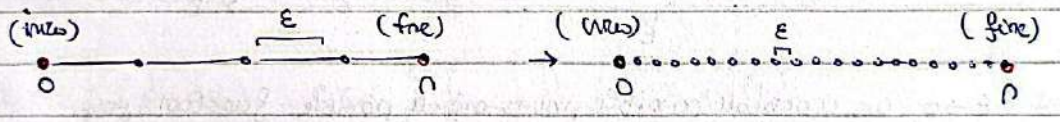
$$= \int dp_n \; \exp\left\{ -i\varepsilon \left[H(p_{n+1}, q_n) - p_{n+1} \left(\frac{q_{n+1} - q_n}{\varepsilon} \right) \right] \right\} \qquad (10.5)$$

b) Hence substituting for all k eq (10.5) into (10.3) we get

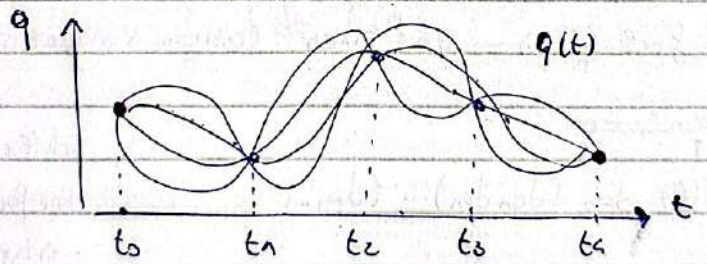
$$A = \int dp_0 \, (dp_1 \, dq_1) \cdots (dq_{n-1} \, dp_{n-1}) \exp\left\{ -i\varepsilon \sum_{n=0}^{n-1} \left[H(p_n, q_n) - p_n \left(\frac{q_{n+1} - q_n}{\varepsilon} \right) \right] \right\} \qquad (10.6)$$

with $\varepsilon = \frac{t_n - t_0}{n}$ evenly spaced

Looking at this equation we see that we are integrating over $q_1 p_1 \cdots q_{n-1} p_{n-1}$ points and $(q_0 p_0)$ & $(q_n p_n)$ are FIXED. Now we have a discrete number of points but nulla ci vieta to take the limit of $\varepsilon \to 0$ hence $n \to \infty$ hence taking the continuum limit of integration points



Consider a set of functions $q(t)$ with points $q_0 = q(t_0)$, $q_1 = q(t_1)$ ..... $q_n = q(t_n)$ that are boundaries = all $q(t)$ need to pass through such points



Now consider to INTEGRATE hence VARY all intermediate points = visually you can imagine to take one intermediate dot with your fingers and move up and down along the vertical line of t_k. Doing this, considering all possible combinations of points, the functions passing through that points are a large set (class) of functions but exact <u>all</u> possible functions going from fixed q_0 and q_n.

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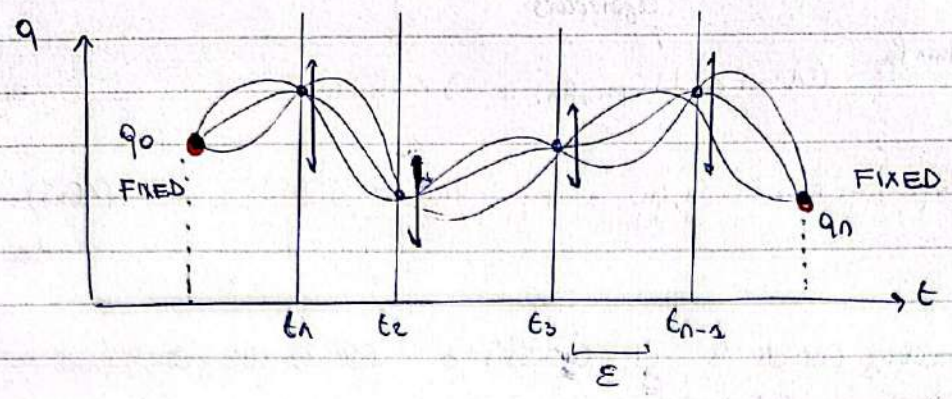
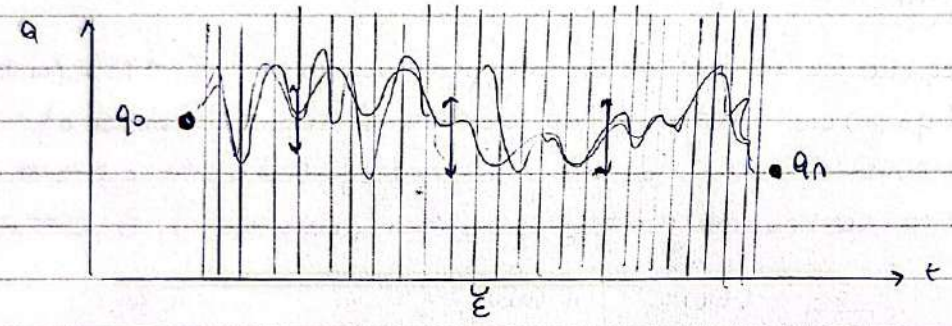


We want ALL POSSIBLE FUNCTION $q(t)$ with FIXED initial and final points $q(t_0)$ $q(t_n)$ = we need to take the LIMIT $n \to \infty$ $\varepsilon \to 0$
Hence infinitely many q_n intermediate point to integrate on



In the limit $\varepsilon \to 0$ we reach all possible points and all possible functions going from fixed q_0 & q_n

$\Rightarrow$ we obtain a FUNCTIONAL INTEGRAL = an integral with variable $q(t)$ (and $p(t)$) which runs over ALL POSSIBLE FUNCTIONS with initial $q(t_i) =: q_i$ & final $q(t_n) =: q_n$ FIXED (estremi d'integrazione)

$$[dp\,dq] := \lim_{\varepsilon \to 0} N(\varepsilon) \, dp_0 \, (dq_1 dp_1) \cdots (dp_{n-1} dq_{n-1})$$

normalization

$[dp\,dq]$: functions

differential of the functional integral

$$\exp\left\{-i\varepsilon \sum_{n=0}^{n-1} \left[H(p_n q_n) - p_n \left(\frac{q_{n+1} - q_n}{\varepsilon}\right)\right]\right\} \xrightarrow[\varepsilon \to 0]{} \exp\left\{-i \int_{t_i}^{t_f} dt \left(H(p(t), q(t)) - p(t)\dot{q}(t)\right)\right\}$$

$\frac{q_{n+1} - q_n}{\varepsilon}$ becomes a derivate on time

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Hence taking the limit for $\varepsilon \to 0$ of eq(10.6) we get:

$$A = \int_{q(t_i)=q_i}^{q(t_f)=q_f} [dp\, dq] \exp\left\{ \frac{i}{\hbar} \int_{t_i}^{t_f} dt\, [p\dot{q} - H(pq)] \right\} \qquad (10.7)$$

with $\hbar$ now expressed explicitly (before was considered $\hbar = 1$).

3) Now consider a PHYSICAL HAMILTONIAN $H = \frac{p^2}{2} + V(q) = \dot{q}p - \mathcal{L} = p^2 - \frac{p^2}{2} + V$

the corresponding Lagrangian is

$$L = \frac{p^2}{2} - V(q) = \frac{\dot{q}^2}{2} - V(q) \qquad \text{with } \dot{q} = p$$

Hence the argument of exponential in 10.7 is the Lagrangian

$$A = \int [dp\, dq] \exp\left\{ \frac{i}{\hbar} \int dt \left(\frac{p^2}{2} - V(q) \right) \right\}$$

Integrating over $[dp]$ means over a function $e^{\frac{i}{\hbar}\frac{p^2}{2}}$ which is a gaussian and gives a number which I absorb into $[dq]$. Then:

$$A = \int_{q_i}^{q_f} [dq] \exp \frac{i}{\hbar} \int_{t_i}^{t_f} dt\, L(q\dot{q})$$

Hence

$$A = \int_{q(t_i)=q_i}^{q(t_f)=q_f} [dq]\, e^{\frac{i}{\hbar} S} = A(q_i t_i, q_f t_f) \qquad (10.8)$$

$$S = \int_{t_i}^{t_f} dt\, L(q, \dot{q}) \qquad \text{action}$$

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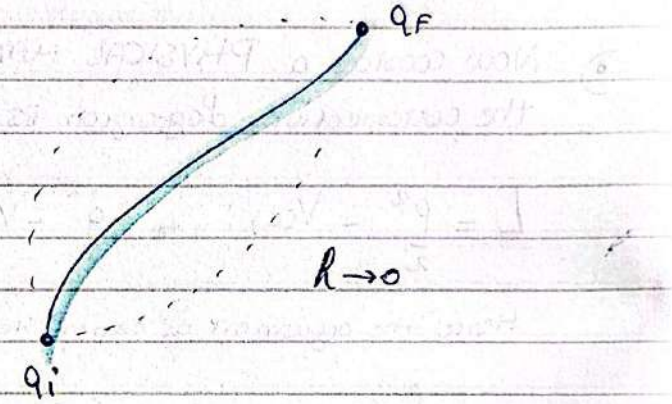
L'ampiezza di propagazione dal punto $q_i t_i = x_i$ dello spaziotempo al punto finale $q_f t_f = x_f$ dello spaziotempo è la SOMMA di tutte le TRAIETTORIE $q(t)$ da q_0 o q_f PESATE per una fase che dip dall'AZIONE CLASSICA $S(q(t))$ per quella traiettoria.

Mentre in MC la propagazione avviene su una sola traiettoria in MQ la prop è una sovrapposizione di tutte le traiettorie, si ha interferenza tra esse

CASO CLASSICO

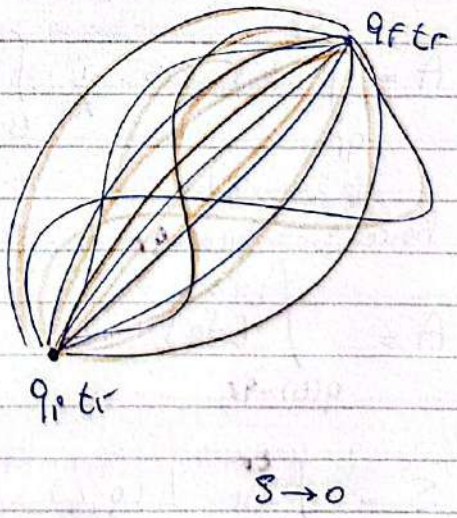
Quando S è GRANDE rispetto ad $\hbar$ o quando $\hbar \to 0$ allora c'è una traiettoria che contribuisce è quella che MINIMIZZA L'AZIONE. Questo perché se S grande la fase $e^{iS/\hbar}$ oscilla MOLTO rapidamente e qualsiasi integrale da q_i o q_f sarà nullo perché l'integrale di un periodo è nullo. L'unico termine con integrale non nullo è quello con S min.



CASO QUANTISTICO

Se piccolo oppure $\hbar$ grande $\gtrsim 1$ = non tal caso più cammini danno contributo alla propagazione.

S piccolo cosa confrontabile con $\hbar$ significa pochi gdl. Perché? $\hbar \gtrsim 1$ sistema di unità notevoli significa eguagliare l'unità di misura di tempo e spazio poiché $c=1$ (unico U.M per x e t, non più metri e secondi), tratto x e t equamente. S confrontabile con $\hbar$ significa una piccola porzione di spaziotempo cioè una piccola porzione di gdl.



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10.1.2 Matrix element of $\hat{q}_H(t)$

Our aim now is to find the matrix element of Heisemberg position operator

$$\langle q_F, t_F | \hat{q}_H(t) | q_i, t_i \rangle$$

and to relate it to the path integral.

a) we start by inserting a factor $q(t)$ with $t_i < t < t_F$ in the path integral

$$\int_{q_i}^{q_F} [dq]\, q(t)\, e^{iS}$$

Consider intermediate value $\bar{q}$ i.e $q_i < \bar{q} < q_F$ then with $q(t) = \bar{q}$

$$\int_{q_i}^{q_F} [dq] = \int_{q_i}^{\bar{q}} [dq] + \int_{\bar{q}}^{q_F} [dq] = \int_{-\infty}^{\infty} d\bar{q} \left(\int_{q_i}^{\bar{q}} [dq] \cdot \int_{\bar{q}}^{q_F} [dq] \right) \quad (1)$$

SUM — integration (sum) over all possible $\bar{q}$ — multiplication

write

$$e^{iS} = e^{i\int_{t_i}^{t_F} dt' L} = e^{i\int_{t_i}^{t} dt' L + i\int_{t}^{t_F} dt' L} = \left(e^{i\int_{t_i}^{t} dt' L} \right) \left(e^{i\int_{t}^{t_F} dt' L} \right) \quad (2)$$

Hence

$$\int_{q(t_i)=q_i}^{q(t_f)=q_F} [dq]\, q(t)\, e^{iS} \overset{(1)}{\underset{(2)}{=}} \int_{-\infty}^{\infty} d\bar{q} \left(\int_{q(t_i)=q_i}^{q(t)=\bar{q}} [dq]\, e^{i\int_{t_i}^{t} dt' L} \right) \bar{q} \left(\int_{q(t)=\bar{q}}^{q(t_f)=q_F} [dq]\, e^{i\int_{t}^{t_F} dt' L} \right) \quad (1a)$$

put $\bar{q} = q(t)$

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•) Using $\int_{q(t_i)=q_i}^{q(t)=\bar{q}} [dq]\, e^{i\int_{t_i}^{\bar{t}} dt' L} = \langle \bar{q}\, \bar{t} | q_i\, t_i \rangle$ we get

$$\int_{q(t_i)=q_i}^{q(t_F)=q_F} [dq]\, q(t)\, e^{iS} = \int_{-\infty}^{\infty} d\bar{q}\, \langle q_F t_F | \bar{q}\, t \rangle\, \bar{q}\, \langle \bar{q}\, t | q_i t_i \rangle =$$

•) Using that in the Heisenberg picture $\hat{q}_H(t) | \bar{q}\, t \rangle = \bar{q}\, | \bar{q}\, t \rangle$

$$= \int_{-\infty}^{\infty} d\bar{q}\, \langle q_F t_F | \hat{q}(t) | \bar{q}\, t \rangle \langle \bar{q}\, t | q_i t_i \rangle = \langle q_F t_F | \hat{q}(t) | q_i t_i \rangle$$

$\int d\bar{q}\, | \bar{q}\, t \rangle \langle \bar{q}\, t | = \mathbb{1}$

Then

$$\langle q_F t_F | \hat{q}_H(t) | q_i t_i \rangle = \int_{q(t_i)=q_i}^{q(t_F)=q_F} [dq]\, q(t)\, e^{iS} \qquad 10.10$$

Now consider n functions $q(t_1)\, q(t_2) \cdots q(t_n)$ points

$$\int_{q_i}^{q_F} [dq]\, q(t_n)\, q(t_2) \cdots q(t_0)\, e^{iS}$$

since they are functions they commute and I can always reorder them in order to have $t_F > t_1 > t_2 > t_3 \cdots > t_n > t_i$

$$t_F > t_n > t_{n-1} > t_1 > t_i$$

repeating the above argument with intervals (t_i, t_n) $(t_n, t_{n-1}) \ldots (t_1, t_F)$ we get

$$\int_{q(t_i)=q_i}^{q(t_F)=q_F} [dq]\, q(t_1)\, q(t_2) \cdots q(t_n)\, e^{iS} = \langle q_F t_F | T\{ \hat{q}(t_n) \cdots \hat{q}(t_0) \} | q_i t_i \rangle \qquad (10.11)$$

$q(t_n) \cdots q(t_2) q(t_1)$

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10.2 PATH INTEGRAL FOR SCALAR FIELDS

PBS 9.2

In chapter 10.1 we always only considered just one particle $q(t)$ propagating, maybe at different points $q(t_n)$. Now consider a multi dimensional space of more particles q_i, in eq 10.1 the functional $[dq] \longrightarrow$ replaced by $\prod_i [dq_i]$

As already done in canonical quantization, we know replace q with fields

$$q_i(t) \longrightarrow \phi(\vec{x}, t)$$

($q_i(t)$: discrete index; $\phi(\vec{x}, t)$: continuous index)

Hence now we have a function of two continuous variable $\vec{x}$ and t. The functional integral is:

$$\prod_i [dq_i] \longrightarrow D\phi$$

$[dq]$ varying in t

$\prod_i [dq_i]$ varying in i and t

$D\phi$ varying in x^μ cioè $\vec{x}$ and t

Since $D\phi$ integrate over all function varying in both t and $\vec{x}$, that's way we need to account for $\prod_i [dq_i]$ and not only $[dq]$

Hence the amplitude for a transition between a field configuration $\phi_i(\vec{x})$ at time T_i and $\phi_F(\vec{x})$ at time T_F simply generalizes 10.8 as:

$$\langle \phi_F(\vec{x}), T_F \mid \phi_i(\vec{x}), T_i \rangle = \int_{\phi(\vec{x}, T_i) = \phi_i(\vec{x})}^{\phi(\vec{x}, T_F) = \phi_F(\vec{x})} D\phi \; e^{i \int_{T_i}^{T_F} d^4x \, \mathcal{L}}$$

with $\int_{T_i}^{T_F} d^4x = \int_{T_i}^{T_F} dt \int_{-\infty}^{+\infty} d^3\vec{x}$

($\Rightarrow$ $D\phi$ is the integration over all possible configuration of $\phi(\vec{x}, t)$ with given boundaries T_i & T_F and unbounded $\vec{x}$)

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$\phi(\vec{x}t)$ are interacting fields. We are interested in modeling the interaction as one initial free field $\phi_i(\vec{x})$ at time $-\infty$ and final $\phi_F(\vec{x})$ at $+\infty$, hence to a interacting field which at beginn. and end of interaction has a free boundary fields. Setting $T_i \to -\infty$ and $T_F \to +\infty$ we get

$$\langle \phi_F(\vec{x}), +\infty \mid \phi_i(\vec{x}), -\infty \rangle = \int_{\phi_i}^{\phi_F} D\phi \; e^{i\int d^4x \mathcal{L}} \qquad (10.12)$$

where now also time is unbounded, $D\phi$ is integration over all $\phi(\vec{x}t)$ interacting with boundary condition $\phi_F(\vec{x})$ at $T \to \infty$ and ϕ_i at $T \to -\infty$

[FOCUS = This is what in chapter 7 book 3 we called $\langle p_1 T_F \mid k_1 T_i \rangle = \langle p_1 \mid S \mid k_1 \rangle$ $\lim_{T_F - T_i \to \infty}$
which is the element of S matrix]

As found in chapter 7, that amplitude is also given by LSZ formula:

$$\int D\phi \; e^{i\int d^4x \mathcal{L}} = \langle \phi_F(\vec{x}) +\infty \mid \phi_i(\vec{x}) -\infty \rangle \underset{\text{LSZ}}{=} \int\int \cdots \langle \Omega \mid T\{\phi(x_1) \cdots \phi(x_n)\} \mid \Omega \rangle$$

So we would like to compute $\langle \Omega \mid T\{\cdots\} \mid \Omega \rangle$. In chapter 8 we saw how to do it perturbatively in the canonical way. Now we see a general non-perturbative way to compute $\langle \Omega \mid \cdots \mid \Omega \rangle$ thanks to path integral

We start to notice that we can generalize for ϕ fields equation 10.11 written for q obtaining:

$$\langle \phi_F(\vec{x}) +\infty \mid T\{\phi(x_1) \cdots \phi(x_n)\} \mid \phi_i(\vec{x}), -\infty \rangle = \int_{i}^{f} D\phi \; \phi(x_1) \cdots \phi(x_n) \, e^{i\int d^4x \mathcal{L}} \qquad (10.13)$$

all these interacting fields are fields with boundary condition to be in $\phi_F(\vec{x})$ at final time and $\phi_i(\vec{x})$ at initial time

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projecting

As done in chapter 7 when ~~projecting~~ interacting $|\Omega\rangle$ vacuum to $|0\rangle$ free vacuum we could observe, that doing that projecting $|\Omega\rangle$ to $|\phi\rangle$ we take a phase

$$|\Omega\rangle \sim \lim_{T\to\infty(1-i\varepsilon)} \frac{1}{\exp\{\dots\}} |\phi\rangle \qquad \langle\Omega|T\{\;\}|\Omega\rangle = \frac{\langle d_F|T\{\phi\phi\}|\phi_i\rangle}{\langle d_F|\phi_i\rangle}$$

We could not repeat calculations but this gives:

$$\langle\Omega|T\{\phi(x_1)\dots\phi(x_n)\}|\Omega\rangle = \lim_{T\to\infty(1-i\varepsilon)} \left(\frac{\int D\phi\, \phi(x_1)\dots\phi(x_n)\, e^{i\int d^4x \mathcal{L}}}{\int D\phi\, e^{i\int d^4x\mathcal{L}}} \right) \tag{10.14}$$

This is exactly the term which enters in the LSZ formula! We found a way to compute LSZ and S matrix in a COMPLETELY NON PERTURBATIVE WAY VIA PATH INTEGRAL!

- all fields in eq 10.14 both left and right side are interacting fields
- until now we have not been careful in collecting all numerical factors in $D\phi$... because at the end it doesn't matter! these factors cancel between the numerator and denominators

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In principle I could also write amplitude with the boundaries at both T_i & T_F and $\vec{x}_i$ & $\vec{x}_f$ fixing initial time and space, fixing final time and space, (Fixing cioè anche il punto spaziale di partenza e di arrivo, e non solo i tempi)

$$\langle \phi_F, x_F^{\vec{}} \mid \phi_i, x_i^{\vec{}} \rangle = \langle \phi_F | T_F | \phi_i \rangle = \int_{\phi_i = \phi(x_i^{\vec{}}) = \phi(\vec{x}_i, T_i)}^{\phi_F = \phi(x_F^{\vec{}}) = \phi(\vec{x}_F, T_F)} D\phi \; e^{i \int_{T_i}^{T_F} dt \int_{\vec{x}_i}^{\vec{x}_F} \mathcal{L}} \qquad (1)$$

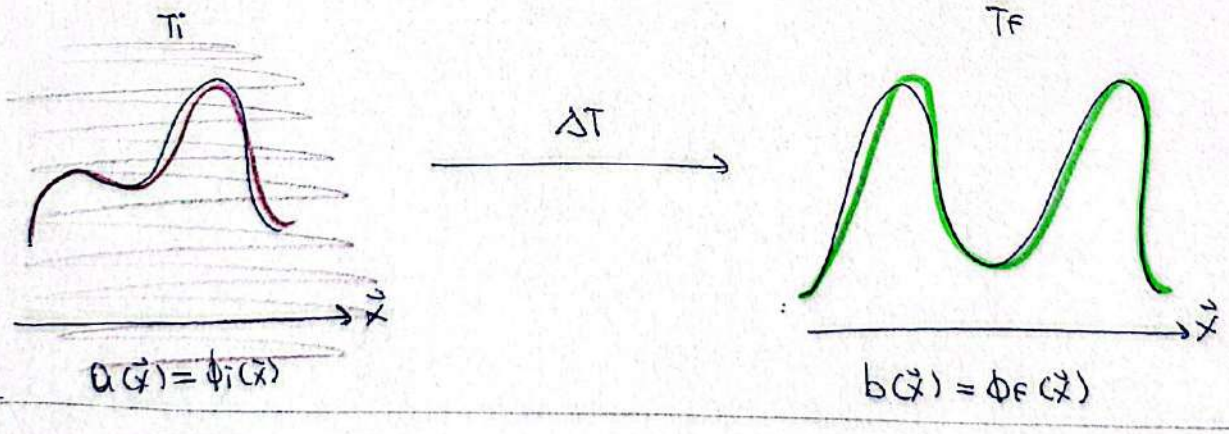
oppure posso decidere di valutare il propagatore escludendo (libero tutto lo spazio $\vec{x}$, cioè di lasciare $\vec{x}$ non bounded, libero, non fisso il punto iniziale e finale dello spazio ma solo il tempo. Vado cioè a sommare su tutte le possibili traiettorie da T_i a T_F escludendo $\phi(\vec{x})$ funzione di $\vec{x}$

$$\langle \phi_F(\vec{x}), T_F \mid \phi_i(\vec{x}), T_i \rangle = \int_{\phi(\vec{x}, T_i) = \phi_i(\vec{x})}^{\phi(\vec{x}, T_F) = \phi_F(\vec{x})} D\phi \; e^{i \int_{T_i}^{T_F} dt \int_{-\infty}^{+\infty} d\vec{x}^3 \mathcal{L}} \qquad (2)$$

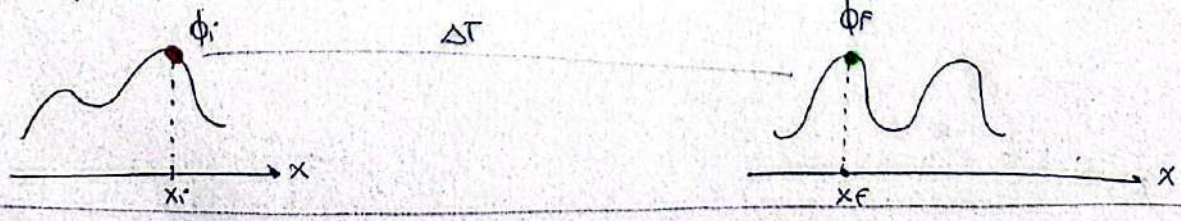
Posso fare entrambe le cose! Sono solo due oggetti diversi! Dipende da cosa voglio guardare!

NB $\phi_i(\vec{x})$ è una funzione di $\vec{x}$ a tempo fissato, è una ~~FOTOGRAFIA~~ del campo a $t = T_i$.

il 2° propagatore mi dice l'ampiezza di passare da una certa configurazione FOTO $\phi_i(\vec{x})$ $\equiv a(\vec{x})$ ad un'altra CONFIGURAZIONE (FOTO) funzione $\phi_f(\vec{x}) \equiv b(\vec{x})$ in tempo $\Delta t = T_f - T_i$



Mentre il (1) mi direbbe come passare da un punto x_i di ϕ_i a tempo t_i ad un Altro punto x_f di ϕ_f a tempo t_f



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When we take the limit of $T_f - T_i \to \infty$ cioè $T_f \to +\infty$ $T_i \to -\infty$ we are watching at $\phi_i(\vec{x})$ and $\phi_f(\vec{x})$ functions of FREE FIELDS ; they can no more interact if infinite time gap separates them!

$$\langle \phi_f(\vec{x}), +\infty \mid \phi_i(\vec{x}), -\infty \rangle = \int_{\phi_i(\vec{x}) = \phi(\vec{x}, -\infty)}^{\phi_f(\vec{x}) = \phi(\vec{x}, +\infty)} D\phi \; e^{i \int d^4 \mathcal{L}}$$

$\phi_i(\vec{x})$ → free field

$\phi(\vec{x}, -\infty)$ → interacting field

A field $|\phi_i(\vec{x})\rangle$, considered WHOLE, è a whole function on ALL space $\vec{x}$, is like $|p_i\rangle$ a fixed momentum! MASSIMO dispersione spaziale (TUTTA la funzione $\phi_i(\vec{x})$ su TUTTO lo spazio) è un momento fissato $|p_i\rangle \Rightarrow$ campi $\phi_f(\vec{x})$ e $\phi_i(\vec{x})$ sono particelle con momento fissato p_i e p_f !!

Tempo T_i → ho campo libero $\phi_i(\vec{x})$ → particella p_i

Tempo T_f → ho il campo libero $\phi_f(\vec{x})$ → particella p_f

Quindi il nostro propagatore con PATH integrale è PROPRIO la nostra S MATRIX !

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because at the end it doesn't matter! these factors cancel between the numerator, and denominators

10.3 FEYNMAN PROPAGATOR FROM PATH INTEGRAL

(For scalar fields)

In chapter 8 (Feynman diagrams) we studied the perturbative expansion of $\langle \Omega | T\{\phi \cdots\} | \Omega \rangle$ and found an expression in terms of Feynman propagators:

$$D(x_1 - x_2) = \langle 0 | T\{\phi(x_1)\phi(x_2)\} | 0 \rangle$$

↓ Free vacuum ↓ Free fields

Feynman propagator of scalar field ϕ

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We now see how to compute this propagator with the path integral with eq 10.14 . Simply we consider a system where $|\psi\rangle = |0\rangle$ and fields are free , hence a Lagrangian of FREE SCALAR FIELD

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)^2 - \frac{m^2}{2}\phi^2$$

And compute eq 10.14 which coincides with Feynman propagators.

$$D(x_1 - x_2) = \langle 0|T\{\phi(x_1)\phi(x_2)\}|0\rangle = \lim_{T\to\infty(1-i\varepsilon)} \left(\frac{\int D\phi\, \phi(x_1)\phi(x_2) e^{iS}}{\int D\phi\, e^{iS}} \right)$$

In order to compute NUMERATOR and DENOMINATOR we need to compute

$$D\phi \quad , \quad \phi(x_i) \quad , \quad S = \int d^4x\, \mathcal{L} \quad , \quad \lim_{T\to\infty(1-i\varepsilon)}$$

1 STEP = DISCRETIZATION

We discretize the continuum x in a lattice of x_i points with even spacing ε^μ in all coordinates of x_i^μ

$$D\phi = \prod_i d\phi(x_i) \qquad (10.15)$$

$\phi(x_i)$

We take the Fourier serie (equivalent of Fourier transform but for lattices)

$$\phi(x_i) = \frac{1}{V} \sum_n e^{-i k_n \cdot x_i} \phi(k_n) \qquad (10.16)$$

where $|k_n^\mu| < \frac{\pi}{\varepsilon_\mu}$ and $V = L^4$ volume of integration

as $\phi(x_i)$ is real then $\phi^*(k_n) = \phi(-k_n)$

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Dφ — MOMENTA SPACE

Considering $\mathrm{Re}\,\phi(k_n)$ and $\mathrm{Im}\,\phi(k_n)$ as independent integration variables with energy positive $k_n^0 > 0$:

$$D\phi = \prod_{k_n^0>0} d\mathrm{Re}\phi_n \, d\mathrm{Im}\phi_n \qquad \text{with } \phi_n = \phi(k_n)$$

$$d\phi(x^\mu) = \prod_n d\phi(k_n^\mu) \quad \text{since} \quad \phi(x^\mu) = \sum_n (\text{const})\,\phi(k_n)$$

($\phi(x)$ spatial is the SUM of MANY $\phi(k)$ momenta so $d\phi_x$ spatial is the PRODUCT of $d\phi_k$ momenta)

S

- $\mathcal{L} = \frac{1}{2}(\partial_\mu\phi)^2 - \frac{m^2}{2}\phi^2$

$$\partial_\mu\phi = \frac{1}{V}\sum_n (-ik_n)\, e^{-ik_n x}\,\phi(k_n)$$

$$(\partial_\mu\phi)^2 = \frac{1}{V^2}\sum_n\sum_m (-ik_n)(-ik_m)\, e^{-i(k_n+k_m)x}\,\phi(k_n)\,\phi(k_m)$$

$$\phi^2 = \frac{1}{V^2}\sum_n\sum_m e^{-i(k_n+k_m)x}\,\phi(k_n)\,\phi(k_m)$$

$$\mathcal{L} = \frac{1}{2}\frac{1}{V^2}\sum_n\sum_m e^{-i(k_n+k_m)x}\,\phi(k_n)\phi(k_m)\left((-ik_n)(-ik_m) - m^2\right)$$

- $S = \int dx^4\,\mathcal{L} = \frac{1}{2}\sum_n\sum_m \frac{1}{V^2}\underbrace{\int dx^4\, e^{-i(k_n+k_m)x}}_{V\,\delta(k_n+k_m)}\phi_n\phi_m(-k_n k_m - m^2) =$

$$= \frac{1}{2}\sum_n\sum_m \frac{V}{V^2}\,\delta(k_n+k_m)\,\phi_n\phi_m(-k_n k_m - m^2) =$$

$$= \frac{1}{2}\sum_n \frac{1}{V}\phi_n\phi_{-n}(k_n^2 - m^2) = \quad \rightarrow \phi^*(n) = \phi(-n)$$

$$= \frac{1}{2V}\sum_n |\phi_n|^2 (k_n^2 - m^2)$$

$$\Rightarrow \quad S = -\frac{1}{2V}\sum_n (m^2 - k_n^2)\left(\mathrm{Re}^2\phi_{(k_n)} + \mathrm{Im}^2\phi_{(k_n)}\right) \qquad 10.17$$

$T \to \infty(1-i\varepsilon)$

change $k_n \to k_n + i\varepsilon$ and $k_n^2 \to k_n^2 + i\varepsilon$ (analytical continuation in a non-formal way)

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DENOMINATOR

$$\lim_{T\to\infty(1-i\varepsilon)} \int D\phi\, e^{iS} = \text{(three circles)}$$

$$= \prod_{k_n^0>0} \int d\mathrm{Re}\phi_n\, d\mathrm{Im}\phi_n\, e^{-\frac{i}{2V}\sum_{n/k_n^0>0}(m^2-k_n^2-i\varepsilon)\left[(\mathrm{Re}\phi_n)^2+(\mathrm{Im}\phi_n)^2\right]} =$$

means = sum over n such that $k_n^0>0$

$i\varepsilon$ terms gives the necessary convergence factor for Gaussian integrals: $\int dx\, e^{-ax^2} = \sqrt{\frac{\pi}{a}}$

$$= \prod_{k_n^0>0} \int d\mathrm{Re}\phi_n \exp\left[\frac{-i}{V2}(m^2-k_n^2-i\varepsilon)(\mathrm{Re}\phi_n)^2\right] \cdot$$

$$\cdot \int d\,\mathrm{Im}\phi_n \exp\left[\frac{-i}{2V}(m^2-k_n^2-i\varepsilon)(\mathrm{Im}\phi_n)^2\right] = \Rightarrow$$

evaluating the two integrals, they give the same result

$$= \prod_{k_n^0>0} \sqrt{\frac{-i\pi V2}{m^2-k_n^2-i\varepsilon}} \quad \sqrt{\frac{-i\pi V2}{m^2-k_n^2-i\varepsilon}} \quad (\text{for } k_n^0>0 \text{ and for } k_n^0<0) = \Rightarrow$$

if I take $k_n^0>0$ or $k_n^0<0$ it's the same, so I can remove the constraint of product over $k_n^0>0$ and solving the result in one only $\sqrt{\frac{-i\pi V}{m^2-k_n^2-i\varepsilon}}$

$$= \prod_{k_n} \sqrt{\frac{-i\pi V2}{m^2-k_n^2-i\varepsilon}}$$

Hence

$$\text{Denominator} = \prod_n \sqrt{\frac{-i\pi V2}{m^2-k_n^2-i\varepsilon}} = \prod_{k_n^0>0}\left(\frac{-i\pi V2}{m^2-k_n^2-i\varepsilon}\right) \quad (\text{No. 18})$$

NUMERATOR

FIRST I need to compute (different points)

$$\phi(x_1)\phi(x_2) = \frac{1}{V^2}\sum_{p,e} e^{-ik_p\cdot x_1}\, e^{-ik_e x_2}\, \phi(k_p)\,\phi(k_e)$$

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$$\int dx\, x^2 e^{-\alpha x^2} = \frac{1}{2\alpha}\sqrt{\frac{\pi}{\alpha}} \qquad \int_{ODD} dx\, x e^{-\alpha x^2} = 0$$

Then

$$\lim_{T\to\infty(1-i\varepsilon)} \int D\phi\, \phi(x_1)\phi(x_2)\, e^{iS} = \text{j j j j}$$

$$= \frac{1}{V^2} \sum_{p,l} e^{-ik_p\cdot x_1 - ik_e x_2} \left(\prod_{k_n>0} \int d\mathrm{Re}\phi_n \int d\mathrm{Im}\phi_n \right) (\mathrm{Re}\,\phi_p + i\,\mathrm{Im}\,\phi_p) \cdot$$

(ϕ_p: b_n; ϕ_e: b_e)

$$\cdot (\mathrm{Re}\,\phi_e + i\,\mathrm{Im}\,\phi_e)\, \exp\left\{ \frac{-i}{2V} \sum_{k_n^0>0} (m^2 - k_n^2 - i\varepsilon) \left[(\mathrm{Re}\,\phi_n)^2 + (\mathrm{Im}\,\phi_n)^2 \right] \right\} =$$

observe that:
- if $k_p \neq \pm k_e$ → non so cosa scrivere il momento! integrand has $\phi(k_p)$ which is odd ogives zero (ϕ is the number of x in $\int dx^3\, x =$ → $\int d\Omega\, \phi = 0$)
- if $k_p = k_e$ $\phi_p \phi_e = \phi_p^2$ contributions from $\mathrm{Re}\,\phi_p^2$ and $\mathrm{Im}\,\phi_p^2$ cancel because of $\phi^*(k_p) = \phi(-k_p)$

so only $k_e = -k_p$ gives non-zero term and repeating the integration done for denominator:

$$\text{NUMERATOR} = \frac{1}{V^2} \sum_p e^{-ik_p(x_1-x_2)} \left(\prod_{k_0>0} \left(\frac{-i\pi V 2}{m^2 - k_n^2 - i\varepsilon} \right) \right) \left(\frac{-iV}{m^2 - k_p^2 - i\varepsilon} \right) \qquad (10.19)$$

Putting all together (○ factor is canceled by denominator):

$$\langle 0|T\{\phi(x_1)\phi(x_2)\}|0\rangle = \frac{\text{NUM}}{\text{DEN}} \overset{10.18,\ 10.19}{=} \frac{1}{V^2} \sum_p e^{-ik_p(x_1-x_2)} \left(\frac{-iV}{m^2 - k_p^2 - i\varepsilon} \right)$$

2 STEP = RECOVERING THE CONTINUUM

Recovering the continuum limit $\frac{1}{V}\sum_{k_p} \longrightarrow \int \frac{d^4k}{(2\pi)^4}$ hence:

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$$\frac{1}{V^2}\sum_{pe} e^{-i(k_p x_1 - k_e x_e)} \int D\phi \; (Re\phi_p + i Im\phi_p)(Re\phi_e + i Im_e)\; e^{\frac{-i}{2V}\sum(\,)(Re^2+Im^2)}$$

if $k_p \neq \pm k_e \longrightarrow \int D\phi \; \phi_e \phi_p \; e^{-i(\cdots)(Re^2+Im^2)} =$

$$= \int dRe\phi \; (Re\phi_e + i Im\phi_e)\, e^{-i(\,)(Re^2+Im^2)} \; (Re\phi_p + i Im\phi_p)$$

is prop to $\sim \int_{-\infty}^{\infty} dx \; x\, e^{-Qx^2} = 0 \rightarrow$ ODD function

if $k_p = k_e \longrightarrow \int D\phi \; (Re + iIm)^2 e^{i(\,)\cdots} = \int dRe\, dIm \; (Re^2 - Im^2 + 2i\, Re\, Im)\, e^{-i(\,)\cdots}$

These two terms are the sum of $\int dx\, x^2 e^{-iQx^2} - \int dy\, y^2 e^{-iQy^2} = 0$

$x = Re\phi \qquad y = Im\phi$

and they cancel case they are the same

This term is zero because is $\sim \int dx\, x\, e^{-Qx^2} =>$

if $k_p = -k_e \longrightarrow \int D\phi \; \phi_p \phi_{-p} \; e^{iS} = \int D\phi \; |\phi_p|^2 e^{iS} = \int D\phi \; (Re^2 + Im\phi^2)\, e^{-i(\,)(Re^2+Im^2)}$

$\sim$ is prop to $\sim \int dx \; x^2 e^{-Qx^2} = \frac{1}{2Q}\sqrt{\frac{\pi}{Q}}$

$$\frac{1}{V\chi}\sum_{\ell} e^{-i(k_e)(x_1 - x_2)} \; \frac{iZV}{Z(k^2 - m^2 - i\varepsilon)} \prod_{k_n \neq 0} \left(\frac{i\pi 2V}{(k_n^2 - m^2 - i\varepsilon)}\right)$$

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$$\langle 0|T\{\phi(x_1)\phi(x_2)\}|0\rangle = \int \frac{d^4k}{(2\pi)^4} e^{-ik(x_1-x_2)} \underbrace{\frac{i}{(k^2-m^2+i\varepsilon)}}_{\tilde{D}(k)} = D(x_1-x_2)$$

I got EXACTLY the Feynman propagator for scalar fields I had computed with canonical quantization ! Now we computed with the path integral formalism (for ref see P&S 9.2)

FOCUS = FUNCTIONAL DETERMINANT

Let's consider the Gaussian integral over a vector ξ

$$I = \left(\prod_n \int d\xi_n\right) \exp\left(-\underbrace{\xi_i B_{ij} \xi_j}_{\text{number given by}}\right)$$

$$(\cdots\cdots)[\cdots](\vdots)$$

where B_{ij} is a SYMMETRIC matrix with eigenvalues b_i. Consider O_{ij} the orthogonal matrix which diagonalizes B_{ij}. x_α the coefficient of ξ decomposed over rotation of O_{ij}

$$\xi_i := O_{i\alpha} x_\alpha$$

then $\xi_i B_{ij} \xi_j = \underbrace{O_{i\alpha} x_\alpha}_{\xi_i} B_{ij} \underbrace{O_{j\beta} x_\beta}_{\xi_j} = x_\alpha \underbrace{O_{i\alpha} B_{ij} O_{j\beta}}_{b_\alpha \delta_{\alpha\beta}} x_\beta = b_\alpha x_\alpha^2$

then

$$I = \left(\prod_n \int dx_n\right) e^{-\sum_\alpha b_\alpha x_\alpha^2} = \prod_\alpha \left(\int dx_\alpha e^{-b_\alpha x_\alpha^2}\right) =$$

$$= \prod_\alpha \sqrt{\frac{\pi}{b_\alpha}} = \text{Const.} \cdot (\det B)^{-1/2} \qquad 10.20$$

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Now consider the Lagrangian of free scalar fields

$$\mathcal{L} = \underset{S}{\phi(x)} \underset{B}{(-\partial^2 - m^2)} \underset{I}{\phi(x)}$$

When I integrate

$$\int D\phi \; e^{iS} = \int D\phi \; e^{i\int d^4x \, \phi(x)(-\partial^2 - m^2)\phi(x)} = \text{const} \cdot \left(\det(\partial^2 + m^2)\right)^{-1/2} \quad 10.21$$

(the last equality: 1920)

this object is called a FUNCTIONAL DETERMINANT and will be useful when computing path integrals for fermions, when we will use grassmann variables

10.4 PERTURBATIVE EVALUATION OF THE PATH INTEGRAL – GENERATING FUNCTIONAL

from Peskin & Schroder

Finding the propagator was a very messy calculation and was only for 2 points ... For $x_1 x_2 x_3 \dots x_n$ points it's completely not a thing! I would like an easier way. This is given by the generating functional (funzionale generante o generatrice) The generating functional approach allows to compute (gives a recipe to compute) denominator and numerator path integrals of eq 10.14, hence gives

$$\langle \Omega | T\{\phi(x_1) \dots \phi(x_n)\} | \Omega \rangle = \lim_{T \to \infty(1-i\varepsilon)} \left(\frac{\int D\phi \; \phi(x_1) \dots \phi(x_n) \, e^{iS}}{\int D\phi \, e^{iS}} \right)$$

Generating functionals can be written for every field type and are generally non-perturbative. However, they find their best application in perturbative interaction because usually, it can be evaluated exactly only in free fields, and approximated by perturbative expansion in case of interacting fields. Formally, however, is always valid.

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FOR a SCALAR FIELD the generating functional is:

Def $$Z[J] := \int D\phi \; e^{iS + i\int d^4x\, J(x)\phi(x)} \qquad (10.22)$$

Hence Z is a functional of the field $J(x)$, $S = \int d^4x\, \mathcal{L}$ is the action of the system considered (free or interacting). For a free theory, free scalar field for example $S = \int d^4x\, \frac{1}{2}(\partial_\mu\phi\,\partial^\mu\phi - (m^2 - i\varepsilon)\phi^2)$

The FUNCTIONAL DERIVATIVE is defined Formally as

Def $$\frac{\delta}{\delta J(y)} J(x) := \delta^{(4)}(x-y) \qquad (10.23)$$

Hence

$$\frac{\delta}{\delta J(y)} \int d^4x\, J(x)\phi(x) = \int d^4x\, \delta^{(4)}(x-y)\,\phi(x) = \phi(y) \qquad (10.24)$$

which means also that it extracts a scalar field when acting on $Z[J]$

$$(-i)\frac{\delta}{\delta J(y)} Z[J] = \int D\phi\, \phi(y)\, e^{iS + i\int J(x)\phi(x)d^4x}$$

$$\left\{ \int D\phi\, e^{iS + i\int dx^4 J\phi} \; (-i)\frac{\delta}{\delta J(y)}(i)\int dx^4 J(x)\phi(x) = \right\}$$

repeating n times in n points $x_1 \dots x_n$ we obtain the numerator as

$$\left(\frac{(-i)\delta}{\delta J(x_1)} \frac{(-i)\delta}{\delta J(x_2)} \dots \frac{(-i)\delta}{\delta J(x_n)}\right) Z[J]\Big|_{J=0} = \int D\phi\, \phi(x_1)\dots\phi(x_n)\, e^{iS}$$

one derivation for each field

Setting $J=0$ at the end after all derivations

The denominator is simply $Z[S=0]$ at $J=0$

$$Z[0] = \int D\phi\, e^{iS}$$

$$\langle \Omega | T\{\phi_1\phi_2\}|\Omega\rangle = \frac{1}{Z(0)}\left(\frac{-i\delta}{\delta J_1}\frac{-i\delta}{\delta J_2}\right) Z(J)\Big|_{J=0}$$

always TRUE, but Z depends on the specific S, hence Lagrangian of the system

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FREE THEORY - FEYNMANN PROPAGATORS

Here is the Z of the FREE Lagrangian

Hence 10.14 becomes, assuming FREE SCALAR AND VACUUM ($\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - m^2+i\varepsilon)\phi^2$)

$$\langle 0|T\{\phi(x_1)\cdots\phi(x_n)\}|0\rangle = \frac{1}{Z[0]_{Free}} \left(-i\frac{\delta}{\delta J(x_1)}\right)\cdots\left(-i\frac{\delta}{\delta J(x_n)}\right) Z[J]_{Free}\Big|_{J=0} \qquad 10.25$$

Feynman propagator

This formula is useful because for a free field theory $Z[J]$ can be written in a very explicit form

- For a FREE SCALAR FIELD $\mathcal{L} = \frac{1}{2}\left[\partial_\mu\phi\partial^\mu\phi - (m^2-i\varepsilon)\phi^2\right]$ where $+i\varepsilon$ is added for convergence, we have:

$$Z[J] = \int D\phi \exp\left\{ i\int d^4x \left[\underbrace{\frac{1}{2}(\partial_\mu\partial^\mu\phi - (m^2-i\varepsilon)\phi^2)}_{1^{st}\text{ term}} + J(x)\phi(x)\right]\right\} =$$

integrating by parts the 1st term of the exponent of the exponential, and discarding the boundary terms, we get

$$= \int D\phi \exp\left\{ i\int d^4x \; \frac{1}{2}\phi\underbrace{(-\partial^2 - m^2+i\varepsilon)}_{\text{operator acting on this}}\phi + \phi(x)J(x)\right\}$$

- Looking at the exponential, with $A = (-\partial^2 - m^2+i\varepsilon)$

$$\int dx^4 \left[\frac{1}{2}\phi_{(x)}A\phi_{(x)} + \phi_{(x)}J\right]$$

we want to complete the square in order to obtain a Gaussian integral which I know how to compute

Completing the square for Gaussian integral

Gaussian integral $I = \int_{-\infty}^{+\infty} e^{-x^2}dx = \sqrt{\pi}$

If I have $I = \int e^{-ax^2+bx}$ with $a>0$ then I have to COMPLETE THE SQUARE i.e. to shift x so to obtain again the Gaussian, only one quadratic term.

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$$\partial(\phi\phi) = \partial\phi\cdot\phi + \phi\partial\phi = 2\phi\partial\phi$$

$$\partial^2(\phi\phi) = 2\,\partial\phi\partial\phi + 2\phi\partial^2\phi$$

$$\partial\phi\partial\phi = \frac{1}{2}\partial^2(\phi\phi) - \phi\partial^2\phi$$

↓

integrating this part vanish because $\phi \to 0$ at boundaries

$$\int dx\ \partial\phi\partial\phi = \int dx\ \phi(-\partial^2)\phi$$

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$$-ax^2+bx = -a\underbrace{\left(x-\frac{b}{2a}\right)^2}_{x'}+\frac{b^2}{4a}$$

Hence

$$I = \int_{-\infty}^{+\infty} e^{-a(x-\frac{b}{2a})^2}dx \; e^{\frac{b^2}{4a}} = \sqrt{\frac{\pi}{a}}\, e^{b^2/4a}$$

In our case we have $x \to \phi$ $\quad a \to -\frac{1}{2}A$ $\quad b \to J$ so a non Rigourous completion of the square is :

$$\frac{1}{2}\phi A\phi + J\phi = \frac{1}{2}A\underbrace{(\phi + A^{-1}J)}_{\phi'}{}^2 - \frac{1}{2}A^{-1}J^2 = \frac{1}{2}\phi' A\phi' - \frac{1}{2}JA^{-1}J$$

so the final result is , recalling $A = (-\partial^2 - m^2 - i\varepsilon)$.

$$\int d^4x\left[\frac{1}{2}\phi A\phi + J\phi\right] = \int d^4x\left[\frac{1}{2}\phi' A\phi' - \frac{1}{2}JA^{-1}J\right]$$

ϕ' now only here in quadratic order

constant coefficient

- Since $\phi' = \phi + (-\partial^2 - m^2 + i\varepsilon)^{-1}J(x)$ is just a shift , it gives the same action as ϕ

$$Z[J] = \int D\phi\, e^{i\int d^4x\left(\frac{1}{2}\phi A\phi + J\phi\right)} \overset{S}{=} \int D\phi\, e^{i\int d^4x\left[\frac{1}{2}\phi' A\phi' - \frac{1}{2}JA^{-1}J\right]}$$

$$= \int D\phi\, e^{iS - i\int d^4x \frac{1}{2}JA^{-1}J}$$

Check $S = \int d^4x\, \frac{1}{2}\phi_{(x)} A\phi_{(x)}$ but if $\phi'(x) = \phi(x) + \text{const}$ dora cambiare variabile da $D\phi \to D\phi$ è uguale è non cambia la misura dell'integrazione perché anche $S = \int dx^4 \frac{1}{2}\phi' A\phi'$

- Now I remember that $A = (-\partial^2 - m^2 + i\varepsilon)$ is the Klein Gordon operator Hence we operate the substitution :

$$\phi' = \phi + (\partial^2 - m^2 + i\varepsilon)^{-1}J(x)$$

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see quadappunti n° 4 pink marked pages

It is possible to prove that the substitution is equivalent as
$\phi' = \phi - i\int d^4y\, D(x-y)\, J(y)$

Hence $\bar{A}^{-1} J = -i\int d^4y\, D(x-y)\, J(y)$

and therefore $-i\int dx^4\, J(x)\, \bar{A}^{-1} J = -\frac{1}{2}\int d^4x d^4y\, J(x)\, D(x-y)\, J(y)$

thus

$$Z[J] = \int D\phi\, e^{iS(\phi)} \underbrace{e^{-\frac{1}{2}\int dx^4 dy^4\, J(x)\, D(x-y)\, J(y)}}_{\text{independent from } \phi}$$

thus

$$Z_{\text{free}}[J] = Z[0]\, e^{-\frac{1}{2}\int dx^4 \int dy^4\, J(x) D(x-y) J(y)}$$

this expression of $Z[J]$ is true only for free ϕ fields, hence (10.25) for $\langle 0|T\{\}|0\rangle$ propagator

with $Z[0] = \int D\phi\, e^{iS}$

So now I have my nice expression for $Z[J]$ for a FREE SCALAR FIELD and I can compute my Feynmann propagators!

2 POINTS PROPAGATOR

$$\langle 0|T\{\phi(x_1)\phi(x_2)\}|0\rangle = \frac{1}{Z[0]}\left(\frac{-i\delta}{\delta J(x_1)}\,\frac{-i\delta}{\delta J(x_2)}\right) Z[J]\Big|_{J=0} =$$

$$= \left(\frac{-i\delta}{\delta J(x_1)}\,\frac{-i\delta}{\delta J(x_2)}\right) e^{-\frac{1}{2}\int dx^4 \int dy^4\, J(x)\, D(x-y)\, J(y)}\Big|_{J=0} = \text{acting with } \frac{\delta}{\delta J(x_2)}$$

$$= \frac{\delta}{\delta J(x_1)}\left[-\frac{1}{2}\int d^4y' D(x_2-y') J(y') - \frac{1}{2}\int d^4x' D(x'-x_2) J(x')\right] e^{-\frac{1}{2}\cdots}\Big|_{J=0} = \text{change } y' \to x' \text{ and use that for scalar field } D(x'-x_2) = D(x_2-x')$$

$$= \frac{\delta}{\delta J(x_1)}\left[-\int dx' D(x_2-x') J(x')\right] e^{-\frac{1}{2}\int dx^4 \int dy^4\, J(x) D(x-y) J(y)}\Big|_{J=0} =$$

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Acting with $\frac{\delta}{\delta J(x_1)}$

$$= \left(D(x_1 - x_2) + \dots \text{terms with } J \right) e^{-\frac{1}{2} \dots J \dots} \Big|_{J=0} = \quad \text{(evaluating in } J=0\text{)}$$

$$= D(x_1 - x_2)$$

$\Rightarrow$ CVD! I found that $\langle 0 | T\{\phi(x_1)\phi(x_2)\} | 0 \rangle = D(x_1 - x_2)$ is the Feynmann propagator!

4 POINTS PROPAGATOR — NOTATION

$$\langle 0 | T\{\phi_1 \phi_2 \phi_3 \phi_4\} | 0 \rangle = \frac{1}{Z(0)} \left(\frac{-i\delta}{\delta J_1} \frac{-i\delta}{\delta J_2} \frac{-i\delta}{\delta J_3} \frac{-i\delta}{\delta J_4} \right) Z[J] \Big|_{J=0} =$$

$$= \frac{\delta}{\delta J_1} \frac{\delta}{\delta J_2} \frac{\delta}{\delta J_3} \frac{\delta}{\delta J_4} \exp\left\{ -\frac{1}{2} \int dx \int dy \, J(x) D(x-y) J(y) \right\} \Big|_{J=0} = \text{acting with } \frac{\delta}{\delta J_4}$$

$$= \frac{\delta}{\delta J_1} \frac{\delta}{\delta J_2} \frac{\delta}{\delta J_3} \left(-J_{x'} D_{x'4} \right) e^{-\frac{1}{2} J_x D_{xy} J_y} \Big|_{J=0} = \text{acting with } \frac{\delta}{\delta J_3}$$

$$= \frac{\delta}{\delta J_1} \frac{\delta}{\delta J_2} \left(-D_{34} + J_{x'} D_{x'y} J_{y'} D_{y'3} \right) e^{\dots} \Big|_{J=0} = \text{acting with } \frac{\delta}{\delta J_2}$$

($\frac{\delta}{\delta J_2}$ on $-D_{34}$; $\frac{\delta}{\delta J_2}$ on $J_{x'} D_{x'y} J_{y'} D_{y'3}$; $\frac{\delta}{\delta J_2}$ on $e^{\dots}$)

$$= \frac{\delta}{\delta J_1} \left(D_{34} J_{x'} D_{x'2} + D_{24} J_{y'} D_{y'3} + D_{23} J_{x'} D_{x'4} + \left(J_{x'} D_{x'4} J_{y'} D_{y'3} \right)\left(-J_z D_{z2} \right) \right) e^{\dots} =$$

$$= \left(D_{34} D_{12} + D_{24} D_{13} + D_{23} D_{14} + \dots \text{terms with } J \right) e^{-\frac{1}{2} \dots J \dots} \Big|_{J=0} =$$

$$= D_{34} D_{12} + D_{24} D_{13} + D_{23} D_{14}$$

$\Rightarrow \langle 0 | T\{\phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4)\} | 0 \rangle = D(x_3 - x_4) D(x_1 - x_2) + D(x_2 - x_4) D(x_1 - x_3) + D(x_2 - x_3) D(x_1 - x_4)$

I obtained the same result of Wicks theorem!

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NOTATION

$\phi_i = \phi(x_i)$ $\quad J_i = J(x_i)$ $\quad D_{xy} = D(x-y)$

repeated indices give integration (sum) : $J_x D_{xy} J_y = \int d^4x \int d^4y \, J(x) D(x-y) J(y)$

Let's define

$$W[J] = \log(Z[J]) = W(0) + \frac{1}{2}\int d^4x\, d^4y\, J(x) D(x-y) J(y) \qquad 10.26$$

Therefore

$$\frac{\delta W[J]}{\delta J} = \frac{1}{Z[J]} \frac{\delta Z[J]}{\delta J}$$

quindi:

$$10.25 \rightarrow \left(\frac{-i\delta}{\delta J_1} \cdots \frac{-i\delta}{\delta J_n}\right) W[J]\Big|_{J=0} = \frac{1}{Z[J]} \left(\frac{-i\delta}{\delta J_1} \cdots \frac{-i\delta}{\delta J_n}\right) Z[J]\Big|_{J=0} = \langle 0|T\Sigma \cdots S|0\rangle$$

provides another way to compute propagators (10.25)

EXMPLE OF INTERACTING ϕ^4 SYSTEM — PERTURBATIVELY

If I have an interacting system, for the numerator, I can expand the interaction and the first order of the expansion it's computable with $Z(J)$ of free fields, with Feynman propagators:

$$\int D\phi\, \phi(x_1) \cdots \phi(x_n)\, e^{i\int d^4x \left(\underbrace{\frac{1}{2}(\partial_\mu\phi\partial^\mu\phi - (m^2 - i\varepsilon)\phi^2)}_{\text{free Lagrangian}} - \underbrace{\frac{\lambda}{4!}\phi^4}_{\text{interacting}}\right)} = \downarrow$$

expand the $e^{\int \frac{\lambda}{4!}\phi^4}$ (perturbatively)

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$$= \int D\phi\, \phi(x_1)\cdots\phi(x_n)\, e^{i\int d^4x \frac{1}{2}(\partial^\mu\phi\partial_\mu\phi - (m^2-i\varepsilon)\phi^2)} + \left\{ \text{zero order} \right.$$

$$- \frac{i\lambda}{4!}\int d^4x \int D\phi\, \phi(x_1)\cdots\phi(x_n)\,\phi^4(x)\, e^{i\int d^4x \frac{1}{2}(\bullet)} + \left\{ \text{1st order} \right.$$

$$+ O(\lambda^2)$$

$$= (\text{zero order}) + \frac{-i\lambda}{4!}\int d^4x\, (-i)^{n+4}\, \frac{\delta}{\delta J_1}\cdots\frac{\delta}{\delta J_n}\left(\frac{\delta}{\delta J(x)}\right)^4 Z[J]_{\text{free}}\Big|_{J=0} + O(\lambda)$$

$Z[J]_{\text{free}}$ → OF the FREE FIELD

If I use $W[J]$ instead of $Z[J]$
I obtain directly all the connected diagrams
and not only the numerator

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10.5 FUNCTIONAL QUANTIZATION OF FERMIONS SPINOR FIELDS (P/S 9.5)

The functional methods we used so far allow to compute propagators of fields that obey canonical COMMUTATION relations. Spinor fields obey ANTICOMMUTATION relations. To generalize the functional method to spinor fields we should REPRESENT FIELDS VIA ANTICOMMUTING NUMBERS called GRASSMAN NUMBERS

Grassmann numbers

Grassman numbers ψ_i are numbers which anticommute

$\psi_i \psi_j = -\psi_j \psi_i$ cioè $\{\psi_i, \psi_j\} = 0 \quad \forall i,j$

Hence a grassman number anticommute with itself thus this means:

$\psi\psi = -\psi\psi \longrightarrow \psi^2 = \psi\psi = \emptyset$

Function of grassman numbers

Be $f(\psi)$ a function of ONE grassman number ψ. I can always expand it in Taylor series around $\psi \to 0$

normal numbers

$f(\psi) = a + \psi b + \cancel{\psi^2 c} + \dots = a + \psi b$

all terms from 2nd order and higher are zero because $\psi^2 = 0$

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We WANT $f(\psi)$ and $f(\psi')$ to COMMUTE (because I'm interested at the integral of f that appears in the Lagrangian which has commuting terms) Hence we IMPOSE

$$f(\psi) f(\psi') \overset{!}{=} f(\psi') f(\psi)$$

$$(a+\psi b)(a'+\psi' b') = (a'+\psi' b')(a+\psi b')$$

$$aa' + \psi b a' + a\psi' b' + (\psi b)(\psi' b') = a'a + a'\psi b' + \psi' b' a + (\psi' b')(\psi b)$$

$$(\psi b)(\psi' b') \overset{!}{=} (\psi' b')(\psi b)$$

what happen when I switch $\psi \leftrightarrow b$ or $b \leftrightarrow b'$? do they commute or anticommute? Don't know yet, lets write in general

$$bb' \overset{(*)}{=} \lambda b'b \qquad \psi b \overset{(*)}{=} \lambda b \psi \quad \text{with } \lambda = \pm 1$$

Then

$$\psi' b' \psi b = \lambda^2 \psi' \psi b b' = -\lambda^2 \psi \psi' b b' = -\lambda^3 \psi b \psi' b'$$

($\psi' b' \psi b \overset{(*)}{=} \lambda \psi' \psi b b'$; $\psi \psi' = -\psi' \psi$)

Then it must be $\lambda = -1$! So we found that

$$bb' = -b'b \longrightarrow \{b, b'\} = 0$$

$$\psi b = -b\psi \longrightarrow \{b, \psi\} = 0$$

Hence:

$$f(\psi) = a + \psi b = a - b\psi$$

Right and Left derivative

We define:

Left derivative $\quad \partial_\psi f(\psi) = \partial_\psi (a + \psi b) = b$

Right derivative $\quad f(\psi) \overleftarrow{\partial_\psi} = (a - b\psi)\overleftarrow{\partial_\psi} = -b$

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Integral OVER ONE ψ GRASSMAN VARIABLE

$$\int d\psi\, f(\psi) =: I$$

- We impose it to be a LINER integral:

$$\int d\psi\, c f(\psi) \overset{!}{=} c \int d\psi\, f(\psi)$$ → thus I is a linear function of a and b (not of ψ because we are integrating over ψ)

- we impose it to be INVARIANT under SHIFT $\psi \to \psi + \eta$ with η a real number (→ this shift invariance is a fundamental aspect of the fermionism we are using - see PLS 9,6)

$$\int d\psi\, f(\psi) \overset{!}{=} \int d\psi\, f(\psi+\eta)$$

but $f(\psi+\eta) = a + \eta b + \psi b \to a' + \psi b$ this essentially shift changes a into $a \to a' = a + \eta b$, Thus shift invariance implies that the integral must not be dependent on a but only on b

- In general then $I = \lambda b$ with λ a real normalization that we impose BY CONVENTION to be 1, hence

$$\int d\psi\, f(\psi) = b \qquad \text{for } f = a + \psi b$$

function of N GRASSMANN VARIABLES

Choose the convention of performing the innermost integral first:

$$\int d\psi_i \int d\psi_j\; \psi_j \psi_i = +1$$

Hence $\int d\psi_j \int d\psi_i\; \psi_j \psi_i = -1$

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the function of n grassmann variables can be written as:

$$f(\psi_{i_1} \dots \psi_{i_n}) = a + \psi_i b_i + \frac{1}{2} \psi_{i_1} \psi_{i_2} C_{i_1 i_2} + \dots \frac{1}{n!} \psi_{i_1} \dots \psi_{i_n} d_{i_1 \dots i_n}$$

summing over repeated indeces

factor to compensate for repeated equal terms in the sum

$$\sum_{i_1 < i_2} \psi_{i_1} \psi_{i_2} C_{i_1 i_2} = \frac{1}{2} \sum_{i_1 i_2} \psi_{i_1} \psi_{i_2} C_{i_1 i_2} = \frac{1}{2} \psi_{i_1} \psi_{i_2} C_{i_1 i_2}$$

convention of sum over repeated indeces

When I perform the derivative over one single ψ_j I obtain for each term:

$$\partial_{\psi_j} (\psi_i b_i) = b_j$$

$$\partial_{\psi_j} \left(\frac{1}{2} \psi_{i_1} \psi_{i_2} C_{i_1 i_2} \right) = \frac{1}{2} \psi_{i_2} C_{j i_2} - \frac{1}{2} \psi_{i_1} C_{i_1 j} = \psi_i C_{ji}$$

$$\partial_{\psi_j} \left(\frac{1}{n} \psi_{i_1} \dots \psi_{i_n} d_{i_1 \dots i_n} \right) = \dots = \frac{n}{n!} \psi_{i_2} \dots \psi_{i_n} d_{j i_2 \dots i_n} = \frac{1}{(n-1)!} \psi_{i_2} \dots \psi_{i_n} d_{j i_2 \dots i_n}$$

then

$$\partial_{\psi_j} f(\psi_{i_1} \dots \psi_{i_n}) = b_j + \psi_i C_{ji} + \dots \frac{1}{(n-1)!} \psi_{i_2} \dots \psi_{i_n} d_{j i_2 i_3 \dots i_n}$$

↓

convention of sum over repeated indeces

When performing the integral of grassmann shift invariance, only the term in f with all $\psi_{i_1} \dots \psi_{i_n}$ remain invariant hence the integral does depend only on d coefficient of n all grassmann numbers

$$\int d^n \psi \; f(\psi) = d$$

with $f(\psi) = a + \psi_i b_i + \dots \frac{1}{n!} \psi_{i_1} \dots \psi_{i_n} d_{i_1 \dots i_n}$

with $d\, \varepsilon_{i_1 \dots i_n} = d_{i_1 \dots i_n}$ because $d_{i_1 \dots i_n}$ is zero if only two or more indeces are the same because $\psi^2 = 0$ property of grassmann numbers, hence it can be written as a number d per Levi-Civita tensor $\varepsilon_{i_1 \dots i_n}$

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PROPERTY OF INTEGRAL OVER CHANGE OF VARIABLE

Be $\psi_i = J_{ij}\, \psi_j'$ a transformation, with J_{ij} a matrix of normal numbers. We can prove that

$$\int d^n\psi'\, f(\psi) = \int d^n\psi\, f(\psi) \cdot \det(J)$$

Compare with the the property for normal real numbers $\bar{x} \in R^n$

$$x_i = J_{ij}\, x_j' \rightarrow \int d^n x'\, f(x) = \int d^n x\, f(x)\, \frac{1}{\det J}$$

GAUSSIAN INTEGRAL

Be M a symmetric matrix of normal complex numbers, then is it possible to prove:

$$\int d^n\psi\, e^{\frac{1}{2}\psi^T M \psi} = (\det M)^{1/2}$$

Compare with gaussian integral for normal $\bar{x} \in R^n$

$$\int^n x\, \exp\left(\tfrac{1}{2} x^T M x\right) = \frac{(2\pi)^{n/2}}{(\det M)^{1/2}}$$

$\Rightarrow$ when for normal number we have $\det()^{-1}$, for grassman we have $\det()$

COMPLEX GRASSMAN NUMBER

What said until now is valid for real grassman numbers. We can create COMPLEX grassmann numbers starting from two real $\psi_1\ \psi_2$ as:

$$\chi = \frac{1}{\sqrt{2}}(\psi_1 + i\psi_2) \qquad \bar{\chi} = \frac{1}{\sqrt{2}}(\psi_1 - i\psi_2)$$

Where each is a vector of n components $\psi_1 = (\psi_{1,1}, \psi_{1,2}, \ldots \psi_{1,n})$ ecc ecc

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We have the property of complex conjugation like hermitian conjugation of operators:

$$(\chi_a \chi_b)^* = \chi_b^* \chi_a^* \qquad \left(\text{with the notation before} \rightarrow \overline{(\chi_a \chi_b)} = \bar{\chi}_b \bar{\chi}_a\right)$$

When integrating over grassman complex numbers I can integrate independently over χ and $\bar{\chi}$. the integral properties before are

$$\int d^n\chi\, d^n\bar{\chi}\; e^{\bar{\chi}^T M \chi} = \det M \qquad (a)$$

with $\bar{\chi}^T = \chi^+$ adjoint (complex conjugate and transposed)

other integrals are:

$$\int d^n\chi\, d^n\bar{\chi}\; e^{\bar{\chi}^T M \chi + \bar{\eta}^T\chi + \bar{\chi}^T\eta} = \det M\; e^{-\bar{\eta}^T M^{-1}\eta} \qquad (b)$$

(η: grassman variable too)

and

$$\int d^n\chi\, d^n\bar{\chi}\; \chi\bar{\chi}\, e^{\bar{\chi}^T M \chi} = \det M\; M^{-1} \qquad (c)$$

n components $\chi = (\chi_1 \dots \chi_n)$

$$\int d^n\chi\, d^n\bar{\chi}\; \chi_\mu \bar{\chi}_\epsilon\, e^{-\sum_{ij}\bar{\chi}_i M_{ij}\chi_j} = \det M\; (M^{-1})_{\mu\epsilon}$$

With M being an symmetric (hermitian) complex matrix

! (Factor $\det(iM)$ or $(iM)^{-1} = +i M^{-1}$, the factor i comes from integration over Minkowski signature)

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2 Points Dirac propagator

using grassmann variables ψ_i to quantize a Dirac field $\psi(x)$ as:

$$\psi(x) = \sum_i \psi_i \cdot \phi_i(x)$$

ψ_i: grassmann numbers

$\phi_i(x)$: complex normal function of a basis set. For the Dirac field we take $\phi_i(x)$ as the basis for 4-component spinors.

- The two point correlator is given by:

$$\langle 0|T\{\psi(x)\bar{\psi}(y)\}|0\rangle = \frac{\int D\bar{\psi}\, D\psi\ \psi(x)\bar{\psi}(y)\, e^{i\int d^4x\, \bar{\psi}(i\not{\partial} - m)\psi}}{\int D\bar{\psi}\, D\psi\ e^{i\int d^4x\, \bar{\psi}(i\not{\partial} - m)\psi}} = (*)$$

- using (a) property of grassmann variable denominator became

$$\text{DEN} \overset{(a)}{=} \det{}^{!}(i(i\not{\partial} - m))$$

- using (c) the numerator

$$\text{NUM} \overset{(c)}{=} \det{}^{!}(i(i\not{\partial} - m))\ \left(+i\right)(i\not{\partial} - m)^{-1}(x,y)$$

hence

$$(*) = \frac{\det(i(i\not{\partial} - m))}{\det(i(i\not{\partial} - m))}(-i)(i\not{\partial} - m)^{-1} = +i\,(i\not{\partial} - m)^{-1}(x,y) \Rightarrow$$

- evaluating this inverse in Fourier space we find the familiar result for Feynman propagators of spinors:

$$\Rightarrow \int \frac{d^4k}{(2\pi)^4}\, e^{-ik\cdot(x-y)}\, \underbrace{i\,\frac{(\not{k} + m)}{(k^2 - m^2 + i\varepsilon)}}_{S_F(k)} = S_F(x-y)$$

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I can see that immediately recognizing recally that the propgator as o green function of $(i\not{\partial} - m)$ operator, obeys

$$(i\not{\partial} - m) S_F(x-y) = i\,\delta^4(x-y)$$

hence acting with the inverse on th $\delta^4(x-y)$ gives $S_F(x-y)$

$$(i\not{\partial} - m)^{-1}\, i\,\delta^4(x-y) = (i\not{\partial} - m)^{-1}(i\not{\partial} - m) S_F(x-y) = S_F(x-y)$$

Generating functional for the Dirac Field

In analogy with what done for scalar fields, we define the generating functional for spinor field as

$$Z[\bar\eta, \eta] = \int D\bar\psi\, D\psi\; e^{\,i\int d^4x \left[\bar\psi(i\not{\partial} - m)\psi + \bar\eta\psi + \bar\psi\eta\right]}$$

where η $\bar\eta$ are grassmann variable called "source terms". Completing the square by shifting $\psi(x)$, similarly as done for scalars field, is it pomble to rewrite Z in a simpler form:

$$Z[\bar\eta, \eta] = Z_0\; e^{-\int d^4x \int d^4y\, \bar\eta(x)\, S_F(x-y)\, \eta(y)}$$

where $Z_0 = Z[0,0] = \int D\bar\psi\, D\psi\; e^{\,i\int d^4x\, \bar\psi(i\not{\partial} - m)\psi} = \det\left(i(i\not{\partial} - m)\right)$

Then we can find the two correlators as

$$\langle 0|T\{\psi(x)\bar\psi(y)\}|0\rangle = \frac{1}{Z_0}\left(\frac{-i\,\overleftarrow{\delta}}{\delta\bar\eta(x)}\right) Z(\eta,\bar\eta)\left(\frac{i\,\overleftarrow{\delta}}{\delta\eta(y)}\right)\Bigg|_{\bar\eta,\eta=0}$$

Left derivative of grassmann variable, comes with factor $-i$

Right derivative of grassmann variable, comes with factor $+i$

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C 11 UV DIVERGENCIES

11.1 DIVERGENT AMPLITUDES = LOOP AND TADPOLES
(REGULARIZATION WITH Λ)

- SINGLE LOOP (4point correlator) → WICH'S THEOREM ROTATION
- TADPOLE & DOUBLE LOOP (2points correlator)

11.2 COUNTING UV DIVERGENCIE & RENORMALIZABILITY

11.3 RENORMALIZATION (WITH Λ REGULATOR) → BARE LAGRANGIAN

- 2POINTS CORRELATOR
- 4 POINTS CORRELATOR
- 1PI DIAGRAMS

C 12 DIMENSIONAL REGULARIZATION

12.1 DIMENSIONAL REGULARIZATION (WITH ε)

12.2 RENORMALIZATION → SHIFT LAGRANGIAN

1° anno Fisica Magistrale

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11

UV DIVERGENCIES

Let's go Back to chapter 8 and Feynmann rules to compute scattering amplitude of an interaction in perturbative regime. We said that the $\langle R|T\Sigma\phi\cdots S|R\rangle$ which appears in LSZ is proportal to

$$\langle R|T\Sigma\cdots S|R\rangle = \frac{\langle 0|T\Sigma\phi\cdots e^{-i\int d^4x\mathcal{H}}|0\rangle}{\langle 0|T\Sigma e^{-i\int d^4x\mathcal{H}}|0\rangle} = \frac{NUM}{DEN} = \text{all connected diagrams}$$

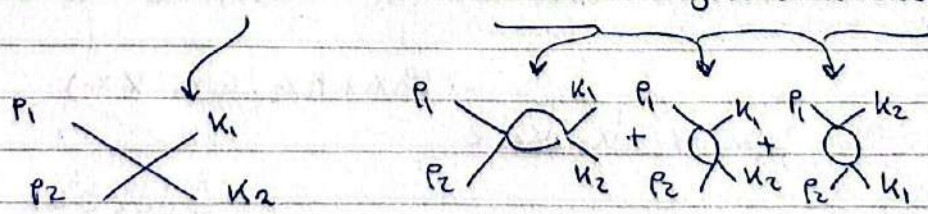
Since NUM = "sum of all diagrams at fixed external points" one can easily just compute NUM assuming to not count disconnected diagrams (which are eliminated canceled by DENOM)

Then we introduced the matrix amplitude $i\mathcal{M}$ which is given by the sum of all diagrams (connected) contributions from Feynman rules in momenta space

$i\mathcal{M}$ = 1st order diagrams + 2nd order diagr + 3d ...

For ϕ^4 theory we observed that 1st order contribution for 4points correlator is $-i\lambda$ hence

$i\mathcal{M}(p_1p_2\to k_1k_2)$ = $-i\lambda$ + 2nd order diagrams connected + ...



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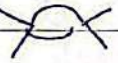
In this chapter we will point out how some diagrams give DIVERGENT CONTRIBUTION To iM amplitude. and present a way to RENORMALIZE this infinite into a finite contribution (when possible)

1) Divergent contributions (Regularization with Λ cut off)
2) counting divergencies
3) Renormalization
4) 2nd way of Renormalization (Chapter 12)

11.1 DIVERGENT AMPLITUDES : LOOPS AND TADPOLES (M 5.5.2)

We will now see where the problem arises, which diagrams give a divergent contribution. Such diagrams are (in ϕ^4 theory)

- loops 
- tadpoles
- two loops

Now we present some example of each divergence :

SINGLE LOOP

Consider ϕ^4 theory and $p_1 + p_2 \longrightarrow k_1 + k_2$ a 4 points process then iM is proportional to Fourier expansion of numerator :

$$iM \sim \int dx_1' dx_2' dx_3' dx_4' \, e^{-i(p_1 x_1 + p_2 x_2 - k_1 x_3 - k_2 x_4)} \cdot$$

$$\cdot \langle 0 | T\{ \phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4) \, e^{-i\frac{\lambda}{4!}\int dx \, \phi^4(x)} \} | 0 \rangle$$

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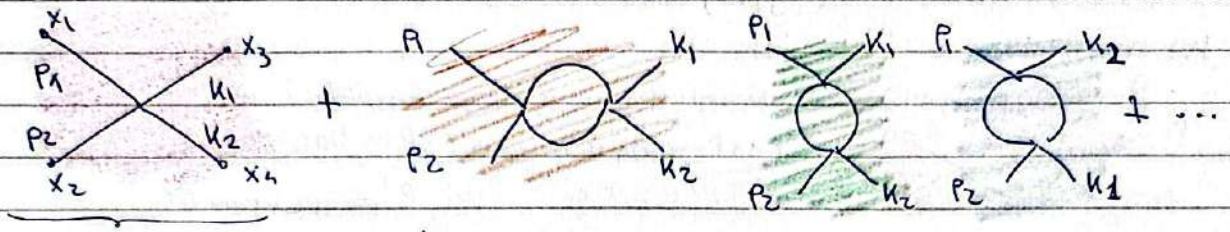
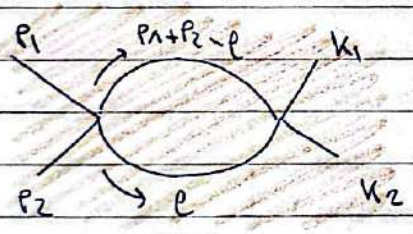
The diagrams (only connected) which contribute to $i\mathcal{U}$ for a 4 points correlator are:



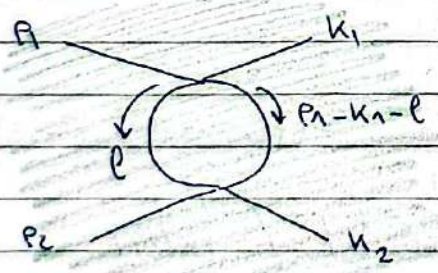
Hence:

$$i\mathcal{U} = -i\mathcal{A} + A_s + A_t + A_u + \text{3d order} + \dots$$

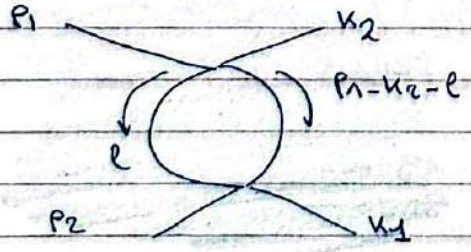
The three possible diagrams of 2nd order contain a single loop.



s-channel $\quad A(p_1+p_2) = A_s$



t-channel $\quad A(p_1-k_1) = A_t$



u-channel $\quad A(p_1-k_2) = A_u$

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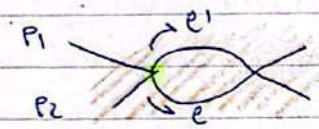
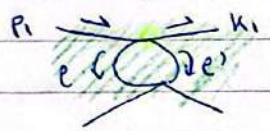
The key points of these loop diagrams is that the momenta of internal lines are not ALL fixed = imposing momenta conservation still some momenta are free, unconstrained = this is what we called the momentum ℓ =



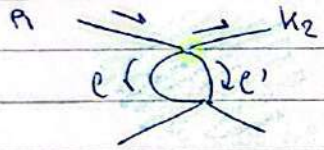
$p_1 + p_2 = \ell + \ell' \rightarrow \ell$ unconstrained

$\ell' = p_1 + p_2 - \ell$



$p_1 - k_1 = \ell + \ell' \rightarrow \ell$ unconstrained

$\ell' = p_1 - k_1 - \ell$

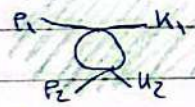


$p_1 - k_2 = \ell + \ell' \rightarrow \ell$ unconstrained

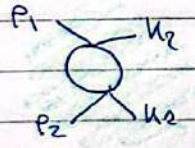
$\ell' = p_1 - k_2 - \ell$

Feynman rules compute : (eq 11.1) :

$$= A(p_1+p_2) = \frac{(-i\lambda)^2}{2} \int \frac{d^4\ell}{(2\pi)^4} \frac{i}{(\ell^2 - m^2 + i\varepsilon)} \frac{i}{((p_1+p_2-\ell)^2 - m^2 + i\varepsilon)}$$



$$= A(p_1 - k_1) = \text{//} \quad \frac{i}{((p_1 - k_1 - \ell)^2 - m^2 + i\varepsilon)}$$



$$= A(p_1 - k_2) = \text{//} \quad \frac{i}{((p_1 - k_2 - \ell)^2 - m^2 + i\varepsilon)}$$

SUCH INTEGRAL IS DIVERGENT! When the integrating ℓ goes to infinity $\ell \rightarrow \infty$ the integral diverges! This is called UV divergence because $\ell \rightarrow \infty$ is ℓ^0 = energy $\rightarrow \infty$ so we are going to "UV" when accounting for visible as starting point

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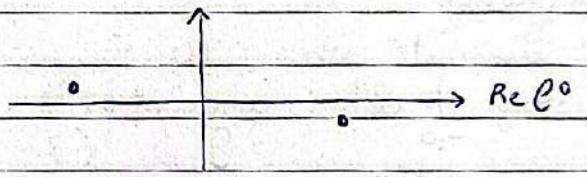
REGULARIZATION (with Λ)

AIM = extract the divergent part in $A(p_1+p_2)$ and separate it from NON-divergent part

1 observation = Integral is performed in Minkoski space over ℓ $\ell = (\ell^0, \vec{\ell})$ with metric $\ell^2 = \ell^{0\,2} - \vec{\ell}^{\,2}$ with poles at $\ell^0 = \pm\sqrt{\vec{\ell}^{\,2} + m^2 - i\varepsilon}$ coming from denominators. The integration path of ℓ^0 from $-\infty$ $+\infty$ in Real line passes near poles and convergence near poles is subtle and not always granted



This divergence can be eliminated by Wicks rotation to euclidean 4 space

this is de UV divergence which remains also after Wicks rotation

Another divergence is for when $\ell^0 \to \infty$ and this is the UV divergence. At high $\ell^0 \to \infty$ then the $(p_1 + p_2 - \ell)^2 \sim \ell^2$ Hence overall the integrand goes like $\frac{1}{\ell^4}$ and from the metric $\ell^3 d\ell \hat{=} d^4\ell$ we obtain $\frac{d\ell}{\ell}$ which is divergent at high $\ell \to \infty$

2 observation = Since the problem, the divergence we want to extract is in the UV limit $\ell \to \infty$, for that limit the $\ell \gg p$ so $(p_1+p_2-\ell)^2 \sim \ell^2$ and the integrand becomes

$$\sim \frac{1}{(\ell^2 - m^2 + i\varepsilon)(\ell^2 - m^2 + i\varepsilon)} = \frac{1}{(\ell^2 - m^2 + i\varepsilon)^2} \sim \frac{1}{\ell^4}$$

which is exactly $A(0)$

→ I can consider $A(0)$ because is sufficient to extract the divergent part since for $\ell \to \infty$ we have $(p_1+p_2-\ell)^2 \to \ell^2$

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- So consider $A(0) = \frac{\lambda^2}{2} \int \frac{d^4 \ell}{(2\pi)^4} \frac{1}{(\ell^2 - m^2 + i\varepsilon)^2}$

In Minkowski space, integrating all ℓ_1 ℓ^2 ℓ^3 and ℓ^0 from $-\infty$ to $+\infty$ I have two poles at:

$$\ell^0 = \pm \sqrt{\vec{\ell}^2 + m^2 - i\varepsilon} = \pm(E_\ell - i\varepsilon')$$

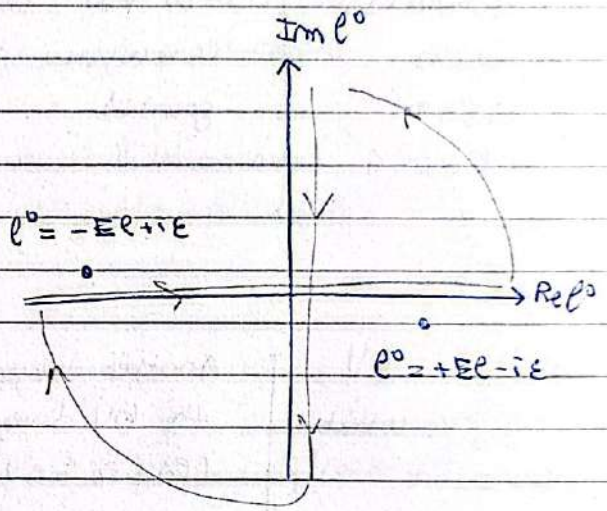
The integration path of ℓ_0 is along the real axis. I observe that if I perform a rotation as shown I don not cross anymore any poles. This is the Wick rotation:

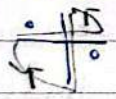
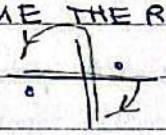
Integrating $\int_{-\infty}^{+\infty} d\ell^0$ is equivalent as

integrating $\int_{-i\infty}^{+i\infty} i\, d\ell_E^0$ when acting

the change of variable $\ell^0 = i\ell_E^0$

$d\ell^0 = i\, d\ell_E^0$



SUBTLE POINT = WICK'S ROTATION IS NOT JUST A SIMPLE CHANGE OF VARIABLE = the key point is that I can change the integration provided into another path of complex integration only if I DO NOT CROSS ANY POLES. So I HAVE TO MAKE THE ROTATION "ANTICLOCKWISE" if  or "CLOCKWISE" if 

- So Wick's rotation consist of $\ell^0 = i\ell_E^0$ and integrate over ℓ_E^0. Re writing the integrand

$$\ell^2 = \ell^{0\,2} - \vec{\ell}^{\,2} = -\ell_E^{0\,2} - \vec{\ell}^{\,2} =: -(\ell_E^{0\,2} + \vec{\ell}^{\,2}) =: -\ell_E^{\,2}$$

Minkowski scalar product → ℓ_E^2 for Euclidean scalar product

with $\ell_E = (\ell_E^0, \vec{\ell})$ is a 4-vector in Euclidean space with norm $\ell_E^2 = \ell_E^{0\,2} + \vec{\ell}^{\,2}$

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WICK'S ROTATION = Integrate over $i\ell_E^4$, since over ℓ_E euclidean vector

$d\ell^0 = i d\ell_E^0$, $d\vec{\ell} = d\vec{\ell}_E$ $\rightarrow$ $d^4\ell = i d^4\ell_E$

- over $\vec{\ell}_E^3$ real part of space $\vec{\ell}_E$
- over $i\ell_E^0$ = imaginary part of ℓ_E euclidean vector

so

$$\rightarrow A(0) = \frac{\lambda^2}{2}\int \frac{i d^4\ell_E}{(2\pi)^4} \frac{1}{(-\ell_E^2 - m^2 + i\epsilon)^2} = \frac{i\lambda^2}{2}\int \frac{d^4\ell_E}{(2\pi)^4}\frac{1}{(\ell_E^2+m^2)^2}$$

$i\epsilon$ $\rightarrow$ no need anymore since no cross poles anymore

- Then in Euclidean space $d^4\ell_E$ can be written as $d^4\ell_E = \ell_E^3\, d\ell_E\, d\Omega$ so (focus, phase)

$$\rightarrow A(0) = \frac{i\lambda^2}{2}\int d\Omega \int d\ell_E\, \ell_E^3 \frac{1}{(\ell_E^2+m^2)^2}$$

- this is $\sim \frac{d\ell_E}{\ell_E^4}\ell_E^3 \sim \frac{d\ell_E}{\ell_E}$ which is still divergent for UV $\ell_E \to \infty$ so I introduce a cut-off Λ and write using $\frac{1}{(\ell_E^2+m^2)^2} = \frac{1}{\ell_E^4}\left(1 - \frac{2m^2}{\ell_E^2}\right)$

$$\rightarrow A(0) = \frac{i\lambda^2}{2}\int \frac{d\Omega}{(2\pi)^4}\int_0^\Lambda d\ell_E \frac{1}{\ell_E} + \{\text{Finite part non divergent}\}$$

- using only the logarithmic integral I get

$$A(0) = \frac{i\lambda^2}{2}\frac{2\pi^2}{(2\pi)^4}\int_0^\Lambda \frac{d\ell_E}{\ell_E} + \{\text{finite}\} = i\lambda^2\left(\frac{1}{16\pi^2}\log(\Lambda) + \{\text{finite}\}\right) \quad (11.4)$$

- all s, t and u channel give same contribution $A(0)$ so

$$i\mathcal{M} = -i\lambda + i\lambda^2\left(\frac{3}{16\pi^2}\log\Lambda + \{\text{finite}\}\right) + \dots \quad (11.5)$$

DIVERGENT for $\Lambda \to \infty$ — FINITE $O(\lambda^3)$ CONTRIBUTION

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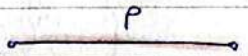
TADPOLES & DOUBLE LOOP - CORRECTIONS TO PROPAGATOR

Tad pole and double loops are higher order corrections to propogator $\tilde{D}(p)$.

(zero order = single propogator $\tilde{D}(p)$. $e^{+i\varepsilon}$ = 

(1st order = single tadpole $\tilde{D}(p)(-iB)\tilde{D}(p) :=$

I decide to call $-iB$ the contribution from the tadpole at 1st order. From Feynmann rules

$$= -iB = -i\lambda \int \frac{d^4\ell}{(2\pi)^4} \frac{i}{\ell^2 - m^2 + i\varepsilon} =$$

this integral diverges! So perform Wicks rotation $d^4\ell \to i d^4\ell_E$ so $\ell^2 \to -\ell_E^2$ and change integration from $-\infty$ to ∞ to $-i\infty$ to $i\infty$ so

$$-i\lambda \int \frac{d^4\ell}{(2\pi)^2} \frac{i}{\ell^2 - m^2 + i\varepsilon} \overset{\text{WICK'S ROTATION}}{\to} -i\lambda \int \frac{d^4\ell_E}{(2\pi)^4} \frac{i^2}{-\ell_E^2 - m^2} = -i\lambda \int \frac{d^4\ell_E}{(2\pi)^4} \frac{1}{\ell_E^2 + m^2} =$$

$$= -i\lambda \int \frac{d\Omega}{(2\pi)^4} \int_0^{\Lambda} d\ell_E \frac{\ell_E^3}{\ell_E^2 + m^2} = -i\lambda \frac{2\pi^2}{(2\pi)^4} \int_0^{\Lambda} d\ell_E \frac{\ell_E^3}{\ell_E^2 + m^2} =$$

$$= \frac{-i\lambda}{16\pi^2} \left[\Lambda^2 - m^2 \log\left(\frac{\Lambda^2 + m^2}{m^2} \right) \right] =: -iB$$

(2 order correction would be $\tilde{D}(p)(-iB)\tilde{D}(p)(-iB)\tilde{D}(p) =$

(3 order is $\tilde{D}(p)(-iB)\tilde{D}(p)(-iB)\tilde{D}(p)(-iB)\tilde{D}(p) =$

summing all correction of all order to propogator I obtain the following:

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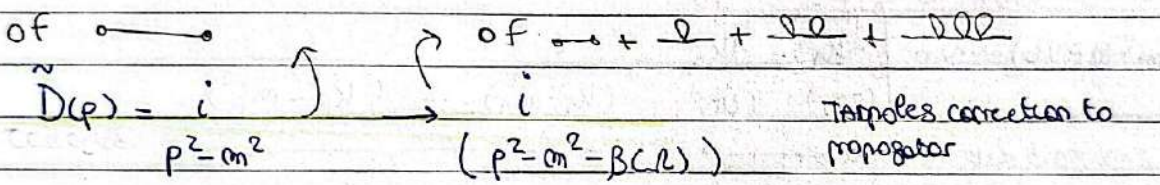
$\tilde{D}(p) + \tilde{D}(p)(-iB)\tilde{D}(p) + \tilde{D}(p)(-iB)\tilde{D}(p)(-iB)\tilde{D}(p) + \dots = \sum x^n =$

$= 1 + x + x^2 + x^3 + \dots =$

$= \tilde{D}(p)\left[1 + (-iB)\tilde{D}(p) + \left((-iB)\tilde{D}(p)\right)^2 + \left((-iB)\tilde{D}(p)\right)^3 + \dots\right] = \frac{1}{1-x}$ serie geometrica

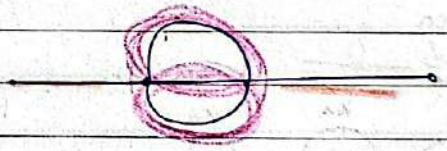
$= \tilde{D}(p)\,\frac{1}{(1+iB\,\tilde{D}(p))} \hookrightarrow \tilde{D}(p) = \frac{i}{p^2-m^2} = \frac{i}{p^2-m^2}\left(\frac{1}{1-\frac{B}{p^2-m^2}}\right) = \frac{i}{(p^2-m^2-B)}$

if I take into account dee tadpoles I obtain a correction which is exactly the same propogator but change the mass from $m^2 \to m^2 + B(\Lambda)$ wher $B(\Lambda)$ is infinite as $\Lambda \to \infty$ UV limit



of $\tilde{D}(p) = \frac{i}{p^2-m^2}$ $\to$ of $\frac{i}{(p^2-m^2-B(\Lambda))}$ = + + + + ...

Tadpoles correction to propogator



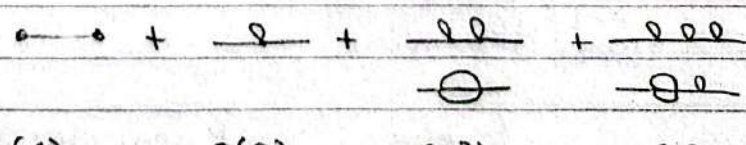
Double loops as: further modify the propogator. We just give the final result:

$$\frac{i}{p^2\,A(\Lambda,p^2) - m^2 - B(\Lambda)}$$

$A(\Lambda,p^2)$ ← from double loops; $B(\Lambda)$ ← from tadpoles

Tadpoles and double loop correction to $\tilde{D}(p)$ in ϕ^4 theory for UV divergence

$= \tilde{D}_{it}(p)$ corrected — it contains the contribution of all corrections ↓



+ + + + ...

$O(1)$ $O(\lambda)$ $O(\lambda^2)$ $O(\lambda^3)$

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11.2 COUNTING UV-DIVERGENCIES

Consider a ϕ^n-theory in a space of dimension d

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)^2 - \frac{m^2}{2}\phi^2 - \frac{\lambda}{n!}\phi^n$$

$\mu = 0, \dots d$; n theory

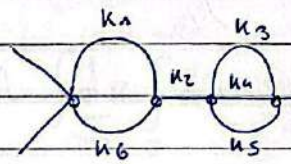
Consider a DIAGRAM with N external legs, P internal propagator, V vertices, L loops, then the contribution to $\mathcal{M}$ of such diagram is =

$$\mathcal{M}(\text{from 'NPVL}) \sim \int \frac{d^d k_1 \dots d^d k_L}{(2\pi)^d \dots (2\pi)^d} \frac{i}{(k_1^2 - m_1^2)} \dots \frac{i}{(k_P^2 - m_P^2)}$$

for each loop I have one unconstrained momentum, where d is the space dimension

one propagator for each internal propagator

ES:



$P = 6$

$L = 3$

$V = 4$

$N = 3$

$$\int \frac{d^d k_1 d^d k_2 d^d k_3}{(2\pi)^{3d}} \frac{i}{(k_1^2 - m_1^2)} \dots \frac{i}{(k_6^2 - m_6^2)}$$

$k_1 k_2 k_3$ are one for each loop

$k_4 k_5 k_6$ are fixed by conservation law and depend on $k_1 k_2 k_3$ and ext momentum

I define:

SUPERFICIAL DEGREE OF DIVERGENCE $D := d \cdot L - 2P$

from $\int d^d k_1 \dots d^d k_L \; \frac{i}{(k_1^2 - m_1^2)} \dots \frac{i}{(e^2 - m^2)}$

integrals in d dimension hence $+dL$

P propagators, each carrying a e^{-2}, hence $-2P$

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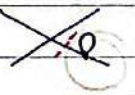
For such diagram NVLP which $D = dL - 2P$ I expect a DEGREE of divergence like:

- $\sim \Lambda^D$ if $D > 0$ where Λ comes from $\int_0^\Lambda dk^4$
- $\sim \log(\Lambda)$ if $D = 0$
- NO divgence if $D < 0$

EXCEPTIONS:

a) Tuwiol diagrams like has $L = P = 0$ hence $D = 0$ but it's not Divergent, there is no loop, nothing to integrate over.

b) A diagram may have overall $D < 0$ but may contain sub diagrams (a part which can be isolated by cutting only one propagator) which are divergent → hence the entire diagram is gonna diverge

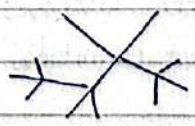
ex:  has $D = 0$ but subdiagram divgent

c) Symmetries (e.g. Gauge symmetry) can reduce the degree of divgence

COUNTING D

Now I want to count Divergencies D in terms of number of vertices V and number # of particles interacting in the vertex hence n for a ϕ^n theory

For a tree graph in ϕ^n theory (no loops) the n° of vertices is:

$V = P + 1$ (for $L = 0$ no loops) 

P = 3
V = 4
(L = 0)

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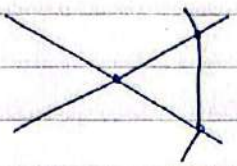
each loop is a vertex joined by two lines hence

$$L = P - V + 1$$



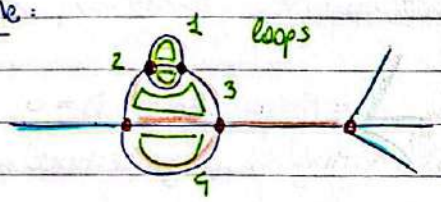
$L = 1$
$P = 3$
$V = 3$

in ϕ^n theory each vertex contains n-lines so

$$nV = N + 2P$$

ext lines ↳ 2 because each propagator connects two vertices

Example:



$V = 5$
$N = 4$
$P = 8$
$L = 4$

So I can rewrite D as:

$$D = dL - 2P = d\underbrace{(P - V + 1)}_{L} - \underbrace{(nV - N)}_{2P} = \dots =$$

$$D = d + V\left(n\frac{d-2}{2} - d\right) - N\left(\frac{d-2}{2}\right)$$

The coefficient $\Delta = n\frac{d-2}{2} - d$ of vertices tells us if a theory is renormalizable or not. To renormalize I want to "eliminate" divergences, but if adding vertices increases the degree of D divergencies, it's not possible to renormalize the whole theory for all order of the expansion!

NON RENORMALIZABLE THEORIES: $\Delta > 0$

if $\Delta > 0$ adding vertices increases D → higher orders (more vertices) would always give divergent amplitudes

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SUPER-RENORMALIZABLE THEORIES $\alpha < 0$

If $\alpha < 0$ adding vertices reduces D. → I have only a finite number of UV divgent diagrams, from 0 vertex order and higher all diagrams are not divgent (D becomes negative)

RENORMALIZABLE THEORIES $\alpha = 0$

If $\alpha = 0$ adding vertices keeps D constant → all order diagrams have same D, D is finite, so a finite number of divgences in all order diagrams

ES: ϕ^4 THEORY IN MINKOWKI SPACE

$n = 4$ $d = 4$ hence $n\frac{d-2}{2} - d = 4\frac{2}{2} - 4 = 0$ is a RENORMALIZABLE THEORY and all orders have some Degeneracy.

$D = d + V - N\left(\frac{d-2}{2}\right) \xrightarrow[d=4]{} D = 4 - N$

D depends only on the number of ext lines N

N = 0 VACUUM $\langle \Omega | \Omega \rangle$

D = 4 hence vacuum diverges as $\sim \Lambda^4$ we found vacuum to vacuum divergence ampetude ... however I can ignore it cause it's renormalizable

N = 2

D = 2 ord for two ext lines, one initial one final, div is $\sim \Lambda^2$

N = 4

D = 0 ord for four ext lines (ex 2 initio and 2 final) $\sim \log(\Lambda)$

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ES:

$d=3$ as $\mathbb{R}^3$ and $n=6$ as ϕ^6 → $\alpha = 0$ renormalizable
$d=3$ as $\mathbb{R}^3$ ad $n=4$ as ϕ^4 → $\alpha < 0$ superrenormalizable
$d=2$ as $\mathbb{R}^2$ ANY ϕ^n IS superrenormalizable

MASS DIMENSION OF COUPLING & RENORMALIZABILITY

In natural unit space is inverse of energy and energy is the same of mass unit $[m]$

$$[x] = [m]^{-1}$$

The Lagrangian is of the unit of $[m]^d$ d = dim of space such that the Action S is DIMENSIONLESS

$$S = \int d^d x \, \mathcal{L} \qquad [S]=1 \qquad [x]=[m]^{-1} \qquad [\mathcal{L}]=[m]^d$$

Also derivative is inverse of space → $[\partial] = \frac{\partial}{\partial [x]} = [m]$
hence is dimension of $[m]$

A typical Lagrangian shows kinetic term and couplings. Let's analyze the dimension of the fields given by different Lagrangian

DIMENSION OF ϕ SCALAR FIELD

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi \partial^\mu \phi - m^2 \phi^2) + \cdots$$

Each piece of the $\mathcal{L}$ should be of dim $[m]^d$ hence

$$[m]^d = [m^2 \phi^2] = [m]^2 [\phi]^2 \rightarrow [\phi] = [m]^{\frac{d-2}{2}}$$

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DIMENSION OF FERMIONIC FIELD $\bar{\psi}\psi$

$$\mathcal{L} = \bar{\psi} i \not{\partial} \psi - m \bar{\psi}\psi + \dots$$

$$[m]^d = [m\bar{\psi}\psi] = [m][\bar{\psi}\psi] \longrightarrow [\bar{\psi}\psi] = [m]^{d-1}$$

DIMENSION OF BOSON FIELD A_μ

$$\mathcal{L} = -\frac{1}{4} F^{\mu\nu} F_{\mu\nu} + \dots \sim [\partial^\mu A_\mu]^2$$

$$[m]^d = [\partial^\mu A_\mu]^2 = [\partial]^2 [A_\mu]^2 \longrightarrow [A_\mu] = [m]^{\frac{d-2}{2}}$$

DIMENSION OF COUPLINGS

A Lagrangian may contain interaction terms with coupling constant ($e.g. \ \frac{\lambda}{4!}\phi^4$ or $g\bar{\psi}\psi\phi$ or $u\bar{\psi}\psi A^2$ ....)

Consider a generic interaction term of coupling g that make a number of f fermionic, s scalars and γ bosonic field to interact

$$\mathcal{L} \sim g (\bar{\psi}\psi)^f \phi^s A_\mu^\gamma$$

we can calculate the mass dimension of fields coupling by imposing:

$$[m]^d = [g][\bar{\psi}\psi]^f [\phi]^s [A_\mu]^\gamma \longrightarrow [g] = [m]^{d - f(d-1) - s\frac{(d-2)}{2} - \gamma\frac{(d-2)}{2}}$$

if $[g] > 0$ → theory is SUPER RENORMALIZABLE

if $[g] = 0$ → RENORMALIZABLE

if $[g] < 0$ → NON RENORMALIZABLE

↓

this $[g]$ number is the $-①$ that occurs as coefficient in V vertex in D, that's why if $[g] > 0$ is superrenormalizable ← because $① < 0$!

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11.3 RENORMALIZATION (M. 5.6)

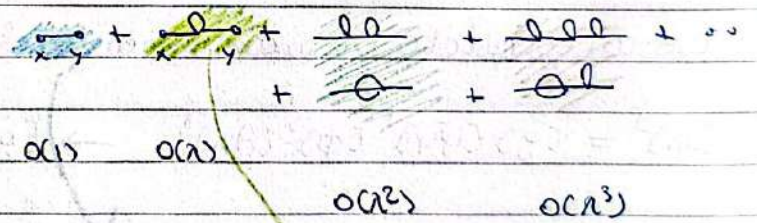
Renormalization is a procedure to eliminate the divergence of some diagrams (loops, tadpole etc etc). A way to proceed is to introduce a uv cutoff Λ over the modulus of euclidean momenta of the loop integral. Sending $\Lambda \to \infty$ the diagram diverges but as long as we consider Λ finite our formulas have a dependence on Λ. We will then first define coupling, mass and field depending on Λ then re-define them eliminating the Λ dependence, ie eliminating the divergence. → FOR BETTER EXPLANATION, SEE INTRODUCTION TO CHAPTER 12 (FEW PAGES AHEAD)

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi_0 \partial^\mu \phi_0 - m_0^2 \phi_0^2) - \frac{\lambda_0}{4!}\phi_0^4$$

where $\lambda_0 = \lambda_0(\Lambda)$ $m_0 = m_0(\Lambda)$ $\phi_0 = \phi_0(x, \Lambda)$ are "BARE" coupling, mass, field

2 POINTS CORRELATOR

In chapter 11.1 I calculated the fourier transform of $\langle \Omega | T\{\phi_x \phi_y\} | \Omega \rangle$ (which is the sum of all corrections — + —●— + —●●— + ···) can be written as a single "corrected" propagator

$\langle \Omega | T\{\phi(x)\phi(y)\} | \Omega \rangle \sim$ 

$$\int dx\, e^{-ixp} \int dy\, e^{iky} \langle \Omega | T\{\phi_x \phi_y\} | \Omega \rangle = \tilde{D}(p) + \tilde{D}(p)(-iB)\tilde{D}(p) + \cdots =$$

Fixing a certain value of Λ finite I get

$$= \tilde{D}_\Lambda(p) = \frac{i}{p^2 A(\Lambda, p^2) - m_0^2 - B(\Lambda)}$$

↓ double loops ↓ tadpole

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Identità picche
PVO con P

Such expression enters in LSZ formula:

$$\int dx^4 e^{-ikx} \int dy^4 e^{iky} \langle \Omega | T\{\phi_x \phi_y\} | \Omega \rangle = \frac{i\sqrt{Z}\, i\sqrt{Z}}{(k^2-m^2)(p^2-m^2)} \langle k | iT | p \rangle$$

where for $k=p$ for sufficiently small p.

$$\int dx^4 e^{-i(x-y)p} \langle \Omega | T\{\phi_x \phi_y\} | \Omega \rangle = \frac{iZ}{(p^2-m^2)} \quad \langle p | \mathcal{M} | p \rangle$$

Hence the vacuum expectation of $T\{\phi_0(x)\phi_0(y)\}$ at fixed Λ is:

$$\langle \Omega | T\{\phi_0(x)\phi_0(y)\} | \Omega \rangle = \int \frac{d^4p}{(2\pi)^4} e^{-ip(x-y)} \tilde{D}_\Lambda(p) = D_\Lambda(x-y)$$

(inverse Fourier)

M.3

where $\tilde{D}_\Lambda(p) = \dfrac{i}{p^2 A(p^2,\Lambda) - m_0^2 - B(\Lambda)}$ } contains all — ∘ ∘∘ ∘∘∘ ...

$$A(p^2,\Lambda) = 1 + \lambda_0^2 (\log \Lambda^2) + (\text{finite parts non dependy on } \Lambda)$$

$$B(\Lambda) = \frac{\lambda_0}{32\pi^2} (\Lambda^2 + \log \Lambda + \text{finite parts}) + O(\lambda_0^2)$$

(observe that A e B have also an implicit dependence on Λ by $\lambda_0(\Lambda)$)

$O(\lambda_0)$ → TADPOLE

At $O(\lambda_0)$ $A=1$ and $B(\Lambda) = \frac{\lambda_0}{32\pi^2}(\Lambda^2 + \log\Lambda + \text{finite})$ and it's the case of a single tadpole. In such case

$$\tilde{D}_\Lambda(p) = \frac{i}{p^2 - m_0^2 - B(\Lambda)} = \text{—}\circ\text{—} + \text{—}\circlearrowright\text{—}$$

We observe that if I define $m_R^2 = m_0^2 + B(\Lambda)$ I obtain the form of normal single propagator of mass m. I call m the REAL MASS. Hence $p^2 = m_R^2$ is the pole of denominator $= 0$

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m is the real mass and it's FINITE, $m_0(\Lambda)$ is a parameter completely in our hand, I define on it in order to compensate for $B(\Lambda)$ divergence so that m^2 is FINITE (. m_0^2 not observable, m observable)

What have I done? I RE-Defined the Lagrangian with a mo mass parameter and noticed that defining $m^2 = m_0^2 + B(\Lambda)$ I can write the Whole propagator of $O(\lambda_0)$ (as $\longrightarrow + \longrightarrow\!\!\bullet\!\!\longrightarrow$) as the normal propagator ($\circ\!\!\longrightarrow\!\!\bullet$) of mass m

(cosè ho cancellato la divergenza partendo da una di quelle che dopo mo scelto a priori per far sparire la divergenza)

$O(\lambda_0^2)$ → DOUBLE LOOP

At $O(\lambda_0^2)$ I'm considering also double loops. I get additional divergence from $A(p^2, \Lambda)$

$$\frac{i}{A(p^2,\Lambda)p^2 - m_0^2 - B(\Lambda)} = \circ\!\!\longrightarrow\!\!\bullet + \longrightarrow\!\!\bullet\!\!\longrightarrow + \longrightarrow\!\!\bullet\!\!\circlearrowright\!\!\bullet\!\!\longrightarrow + \circlearrowright$$

I still want to determine the pole for which the denominator goes to zero in order to define m^2 real mass (. ie m^2 è quel valore per cui quando $p^2 = m^2$ il denominatore si annulla → così ritrovo la condizione di "singolo" normale propagatore)

I define the Real mass as:

$$A\,p^2 - m_0^2 - B \Big|_{p^2 = m_R^2} \overset{!}{=} 0 \quad \Longrightarrow \quad A(m_R^2, \Lambda)\, m_R^2 - m_0^2 - B \overset{!}{=} 0$$

(1st imposition) On mass

↓ A depends on $p^2 \overset{!}{=} m_R^2$

however this unique condition is not enough to eliminate the two divergences of B and A, because A depends on p^2 too.

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I can expand $f(x) = f(x_0) + f'(x)|_{x_0}(x-x_0) + \cdots$
with $x = p^2$ $x_0 = m^2$ the denominator hence

$$A(p^2)\,p^2 - m_0^2 - B = \emptyset + \frac{d}{dp^2}\left(A(p^2)\,p^2\right)\Big|_{p^2=m^2}(p^2-m^2) + \cdots$$

hence

$$\tilde{D}_\Lambda(p) = \frac{i}{A p^2 - m_0^2 - B} = \frac{i\,Z}{p^2 - m^2} + \cdots$$

with $$Z = \frac{1}{\frac{d}{dp^2}\left(A(\Lambda, p^2)\,p^2\right)\Big|_{p^2=m^2}} = Z\left(\lambda_0(\Lambda), \frac{\Lambda}{m}\right)$$

Z is a factor depends only on $\frac{\Lambda}{m}$, is dimensionless. (a meglio, essendo che Z deve essere dimensionless può dipendere da m e Λ solo come in Λ/m)

I see that if I impose Z to be same factor present in LSZ formula I can rewrite a renormalized field as.

⊙ $\phi_R(x,\Lambda) \overset{!}{=} \phi_R(x)\sqrt{Z(\Lambda)}$ (2nd important) (on field)

ϕ_R doesn't depend on Λ, the whole Λ dependence of ϕ_0 is inside Z. I see the with this field, in the LSZ formula, Z cancel out!

$$LSZ: \underbrace{\int dx^4\, e^{-i(x-y)p}\, \underbrace{\langle \Omega | T\{\phi_0(x)\phi_0(y)\} | \Omega\rangle}_{D_\Lambda(x-y)}}_{\tilde{D}_\Lambda(p)} = \frac{i\,Z}{(p^2-m^2)}$$

with ϕ_R is:

$$\int dx^4\, e^{-i(x-y)p}\, \langle \Omega | T\{\phi_R(x)\phi_R(y)\} | \Omega\rangle\, Z = \frac{i\,Z}{(p^2-m^2)}$$

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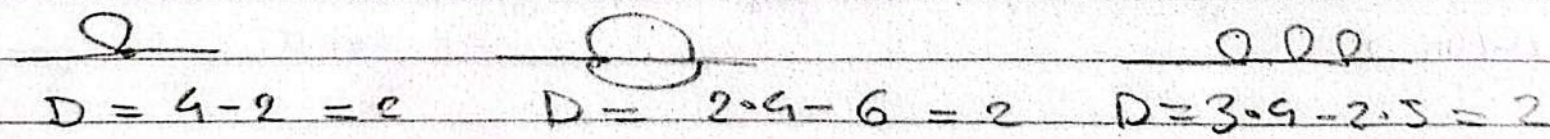
Using ϕ_R I obtain exactly the propagator of free field with mass m^2!

$\tilde{D}(p)$ of ϕ_R $= \dfrac{i}{p^2 - m_R^2}$ accounts for all $\bullet\!\!-\!\!\bullet + \bullet\!\!-\!\!\bigcirc\!\!-\!\!\bullet + \bullet\!\!-\!\!\bigcirc\!\!-\!\!\bigcirc\!\!-\!\!\bullet \ldots$

$\Rightarrow$ So we have removed UV Divergences by defining m_R^2 and ϕ_R^2 from ϕ_0 and m_0 as depicted by the two insertions = Renormalization conditions marked in ●. After that, the theory looks like a normal propagator for m_R^2 and ϕ_R^2

$\Rightarrow$ we calculated for $O(\lambda_0^2)$ but it's valid for all other orders because the $\frac{\lambda}{4!}\phi^4$ theory is renormalizable and each order has some degree of divergence in case of tadpoles and propagator corrections



4 POINTS CORRELATOR

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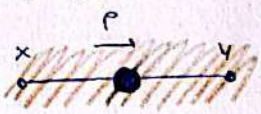
Straightforward derivation of Renormalization conditions on m and ϕ from 2 points correlator

- Whole two points correlator of $\mathcal{L}_0$ theory gives:

$$\tilde{D}_{\lambda_0}(p) = \frac{i}{p^2 A - m_0^2 - B} =$$



for $\mathcal{L}_0$, m_0, ϕ_0, λ_0

- This enters in LSZ, where initial and final p is the same p

$$\int dy^4 \int dx^4\, e^{-ipx} e^{+ipy} \langle \Omega | T\{\phi_0(x)\phi_0(y)\} | \Omega \rangle = \frac{i\, Z}{(p^2 - m^2)} \langle p | S | p \rangle \qquad \text{LSZ}$$

This is exactly the Whole 2 points correlator, is not only zero order propagator, is Whole, contains all expansion = so it's $D_{\lambda_0}(x-y)$ in space dimension

$$\int dx^4 \underbrace{\int dy^4\, e^{-ip(x-y)} e^{-ipx}}_{\tilde{D}_{\lambda_0}(p)} D_{\lambda_0}(x-y) = \frac{i\, Z}{(p^2 - m^2)} \langle p | S | p \rangle$$

$$\int dy^4 \quad \tilde{D}_{\lambda_0}(p) = \frac{i\, Z}{(p^2 - m^2)} \langle p | S | p \rangle$$

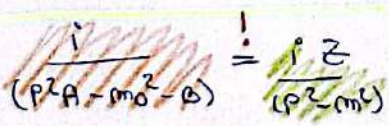
$$\int dy^4\, 1 = \int dy^4\, e^{i0\cdot y} = (2\pi)^4 \delta(0) \;\Big|_{p \to p} \quad (2\pi)^4 \delta(0)\, i\mathcal{M}_{p \to p} \qquad \text{overall momento conservation}$$

$$(2\pi)^4 \delta(0)\, \tilde{D}_{\lambda_0}(p) = \frac{i\, Z}{(p^2 - m^2)} (2\pi)^4 \delta(0)\, i\mathcal{M}_{p \to p}$$

$$\tilde{D}_{\lambda}(p) = \frac{i\, Z}{(p^2 - m^2)}\, i\mathcal{M}_{(p \to p)}$$

- But $i\mathcal{M}_{(p \to p)} \overset{\text{(finite part)}}{=} \Delta$, it's the identity, because $\langle p | S | p \rangle$ do not change particle, is just propagating

$$\tilde{D}_{\lambda}(p) = \frac{i\, Z}{(p^2 - m^2)} + (\text{finite parts})$$



$$\frac{i}{(p^2 A - m_0^2 - B)} \overset{!}{=} \frac{i\, Z}{(p^2 - m^2)}$$

This is our Renormalization condition !

This condition accounts for two conditions, since I would like to have Z to vanish also

① $A p^2 - m_0^2 - B \overset{!}{=} 0 \;\Big|_{p^2 = m^2}$ mass definition at pole

② $\phi_0 = \phi \sqrt{Z}$ To make Z vanish → because if I substitute $\phi\sqrt{Z} = \phi_0$ in my LSZ:

$\int dx \int dy\, e^{\cdots} \langle \Omega | T\{\phi_0 \phi_0\} | \Omega \rangle = \frac{i\, Z}{(\,)} \langle p | S | p \rangle$

$\hookrightarrow \int\int e^{\cdots} \langle \Omega | T\{\phi \phi\} | \Omega \rangle \not{Z} = \frac{i \not{Z}}{(\,)} \langle p | S | p \rangle$

the two Z cancel out! ✓

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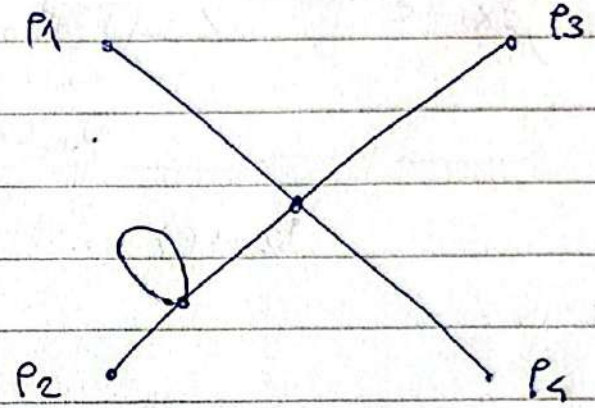
4 POINTS CORRELATOR

At 4 points correlator, at one-loop there are two types of divergent diagrams:

SUBDIAGRAM - TADPOLES



the divergence of tadpole diagrams is automatically cured by the renormalization of the two-point function because is just a subdiagram which is exactly ○—○ the 2 point correlator tadpole... It's NOT a new divergence

→ cured by mass m_R and ϕ_R renormalization

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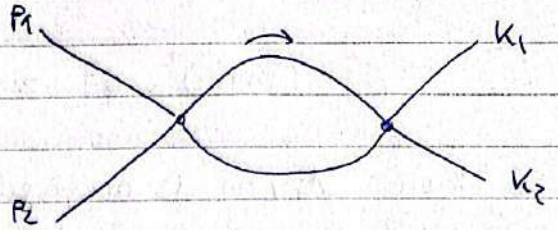
ONE LOOP DIAGRAM



the new divergence is carried by the one loop diagram shown. We already computed the correction on the Matrix element due to the loop:

$$i\mathcal{M}_{(p_1p_2\to k_1k_2)} = -i\lambda_0 + i\lambda_0^2\left(\beta \log(\Lambda) + \{\text{finites}\}\right) =$$

$$= -i\lambda_0\left(1 - \lambda_0\left(\beta \log(\Lambda) + \text{finite}\right)\right)$$

To renormalize it, we have first to be more careful in specifying the kinematical configuration. A simple choice is everything still =

$$(1) \quad p_1 = p_2 = k_1 = k_2 = (m_R, \vec{0})$$

Since the argument of the log must be dimensionless, since $[\Lambda] = [m]$, since only m and Λ are scales (because $\vec{p}$ is set to zero), I know that I can extract from the {finites} well of terms a term $-\beta \log(m_R)$ such that

↑ it was hidden in {finites} term

$$\beta \log(\Lambda) - \beta \log(m) = \beta \log\left(\frac{\Lambda}{m}\right)$$

So that the argument is dimensionless

$$i\mathcal{M}(p_1 = p_2 = k_1 = k_2 = (m, \vec{0})) = -i\lambda_0\left[1 - \lambda_0\left(\beta \log\left(\frac{\Lambda}{m_R}\right) + \text{finite}\right)\right]$$

Renormalization condition simply consists in setting whole $+i\mathcal{M}$ equal to $-i\lambda_R$, which is the real physical observable coupling that can be measured hence is finite

$$\odot (1) \quad -i\lambda_R \overset{!}{=} -i\lambda_0(\Lambda)\left(1 - \lambda_0(\Lambda)\left(\beta \log\left(\frac{\Lambda}{m_R}\right) + \text{finite}\right)\right)$$

λ_0 is fixed by requiring λ_R to be finite and equal to the experimental value one can measure.

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I can also chose a different kinematical regime, as in high energy:

(2) $(p_1+p_2)^2 = q^2 \gg m_R^2$

So I know Λ / m is negligible because in the computation of the loop only $(p_1+p_2)^2 = q^2$ terms dominate and m_R cancel out. So in the log (Λ) I will remain only with Λ^2/q^2 terms (dimensionless) (perché il log deve essere senza dimensione, non avendo più m, ho solo q^2 come scale da usare per compensare Λ)

Recalling the log property $\beta \log(\Lambda) = \frac{\beta}{2} \log(\Lambda^2)$

I know I can write, taking from the {finite} well the term $-\frac{\beta}{2}\log(q^2)$ as:

(2) $-i\lambda_R \overset{!}{=} -i\lambda_0 \left(1 - \lambda_0 \left(\frac{\beta}{2}\log\left(\frac{\Lambda^2}{q^2}\right) + \text{finite}\right)\right)$

$\Rightarrow$ LOOPS ><, TADPOLES — AND DOUBLE LOOPS ⊖ ARE ALL POSSIBLE DIVERGENCIES IN ϕ^4 THEORY in $d=4$, NO OTHER TYPES OF DIVERGENCIES EXIST

$\Rightarrow$ Hence renormalization of $\underbrace{\phi_0\ m_0\ \lambda_0}_{\text{not observable}} \rightarrow \text{into} \rightarrow \underbrace{\lambda_R, m_R, \phi_R}_{\text{real observable}}$ fixes / cures all divergences

(1PI) ONE-PARTICLE-IRREDUCIBLE DIAGRAMS

Why can I affirm that these are the only divergencies in ϕ^4 $d=4$ theory? In principle we could continue to examine 6 points correlators, 8 points correlators and etc etc and we will find (some diagrams which have already known sub diagrams carrying the divergence (—, ><, ⊖)) and some genuinely new divergent diagrams!

BUT ALSO

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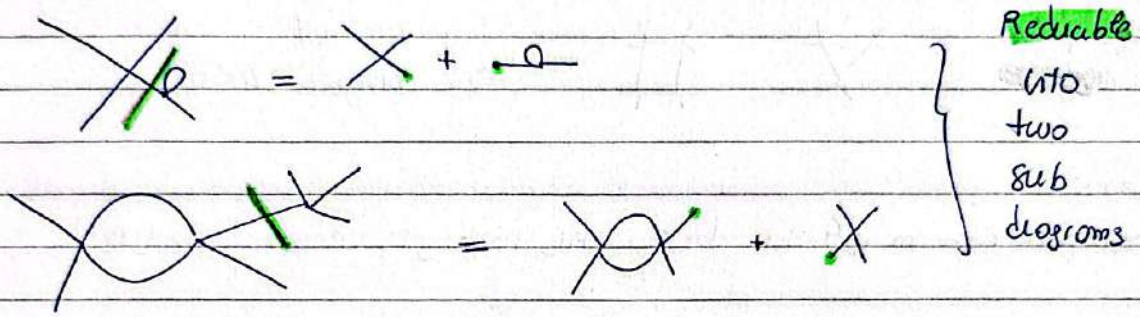
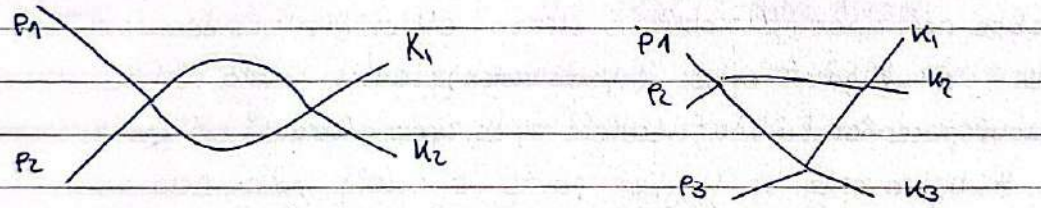
The diagrams which carry already known divergences show sub diagrams. They are called 1-PARTICLE REDUCIBLE diagrams because can be splitted in two independent diagrams by cutting just one propagator. → same procedure as AMPUTATION



DIAGRAMS which CANNOT BE SEPARATED INTO TWO independent diagrams are called 1PI = 1 PARTICLE IRREDUCIBLE And are the only who carry NEW DIVERGENCE



cannot be splitted into two independent diagrams

Note che per 6-point correlator there is a new 1PI diagram .
Does it carries new divergence? NO!

$$D = 4L - 2P = 4 - 6 = -2$$

(1 loop, P propagator)

Fixed point, fixed number of internal and external legs

non è divergente! In generale si può vedere che EACH HIGHER ORDER diagram can be splitted into sub diagrams, and only higher points correlators bring new 1PI diagrams, ma che tutti questi nuovi 1PI, per ϕ^4 teory and $d=4$, NON sono divergenti $\Rightarrow$ PER CUI TUTTE LE POSSIBILI DIVERGENCE in ϕ^4 in $d=4$ sono FISSATE con λ_R ma ϕ_R visto prima

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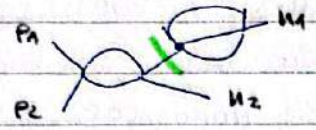
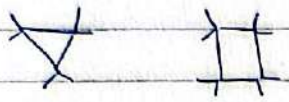
Focus = ϕ^4 theory $d=4$ VS OTHER THEORIES

Higher order diagrams



$\Rightarrow$ can be splitted into already know subdiagrams

Higher points 1PI diagrams



$\Rightarrow$ are not divergent $D<0$

Questo perchè in ϕ^4 in $d=4$ the theory is Renormalizable!

$n \frac{d-2}{2} - d \xrightarrow[d=4]{n=4} 0$ each order has some degree of divergence

Se avessi avuto una teoria diversa ex yukawa $g\phi\bar{\psi}\psi$ o QED $e\bar{\psi}A\!\!\!/\psi$ ecc ecc in d dim diverse non sarebbe detto! magari sarebbe non renormalizzabile e allora all'aumentare di higher points or higher order (n° of vertex) avrei dovuto continuare a definire nuovi parametri da misurare, da renormalizzare -- così all'infinito! At higher order or Higher points I would have new 1PI diagrams which ARE divergent with a new type of divergence! (se la teoria è NON-RINORMALIZZABILE)

renormalizable → At higher order or points → New 1PI but NOT Divget

non renormalizbl → At Higher orde l point → new 1PI AND DIVERGENT!

Super renormalizbl → Hyher orde l point → degree of Divgence Decreases!

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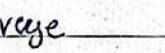
12

DIMENSIONAL REGULARIZATION

In this chapter we present another way to renormalize the theory. Before doing that, it's useful to summarize what was done in chapter 11.

↓

When using a Lagrangian in terms of physical mass coupling and fields
$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi \partial^\mu \phi - m^2\phi^2) - \frac{\lambda}{4!}\phi^4$
we noticed that some diagrams carry a divergent amplitude in the ill motux contribution. Such diagrams are diagrams with loops or holes in general

don't diverge  may diverge

Diagrams with loops bring an integral over at least one unconstrained momentum of the loop that, when taking the UV limit, may diverge

$\int d^4 f(m, \lambda, \ell) \rightarrow$ may diverge in the UV limit

↓

Since we do not WANT Things to diverge because it's not physical, we realize that our theory with such $\mathcal{L}$ is <u>wrong</u>. Then we correct our theory re-defining the Lagrangian in terms of m_0 λ_0 ϕ_0 parameters ("bare" parameters)
$\mathcal{L}_0 = \frac{1}{2}(\partial_\mu \phi_0 \partial^\mu \phi_0 - m_0^2 \phi_0^2) - \frac{\lambda_0}{4!}\phi_0^4$

When computing the ill motux element of diagrams which previously diverged, we SET RENORMALIZATION CONDITION, ie we <u>Defined</u> our REAL m λ and ϕ in order to MAKE the motux element computed from $\mathcal{L}_0$ to be NOT DIVERGENT

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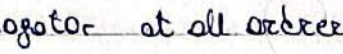
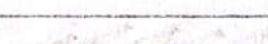
For example computing the propagator at all orders  we found, for $\mathcal{L}_0$, that:

$\frac{i}{Ap^2 - m_0^2 - B(\Lambda)}$ of  in $\mathcal{L}_0$

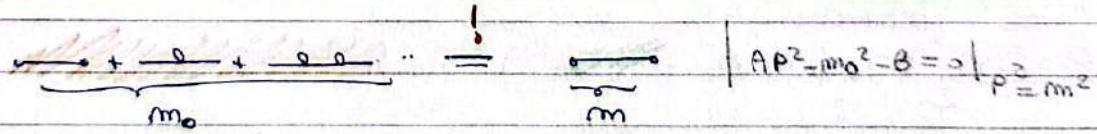
and we imposed that equal to

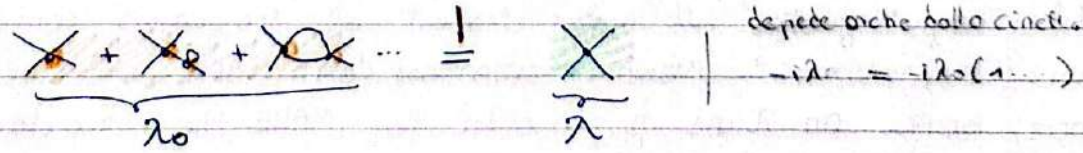
$\frac{i}{Ap^2 - m_0^2 - B(\Lambda)} \overset{!}{=} \frac{i}{p^2 - m^2}$ of in $\mathcal{L}$

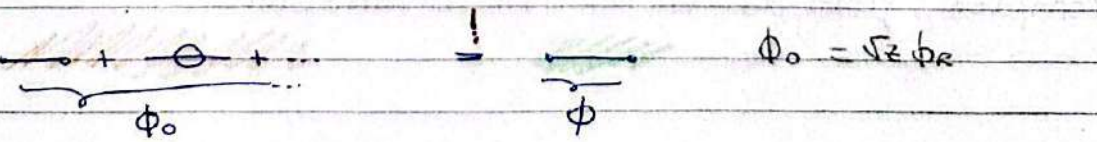
this condition defines our m and re-absorb the divergence! we imposed that our $\mathcal{L}_0$ theory gives COMPLETE $i\mathcal{M}_0$ (complete of all order, hence of loops diagrams also) that are EQUAL to the $i\mathcal{M}$ of JUST NON DIVERGENTE (no loops) diagrams in $\mathcal{L}$

Abbiamo proprio cambiato prospettiva! Prima stavamo considerando una teoria sbagliata! La teoria giusta è quella con $\mathcal{L}_0$ in termini di m_0 λ_0 e ϕ_0 che sono parametri che DIPENDONO dalla DIVERGENZA. Come definisco tali parametri? Con le condizioni di RINORMALIZZAZIONE

condizioni di renormalizzazione = impongo che l'elemento di matrice $i\mathcal{M}(\lambda_0\ \phi_0\ m_0)$ di $\mathcal{L}_0$ (completo di tutti gli ordini e tutti i diagrammi) sia UGUALE all'elemento $r\mathcal{M}(\lambda\ m\ \phi)$ in termini di costanti $m\ \phi\ \lambda$ REALI FINITE MA dei SOLI diagrammi SENZA LOOPS!

m : 

λ : 

ϕ : 

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REGULARIZATION + RENORMALIZATION

We can describe that we can divide our procedure into two steps:

(1) REGULARIZATION

the "Regulator"

We write our divergent amplitude in terms of a finite parameter, the Λ cutoff, until Λ is finite the theory is finite, when $\Lambda \to \infty$ the theory diverges. (the $\mathcal{L}$ theory, the $\mathcal{L}_0$ is corrected)

(2) RENORMALIZATION

imposing the conditions which define (copiate) $\mathcal{L}_0$ ϕ_0 in terms of real on $\lambda \phi$ in order to reabsorb divergence (= making $\mathcal{L}_0$ theory equal to non divergent part of $\mathcal{L}$ theory)

In this chapter we will present a NEW WAY TO REGULARIZE the theory. We had, in chapter 11, used Λ as regulator. However this choice is not optimal, this Λ IS NOT GAUGE INVARIANT (the Λ value depend on the gauge) and can lead to Renormalization condition quite difficult to define (noi abbiamo visto solo la teoria scalare $\phi^4 \lambda$ in cui usando Λ abbiamo ricavato delle condizioni di renormalizzazione, che dipendono nella loro forma da Λ, tutto sommato semplice ... o cmq ci siamo riusciti... per altre teorie non è detto che sia così semplice! Per la QED per esempio!). Another possibility is the DIMENSIONAL REGULARIZATION that uses another regulator ε instead of Λ.

In the following paragraph we will present the dimensional regularization and then the renormalization condition written in terms of this new regulator ε.

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12.1 DIMENSIONAL REGULARIZATION

Consider ϕ^4 theory at $\phi\phi \to \phi\phi$, at 1 loop
$D = d - 2P = d - 4$ (cause $P = 2$ # of propagators and d is the dimension of the space) we have an integral $\int \frac{d^d p_E}{p_E^4}$ that is divergent for $p_E \to \infty$ if $d \geq 4$.
$\Rightarrow$ ϕ^4 theory is divergent for $d \geq 4$

Instead of using a regulator on momentum p (the Λ cutoff) we take $d = 4 - \varepsilon$ and use ε as regulator. When $\varepsilon \to 0$ we approach $d \to 4$ and theory diverges. We intuitively can already expect the divergence to appear as poles $\frac{1}{\varepsilon}$ as $\varepsilon \to 0$

$\Rightarrow$ using a regulator on the dimension of the space is Gauge invariant! We are working on the dimension of the space in a completely general way, we are not referred to a specific Gauge!

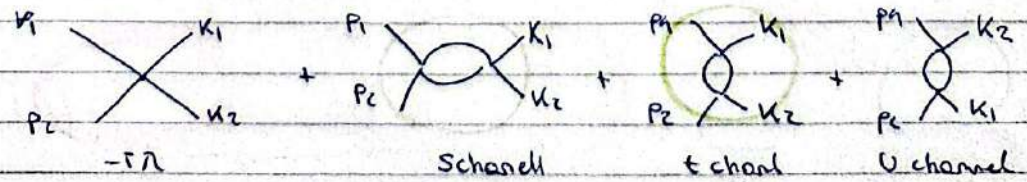
1-Loop in $d=4$ in ϕ^4 theory for $\phi\phi \to \phi\phi$ scattering

Let's compute the matrix element at 1 loop level for $\mathcal{L}$ theory with real mass etc etc (hence the Divergent one) and express the divergence in terms of ε regulator. From eq 11.2 we had:

$$i\mathcal{M}(p_1 p_2 \to k_1 k_2) = -i\lambda + A(p_1+p_2) + A(p_1-k_1) + A(p_1-k_2) =$$

$$= -i\lambda + (-i\lambda)^2 \left[i V(s) + i V(t) + i V(u) \right] \qquad 12.7$$

where $s = (p_1+p_2)^2$ $t = (p_1-k_1)^2$ $u = (p_1-k_2)^2$



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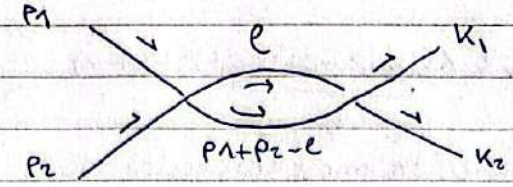
S-CHANNEL



- Consider S-channel

$$A(p_1+p_2) = \frac{(-i\lambda)^2}{2}\int \frac{d^d\ell}{(2\pi)^d}\,\frac{i}{(\ell^2-m^2)}\,\frac{i}{((p_1+p_2-\ell)^2-m^2)} \qquad 12.1$$

(I wrote $A(p_1+p_2)$ with Feynmann prescription on propagators $\frac{i}{p^2-m^2+i\varepsilon}$ sottintese understood — where such $i\varepsilon$ IS NOT the same of our Regulator ε from $d=4-\varepsilon$!)

$$A(p_1+p_2) = (-i\lambda)^2\, i\, V((p_1+p_2)^2) \quad , \quad V((p_1+p_2)^2) = \frac{i}{2}\int \frac{d^d\ell}{(2\pi)^d}\,\frac{1}{\ell^2-m^2}\,\frac{1}{(p_1+p_2-\ell)^2-m^2}$$

[12.2]

To compute the divergent integral $V()$ we use Feynmann parametrization.

FEYNMANN PARAMETRIZATION

$$\frac{1}{A_1\cdot A_2\cdots A_n} = \int_0^1 dx_1\dots dx_n\;\frac{\delta\left(\sum_{i=1}^{n} x_i - 1\right)(n-1)!}{(x_1A_1+\dots+x_nA_n)^n} \qquad 12.3$$

this term simply means that the collection of all x_i must sum to 1 at every step of the integration

For two pieces AB

$$\frac{1}{A\cdot B} = \int_0^1 dx_1\,dx_2\;\frac{\delta(x_1+x_2-1)}{(x_1A+x_2B)^2} = \int_0^1 dx\;\frac{1}{(xA+(1-x)B)^2} \quad *$$

- Then our integrand in $V()$ is:

$$\frac{1}{(\ell^2-m^2)}\cdot\frac{1}{((\ell-p_1-p_2)^2-m^2)} \overset{*}{=} \int_0^1 dx\;\frac{1}{\left[x\,\underbrace{((p_1+p_2-\ell)^2-m^2)}_{B} + (1-x)\,\underbrace{(\ell^2-m^2)}_{A}\right]^2} =$$

$\ell^2-m^2 \Rightarrow A$; $(\ell-p_1-p_2)^2-m^2 \Rightarrow B$

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TIP: $\int_0^1 dx \left(x((p_1+p_2-\ell)^2 - m^2) + (1-x)(\ell^2 - m^2) \right)^{-2}$

As a rule of thumb the most complicated term between A and B should go with the simples x and the simplest close to (1-x) .... then it eventually it's equivalent to swap A with B but in that way computations are easier)

$p = p_1 + p_2$

$$= \int_0^1 dx \left(\ell^2 + xp^2 - 2xp\cdot\ell - m^2 + i\varepsilon \right)^{-2} =$$

↳ restoring feynmann prescription of propagators

$$= \int_0^1 dx \left((\ell - xp)^2 + (1-x)xp^2 - m^2 + i\varepsilon \right)^{-2} =$$

changing variable $\ell' = \ell - xp$ which is just a shift and doesn't change $d\ell' = d\ell$ in the integration measure in the $V(\,) = i\int \frac{d^d\ell}{(2\pi)^d}$ ☐ integral of our term but changes only extremes of $\int d\ell$ .... but I'm integrating from $-\infty$ to $+\infty$ so it doesn't change neither the extremes at the end

$$= \int_0^1 dx \left[\ell'^2 + (1-x)xp^2 - m^2 + i\varepsilon \right)^{-2} =$$ renominating $\ell' = \ell$ because it hasn't changed a thing

$$= \int_0^1 dx \left[\ell^2 - \Delta + i\varepsilon \right]^{-2}$$

with $\Delta = m^2 - x(1-x)p^2$ $\quad p = p_1 + p_2$

- so my $V((p_1+p_2)^2)$ is

$$V((p_1+p_2)^2) = \frac{i}{2} \int \frac{d^d\ell}{(2\pi)^d} \square = \frac{i}{2} \int \frac{d^d\ell}{(2\pi)^d} \int_0^1 dx \left[\ell^2 - \Delta + i\varepsilon \right]^{-2} =$$

- WICK'S ROTATION $\quad d\ell \to i\, d\ell_E \quad \ell^2 \to -\ell_E^2$

$$V(\,) = -\frac{1}{2} \int \frac{d^d\ell_E}{(2\pi)^d} \int_0^1 dx \left[-\ell_E^2 - \Delta + i\varepsilon \right]^{-2} = \frac{-1}{2} \int_0^1 dx \int \frac{d^d\ell_E}{(2\pi)^d} \left(\ell_E^2 + \Delta - \dots \right)^{-2}$$

(12.9)

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- NOW I COMPUTE for generic d dimension

$$\int \frac{d^d \ell_E}{(2\pi)^d} (\ell_E^2 + \Delta)^{-m} = \left[\int \frac{d\Omega}{(2\pi)^d} \int_0^\infty d\ell_E \, \ell_E^{d-1}\right] (\ell_E^2 + \Delta)^{-m} =$$

this part is the unit sphere in d-dim

$$\int d\Omega = \frac{2\pi^{d/2}}{\Gamma(d/2)} \quad \text{hence} \quad \int \frac{d\Omega}{(2\pi)^d} = \frac{2\pi^{d/2}}{(2\pi)^d \Gamma(d/2)}$$

where $\Gamma()$ is Euler gamma function

$$= \frac{2\pi^{d/2}}{(2\pi)^d \Gamma(d/2)} \int_0^\infty d\ell_E \frac{\ell_E^{d-1}}{(\ell_E^2 + \Delta)^m} = \quad \text{change } d\ell_E = \frac{\ell_E^{-1}}{2} d(\ell_E^2)$$

integration variable (same extremes)

$$= \frac{1}{2} \int_0^\infty d(\ell_E^2) \frac{\ell_E^{-1} \ell_E^{d-1}}{(\ell_E^2 + \Delta)^m} = \frac{1}{2} \int_0^\infty d(\ell_E^2) \frac{(\ell_E^2)^{\frac{d}{2}-1}}{(\ell_E^2 + \Delta)^m} =$$

write $y = \frac{\Delta}{(\Delta + \ell_E^2)}$ with $0 \le y \le 1$ for $\ell_E^2 \ge 0$ $\rightarrow \int_0^\infty d(\ell_E^2) \Rightarrow \int_1^0 dy \left(-\frac{\Delta}{y^2}\right)$

and $\ell_E^2 = \Delta\left(\frac{1}{y} - 1\right)$

then

$$= \frac{1}{2} \int_1^0 dy \left(-\frac{\Delta}{y^2}\right) \left[\Delta\left(\frac{1}{y} - 1\right)\right]^{\frac{d}{2}-1} \left(\frac{y}{\Delta}\right)^m =$$

$$= \frac{1}{2} \Delta^{d/2 - m} \int_0^1 dy \, (1-y)^{\frac{d}{2}-1} \, y^{m - \frac{d}{2} - 1} = \text{see behind} \rightarrow$$

This is exactly Euler Beta function!

$$B(a,b) = \int_0^1 dy \, y^{a-1} (1-y)^{b-1} = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

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$$= \bigcirc \frac{1}{2} \Delta^{d/2-m} \frac{\Gamma(n-\frac{d}{2})\Gamma(d/2)}{\Gamma(n)} =$$

$$= \frac{\pi^{d/2}}{(2\pi)^d} \frac{1}{\Gamma(d/2)} \Delta^{d/2-m} \frac{\Gamma(n-d/2)\Gamma(d/2)}{\Gamma(n)} =$$

$$= \frac{\pi^{d/2}}{(2\pi)^d} \Delta^{d/2-n} \frac{\Gamma(n-d/2)}{\Gamma(n)}$$

- so to compute $V()$ using $n=2$

$$V((p_1+p_2)^2) = -\frac{1}{2}\int_0^1 dx \bigcirc = \text{taking out of the integrese the factors which do not depend on } x$$

$$= \frac{-1}{2} \frac{\Gamma(2-d/2)}{(4\pi)^{d/2}\,\Gamma(2)} \int_0^1 dx\, \Delta^{d/2-2}$$

since for positive integer the gamma functions are $\Gamma(n) = (n-1)!$ then
$\Gamma(2) = (2-1)! = 1! = 1$
For $\Gamma(2-\frac{d}{2})$, using $d = 4-\varepsilon$ our regulator ε, we obtain for small $\varepsilon \to 0$, hence approaching divergence, that

$$\Gamma(2-d) = \Gamma(\tfrac{\varepsilon}{2}) \simeq \frac{2}{\varepsilon} - \gamma + O(\varepsilon) \qquad \gamma \text{ Euler-Mascheroni constant}$$

as for $\varepsilon \to 0$ $\Gamma(\varepsilon) \approx \frac{1}{\varepsilon} - \gamma + O(\varepsilon)$ for $\varepsilon \to 0$

hence recalling also $\Delta = m^2 - (x p^2)(1-x)$ we have for $d = 4-\varepsilon$

$$V() = \frac{-1}{32\pi^2} (4\pi)^{\frac{\varepsilon}{2}} \left(\frac{2}{\varepsilon} - \gamma + O(\varepsilon)\right) \int_0^1 dx\, \Delta^{-\frac{\varepsilon}{2}} =$$

- using property $A^{\varepsilon} \underset{\varepsilon \to 0}{=} 1 + \varepsilon \log(A)$ per $\varepsilon \to 0$ si ha

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$\varepsilon \to 0$

$$= -\frac{1}{(32\pi^2)} \left(1+\frac{\varepsilon}{2}\log(4\pi)\right)\left(\frac{2}{\varepsilon}-\gamma+O(\varepsilon)\right)\int_0^1 \left(1-\frac{\varepsilon}{2}\log(\Delta)\right) =$$

- se moltiplico $\left(1+\frac{\varepsilon}{2}\log(4\pi)\right)\cdot\left(\frac{2}{\varepsilon}-\gamma+O(\varepsilon)\right)\left(1-\frac{\varepsilon}{2}\log(\Delta)\right)$ tenendo solo i termini fino a $O(\varepsilon)$, cioè costanti, ho:

$$(\ )(\ )(\ ) = \frac{2}{\varepsilon} - \gamma + \log(4\pi) - \log(\Delta)$$

$$p^2 = s = (p_1+p_2)^2$$

allora

$$V(s) \xrightarrow[\text{cioè } d\to 4]{\varepsilon\to 0} \frac{-1}{32\pi^2}\int_0^1 dx \left(\frac{2}{\varepsilon} - \gamma + \log(4\pi) - \log\left(m^2 - x(1-x)p^2\right) + O(\varepsilon)\right) \quad \text{for s-channel} \qquad (12.9)$$

Hence I have extracted the divergence behaviour expressed by the regulator ε for $\varepsilon \to 0$ // Note that the divergence goes as $\frac{1}{\varepsilon}$ pole and we also have UV finite terms! (the constant ones and $O(\varepsilon)$)

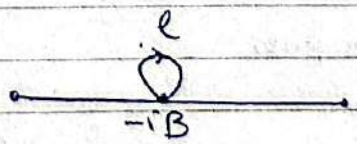
Whole matrix element of the loop was, combing all s t u channel

$$i\mathcal{M}(p_1p_2 \to k_1k_2) = -i\lambda + \overbrace{(-i\lambda)^2\left(iV(s) + iV(t) + iV(u)\right)}^{A(\varepsilon)}$$

each term $V(s)$ $V(u)$ $V(t)$ has a $\frac{1}{\varepsilon}$ pole divergence.

TADPOLES UV DIVERGENCE

For a tadpole the matrix element of the loop was



$$-iB = -i\lambda \int_{(2\pi)^d} d^d\ell \, \frac{i}{\ell^2 - m^2} \xrightarrow{\text{Wick's}} -i^2\lambda \int_{(2\pi)^d} d^d\ell_E \, \frac{1}{(\ell_E^2 + m^2)}$$

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following same result is easy to prove :

$$= \frac{-i\lambda}{2}\frac{1}{(4\pi)^{d/2}}(m^2)^{d/2-1}\Gamma\left(1-\frac{d}{2}\right) =$$

then for $d = 4-\varepsilon$ and $\varepsilon \to 0$ it becomes

$$-i\mathcal{B} \underset{\varepsilon\to 0}{=} \frac{i\lambda}{32\pi^2}\frac{m^2}{\varepsilon} + O(\varepsilon) \quad \Longrightarrow \mathcal{B}(\varepsilon)$$

Still UV divgence as $\frac{1}{\varepsilon}$ pole plus finite terms ($O(\varepsilon)$)

$\Rightarrow$ So I have regulard $\mathcal{B}(\varepsilon)$ and $A(\varepsilon)$ as function of ε this time and not of Λ as before

12.2 RENORMALIZATION

As discussed, the correct Lagrangian to use in computation to get finite amplitude is the bare $\mathcal{L}_0$:

$$\mathcal{L}_0 = \frac{1}{2}(\partial_\mu\phi_0\partial^\mu\phi_0 - m_0^2\phi_0^2) - \frac{\lambda_0}{4!}\phi_0^4$$

<u>With RENORMALIZATION CONDITIONS =</u>

(1+2) $$\frac{i}{A p^2 - m_0^2 - B} \overset{!}{=} \frac{iZ}{p^2 - m^2} + \{\text{regular part at } p^2 \equiv m^2\}$$

Which gives

- pole prescription = m^2 is defined as the position of the pole $\quad Ap^2 - m_0^2 - B \overset{!}{=} 0 \Big|_{p^2 = m^2}$
- Residue should be without Z Hence $\quad \phi_0 \overset{!}{=} \phi\sqrt{Z}$

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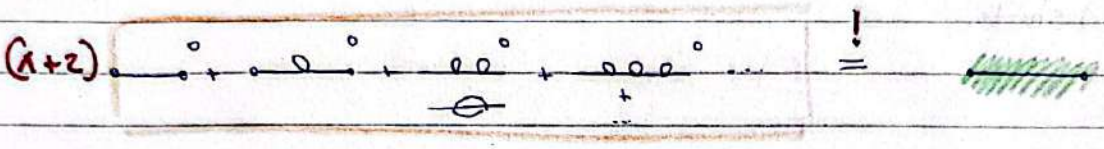
(3) $-i\lambda_0 + A(s) + A(t) + A(u) + \cdots \overset{!}{=} -i\lambda$

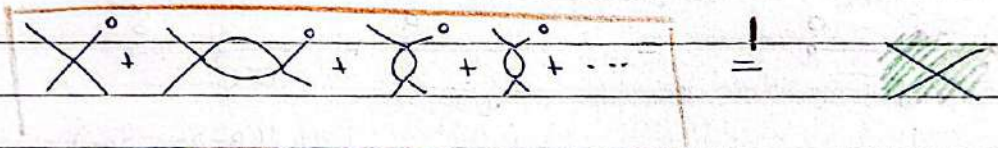
λ is equal to the magnitude of 1 loop diagram = (this condition depends on the kinetic of particle-like process)

Diagramatically

(1+2) 

whole
in $m_0 \lambda_0 \phi_0$

propagator
in $m \phi \lambda$

(3) 

whole
in $\lambda_0 \phi_0 m_0$

zero order in
ϕ in λ

When using ξ as regulator is more convenient to express $\mathcal{L}_0$ in terms of $\mathcal{L}$ and a $\mathcal{L}$ of shift. First we explicit $\phi_0 = \phi\sqrt{Z}$

$$\mathcal{L}_0 = \frac{1}{2}(\partial_\mu \phi_0 \partial^\mu \phi_0 - m_0^2 \phi_0^2) - \frac{\lambda_0}{4!}\phi_0^4 = \frac{1}{2} Z(\partial_\mu \phi)^2 - \frac{m_0^2}{2} Z\phi^2 - \frac{\lambda_0}{4!} Z^2 \phi^4$$

Then we define the SHIFTS:

$$\delta Z := Z - 1 \qquad \delta m = m_0^2 Z - m^2 \qquad \delta\lambda = \lambda_0 Z^2 - \lambda$$

where m λ are physically real mass and coupling, hence the divergence is inside $\delta_Z(\xi)$ $\delta_m(\xi)$ $\delta_\lambda(\xi)$ the shifts (since $m_0(\xi)$, $\lambda_0(\xi)$ depends on the regulator). The $\mathcal{L}_0$ then rewrites as:

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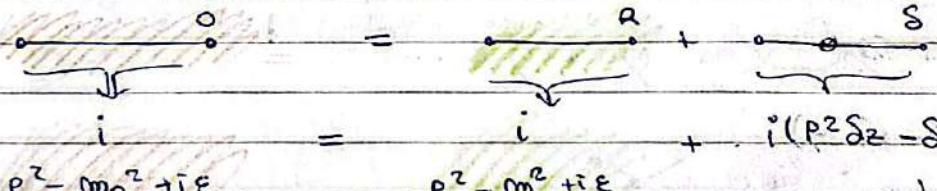
$$\mathcal{L}_0 = \frac{1}{2}(\partial_\mu \phi_r)^2 - \frac{1}{2} m^2 \phi_r^2 - \frac{\lambda}{4!}\phi_r^4 + \frac{1}{2}\delta_Z(\partial_\mu \phi_r)^2 - \frac{1}{2}\delta_m \phi_r^2 - \frac{\delta_\lambda}{4!}\phi_r^4$$

$$\mathcal{L}_0 = \mathcal{L}_R + \mathcal{L}_S$$

We splitted the bare Lagrangian into a real Lagrangian $\mathcal{L}_R$ and a "stuff" Lagrangian $\mathcal{L}_S$. Since $\mathcal{L}_0$ is NOT divgent while $\mathcal{L}_R$ is, then $\mathcal{L}_S$ compensates for the divgence of $\mathcal{L}_R$, hence they are infinite (but unobservable) shifts.

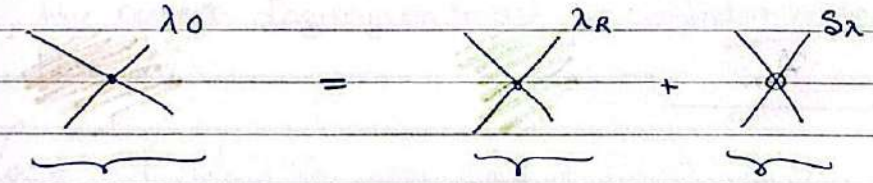
Each piece of Lagrangian gives their Feynmann rules as follows:

PROPAGATOR:



$$\frac{i}{p^2 - m_0^2 + i\varepsilon} = \frac{i}{p^2 - m^2 + i\varepsilon} + i(p^2\delta_Z - \delta_m)$$

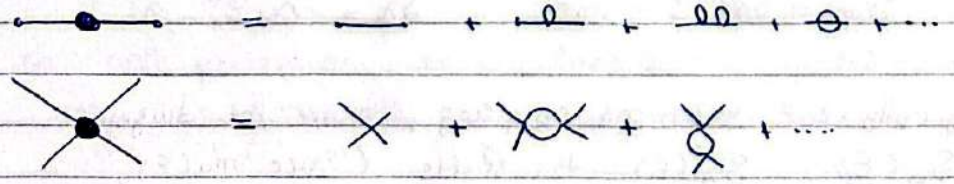
VERTEX:



$$-i\lambda_0 = (-i\lambda_R) + (-i\delta_\lambda)$$

EXT LEGS: always 1 for both ϕ_0 and ϕ_R of $\mathcal{L}_R$ and $\mathcal{L}_S$

NOTATION:



symbols for whole expansion of propagator and 4 point correlator

we are treating $\frac{1}{2}\delta_Z(\partial_\mu\phi)^2 - \frac{1}{2}\delta_m\phi^2$ as an interaction, so Feynmann rules are the ones I use for vertex! (In fact δ_m and δ_Z are proportional to λ or λ^2 as Flavio said)

$\langle p | -\frac{1}{2}\delta_m \phi^2 | p \rangle = -i\delta_m$

$\langle p | \frac{1}{2}\delta_Z \partial_\mu\phi\partial_\mu\phi | p \rangle \Rightarrow i\delta_Z p^2$

with multiplicity factor of ...

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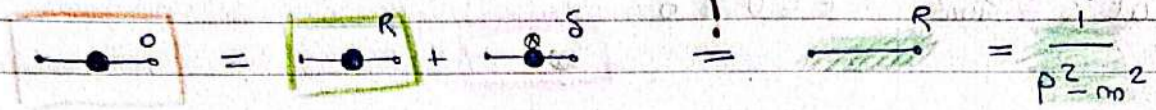
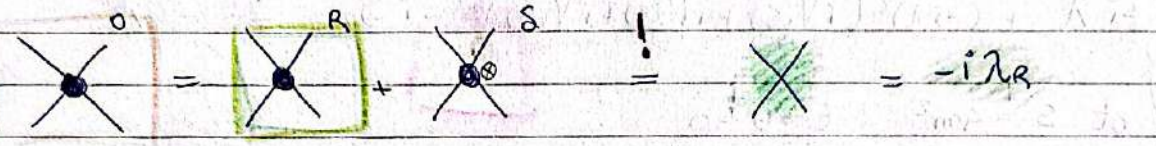
RENORMALIZATION CONDITIONS

(1+2) 

Whole $\mathcal{L}_0$ prop. | Whole $\mathcal{L}_R$ prop | Whole $\mathcal{L}_\delta$ prop | Single zero order propagator $\mathcal{L}_R$

(3) 

Whole 4 points correlator $\mathcal{L}_0$ | Whole // $\mathcal{L}_R$ | Whole // $\mathcal{L}_\delta$ | Single vertex $\mathcal{L}_R$

at $s \overset{!}{=} 4m^2$ $t = u \overset{!}{=} 0$ (Hence for kinetic $p_1 = p_2 = k_1 = k_2 = (m, \vec{0})$)

Now we use these renorm conditions to find explicit expressions of shifts δ_λ, δ_2 and δ_m.

$\delta\lambda$ SHIFT = (3) RENORM. condition ON 4-Point correlator

NOTA = i propagatori / diagrammi in δ come sono di un ordine in λ successivo ! $O(\lambda)$ è un loop per $\mathcal{L}_R$ mentre $O(\lambda)$ è per δ !! Lo δ_λ è un $\sim\lambda^2$!! Sarà chiaro subito come mai nei seguenti calcoli:

Let's use (3) Renorm condition but stopping at $O(\lambda^2)$ hence at 1-loop order (because it contains all the divergence already). So we compute:

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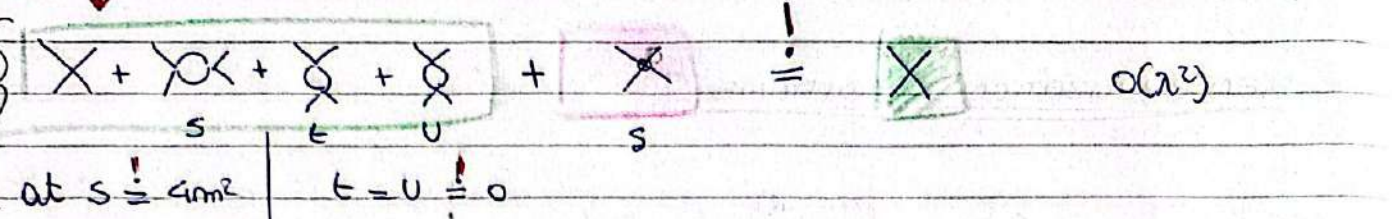
(3)



write this part with ε regulator

$$-i\lambda + (-i\lambda)^2\left[iV(s)+iV(t)+iV(u)\right] - i\delta\lambda \overset{!}{=} -i\lambda$$

at $s \overset{!}{=} 4m^2 \quad t = u \overset{!}{=} 0$

$$(-i\lambda)^2\left[iV(4m^2)+2iV(0)\right] \overset{!}{=} i\delta\lambda$$

(Già qui vedo che $\delta\lambda$ è $\sim\lambda^2$, ecco perché non ho meno 1 loop diagrams per δ)

which gives: (substitute in $V(p^2)$ p^2 with s u and t in equation 12.4)

$$\delta\lambda = \frac{+\lambda^2}{32\pi^2}\int_0^1 dx\left[\underbrace{\frac{6}{\varepsilon} - 3\gamma_\varepsilon + 3\log(4\pi)}_{\text{Divergent PART e generico } \forall s,t,u} - \underbrace{\log\left(m^2 - x(1-x)4m^2\right) - 2\log(m^2)}_{\text{NON DIVergent, tipico di } s=4m,\ t=u=0}\right] \quad (12.5)$$

I found any expression for $\delta\lambda$!! And it's of order of $O(\lambda^2)$

Now If I want to compute the matrix element for $p_1 p_2 \to k_1 k_2$ scattering process at $O(\lambda^2)$ for generic s and u and t I have

$$i\mathcal{M}_0(p_1p_2\to k_1k_2) = -i\lambda + (-i\lambda)^2\left[iV(s)+iV(t)+iV(u)\right] - i\delta\lambda + O(\lambda^2) =$$

(1 calculated for $s=4m^2$, $t=u=0$)

$$= -i\lambda \underbrace{-\frac{i\lambda^2}{32\pi^2}\int_0^1 dx\left[\log\left(\frac{m^2-x(1-x)s}{m^2-x(1-x)4m^2}\right)+\log\left(\frac{m^2-x(1-x)t}{m^2}\right)+\log\left(\frac{m^2-x(1-x)u}{m^2}\right)\right]}_{\text{using 12.5}}$$

which is NOT DIVergent because $\delta\lambda$ canceled all divergent part in $\sim\frac{1}{\varepsilon}$!!

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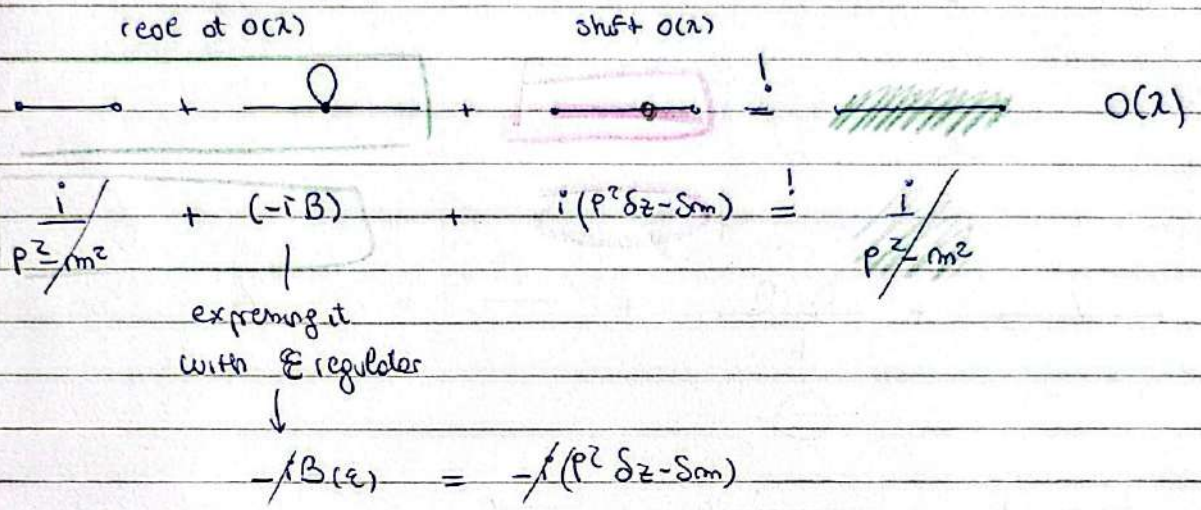
δm and δz = (1) (2) Renorm condition of 2 point correlator

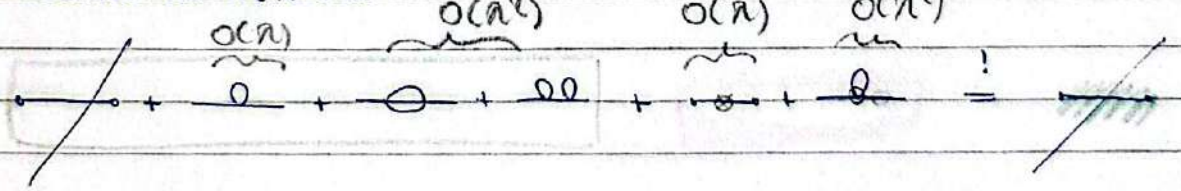
From 2-points correlator condition I can fix δm and δz. First consider the 2 point propagator at $O(\lambda)$ hence only 1 loop diagram contribute



if $\delta z = 0$ (we set it to be zero ... and this is always true for 1 loop order diagram) then

$$\delta m = -B = -\frac{\lambda}{2(4\pi)^{d/2}} (m^2)^{\frac{d}{2}-1} \Gamma\left(1-\frac{d}{2}\right) \xrightarrow[d\to 4]{\varepsilon\to 0} \frac{\lambda}{32\pi^2}\frac{m^2}{\varepsilon} + O(\varepsilon)$$

However is not always true that $\delta z = 0$, to find expression for δz and δm in case $\delta z \neq 0$ we should expand the propagator at $O(\lambda^2)$. Renorm condition at $O(\lambda^2)$ is



→ This ○ diagram contains $\delta\lambda$ counterterm that we already computed and compensate for UV divergences coming for ⊖ + ○○ when just only one of the two loop momenta is large

→ ⊖ term contain δz and δm and is fixed at $O(\lambda^2)$ requiring that pole of propagator is $p^2 = m^2$ and that the removing divergences (when both loop momenta in ⊖ and ○○ are large) cancel

~~DI solit di $O(\lambda)$ $\delta z = 0$ per ϕ^4 theory / però $\delta z = O(\lambda^2)$ di solito~~

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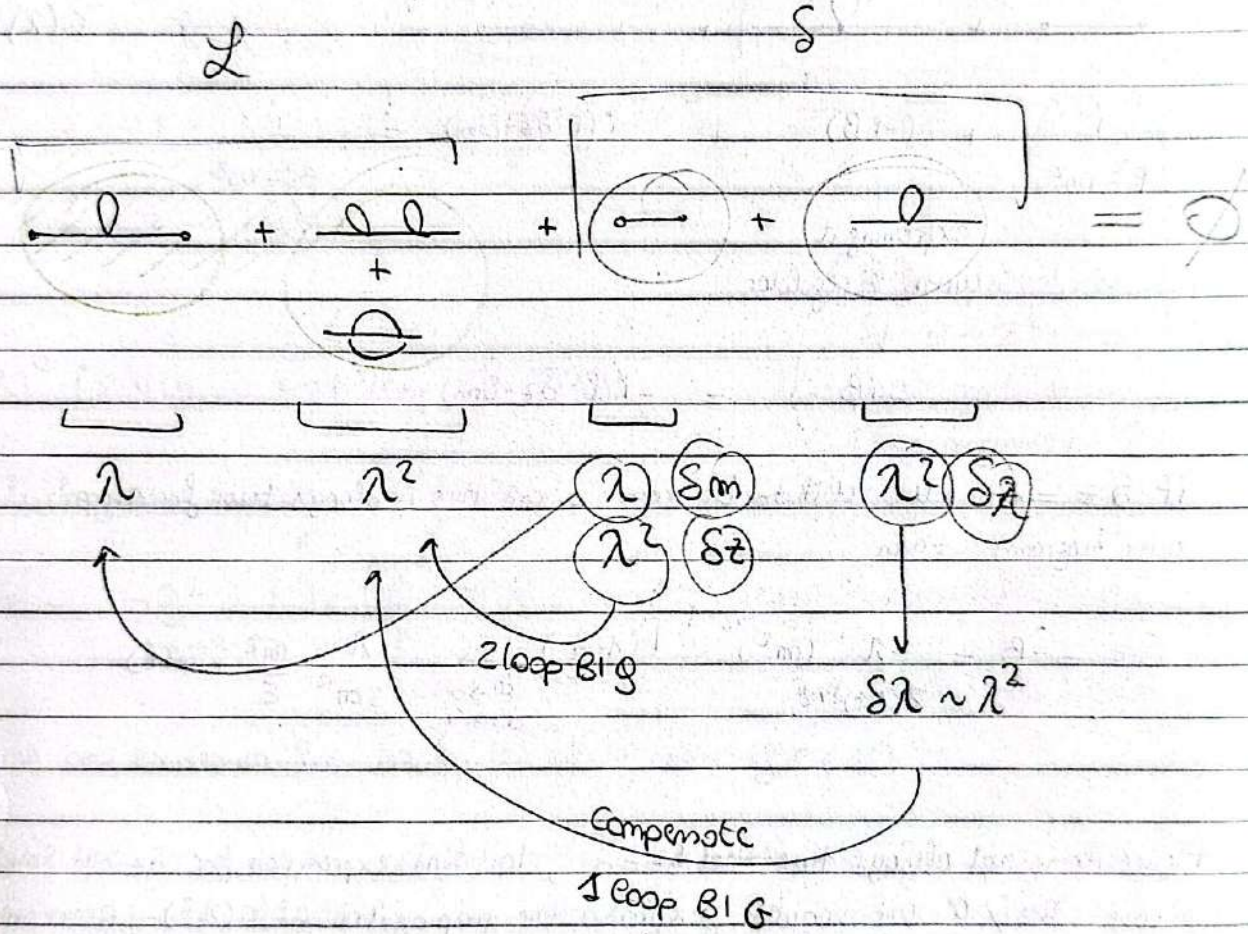
$\delta\lambda \sim \lambda^2$

$\delta m \sim \lambda$

$\delta z \sim \lambda^2$



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